

Theory of Mathematics

Volume VII: Differential, Riemannian and Hyperbolic Geometry

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Part I

Smooth Manifolds

Chapter 1

Topological Manifolds

Volume VII begins by separating the topological notion of a manifold from the smooth structures added in VII/02. The basic definition has three independent ingredients:

locally Euclidean + Hausdorff + second countable.

The mapped legacy manuscripts emphasize that each condition rules out a different pathology. A cone apex can fail local Euclidean structure, a plane with two origins can be locally Euclidean but non-Hausdorff, and an uncountable disjoint union of ordinary manifolds can be Hausdorff and locally Euclidean but fail second countability.

Standing conventions

All spaces are topological spaces. By a topological n -manifold we mean a Hausdorff, second-countable space locally homeomorphic to open subsets of $\mathbb{R}^n$. Smooth compatibility of charts is deliberately postponed to VII/02.

Learning goals

By the end of the chapter, the reader should be able to:

- state and test the three manifold axioms;
- work with local Euclidean neighborhoods and topological charts;
- distinguish local failures from separation and countability failures;
- prove that open subsets, products, and disjoint unions under suitable countability hypotheses are manifolds;
- recognize spheres, tori, projective spaces, and quotient surfaces as manifolds;
- analyze quotient constructions such as the torus and Klein bottle;
- use local invariants such as punctured-neighborhood connectivity to detect nonmanifold points;
- explain why second countability implies useful global regularity such as a countable atlas;
- prepare for smooth atlases without using differentiability prematurely.

1.1 Locally Euclidean spaces

Definition 1.1 (Locally Euclidean space). A space X is *locally Euclidean of dimension n* if every point $p \in X$ has an open neighborhood U homeomorphic to an open subset $V \subseteq \mathbb{R}^n$.

Definition 1.2 (Topological chart). A *topological chart* is a pair (U, φ) , where $U \subseteq X$ is open and

$$\varphi : U \xrightarrow{\sim} V \subseteq \mathbb{R}^n$$

is a homeomorphism onto an open set. The functions supplied by φ are local coordinates. No differentiability is required in this chapter.

Proposition 1.3 (Dimension is locally constant). *If X is locally Euclidean and connected, all charts have the same Euclidean dimension. Hence one may speak of the dimension of a connected topological manifold.*

Proof. This is the invariance-of-domain/invariance-of-dimension principle: nonempty open subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$ cannot be homeomorphic when $m \neq n$. Overlapping chart domains therefore have the same dimension, and connectedness propagates the equality. $\square$

1.2 Hausdorff and second-countable conditions

Definition 1.4 (Hausdorff). A space X is *Hausdorff* if distinct points admit disjoint open neighborhoods.

Definition 1.5 (Second countable). A space X is *second countable* if its topology has a countable basis.

Definition 1.6 (Topological manifold). A *topological n -manifold* is a Hausdorff, second-countable space locally Euclidean of dimension n . A topological 2-manifold is also called a *topological surface*.

Proposition 1.7 (Countable atlas). *Every second-countable locally Euclidean space admits a countable atlas.*

Proof. Choose a countable basis $\mathcal{B}$. For every point choose a chart and then a basis element contained in its chart domain. Only countably many basis elements are needed, and restricting the corresponding charts yields a countable cover by chart domains. $\square$

Remark 1.8 (Why the hypotheses are separate). Local Euclidean structure is a local condition. Hausdorffness controls global separation. Second countability rules out spaces that are locally ordinary but globally too large. None of these roles should be hidden inside the others.

1.3 Three diagnostic counterexamples

Example 1.9 (The double cone). Let

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}.$$

Away from the apex, C is locally a surface. At the apex, a sufficiently small punctured neighborhood has two connected components, whereas a punctured open disc in $\mathbb{R}^2$ is connected. Hence the apex has no surface chart.

Example 1.10 (The plane with two origins). Take two copies of $\mathbb{R}^2$ and identify corresponding nonzero points, leaving the two origins distinct. Every point has a neighborhood homeomorphic to $\mathbb{R}^2$, but the two origins cannot be separated by disjoint neighborhoods. Thus local Euclidean structure does not imply Hausdorffness.

Example 1.11 (Too many components). If Σ is a surface and $\mathbb{R}_\delta$ is the real line with the discrete topology, then

$$\Sigma \times \mathbb{R}_\delta$$

is an uncountable disjoint union of copies of Σ . It is Hausdorff and locally Euclidean of dimension two, but not second countable.

1.4 Basic constructions and examples

Proposition 1.12 (Open subsets). *Every open subset of a topological n -manifold is a topological n -manifold.*

Proof. Hausdorffness and second countability pass to open subspaces, and restricting charts gives the local Euclidean property. $\square$

Proposition 1.13 (Products). *If M is an m -manifold and N an n -manifold, then $M \times N$ is an $(m + n)$ -manifold.*

Proof. Products of Hausdorff spaces are Hausdorff, products of countable bases are countable bases, and products of charts give homeomorphisms to open subsets of $\mathbb{R}^{m+n}$. $\square$

Example 1.14 (Spheres). The sphere $S^n \subset \mathbb{R}^{n+1}$ is an n -manifold. One may use stereographic projection from the north and south poles as two charts.

Example 1.15 (The torus). The torus can be represented as $S^1 \times S^1$ or as the quotient

$$T^2 = \mathbb{R}^2 / \mathbb{Z}^2.$$

The product description immediately shows that it is a topological surface.

1.5 Quotients and identifications

Definition 1.16 (Quotient topology). If Y is a topological space and $\sim$ an equivalence relation, the quotient $Y/\sim$ has the topology in which $U \subseteq Y/\sim$ is open exactly when its inverse image under the quotient map is open in Y .

Proposition 1.17 (Free properly discontinuous quotients). *If a discrete group G acts freely and properly discontinuously by homeomorphisms on an n -manifold M , and the quotient is second countable, then M/G is an n -manifold and $M \rightarrow M/G$ is a local homeomorphism.*

Proof. Proper discontinuity gives, near each p , an open set whose distinct translates do not meet. The quotient map is therefore a homeomorphism from that neighborhood onto its image. The remaining manifold axioms descend under the stated hypotheses. $\square$

Example 1.18 (Torus and Klein bottle). Identifying opposite sides of a square with parallel orientations produces T^2 ; reversing one pair produces the Klein bottle K . Both are topological surfaces. Their quotient-square descriptions are the first important examples where global topology is encoded by edge identifications.

1.6 Covering maps

Definition 1.19 (Covering map). A map $p : E \rightarrow B$ is a *covering map* if every $b \in B$ has an open neighborhood U whose inverse image is a disjoint union of opens, each mapped homeomorphically onto U .

Example 1.20 (A double cover of the torus). Writing $T^2 = S^1 \times S^1$,

$$p(z, w) = (z^2, w)$$

is a two-sheeted covering $T^2 \rightarrow T^2$.

Example 1.21 (Torus covering the Klein bottle). The translation lattice of the torus can be realized as an index-two subgroup of the standard deck group defining the Klein bottle. The induced quotient map $T^2 \rightarrow K$ is therefore a two-sheeted covering.

1.7 Exercises

Exercise 1.22. Verify that $\mathbb{R}^n$ is an n -manifold.

Hint. Use the identity chart and the standard countable basis. □

Solution. Euclidean space is Hausdorff, has the countable basis of rational balls, and the identity gives a global chart. □

Exercise 1.23. Show that every open subset of $\mathbb{R}^n$ is an n -manifold.

Hint. Restrict the Euclidean basis. □

Solution. Hausdorffness and second countability are inherited, and the inclusion coordinates are local Euclidean charts. □

Exercise 1.24. Show that S^1 is a 1-manifold.

Hint. Use two stereographic or angle charts. □

Solution. Deleting one point from the circle gives an open arc homeomorphic to $\mathbb{R}$; two such opens cover S^1 . □

Exercise 1.25. Show that S^n is an n -manifold.

Hint. Use stereographic projection. □

Solution. North- and south-pole stereographic projections give two charts onto $\mathbb{R}^n$; the sphere is Hausdorff and second countable as a subspace of Euclidean space. □

Exercise 1.26. Prove that the product of an m -manifold and an n -manifold is an $(m + n)$ -manifold.

Hint. Take products of charts and bases. □

Solution. Chart products map to open subsets of $\mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$; the separation and countability axioms are preserved. □

Exercise 1.27. Explain why the double cone fails to be a surface at its apex.

Hint. Remove the apex from a small neighborhood. □

Solution. The punctured cone neighborhood has two components, whereas a sufficiently small punctured planar disc is connected, so no local homeomorphism exists. $\square$

Exercise 1.28. Prove that the plane with two origins is locally Euclidean.

Hint. Use either original plane copy near each origin. $\square$

Solution. Each origin has a neighborhood represented by one copy of a Euclidean disc, while every nonzero point has the usual Euclidean neighborhood. $\square$

Exercise 1.29. Prove that the plane with two origins is not Hausdorff.

Hint. Try to separate the two origins. $\square$

Solution. Any neighborhoods of the two origins contain common nonzero points after quotienting, so they cannot be disjoint. $\square$

Exercise 1.30. Show that an uncountable discrete space is not second countable.

Hint. Each singleton is open. $\square$

Solution. A basis must contain, for each point, a basis element contained in its singleton; hence it must contain all singletons and is uncountable. $\square$

Exercise 1.31. Show that $\Sigma \text{imes } \mathbb{R}_\delta$ is not second countable.

Hint. Each slice is open and the slices are disjoint. $\square$

Solution. A countable basis cannot meet an uncountable family of pairwise disjoint nonempty open components with distinct basis elements. $\square$

Exercise 1.32. Show that a second-countable space is separable.

Hint. Choose one point from each nonempty basis element. $\square$

Solution. The chosen countable set meets every nonempty open set, hence is dense. $\square$

Exercise 1.33. Construct a countable atlas from a countable basis.

Hint. Refine chart neighborhoods by basis elements. $\square$

Solution. For each needed basis element choose one chart containing it and restrict; only countably many domains occur. $\square$

Exercise 1.34. Show that a countable disjoint union of n -manifolds is an n -manifold.

Hint. Combine countable bases and atlases. $\square$

Solution. The union is Hausdorff, a countable union of countable bases is countable, and each point retains its original chart. $\square$

Exercise 1.35. Explain why an uncountable disjoint union of nonempty n -manifolds is not a manifold under the adopted definition.

Hint. Use pairwise disjoint open components. $\square$

Solution. Second countability fails because each open component needs a distinct basis element. $\square$

Exercise 1.36. Verify that $T^2 = S^1 \times S^1$ is a surface.

Hint. Apply the product theorem. □

Solution. The product of two 1-manifolds is a 2-manifold. □

Exercise 1.37. Show that $\mathbb{RP}^n$ is locally Euclidean.

Hint. Use the standard projective coordinate opens. □

Solution. On the open where $x_i \neq 0$, normalize $x_i = 1$; the remaining n coordinates give a homeomorphism to $\mathbb{R}^n$. □

Exercise 1.38. Check that the quotient map $\mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$ is a local homeomorphism.

Hint. Use a disc smaller than half the lattice spacing. □

Solution. Such a disc has disjoint integer translates, so quotienting is injective and open on it. □

Exercise 1.39. Verify that $p(z, w) = (z^2, w)$ has two points in every fiber.

Hint. Solve $z^2 = u$ on S^1 . □

Solution. Each $u \in S^1$ has exactly two square roots $z, -z$, while w is fixed. □

Exercise 1.40. Prove that a covering map is a local homeomorphism.

Hint. Read the definition on an evenly covered neighborhood. □

Solution. Each sheet over an evenly covered neighborhood is mapped homeomorphically onto the base open. □

Exercise 1.41. Show that a local homeomorphism preserves local manifold dimension.

Hint. Compose a chart with the local inverse. □

Solution. A chart on the source transports across the local homeomorphism to a chart of the same Euclidean dimension on the target. □

Exercise 1.42. Give a topological reason the cone with its apex removed is a surface.

Hint. Use the two graph descriptions $z = \pm\sqrt{x^2 + y^2}$. □

Solution. Every remaining point belongs to one smooth graph sheet and has a neighborhood homeomorphic to an open planar disc. □

Exercise 1.43. Explain why Hausdorffness is not a local-Euclidean consequence.

Hint. Use the two-origin example. □

Solution. The plane with two origins is locally Euclidean everywhere but fails to separate its two origins. □

Exercise 1.44. Explain why second countability is not a local-Euclidean consequence.

Hint. Use the uncountable disjoint union example. □

Solution. Each component is locally Euclidean, but the uncountable family of open components forces every basis to be uncountable. □

Exercise 1.45. State exactly what new condition VII/02 will add to the present charts.

Hint. Do not change the topological manifold definition. $\square$

Solution. VII/02 imposes smooth compatibility on chart transition maps and organizes charts into smooth atlases; no such differentiability condition is part of VII/01. $\square$

1.8 Solved problem dossiers

Problem 1.46 (Legacy Problem 1(a): the cone). Prove that

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$$

is not a topological surface.

Solution. All non-apex points lie on one of the two graph sheets and are locally Euclidean. Let o be the apex. A sufficiently small punctured neighborhood of o in C has exactly two connected components, the upper and lower cones. A sufficiently small punctured disc in $\mathbb{R}^2$ is connected. A homeomorphism would preserve connected components, so no neighborhood of o can be homeomorphic to an open subset of $\mathbb{R}^2$. $\square$

Problem 1.47 (Legacy Problem 1(b): plane with two origins). Form a quotient of two copies of $\mathbb{R}^2$ by identifying corresponding nonzero points and leaving the origins distinct. Show that every point is locally Euclidean but the space is not a surface.

Solution. Each origin has a neighborhood coming from its own copy of a Euclidean disc, and nonzero points have their ordinary disc neighborhoods, so the quotient is locally Euclidean. However any neighborhood of the first origin and any neighborhood of the second contain identified nonzero points. Hence the two origins cannot be separated by disjoint opens. The quotient is non-Hausdorff and therefore is not a topological manifold under the standing definition. $\square$

Problem 1.48 (Legacy Problem 1(c): uncountably many surface components). Let Σ be a surface and $\mathbb{R}_\delta$ the real line with the discrete topology. Show that $\Sigma \times \mathbb{R}_\delta$ is locally a surface but is not a topological surface.

Solution. Each point lies in $U \times \{t\}$, where U is a surface chart and $\{t\}$ is open, so the product is locally Euclidean of dimension two. It is an uncountable disjoint union of nonempty open copies of Σ . In a second-countable space, every family of pairwise disjoint nonempty opens is countable: assign to each one a distinct basis element contained in it. Thus second countability fails. $\square$

Problem 1.49 (Legacy Problem 2(a): a double cover of the torus). Construct a two-sheeted covering $p : T^2 \rightarrow T^2$.

Solution. Use $T^2 = S^1 \times S^1$ and define

$$p(z, w) = (z^2, w).$$

Every (u, v) has exactly the two preimages (z, v) and $(-z, v)$. On a sufficiently short arc $A \subset S^1$, the squaring map is a homeomorphism onto its image; hence on $A \times B$ the product map is a homeomorphism. Therefore p is a two-sheeted covering. $\square$

Problem 1.50 (Legacy Problem 2(b): torus to Klein bottle). Construct a two-sheeted covering $T^2 \rightarrow K$ of the Klein bottle.

Solution. Write the torus as $\mathbb{R}^2/\Lambda$ for the translation lattice and the Klein bottle as $\mathbb{R}^2/\Gamma$, where Γ is generated by a horizontal translation and a glide/reflection generator. The translation subgroup defining the torus has index two in Γ . The induced quotient

$$\mathbb{R}^2/\Lambda \longrightarrow \mathbb{R}^2/\Gamma$$

is therefore a two-sheeted covering and local homeomorphism. $\square$

Problem 1.51 (Legacy Problem 2(c): nonconjugate involutions of T^2). Give three pairwise nonconjugate subgroups $\mathbb{Z}/2 \leq \text{Homeo}(T^2)$.

Solution. In coordinates $[x, y] \in \mathbb{R}^2/\mathbb{Z}^2$, take

$$\tau[x, y] = [x + \tfrac{1}{2}, y], \quad \rho[x, y] = [-x, y], \quad \sigma[x, y] = [-x, -y].$$

Their fixed sets are respectively empty, two circles, and four isolated points. Conjugate homeomorphisms have homeomorphic fixed-point sets, so the three generated order-two subgroups are pairwise nonconjugate. $\square$

Problem 1.52 (Local dimension test). Why can a connected space not be simultaneously a topological m -manifold and a topological n -manifold with $m \neq n$?

Solution. Choose overlapping charts of the two alleged dimensions. Restrict to a nonempty overlap; composing one chart with the inverse of the other would produce a homeomorphism between nonempty open subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$. Invariance of dimension forbids this when $m \neq n$. Connectedness propagates one dimension throughout the space. $\square$

Problem 1.53 (Projective space charts). Prove that $\mathbb{RP}^n$ is an n -manifold.

Solution. Let U_i be the set of projective points with $x_i \neq 0$. Normalize $x_i = 1$; the remaining coordinates give a homeomorphism $U_i \cong \mathbb{R}^n$. The U_i cover $\mathbb{RP}^n$. As a quotient of the compact Hausdorff sphere by the finite antipodal action, $\mathbb{RP}^n$ is Hausdorff and second countable, completing the manifold axioms. $\square$

Problem 1.54 (A quotient manifold criterion). Let a discrete group act freely and properly discontinuously on an n -manifold. Explain why the quotient is locally Euclidean.

Solution. For each p , proper discontinuity provides an open neighborhood U whose nontrivial translates are disjoint from U . The quotient map is injective on U , and because quotient maps are open for group actions, U maps homeomorphically to its image. Composing with a chart on U gives a quotient chart of dimension n . $\square$

Problem 1.55 (Countability versus components). Prove that a second-countable space has at most countably many pairwise disjoint nonempty open subsets.

Solution. Let $\mathcal{B}$ be a countable basis. For each nonempty open U_α , choose $x_\alpha \in U_\alpha$ and a basis element B_α with $x_\alpha \in B_\alpha \subseteq U_\alpha$. Pairwise disjointness forces the B_α to be distinct. Thus the index set injects into the countable set $\mathcal{B}$. $\square$

Problem 1.56 (Why the two-origin quotient is the right Hausdorff counterexample). Show explicitly that the two-origin quotient has all the desired local charts even though no Hausdorff separation exists.

Solution. Around either origin, take a disc in the corresponding source copy small enough to serve as the chosen representative; its quotient image is open and the quotient map restricts to a homeomorphism. Around any nonzero point, choose a disc avoiding the origin; the two copies have been identified there, producing one ordinary disc chart. Thus local charts do not detect the failure: only the impossibility of separating the two origins violates the manifold definition. $\square$

Problem 1.57 (Boundary of this chapter). Separate the topological data needed in VII/01 from the smooth-atlas data reserved for VII/02.

Solution. VII/01 needs only open chart domains and homeomorphisms to Euclidean opens, together with Hausdorffness and second countability. It may discuss overlap homeomorphisms topologically, but does not demand differentiability. VII/02 adds smooth compatibility of transition maps, maximal smooth atlases, and equivalence of smooth structures. This boundary prevents the legacy smooth-atlas material from being migrated one chapter too early. $\square$

1.9 Challenges

Problem 1.58 (Challenge). Give a rigorous proof that the long line is locally Euclidean and Hausdorff but not second countable.

Solution. Construct the long ray from $\omega_1 \times [0, 1)$ with lexicographic order topology and double it to obtain the long line. Order intervals give local coordinates homeomorphic to ordinary real intervals, and linearly ordered topological spaces are Hausdorff. An uncountable cofinal family of ordinal stages prevents any countable basis from reaching all local neighborhoods, so second countability fails. $\square$

Problem 1.59 (Challenge). Classify which of the three manifold axioms fail for the cone, plane with two origins, and $\Sigma \times \mathbb{R}_\delta$.

Solution. The cone is Hausdorff and second countable as a Euclidean subspace but fails local Euclidean structure at the apex. The plane with two origins is locally Euclidean and second countable but fails Hausdorffness. The product with the uncountable discrete line is locally Euclidean and Hausdorff but fails second countability. Together the examples show the logical independence and geometric purpose of the three axioms. $\square$

Problem 1.60 (Challenge). Develop a polygon-identification criterion sufficient to produce a compact topological surface.

Solution. Require edge pairings by homeomorphisms with each edge paired exactly once, verify that interior points have disc neighborhoods, paired edge interiors have two half-discs glued to a disc, and analyze each vertex equivalence class to ensure the cyclic collection of sectors glues to a disc rather than a cone or branching point. The finite quotient is compact and second countable; Hausdorffness follows from the standard closed-equivalence-relation criterion in this finite polygon setting. $\square$

Problem 1.61 (Challenge). Relate finite-sheeted covering maps of manifolds to subgroup inclusions in the torus/Klein-bottle examples without developing the full fundamental-group classification.

Solution. When a common simply connected model carries properly discontinuous deck-group actions, an inclusion $\Lambda \leq \Gamma$ induces $M/\Lambda \rightarrow M/\Gamma$. If the index is finite, each target fiber has

that many cosets and the quotient map is a finite covering. The torus-to-Klein-bottle map is the index-two case, illustrating the algebraic shadow of covering theory before the later algebraic-topology volume. $\square$

Problem 1.62 (Challenge). Audit a candidate “manifold” by designing a decision tree based on local Euclidean structure, Hausdorffness, and second countability.

Solution. First test local models: singular/branch points already rule out a manifold. If all local models pass, test separation of suspicious duplicated or quotient points. Finally test countability by seeking a countable atlas/basis or an uncountable family of disjoint opens. The order is efficient because each stage isolates a distinct failure mode and produces a concrete certificate of nonmanifold behavior. $\square$

Bridge to VII/02

A topological manifold comes with many possible charts, but so far chart changes are only homeomorphisms. VII/02 adds the differentiable compatibility condition, defines smooth atlases and maximal smooth structures, and asks when two topological charts belong to the same smooth geometry.

Chapter 2

Smooth Structures and Atlases

VII/01 supplied topological manifolds. Smooth geometry begins by requiring changes of coordinates to be smooth:

$$\boxed{\text{smooth structure} = \text{topological atlas} + \text{smooth transition maps.}}$$

Standing conventions

M is a Hausdorff, second-countable topological n -manifold. Euclidean smoothness means C^∞ .

2.1 Transition maps

Definition 2.1 (Transition map). For charts (U, φ) , (V, ψ) , the transition is

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V).$$

Definition 2.2 (Smooth compatibility). Two charts are smoothly compatible when the overlap transition is a diffeomorphism of Euclidean open sets.

Proposition 2.3 (Restriction). *Restrictions of compatible charts remain compatible.*

2.2 Atlases and smooth structures

Definition 2.4 (Smooth atlas). A smooth atlas is a covering atlas of pairwise smoothly compatible charts.

Definition 2.5 (Equivalent atlases). Two smooth atlases are equivalent when every chart of one is compatible with every chart of the other.

Theorem 2.6 (Maximal atlas). *Every smooth atlas lies in a unique maximal smooth atlas consisting of all charts compatible with it.*

Definition 2.7 (Smooth structure). A smooth structure is a maximal smooth atlas.

2.3 Smooth functions

Definition 2.8 (Smooth function). For open $W \subset M$, $f : W \rightarrow \mathbb{R}$ is smooth when $f \circ \varphi^{-1}$ is smooth in every compatible chart.

Theorem 2.9 (Atlas equivalence from smooth functions). *Two smooth atlases are equivalent iff they determine the same smooth real-valued functions on every open set.*

Proof. Equivalent atlases transfer coordinate expressions through smooth transition maps. Conversely, coordinate functions of charts detect the component functions of cross-transitions. $\square$

2.4 Standard structures

Example 2.10 (Euclidean structure). The identity chart generates the standard smooth structure on every Euclidean open set.

Example 2.11 (Stereographic sphere atlas). The north/south stereographic charts form a smooth atlas on S^n ; their transition is inversion away from the origin.

2.5 Projective spaces

Definition 2.12 (Homogeneous charts). On $\mathbb{RP}^n$, $U_i = \{x_i \neq 0\}$ is identified with $\mathbb{R}^n$ by normalizing $x_i = 1$. The analogous complex chart identifies the standard open in $\mathbb{CP}^n$ with $\mathbb{C}^n$.

Theorem 2.13 (Real projective space). *The homogeneous charts make $\mathbb{RP}^n$ a smooth n -manifold.*

Theorem 2.14 (Complex projective space). *The homogeneous charts make $\mathbb{CP}^n$ a smooth real $2n$ -manifold.*

Warning 2.15 (Pseudo-atlas warning). Compatible local formulas must still be accompanied by Hausdorffness and second countability.

2.6 Atlas versus structure

Remark 2.16 (Generators are not intrinsic). Restricting charts or adding compatible charts changes the atlas but not the maximal smooth structure.

2.7 Exercises

Exercise 2.17. Let (U, φ) and (V, ψ) be charts on a topological n -manifold. Write the transition map and its domain precisely.

Hint. Restrict to $U \cap V$ before composing. $\square$

Solution. The transition map is $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$, between open subsets of $\mathbb{R}^n$. $\square$

Exercise 2.18. Show that smooth compatibility of charts is symmetric.

Hint. Use the inverse transition map. $\square$

Solution. If $\psi \circ \varphi^{-1}$ is a diffeomorphism, its inverse is $\varphi \circ \psi^{-1}$, hence smooth. $\square$

Exercise 2.19. Show that restricting a smooth chart to a smaller open set gives a compatible chart.

Hint. Transitions restrict. □

Solution. Every new transition is a restriction of an old smooth transition, hence smooth. □

Exercise 2.20. Prove that every smooth atlas is contained in a unique maximal smooth atlas.

Hint. Take all compatible charts. □

Solution. Collect all charts compatible with the atlas. They are mutually compatible locally through generating charts; maximality and uniqueness follow immediately. □

Exercise 2.21. Show that two atlases define the same smooth structure iff their union is a smooth atlas.

Hint. Use cross-compatibility. □

Solution. Equal maximal atlases imply all cross-transitions are smooth. Conversely a smooth union generates one maximal atlas containing both. □

Exercise 2.22. Verify that the identity chart generates the usual smooth structure on an open $U \subseteq \mathbb{R}^n$.

Hint. Compare all charts to Euclidean coordinates. □

Solution. The maximal extension consists precisely of charts whose coordinate changes with the identity chart are smooth diffeomorphisms. □

Exercise 2.23. Show affine coordinate changes $x \mapsto Ax + b$, $A \in GL(n, \mathbb{R})$, are compatible with the standard structure.

Hint. Write the inverse. □

Solution. Both the affine map and $y \mapsto A^{-1}(y - b)$ are smooth. □

Exercise 2.24. Prove the north/south stereographic charts form a smooth atlas on S^n .

Hint. Use the overlap inversion. □

Solution. The overlap transition is $x \mapsto x/\|x\|^2$, up to convention, smooth on $\mathbb{R}^n \setminus \{0\}$. □

Exercise 2.25. Explain why the maximal atlas, rather than a chosen finite atlas, is the intrinsic smooth structure.

Hint. Refinements change generators but not compatibility. □

Solution. Restricting or adding compatible charts changes the written atlas while leaving all allowed coordinate changes and smooth functions unchanged. □

Exercise 2.26. Show equivalent atlases define the same smooth real-valued functions.

Hint. Insert a transition. □

Solution. Locally $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$, a composition of smooth maps. □

Exercise 2.27. Prove the converse using coordinate functions.

Hint. Each chart component is a smooth function. □

Solution. If both atlases give the same smooth functions, the components of every cross-transition are smooth; symmetry gives smooth inverse transitions. $\square$

Exercise 2.28. Write the standard affine chart on $\mathbb{RP}^n$ where $x_i \neq 0$.

Hint. Normalize $x_i = 1$. $\square$

Solution. Send $[x]$ to all ratios $x_j/x_i, j \neq i$. $\square$

Exercise 2.29. Compute the transition map between two standard $\mathbb{RP}^n$ charts.

Hint. Divide the normalized coordinates by the new pivot coordinate. $\square$

Solution. The components are rational functions with nonzero denominator on the overlap, so they are smooth. $\square$

Exercise 2.30. Explain why $\mathbb{CP}^n$ has real dimension $2n$.

Hint. Each affine chart is $\mathbb{C}^n$. $\square$

Solution. Since $\mathbb{C}^n \cong \mathbb{R}^{2n}$, the real manifold dimension is $2n$. $\square$

Exercise 2.31. Show the homogeneous projective charts cover projective space.

Hint. A nonzero vector has a nonzero coordinate. $\square$

Solution. Every projective point lies in some chart $U_i = \{x_i \neq 0\}$. $\square$

Exercise 2.32. Why must Hausdorffness and second countability be checked for a pseudo-atlas?

Hint. Local coordinate data do not imply global axioms. $\square$

Solution. Locally Euclidean spaces can be non-Hausdorff or fail second countability, as VII/01 showed. $\square$

Exercise 2.33. Show a finite atlas with Euclidean chart images gives second countability.

Hint. Pull back rational-ball bases. $\square$

Solution. Each chart image has a countable basis; a finite union of their pullbacks is countable and forms a basis. $\square$

Exercise 2.34. Verify Hausdorffness of $\mathbb{RP}^n$ from $S^n/\{\pm 1\}$.

Hint. Use the closed antipodal relation. $\square$

Solution. The finite group quotient of compact Hausdorff S^n by a closed equivalence relation is Hausdorff. $\square$

Exercise 2.35. Show adding an already compatible chart does not change the smooth structure.

Hint. It already lies in the maximal atlas. $\square$

Solution. The maximal extension before and after adding the chart is identical. $\square$

Exercise 2.36. Show transition-map smoothness can be checked componentwise.

Hint. Use Euclidean vector-valued smoothness. $\square$

Solution. A Euclidean map is smooth iff all coordinate components are smooth. □

Exercise 2.37. Give a smooth atlas on S^1 using two stereographic charts.

Hint. Delete opposite points. □

Solution. Each deletion is homeomorphic to $\mathbb{R}$, and the rational overlap transition is smooth. □

Exercise 2.38. Distinguish a topological atlas from a smooth atlas.

Hint. Compare overlap requirements. □

Solution. A topological atlas only requires homeomorphic charts; a smooth atlas also requires smooth transition diffeomorphisms. □

Exercise 2.39. Relate compatibility of two smooth structures to smoothness of the identity map.

Hint. Write its coordinate representative. □

Solution. The identity from one structure to the other is represented by cross-transition maps; it is a diffeomorphism exactly when the structures are equivalent. □

Exercise 2.40. State the conceptual boundary between VII/02 and VII/03.

Hint. Structure first, maps second. □

Solution. VII/02 decides which coordinates are smooth-compatible. VII/03 fixes those structures and studies smooth maps between manifolds. □

2.8 Solved problem dossiers

Problem 2.41 (Legacy Problem 1(a): equivalent atlases and smooth functions). Let $\mathcal{A}, \mathcal{B}$ be smoothly equivalent atlases. Prove they determine the same smooth real-valued functions.

Solution. Factor coordinate expressions through smooth cross-transition maps. The argument is symmetric. □

Problem 2.42 (Legacy Problem 1(b): smooth functions determine atlas equivalence). Assume two atlases define exactly the same smooth real-valued functions. Prove they are equivalent.

Solution. Each coordinate function of a chart from one atlas is smooth for the other. Their coordinate expressions are the components of the cross-transition, hence smooth; reverse roles for the inverse. □

Problem 2.43 (Legacy Problem 2(a): homogeneous charts on $\mathbb{RP}^n$). Prove the homogeneous coordinate maps define a smooth atlas on real projective space.

Solution. Normalize one nonzero coordinate. Overlap maps are rational with nonzero denominators, so smooth. □

Problem 2.44 (Legacy Problem 2(b): homogeneous charts on $\mathbb{CP}^n$). Prove the homogeneous coordinate maps define a smooth atlas on complex projective space.

Solution. Each standard open is $\mathbb{C}^n$; transitions are holomorphic rational maps, hence smooth as real maps. $\square$

Problem 2.45 (Projective-space global axioms). Complete the pseudo-atlas proof by showing $\mathbb{RP}^n, \mathbb{CP}^n$ are Hausdorff and second countable.

Solution. Use compact group quotient models for Hausdorffness and the finite standard atlas for second countability. $\square$

Problem 2.46 (Maximal atlas theorem). Construct the maximal atlas generated by a smooth atlas and prove uniqueness.

Solution. Take all charts compatible with the atlas. Mutual compatibility follows locally through generating charts; maximality and uniqueness are immediate. $\square$

Problem 2.47 (Stereographic smooth structure). Show the two stereographic charts generate a smooth structure on S^n .

Solution. The overlap inversion $x/\|x\|^2$ is smooth away from zero. $\square$

Problem 2.48 (Atlas refinement invariance). Prove an open refinement of a smooth atlas generates the same maximal atlas.

Solution. Restricted charts are compatible with originals, and each original chart is locally covered by restrictions. $\square$

Problem 2.49 (Open-submanifold smooth structure). Construct the canonical smooth structure on an open subset $U \subset M$.

Solution. Restrict the maximal atlas of M to domains intersected with U . $\square$

Problem 2.50 (Unequal atlases, same structure). Give two unequal atlases on $\mathbb{R}$ that define the standard structure.

Solution. Use the identity chart and a countable cover by restrictions of that chart. $\square$

Problem 2.51 (A topological chart outside the standard smooth structure). Show $x \mapsto x^3$ defines a topological chart on $\mathbb{R}$ not compatible with the identity chart.

Solution. The inverse transition $x^{1/3}$ is not smooth at zero. $\square$

Problem 2.52 (Smooth structures from smooth functions). Explain why the assignment $U \mapsto C^\infty(U)$ determines the maximal smooth atlas.

Solution. Coordinate functions characterize cross-transition smoothness, exactly as in the converse legacy atlas problem. $\square$

2.9 Challenges

Problem 2.53 (Challenge). Construct two topological atlases on $\mathbb{R}$ that are not smoothly equivalent.

Solution. Use the identity and $x \mapsto x^3$; the inverse transition $x^{1/3}$ fails smoothness at zero. $\square$

Problem 2.54 (Challenge). Prove $\mathbb{RP}^n$ is a smooth quotient of S^n by the antipodal action using local slices.

Solution. The free finite action gives neighborhoods disjoint from nontrivial translates; quotient them to charts and transfer the sphere smooth structure. $\square$

Problem 2.55 (Challenge). Compare stereographic and graph atlases on S^2 and prove equivalence.

Solution. On common domains their transitions are explicit rational/square-root formulas with positive denominators, smooth both ways. $\square$

Problem 2.56 (Challenge). Explain precisely what an exotic smooth structure would mean.

Solution. It is a maximal smooth atlas on a fixed topological manifold not equivalent to another; the identity homeomorphism is not a diffeomorphism. $\square$

Problem 2.57 (Challenge). Design a complete finite pseudo-atlas audit.

Solution. Check cover, open chart domains/images, homeomorphism property, all overlap transitions, Hausdorffness, and second countability. $\square$

Bridge to VII/03

With smooth structures fixed, VII/03 defines smooth maps by smoothness of their coordinate representatives.

Chapter 3

Smooth Maps and Diffeomorphisms

With smooth structures fixed, a map is smooth when its Euclidean coordinate representatives are smooth:

$$F \text{ smooth} \iff \psi \circ F \circ \varphi^{-1} \text{ smooth locally.}$$

3.1 Smooth maps

Definition 3.1 (Smooth map at a point). $F : M \rightarrow N$ is smooth at p if some compatible source and target charts make $\psi \circ F \circ \varphi^{-1}$ smooth near $\varphi(p)$.

Theorem 3.2 (Chart independence). *Smoothness in one compatible chart pair implies smoothness in every compatible chart pair.*

Proof. New coordinate representatives differ by pre- and post-composition with smooth transition maps. $\square$

Definition 3.3 (Smooth map). A map is smooth if it is smooth at every point.

3.2 Local criteria

Theorem 3.4 (Locality). *Smoothness can be checked on an arbitrary open cover of the source.*

Proposition 3.5 (Componentwise criterion). *A map $M \rightarrow \mathbb{R}^m$ is smooth iff all its components are smooth real-valued functions.*

3.3 Identity and composition

Proposition 3.6 (Identity). *Identity maps are smooth.*

Theorem 3.7 (Composition). *Compositions of smooth maps are smooth.*

3.4 Diffeomorphisms

Definition 3.8 (Diffeomorphism). A diffeomorphism is a bijective smooth map with smooth inverse.

Warning 3.9 (Smooth bijection is insufficient). $x \mapsto x^3$ is smooth and bijective on $\mathbb{R}$, but its inverse is not smooth at zero.

3.5 An explicit model diffeomorphism

Theorem 3.10 (Ball to Euclidean space). *For $B_r \subset \mathbb{R}^k$,*

$$F(x) = \frac{rx}{\sqrt{r^2 - \|x\|^2}}$$

is a diffeomorphism onto $\mathbb{R}^k$, with inverse

$$G(y) = \frac{ry}{\sqrt{r^2 + \|y\|^2}}.$$

Corollary 3.11 (Full-space charts). *Every point of a smooth k -manifold has a chart onto all of $\mathbb{R}^k$.*

3.6 Dimension invariance

Theorem 3.12 (Euclidean dimension obstruction). *Nonempty Euclidean opens can be diffeomorphic only when their dimensions agree.*

Corollary 3.13 (Manifold dimension). *Diffeomorphic smooth manifolds have the same local dimension.*

3.7 Category viewpoint

Remark 3.14 (Smooth category). Smooth manifolds and smooth maps form a category; its isomorphisms are diffeomorphisms.

3.8 Exercises

Exercise 3.15. State smoothness of $F : M \rightarrow N$ at p in coordinates.

Hint. Choose source and target charts. □

Solution. Require $\psi \circ F \circ \varphi^{-1}$ to be smooth near $\varphi(p)$. □

Exercise 3.16. Prove chart-independence of smoothness.

Hint. Factor through transition maps. □

Solution. Any new representative equals a smooth target transition composed with the old representative and a smooth source transition. □

Exercise 3.17. Show the identity map is smooth.

Hint. It is a transition map in coordinates. □

Solution. Its coordinate representation is $\psi \circ \varphi^{-1}$, smooth by compatibility. □

Exercise 3.18. Show constant maps are smooth.

Hint. Coordinate representatives are constant. □

Solution. Constant Euclidean maps are smooth. □

Exercise 3.19. Prove compositions of smooth maps are smooth.

Hint. Choose an intermediate chart. □

Solution. The coordinate representative of a composite is the composite of Euclidean representatives. □

Exercise 3.20. Prove smoothness is local on an open cover.

Hint. Check each point in one cover member. □

Solution. Every point has a neighborhood on which the restriction is smooth, hence the map is smooth at every point. □

Exercise 3.21. Prove $F : M \rightarrow \mathbb{R}^k$ is smooth iff each component is smooth.

Hint. Use the Euclidean componentwise criterion. □

Solution. In source coordinates the representative is vector-valued and is smooth exactly when all components are smooth. □

Exercise 3.22. Relate VII/02 smooth functions to smooth maps $M \rightarrow \mathbb{R}$.

Hint. Use the identity target chart. □

Solution. The two coordinate definitions are identical. □

Exercise 3.23. Define diffeomorphism and prove it is an equivalence relation.

Hint. Use identity, inverse, composition. □

Solution. All three properties follow from smooth identity maps and closure of smooth maps under composition. □

Exercise 3.24. Give a smooth bijection that is not a diffeomorphism.

Hint. Use x^3 . □

Solution. $x \mapsto x^3$ is smooth and bijective, but $x^{1/3}$ is not smooth at zero. □

Exercise 3.25. Show affine isomorphisms of $\mathbb{R}^n$ are diffeomorphisms.

Hint. Write the affine inverse. □

Solution. $x \mapsto Ax + b$ has smooth inverse $y \mapsto A^{-1}(y - b)$. □

Exercise 3.26. Show restrictions of smooth maps to open subsets are smooth.

Hint. Restrict coordinate representatives. □

Solution. Restrictions of smooth Euclidean maps are smooth. □

Exercise 3.27. Show corestriction to an open submanifold is smooth.

Hint. Use induced target charts. □

Solution. Target charts are restrictions of ambient charts, giving the same coordinate expressions. □

Exercise 3.28. Verify $F(x) = rx/\sqrt{r^2 - \|x\|^2}$ is smooth on B_r .

Hint. The denominator is positive. □

Solution. It is built from polynomial, square-root and reciprocal operations on a positive function. □

Exercise 3.29. Find the inverse of the ball map.

Hint. Solve radially. □

Solution. $G(y) = ry/\sqrt{r^2 + \|y\|^2}$. □

Exercise 3.30. Prove that inverse is smooth everywhere.

Hint. The denominator never vanishes. □

Solution. $r^2 + \|y\|^2 > 0$ for all y . □

Exercise 3.31. Use the ball diffeomorphism to get a chart onto all $\mathbb{R}^n$.

Hint. Shrink to a coordinate ball. □

Solution. Restrict a chart to the preimage of a ball and compose with the ball-to-space diffeomorphism. □

Exercise 3.32. Prove diffeomorphic Euclidean opens have equal dimension.

Hint. Differentiate inverse identities. □

Solution. The derivative of a diffeomorphism has a two-sided linear inverse, forcing equal dimensions. □

Exercise 3.33. Show every diffeomorphism is a homeomorphism.

Hint. Smooth maps are continuous. □

Solution. A diffeomorphism and its smooth inverse are continuous inverse maps. □

Exercise 3.34. Show changing the target chart preserves smoothness.

Hint. Post-compose by transition. □

Solution. The new representative is a smooth transition composed with the old representative. □

Exercise 3.35. Show changing the source chart preserves smoothness.

Hint. Pre-compose by transition. □

Solution. The new representative is the old one composed with a smooth source transition. □

Exercise 3.36. Explain why a generating atlas suffices to check smoothness.

Hint. All maximal charts are compatible with it. □

Solution. Chart-independence propagates smoothness from generating charts to every chart in the maximal atlas. □

Exercise 3.37. Explain why a homeomorphism need not be a diffeomorphism.

Hint. Continuity is weaker than differentiability. □

Solution. The x^3 example is a homeomorphism whose inverse is not smooth. □

Exercise 3.38. State the bridge to tangent spaces.

Hint. Smooth maps have first-order linearizations. □

Solution. VII/04 defines $dF_p : T_p M \rightarrow T_{F(p)} N$, the coordinate-free derivative. □

3.9 Solved problem dossiers

Problem 3.39 (Legacy Problem 3: local chart criterion). Prove it suffices to cover M by open sets on which one chosen coordinate representative of F is smooth.

Solution. At each point use the representative supplied by its cover member; chart-independence upgrades it to all compatible coordinates. □

Problem 3.40 (Legacy diffeomorphism problem: ball to space). Prove $F(x) = rx/\sqrt{r^2 - \|x\|^2}$ is a diffeomorphism $B_r \rightarrow \mathbb{R}^k$.

Solution. The inverse is $G(y) = ry/\sqrt{r^2 + \|y\|^2}$; both are smooth and direct substitution gives both identities. □

Problem 3.41 (Legacy consequence: full-space charts). Show every point of a smooth k -manifold admits a chart onto all of $\mathbb{R}^k$.

Solution. Shrink any chart to a coordinate ball and compose with the legacy ball diffeomorphism. □

Problem 3.42 (Legacy dimension obstruction). Prove nonempty opens in $\mathbb{R}^n$ and $\mathbb{R}^m$ cannot be diffeomorphic when $n \neq m$.

Solution. Differentiate the inverse identities; the derivative must be a linear isomorphism. □

Problem 3.43 (Chart-independence theorem). Prove the definition of smooth map is independent of both source and target charts.

Solution. Factor a new representative through the old representative and two smooth transition maps. □

Problem 3.44 (Composition theorem). Prove composition of smooth manifold maps is smooth.

Solution. Choose compatible source, middle and target charts so the coordinate representative is a Euclidean composition. □

Problem 3.45 (Smooth bijection warning). Give a smooth bijection with nonsmooth inverse.

Solution. Use $x \mapsto x^3$, whose inverse $x^{1/3}$ is not differentiable at zero. □

Problem 3.46 (Identity between two smooth structures). Characterize when the identity homeomorphism between two smooth structures on one topological manifold is a diffeomorphism.

Solution. Exactly when the structures are equivalent, because its coordinate representatives are the cross-transition maps. □

Problem 3.47 (Componentwise criterion). Prove $F : M \rightarrow \mathbb{R}^m$ is smooth iff all coordinate components are smooth functions.

Solution. Apply the Euclidean componentwise smoothness theorem in any source chart. $\square$

Problem 3.48 (Open-cover locality). If $M = \bigcup U_\alpha$ and every $F|_{U_\alpha}$ is smooth, prove F smooth.

Solution. Smoothness at a point is witnessed in any cover member containing that point. $\square$

Problem 3.49 (Dimension invariance). Show diffeomorphic connected smooth manifolds have equal dimension.

Solution. Choose charts at corresponding points; the induced Euclidean diffeomorphism forces equal chart dimensions. $\square$

Problem 3.50 (Category viewpoint). Show smooth manifolds and smooth maps form a category and identify its isomorphisms.

Solution. Identity and composition are smooth; categorical isomorphisms are exactly diffeomorphisms. $\square$

3.10 Challenges

Problem 3.51 (Challenge). Give a smooth homeomorphism that is not a diffeomorphism and diagnose failure of the inverse.

Solution. $x \mapsto x^3$; the inverse cube root is not differentiable at zero. $\square$

Problem 3.52 (Challenge). Prove $F : M \rightarrow N$ is smooth iff every local smooth real-valued g on N pulls back to a smooth $g \circ F$.

Solution. The forward direction is composition. For the converse choose target coordinate functions; their pullbacks are the components of the coordinate representative. $\square$

Problem 3.53 (Challenge). Analyze the radial behavior of the ball diffeomorphism near ∂B_r .

Solution. For $x = \rho u$, the radial factor $r\rho/\sqrt{r^2 - \rho^2}$ tends to $+\infty$ as $\rho \rightarrow r^-$. $\square$

Problem 3.54 (Challenge). Prove a gluing principle for smooth maps on an open cover.

Solution. Topological gluing gives the map; smoothness follows locally because the glued map agrees with a smooth piece near every point. $\square$

Problem 3.55 (Challenge). Design a practical diffeomorphism checklist.

Solution. Prove bijectivity, smoothness locally in convenient charts, and smoothness of the inverse; do not infer inverse smoothness from bijectivity. $\square$

Bridge to VII/04

VII/04 constructs tangent spaces and the differential dF_p , turning the first derivative of a coordinate representative into an intrinsic linear map.

Chapter 4

Tangent Spaces

A smooth manifold has no preferred ambient vector space, but every point carries an intrinsic first-order linear space. The tangent space turns derivatives of coordinate expressions into coordinate-free geometry:

$$T_p M = \{\text{first-order derivations at } p\}, \quad dF_p : T_p M \rightarrow T_{F(p)} N.$$

4.1 Derivations at a point

Definition 4.1 (Derivation at a point). A derivation at $p \in M$ is a linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

Definition 4.2 (Tangent space). The tangent space $T_p M$ is the vector space of all derivations at p .

Proposition 4.3 (Constants vanish). *Every $v \in T_p M$ annihilates constant functions.*

Proof. Since $v(1) = v(1 \cdot 1) = 2v(1)$, one has $v(1) = 0$; linearity gives the result. $\square$

Proposition 4.4 (First-order locality). *A derivation depends only on the germ of a smooth function at p .*

Proof. If f vanishes near p , choose a smooth cutoff χ equal to 1 near p and supported where $f = 0$. Then $0 = v(f\chi) = v(f)$. $\square$

4.2 Coordinate tangent vectors

Let (U, x) be a chart around p , with coordinates $x^1, \dots, x^n$.

Definition 4.5 (Coordinate derivations).

$$\left. \frac{\partial}{\partial x^i} \right|_p f = \left. \frac{\partial(f \circ x^{-1})}{\partial u^i} \right|_{u=x(p)}.$$

Theorem 4.6 (Coordinate basis theorem). *The vectors*

$$\left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p$$

form a basis of $T_p M$, and every $v \in T_p M$ has the unique expansion

$$v = \sum_{i=1}^n v(x^i) \left. \frac{\partial}{\partial x^i} \right|_p.$$

Consequently $\dim T_p M = n$.

Proof. In coordinates, subtract the affine first-order part of a smooth function. The remainder has zero value and first derivatives at $x(p)$, hence is a sum of products of functions vanishing at p . The Leibniz rule kills such products. Thus a derivation is determined by its values on the coordinate functions. Evaluation on x^j proves linear independence. $\square$

Corollary 4.7 (Euclidean tangent space). *The standard coordinates identify $T_p \mathbb{R}^n$ canonically with $\mathbb{R}^n$.*

4.3 Change of coordinates

Theorem 4.8 (Transformation law). *If x and y are charts at p , then*

$$\left. \frac{\partial}{\partial x^i} \right|_p = \sum_{a=1}^n \frac{\partial y^a}{\partial x^i}(p) \left. \frac{\partial}{\partial y^a} \right|_p.$$

Proof. Apply both sides to an arbitrary smooth function and use the Euclidean chain rule. $\square$

Remark 4.9 (Intrinsic vector, coordinate components). A tangent vector is not a coordinate list. Its components transform by a transition Jacobian while the derivation itself is unchanged.

4.4 Velocities of curves

Definition 4.10 (Velocity derivation). For a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = p$, define

$$\dot{\gamma}(0)(f) = \left. \frac{d}{dt} \right|_0 f(\gamma(t)).$$

Theorem 4.11 (Curve realization). *Every $v \in T_p M$ is the velocity of a smooth curve through p .*

Proof. Write $v = \sum v^i \partial_{x^i}|_p$. For small t , put

$$\gamma(t) = x^{-1}(x(p) + t(v^1, \dots, v^n)).$$

Its coordinate velocity is $(v^1, \dots, v^n)$. $\square$

Corollary 4.12 (Equivalent curve model). *Two smooth curves through p determine the same tangent vector exactly when their coordinate velocities agree in one, equivalently every, chart.*

4.5 The differential of a smooth map

Definition 4.13 (Differential or pushforward). For a smooth map $F : M \rightarrow N$, define

$$(dF_p v)(g) = v(g \circ F), \quad dF_p : T_p M \rightarrow T_{F(p)} N.$$

Proposition 4.14 (Well-defined linear map). *The differential is linear and sends derivations at p to derivations at $F(p)$.*

Proof. Linearity is immediate. For a product gh , pull back to $(g \circ F)(h \circ F)$ and apply Leibniz. $\square$

Theorem 4.15 (Jacobian formula). *For source coordinates x^i , target coordinates y^a , and $F^a = y^a \circ F \circ x^{-1}$,*

$$dF_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \sum_{a=1}^m \frac{\partial F^a}{\partial x^i}(p) \frac{\partial}{\partial y^a} \Big|_{F(p)}.$$

Thus the coordinate matrix of dF_p is the ordinary Jacobian.

Theorem 4.16 (Compatibility with curves). *For every smooth curve γ through p ,*

$$dF_p(\dot{\gamma}(0)) = (F \circ \gamma)'(0).$$

Proof. Apply both derivations to a test function $g \in C^\infty(N)$. $\square$

4.6 Functorial rules

Theorem 4.17 (Chain rule). *For smooth $F : M \rightarrow N$ and $G : N \rightarrow P$,*

$$d(G \circ F)_p = dG_{F(p)} \circ dF_p.$$

Proof. Both sides, after evaluation on v and $h \in C^\infty(P)$, give $v(h \circ G \circ F)$. $\square$

Proposition 4.18 (Identity and constants).

$$d(\text{id}_M)_p = \text{id}_{T_p M}, \quad dC_p = 0$$

for a constant map C .

Corollary 4.19 (Diffeomorphisms). *If $F : M \rightarrow N$ is a diffeomorphism, then dF_p is a linear isomorphism and*

$$(dF_p)^{-1} = d(F^{-1})_{F(p)}.$$

4.7 Rank and first-order geometry

Definition 4.20 (Rank). The rank of F at p is $\text{rank}(dF_p)$.

Definition 4.21 (Immersion and submersion at a point). A smooth map is an immersion at p if dF_p is injective and a submersion at p if dF_p is surjective.

Warning 4.22 (Local versus global). Injectivity or surjectivity of dF_p is first-order local information; it does not imply global injectivity or surjectivity of F .

4.8 Examples

Example 4.23 (Projection). For $\pi : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$, $d\pi(v, w) = v$, so π is a submersion.

Example 4.24 (Polar map). For $F(r, \theta) = (r \cos \theta, r \sin \theta)$,

$$[dF] = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

whose determinant is r . Rank drops at $r = 0$.

Example 4.25 (A Euclidean curve). If $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$, then

$$\dot{\gamma}(t) = \sum_i \dot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}.$$

4.9 Exercises

Exercise 4.26. Prove that every derivation annihilates constants.

Hint. Apply Leibniz to $1 \cdot 1$. □

Solution. $v(1) = 2v(1)$, so $v(1) = 0$; linearity handles all constants. □

Exercise 4.27. Show sums and scalar multiples of derivations are derivations.

Hint. Check linearity and Leibniz. □

Solution. Both defining identities are preserved by linear combinations. □

Exercise 4.28. Verify that $\partial/\partial x^i|_p$ is a derivation.

Hint. Use the ordinary product rule. □

Solution. The Euclidean partial derivative is linear and satisfies the product rule after evaluation at the point. □

Exercise 4.29. Compute $(\partial/\partial x^i|_p)(x^j)$.

Hint. Differentiate a coordinate function. □

Solution. It equals δ_i^j . □

Exercise 4.30. Prove uniqueness in the coordinate expansion of a tangent vector.

Hint. Evaluate on every x^j . □

Solution. If $\sum a^i \partial_{x^i}|_p = 0$, evaluation on x^j gives $a^j = 0$. □

Exercise 4.31. Show $\dim T_p M = \dim M$.

Hint. Use a coordinate basis. □

Solution. An n -chart supplies exactly n basis derivations. □

Exercise 4.32. Derive the coordinate-change formula for tangent bases.

Hint. Differentiate the transition map. □

Solution. The chain rule gives $\partial_{x^i} = \sum_a (\partial y^a / \partial x^i) \partial_{y^a}$ at p . □

Exercise 4.33. Show equality of curve velocities in one chart is chart independent.

Hint. Differentiate the transition map. □

Solution. Both velocities are multiplied by the same invertible transition Jacobian. □

Exercise 4.34. Prove that a smooth curve velocity defines a derivation.

Hint. Differentiate $(fg) \circ \gamma$. □

Solution. The one-variable product rule gives the Leibniz identity. □

Exercise 4.35. Construct a curve realizing $v = a^i \partial_{x^i}|_p$.

Hint. Use a straight line in coordinates. □

Solution. For small t , take $\gamma(t) = x^{-1}(x(p) + ta)$. □

Exercise 4.36. For $F(x, y) = (x^2 - y, xy)$, compute $dF_{(1,2)}$.

Hint. Compute the Jacobian. □

Solution. The matrix is $\begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}$. □

Exercise 4.37. Prove directly from the definition that dF_p is linear.

Hint. Evaluate on a test function. □

Solution. $dF_p(av + bw)(g) = (av + bw)(g \circ F)$, which splits linearly. □

Exercise 4.38. Prove that $dF_p v$ satisfies Leibniz.

Hint. Pull products back by F . □

Solution. Apply Leibniz to $(g \circ F)(h \circ F)$. □

Exercise 4.39. Prove the chain rule without coordinates.

Hint. Test both sides on a smooth function. □

Solution. Both sides evaluate to $v(h \circ G \circ F)$. □

Exercise 4.40. Show the differential of a constant map is zero.

Hint. Pulled-back functions are constant. □

Solution. Derivations annihilate every pulled-back test function. □

Exercise 4.41. Show $d(\text{id}_M)_p$ is the identity.

Hint. Evaluate on test functions. □

Solution. $(d \text{id}_p v)(f) = v(f)$. □

Exercise 4.42. Prove a diffeomorphism induces tangent-space isomorphisms.

Hint. Differentiate both inverse identities. □

Solution. The chain rule gives mutually inverse differentials. □

Exercise 4.43. Compute the rank of the polar map.

Hint. Use $\det[dF] = r$. □

Solution. The rank is 2 for $r \neq 0$ and 1 at $r = 0$. □

Exercise 4.44. Show $i : \mathbb{R}^k \hookrightarrow \mathbb{R}^n$, $i(x) = (x, 0)$, is an immersion.

Hint. Differentiate the inclusion. □

Solution. $di(v) = (v, 0)$, which is injective. □

Exercise 4.45. Show $\pi : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ is a submersion.

Hint. Differentiate the projection. □

Solution. $d\pi(v, w) = v$, which is surjective. □

Exercise 4.46. Give an immersion that is not globally injective.

Hint. Use a periodic parametrization. □

Solution. $t \mapsto (\cos t, \sin t)$ has nonzero derivative everywhere but is periodic. □

Exercise 4.47. If $dF_p = 0$, show every curve through p maps to a curve with zero velocity.

Hint. Use curve compatibility. □

Solution. $(F \circ \gamma)'(0) = dF_p \dot{\gamma}(0) = 0$. □

Exercise 4.48. For $f : M \rightarrow \mathbb{R}$, identify $df_p(v)$ under $T_{f(p)}\mathbb{R} \cong \mathbb{R}$.

Hint. Use the differential definition. □

Solution. $df_p(v) = v(f)$. □

Exercise 4.49. Explain why the coordinate matrix of dF_p depends on charts but its rank does not.

Hint. Coordinate changes give invertible factors. □

Solution. Left and right multiplication by invertible transition Jacobians preserves rank. □

4.10 Solved problem dossiers

Problem 4.50 (Legacy tangent-space problem: derivations). Starting from Leibniz, prove that derivations at a point of an n -manifold form an n -dimensional vector space.

Solution. A chart gives coordinate derivations. First-order Taylor decomposition shows every derivation is $\sum_i v(x^i) \partial_{x^i}|_p$, and evaluation on coordinates proves uniqueness. □

Problem 4.51 (Legacy tangent-space problem: curve classes). Identify tangent vectors with equivalence classes of curves through p having equal coordinate velocities.

Solution. Each curve differentiates test functions to give a derivation; equal coordinate velocities give equal derivations. Every derivation is realized by a coordinate straight line. □

Problem 4.52 (Legacy differential problem). Construct dF_p intrinsically and prove its coordinate matrix is the Jacobian.

Solution. Define $(dF_p v)(g) = v(g \circ F)$ and apply it to target coordinates and source coordinate basis vectors. □

Problem 4.53 (Coordinate basis reconstruction). Prove a tangent vector is determined by its values on local coordinate functions.

Solution. Modulo products of germs vanishing at the point, every germ is its value plus its first-order coordinate part; derivations kill the constant and quadratic remainder. □

Problem 4.54 (Change-of-chart covariance). Derive the tangent-component transformation law.

Solution. Differentiate the chart transition; the inverse transition supplies the inverse matrix. □

Problem 4.55 (Differential chain rule). Prove the chain rule without coordinates.

Solution. After evaluation on v and a test function h , both sides are $v(h \circ G \circ F)$. $\square$

Problem 4.56 (Diffeomorphism invariance). Show a diffeomorphism induces tangent-space isomorphisms.

Solution. Differentiate the two inverse identities. $\square$

Problem 4.57 (Rank in coordinates). Show Jacobian rank is independent of the chosen charts.

Solution. Coordinate matrices differ by invertible transition Jacobians on the left and right. $\square$

Problem 4.58 (Velocity and pushforward). Prove $dF_p \dot{\gamma}(0) = (F \circ \gamma)'(0)$.

Solution. Both derivations return $\frac{d}{dt}|_0 g(F(\gamma(t)))$ on each test function g . $\square$

Problem 4.59 (Critical point criterion). For $f : M \rightarrow \mathbb{R}$, prove $df_p = 0$ iff $v(f) = 0$ for every $v \in T_p M$.

Solution. Under $T_{f(p)} \mathbb{R} \cong \mathbb{R}$, df_p is precisely $v \mapsto v(f)$. $\square$

Problem 4.60 (Product preview). For Euclidean product projections, recover $T_{(p,q)}(\mathbb{R}^m \times \mathbb{R}^n) \cong T_p \mathbb{R}^m \oplus T_q \mathbb{R}^n$.

Solution. In standard coordinates this is $\mathbb{R}^{m+n} = \mathbb{R}^m \oplus \mathbb{R}^n$; VII/06 globalizes it. $\square$

Problem 4.61 (Immersion versus injection). Explain why $t \mapsto (\cos t, \sin t)$ is an immersion but not injective.

Solution. Its derivative never vanishes, but the map is periodic. $\square$

4.11 Challenge problems

Problem 4.62 (Tangent space from the maximal ideal). Let $\mathfrak{m}_p$ be germs vanishing at p . Show $T_p M \cong (\mathfrak{m}_p / \mathfrak{m}_p^2)^*$.

Solution. A derivation kills constants and products of two germs vanishing at the point, hence descends to the quotient. Conversely a functional on the quotient defines a first-order derivation; coordinate germs prove the constructions are inverse. $\square$

Problem 4.63 (Naturality of tangent spaces). Show identities and compositions are preserved by the tangent construction.

Solution. The identity rule and chain rule are exactly the functorial axioms pointwise. $\square$

Problem 4.64 (Rank under diffeomorphic conjugacy). If $\tilde{F} = H \circ F \circ K^{-1}$ with H, K diffeomorphisms, prove corresponding ranks agree.

Solution. $d\tilde{F} = dH \, dF \, (dK)^{-1}$, and the outer factors are isomorphisms. $\square$

Problem 4.65 (Zero differential need not mean local constancy). Give a smooth function with zero differential at one point but not locally constant there.

Solution. $f(x) = x^2$ has $df_0 = 0$ and is not locally constant at 0. $\square$

Problem 4.66 (First-order equivalence of maps). For $F, G : M \rightarrow N$ with $F(p) = G(p)$, give a coordinate-independent criterion for equal first-order behavior.

Solution. The criterion is $dF_p = dG_p$. In one chart pair this is equality of Jacobians; transition Jacobians make the statement intrinsic. $\square$

Bridge to VII/05

For a real-valued function, df_p is a linear functional on T_pM . VII/05 abstracts these functionals as T_p^*M , develops coordinate coframes, pullbacks, annihilators, and the intrinsic meaning of differentials of functions.

Chapter 5

Cotangent Spaces

Tangent vectors differentiate functions. Cotangent vectors are the linear measurements that can be applied to tangent vectors:

$$T_p^*M = (T_pM)^*, \quad df_p(v) = v(f).$$

They are the natural home of differentials, constraints, and the covariant side of coordinate calculus.

5.1 Covectors and the cotangent space

Definition 5.1 (Cotangent space). The cotangent space at p is the dual vector space

$$T_p^*M = \text{Hom}_{\mathbb{R}}(T_pM, \mathbb{R}).$$

Its elements are called cotangent vectors or covectors.

Proposition 5.2 (Dimension). *If M is n -dimensional, then $\dim T_p^*M = n$.*

Proof. The dual of an n -dimensional vector space has dimension n . □

5.2 Coordinate coframes

Let $x = (x^1, \dots, x^n)$ be local coordinates near p .

Definition 5.3 (Coordinate covectors). Define $dx^i|_p \in T_p^*M$ by

$$dx^i|_p(v) = v(x^i).$$

Theorem 5.4 (Dual coordinate basis). *The family*

$$dx^1|_p, \dots, dx^n|_p$$

is dual to

$$\partial_{x^1}|_p, \dots, \partial_{x^n}|_p,$$

so

$$dx^i|_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i.$$

*Every $\alpha \in T_p^*M$ has the unique expansion*

$$\alpha = \sum_i \alpha \left(\frac{\partial}{\partial x^i} \Big|_p \right) dx^i|_p.$$

5.3 Differentials of functions

Definition 5.5 (Differential of a function). For $f \in C^\infty(M)$, define

$$df_p \in T_p^*M, \quad df_p(v) = v(f).$$

Theorem 5.6 (Coordinate formula for df). In local coordinates,

$$df_p = \sum_{i=1}^n \frac{\partial(f \circ x^{-1})}{\partial x^i}(x(p)) dx^i|_p.$$

Proof. Evaluate both sides on the coordinate basis vectors of T_pM . □

Proposition 5.7 (Algebra rules). For smooth functions f, g and $a, b \in \mathbb{R}$,

$$d(af + bg) = a df + b dg, \quad d(fg) = f dg + g df$$

pointwise.

Proof. These identities are the linearity and Leibniz rule for tangent derivations. □

5.4 Pullback of covectors

Definition 5.8 (Pullback at a point). For smooth $F : M \rightarrow N$ and $p \in M$, define

$$F_p^* : T_{F(p)}^*N \rightarrow T_p^*M, \quad (F_p^*\alpha)(v) = \alpha(dF_p v).$$

Theorem 5.9 (Pullback functoriality). For $F : M \rightarrow N$, $G : N \rightarrow P$,

$$(G \circ F)_p^* = F_p^* \circ G_{F(p)}^*, \quad (\text{id}_M)_p^* = \text{id}_{T_p^*M}.$$

Proof. This is dual to the tangent-space chain rule; evaluate on an arbitrary tangent vector. □

Theorem 5.10 (Pulling back differentials). For $g \in C^\infty(N)$,

$$F_p^*(dg_{F(p)}) = d(g \circ F)_p.$$

Proof. For $v \in T_pM$, both sides evaluate to $v(g \circ F)$. □

5.5 Coordinate transformation law

Theorem 5.11 (Coframe transformation). For two coordinate systems x and y ,

$$dy^a|_p = \sum_i \frac{\partial y^a}{\partial x^i}(p) dx^i|_p.$$

Equivalently, covector components transform by the inverse transpose relative to the transformation law for tangent-vector components.

Proof. Since $dy^a = d(y^a)$, apply the coordinate formula for the differential of the function y^a . □

Remark 5.12 (Covariant versus contravariant). The basis covectors and their component columns transform oppositely so that the scalar pairing $\alpha(v)$ remains invariant.

5.6 Annihilators

Definition 5.13 (Annihilator). For a linear subspace $W \subset T_p M$, its annihilator is

$$W^\circ = \{\alpha \in T_p^* M : \alpha(w) = 0 \text{ for every } w \in W\}.$$

Theorem 5.14 (Dimension of the annihilator). If $\dim T_p M = n$ and $\dim W = k$, then

$$\dim W^\circ = n - k.$$

Proof. Extend a basis of W to one of $T_p M$ and inspect the dual basis. $\square$

Proposition 5.15 (Kernel annihilator). For a linear map $A : V \rightarrow W$,

$$\text{im}(A^*) = (\ker A)^\circ.$$

In finite dimensions, A is surjective iff A^* is injective, and A is injective iff A^* is surjective.

5.7 Differentials and level sets

If $f : M \rightarrow \mathbb{R}$, the covector df_p measures first-order change of f . When $df_p \neq 0$, its kernel is a hyperplane:

$$\ker df_p = \{v \in T_p M : v(f) = 0\}.$$

VII/06 identifies this kernel with the tangent space to a regular level set.

Definition 5.16 (Critical and regular points). A point p is critical for f when $df_p = 0$; otherwise it is regular.

5.8 Differential versus gradient

Warning 5.17 (No gradient without extra structure). The differential df_p is defined on every smooth manifold. A gradient vector $\nabla f(p)$ requires a Riemannian metric g to identify covectors with vectors through

$$g_p(\nabla f(p), v) = df_p(v).$$

Without a metric, df_p and a tangent vector are different kinds of objects.

5.9 Examples

Example 5.18 (Euclidean differential). For $f(x, y) = x^2 y + \sin y$,

$$df = (2xy) dx + (x^2 + \cos y) dy.$$

Example 5.19 (Constraint covector). For $F(x) = \|x\|^2$ on $\mathbb{R}^n$,

$$dF_x = 2 \sum_i x^i dx^i.$$

At $x \neq 0$, $\ker dF_x$ is the Euclidean hyperplane orthogonal to x ; VII/06 recognizes it as $T_x S^{n-1}$ on a sphere level.

5.10 Exercises

Exercise 5.20. Prove the cotangent space has the same dimension as the tangent space.

Hint. Use finite-dimensional duality. □

Solution. A vector space and its dual have the same finite dimension. □

Exercise 5.21. Verify $dx^i|_p(\partial_{x^j}|_p) = \delta_j^i$.

Hint. Use the definition of coordinate derivations. □

Solution. $\partial x^i / \partial x^j = \delta_j^i$. □

Exercise 5.22. Expand an arbitrary covector in the coordinate coframe.

Hint. Use the dual-basis theorem. □

Solution. $\alpha = \sum_i \alpha(\partial_{x^i}|_p) dx^i|_p$. □

Exercise 5.23. Show $df_p(v) = v(f)$ agrees with VII/04's differential $df_p : T_p M \rightarrow T_{f(p)} \mathbb{R}$.

Hint. Identify $T_{f(p)} \mathbb{R}$ with $\mathbb{R}$. □

Solution. The two definitions give the same scalar on every tangent vector. □

Exercise 5.24. Compute df for $f(x, y, z) = xe^{yz}$.

Hint. Take partial derivatives. □

Solution. $df = e^{yz} dx + xze^{yz} dy + xye^{yz} dz$. □

Exercise 5.25. Prove linearity of the differential in the function.

Hint. Apply a tangent derivation. □

Solution. $d(af + bg)_p(v) = v(af + bg) = a v(f) + b v(g)$. □

Exercise 5.26. Prove the product rule for differentials.

Hint. Use the derivation Leibniz rule. □

Solution. $d(fg) = f dg + g df$ pointwise. □

Exercise 5.27. Derive the coordinate transformation law for coframes.

Hint. Apply the formula for $d(y^a)$. □

Solution. $dy^a = \sum_i (\partial y^a / \partial x^i) dx^i$. □

Exercise 5.28. Show the scalar pairing between a covector and tangent vector is chart independent.

Hint. Combine the dual transformation laws. □

Solution. The transition Jacobian and its inverse cancel in the pairing. □

Exercise 5.29. Prove F_p^* is linear.

Hint. Use linearity of a covector. □

Solution. $F_p^*(a\alpha + b\beta)(v) = a\alpha(dF_p v) + b\beta(dF_p v)$. □

Exercise 5.30. Prove pullback reverses composition.

Hint. Evaluate on a tangent vector. □

Solution. $(G \circ F)^* = F^*G^*$ follows from $d(G \circ F) = dG dF$. □

Exercise 5.31. Prove $F^*(dg) = d(g \circ F)$ at each point.

Hint. Evaluate on a tangent vector. □

Solution. Both sides give the directional derivative of the composite. □

Exercise 5.32. For $F(x, y) = (x + y, x - y)$ and $\alpha = du + 2 dv$, compute $F^*\alpha$.

Hint. Compute $d(u \circ F)$ and $d(v \circ F)$. □

Solution. $F^*\alpha = (dx + dy) + 2(dx - dy) = 3dx - dy$. □

Exercise 5.33. Find the annihilator of $W = \text{span}(e_1, \dots, e_k) \subset \mathbb{R}^n$.

Hint. Use the dual standard basis. □

Solution. $W^\circ = \text{span}(dx^{k+1}, \dots, dx^n)$. □

Exercise 5.34. Prove the annihilator dimension theorem.

Hint. Extend a basis and dualize. □

Solution. If W has basis $e_1, \dots, e_k$, its annihilator is spanned by the remaining dual basis vectors. □

Exercise 5.35. Show $\text{im}(A^*) \subseteq (\ker A)^\circ$.

Hint. Evaluate $A^*\alpha$ on $v \in \ker A$. □

Solution. $(A^*\alpha)(v) = \alpha(Av) = 0$. □

Exercise 5.36. Prove equality $\text{im}(A^*) = (\ker A)^\circ$ in finite dimensions.

Hint. Compare dimensions after proving inclusion. □

Solution. Both dimensions equal $\text{rank } A$. □

Exercise 5.37. Show a surjective dF_p gives an injective pullback F_p^* .

Hint. Use linear duality. □

Solution. The dual of a surjective finite-dimensional linear map is injective. □

Exercise 5.38. Show an injective dF_p gives a surjective pullback F_p^* .

Hint. Use linear duality. □

Solution. The dual of an injective finite-dimensional linear map is surjective. □

Exercise 5.39. For $f(x, y) = x^2 + y^2$, determine critical points.

Hint. Solve $df = 0$. □

Solution. $df = 2x dx + 2y dy$, so only $(0, 0)$ is critical. □

Exercise 5.40. At $p \neq 0$, compute $\ker d(\|x\|^2)_p$.

Hint. Write the differential. □

Solution. It is $\{v : p \cdot v = 0\}$. □

Exercise 5.41. Explain why a gradient cannot be defined canonically on a smooth manifold from its smooth structure alone.

Hint. A covector-to-vector identification is missing. □

Solution. The smooth structure gives the dual space but no preferred isomorphism between a vector space and its dual. □

Exercise 5.42. Given a Riemannian metric g , show the equation $g(\nabla f, \cdot) = df$ determines ∇f uniquely.

Hint. Use nondegeneracy of the inner product. □

Solution. The musical map from tangent vectors to covectors is an isomorphism. □

Exercise 5.43. State the bridge from annihilators to regular level sets.

Hint. Think of $\ker dF_p$. □

Solution. For a regular level set $S = F^{-1}(c)$, VII/06 proves $T_p S = \ker dF_p$; equivalently the conormal space is generated by the pulled-back coordinate differentials of the constraints. □

5.11 Solved problem dossiers

Problem 5.44 (Legacy cotangent problem: dual basis). Construct the coordinate coframe and prove it is dual to the coordinate tangent basis.

Solution. Define $dx^i(v) = v(x^i)$; evaluation on ∂_{x^j} gives δ_j^i , so dual-basis theory completes the proof. □

Problem 5.45 (Legacy differential problem: df). Show that the differential of a smooth real-valued function is intrinsically a cotangent vector.

Solution. Define $df_p(v) = v(f)$. This is linear in v , hence belongs to $T_p^* M$, and its coordinate formula follows by evaluating on the coordinate basis. □

Problem 5.46 (Legacy covector problem: transformation law). Derive the coordinate change law for covectors.

Solution. Use $dy^a = d(y^a) = \sum_i (\partial y^a / \partial x^i) dx^i$, then dualize to obtain the component transformation law. □

Problem 5.47 (Pullback construction). Construct the pullback of a covector under a smooth map and prove it is coordinate free.

Solution. Set $F_p^* \alpha = \alpha \circ dF_p$. Both ingredients are intrinsic, so no chart enters the definition. □

Problem 5.48 (Pullback chain rule). Prove pullback reverses composition.

Solution. Dualizing $d(G \circ F) = dG \circ dF$ gives $(G \circ F)^* = F^* \circ G^*$. $\square$

Problem 5.49 (Differential of a composite). Prove $d(g \circ F) = F^*(dg)$.

Solution. Evaluate both covectors on an arbitrary tangent vector and use the definition of the differential. $\square$

Problem 5.50 (Annihilator dimension). For $W \subset T_p M$, prove $\dim W + \dim W^\circ = \dim M$.

Solution. Extend a basis of the subspace to a tangent basis and inspect the dual basis. $\square$

Problem 5.51 (Kernel-image duality). Prove $\text{im}(A^*) = (\ker A)^\circ$.

Solution. Inclusion is immediate; rank-nullity gives equal dimensions in finite dimensions. $\square$

Problem 5.52 (Critical-point criterion). Show p is critical for f iff every tangent direction has zero first-order change.

Solution. This is exactly $df_p = 0 \iff df_p(v) = v(f) = 0$ for all v . $\square$

Problem 5.53 (Constraint hyperplane). For a regular scalar constraint $f = c$, explain why $\ker df_p$ has codimension one.

Solution. A nonzero linear functional on an n -dimensional vector space has an $(n - 1)$ -dimensional kernel. $\square$

Problem 5.54 (Differential versus gradient). Separate the notions of differential and gradient and state the extra datum needed for the latter.

Solution. The differential is a covector defined from smooth structure alone. A Riemannian metric supplies the isomorphism $v \mapsto g(v, \cdot)$ needed to turn df into ∇f . $\square$

Problem 5.55 (Diffeomorphism and cotangent spaces). Show a diffeomorphism induces cotangent-space isomorphisms in the reverse direction.

Solution. If F is a diffeomorphism, dF_p is an isomorphism; its dual F_p^* is therefore an isomorphism with inverse $(F^{-1})_{F(p)}^*$. $\square$

5.12 Challenge problems

Problem 5.56 (Cotangent space from first-order function germs). Show $\mathfrak{m}_p/\mathfrak{m}_p^2 \cong T_p^* M$.

Solution. Coordinate germs $x^i - x^i(p)$ form a basis of the quotient and map to $dx^i|_p$. This is dual to the tangent-space description from VII/04. $\square$

Problem 5.57 (Double annihilator). For $W \subset T_p M$, prove $(W^\circ)^\circ = W$ under the canonical identification with the double dual.

Solution. The inclusion is immediate. Dimensions agree: the double annihilator has dimension $n - (n - \dim W) = \dim W$. $\square$

Problem 5.58 (Conormal space of several constraints). If $F = (f^1, \dots, f^k) : M \rightarrow \mathbb{R}^k$ has rank k at p , show $df_p^1, \dots, df_p^k$ are linearly independent.

Solution. They are the pullbacks under dF_p^* of the standard dual basis. Surjectivity of dF_p makes the dual map injective. $\square$

Problem 5.59 (Invariant pairing). Prove directly that tangent and cotangent coordinate transformation laws leave the pairing invariant.

Solution. Write the transformations with a Jacobian and inverse Jacobian; their matrix product is the identity. $\square$

Problem 5.60 (Metric identification is additional structure). Show two different inner products on one vector space can send the same covector to different gradient vectors.

Solution. On $\mathbb{R}^2$, take $df = dx$. The Euclidean metric gives gradient $(1, 0)$, while the metric with matrix $\text{diag}(2, 1)$ gives $(1/2, 0)$. $\square$

Bridge to VII/06

Tangent kernels and cotangent annihilators now describe constraints intrinsically. VII/06 uses the inverse and implicit function principles to turn regular constraints into embedded submanifolds and proves $T_p F^{-1}(c) = \ker dF_p$, while product manifolds give the canonical splitting $T_{(p,q)}(M \times N) \cong T_p M \oplus T_q N$.

Chapter 6

Submanifolds and Products

Submanifolds are manifolds sitting smoothly inside other manifolds; product manifolds combine independent geometric variables. Their tangent spaces are controlled by two basic linear patterns:

$$T_p F^{-1}(c) = \ker dF_p, \quad T_{(p,q)}(M \times N) \cong T_p M \oplus T_q N.$$

6.1 Embedded submanifolds

Definition 6.1 (Embedded submanifold). A subset $S \subset M^n$ is an embedded k -dimensional submanifold if every $p \in S$ has a chart

$$x : U \rightarrow x(U) \subset \mathbb{R}^n$$

such that

$$x(U \cap S) = x(U) \cap (\mathbb{R}^k \times \{0\}).$$

Such a chart is called adapted to S .

Proposition 6.2 (Induced smooth structure). *An embedded submanifold inherits a unique smooth k -manifold structure for which the inclusion $i : S \hookrightarrow M$ is a smooth embedding.*

Proof. Restrict the first k coordinate functions of adapted charts to S . Their transition functions are restrictions of smooth ambient transitions. $\square$

Definition 6.3 (Embedding). A smooth map $F : N \rightarrow M$ is an embedding if it is an immersion and a homeomorphism onto its image with the subspace topology.

Warning 6.4 (Immersed is not embedded). An injective immersion need not be an embedding unless its topology agrees with the subspace topology. Likewise an immersion may have self-intersections and fail even to be injective.

6.2 Tangent spaces of embedded submanifolds

Theorem 6.5 (Tangent inclusion). *For an embedded submanifold $S \subset M$, the differential of inclusion*

$$di_p : T_p S \rightarrow T_p M$$

is injective. We therefore identify $T_p S$ with its image, a linear subspace of $T_p M$.

Proof. In adapted coordinates the inclusion is $u \mapsto (u, 0)$, whose derivative is injective. $\square$

Theorem 6.6 (Curve characterization). *Under this identification,*

$$T_p S = \{\dot{\gamma}(0) : \gamma \text{ is a smooth curve in } S, \gamma(0) = p\} \subset T_p M.$$

6.3 Regular level sets

Definition 6.7 (Regular value). For $F : M^n \rightarrow N^m$, a point $q \in N$ is a regular value if dF_p is surjective for every $p \in F^{-1}(q)$.

Theorem 6.8 (Regular level-set theorem). *If q is a regular value of $F : M^n \rightarrow N^m$, then*

$$F^{-1}(q)$$

is an embedded submanifold of dimension $n - m$, and for each $p \in F^{-1}(q)$,

$$T_p F^{-1}(q) = \ker dF_p.$$

Proof. Choose charts near p and q . Surjectivity of the differential gives rank m . The implicit-function/constant-rank normal form supplies local coordinates in which F is projection onto the last m coordinates. The level set is therefore a coordinate slice of dimension $n - m$, and its tangent directions are precisely the kernel of the projection derivative. $\square$

Corollary 6.9 (Hypersurfaces). *If $f : M^n \rightarrow \mathbb{R}$ and c is a regular value, then $f^{-1}(c)$ is an embedded hypersurface and*

$$T_p f^{-1}(c) = \ker df_p.$$

Example 6.10 (Sphere). For $F(x) = \|x\|^2$ on $\mathbb{R}^n$, every $r^2 > 0$ is a regular value. Hence

$$S_r^{n-1} = F^{-1}(r^2), \quad T_p S_r^{n-1} = \{v : p \cdot v = 0\}.$$

Example 6.11 (Special linear group). The determinant map $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^\times$ has differential

$$d(\det)_A(H) = \det(A) \operatorname{tr}(A^{-1}H).$$

At $A \in SL_n(\mathbb{R})$, this is nonzero, so $SL_n(\mathbb{R})$ is a codimension-one embedded submanifold of $GL_n(\mathbb{R})$ with

$$T_A SL_n = \{H : \operatorname{tr}(A^{-1}H) = 0\}.$$

6.4 Graphs of smooth maps

Theorem 6.12 (Graph theorem). *For smooth $F : M \rightarrow N$, the graph*

$$\Gamma_F = \{(p, F(p)) : p \in M\}$$

is an embedded submanifold of $M \times N$, diffeomorphic to M .

Proof. The map $p \mapsto (p, F(p))$ is an embedding; locally the product chart sends the graph to the graph of a Euclidean smooth function, which can be straightened by $(x, y) \mapsto (x, y - F(x))$. $\square$

6.5 Product manifolds

Theorem 6.13 (Product smooth structure). *If M^m and N^n are smooth manifolds, then $M \times N$ has a canonical smooth structure of dimension $m + n$, generated by product charts*

$$(U \times V, x \times y), \quad (x \times y)(p, q) = (x(p), y(q)).$$

Proof. Product-chart transitions are products of smooth transition maps, hence smooth. $\square$

Proposition 6.14 (Product projections). *The projections*

$$\pi_M : M \times N \rightarrow M, \quad \pi_N : M \times N \rightarrow N$$

are smooth submersions.

Theorem 6.15 (Tangent splitting). *The map*

$$T_{(p,q)}(M \times N) \rightarrow T_p M \oplus T_q N, \quad v \mapsto (d\pi_M(v), d\pi_N(v))$$

is a canonical linear isomorphism.

Proof. In a product chart it is the standard identification $\mathbb{R}^{m+n} \cong \mathbb{R}^m \oplus \mathbb{R}^n$. Chart independence follows from the intrinsic projection formula. $\square$

Corollary 6.16 (Differential of a product map). *For smooth $F : P \rightarrow M$ and $G : P \rightarrow N$, the map*

$$(F, G) : P \rightarrow M \times N$$

satisfies

$$d(F, G)_r(v) = (dF_r(v), dG_r(v)).$$

6.6 Diagonal and slices

Proposition 6.17 (Diagonal). *The diagonal*

$$\Delta_M = \{(p, p) : p \in M\} \subset M \times M$$

is an embedded submanifold diffeomorphic to M , and

$$T_{(p,p)}\Delta_M = \{(v, v) : v \in T_p M\}.$$

Proposition 6.18 (Product slices). *For fixed $q \in N$, the subset $M \times \{q\}$ is an embedded submanifold with tangent space*

$$T_{(p,q)}(M \times \{q\}) = T_p M \oplus \{0\}.$$

6.7 Transverse intersections

Definition 6.19 (Transverse submanifolds). Embedded submanifolds $A, B \subset M$ meet transversely at $p \in A \cap B$ if

$$T_p A + T_p B = T_p M.$$

Theorem 6.20 (Transverse intersection). *If A^a and B^b are embedded submanifolds of M^n and meet transversely, then $A \cap B$ is locally an embedded submanifold of dimension $a + b - n$, with*

$$T_p(A \cap B) = T_p A \cap T_p B.$$

Proof. Locally describe one submanifold as a regular level set and restrict its defining submersion to the other. Transversality is exactly the surjectivity needed to apply the regular level-set theorem to that restriction. $\square$

6.8 Examples

Example 6.21 (Torus as a product). The torus $T^2 = S^1 \times S^1$ is a smooth 2-manifold, and

$$T_{(p,q)}T^2 \cong T_pS^1 \oplus T_qS^1.$$

Example 6.22 (Cylinder). $S^1 \times \mathbb{R}$ is a 2-manifold. Its tangent space splits into angular and axial directions.

6.9 Exercises

Exercise 6.23. Show an adapted chart induces a smooth chart on an embedded submanifold.

Hint. Restrict the first coordinates. □

Solution. The first k adapted coordinates give a homeomorphism of the submanifold slice onto an open subset of $\mathbb{R}^k$, and transitions are smooth restrictions. □

Exercise 6.24. Prove the inclusion of an embedded submanifold is an immersion.

Hint. Use adapted coordinates. □

Solution. Locally the inclusion is $u \mapsto (u, 0)$, whose derivative is injective. □

Exercise 6.25. Identify T_pS for the linear subspace $S = \mathbb{R}^k \times \{0\} \subset \mathbb{R}^n$.

Hint. Differentiate curves in the slice. □

Solution. It is $\mathbb{R}^k \times \{0\}$. □

Exercise 6.26. Prove the curve characterization of tangent vectors to an embedded submanifold.

Hint. Use an adapted chart. □

Solution. Curves in the coordinate slice have zero normal velocity; conversely every vector with zero normal components is realized by a straight-line curve in the slice. □

Exercise 6.27. State the regular-value condition for $F : M^n \rightarrow N^m$.

Hint. Use surjectivity. □

Solution. For every $p \in F^{-1}(q)$, the map $dF_p : T_pM \rightarrow T_qN$ must be surjective. □

Exercise 6.28. Compute the dimension of a regular level set.

Hint. Apply rank-nullity. □

Solution. It is $n - m$. □

Exercise 6.29. Prove $T_pF^{-1}(q) \subseteq \ker dF_p$.

Hint. Differentiate the constant curve $F \circ \gamma = q$. □

Solution. If $\dot{\gamma}(0) \in T_pF^{-1}(q)$, then $dF_p\dot{\gamma}(0) = 0$. □

Exercise 6.30. Explain why equality $T_pF^{-1}(q) = \ker dF_p$ follows after regularity.

Hint. Compare dimensions. □

Solution. Both spaces have dimension $n - m$, so the inclusion is equality. □

Exercise 6.31. Show $S^{n-1} \subset \mathbb{R}^n$ is an embedded submanifold.

Hint. Use $F(x) = \|x\|^2$. □

Solution. On $F^{-1}(1)$, $dF_x(v) = 2x \cdot v$ is nonzero, hence surjective onto $\mathbb{R}$. □

Exercise 6.32. Compute $T_p S^{n-1}$.

Hint. Take the kernel of the constraint differential. □

Solution. $T_p S^{n-1} = \{v : p \cdot v = 0\}$. □

Exercise 6.33. Show the hyperboloid $x^2 + y^2 - z^2 = 1$ is a smooth surface.

Hint. Check the gradient of the constraint on the level. □

Solution. The differential $(2x, 2y, -2z)$ cannot vanish on the level 1, so the level is regular. □

Exercise 6.34. Show the graph of a smooth function between Euclidean spaces is an embedded submanifold.

Hint. Straighten $(x, y) \mapsto (x, y - F(x))$. □

Solution. This local diffeomorphism sends the graph to a coordinate slice. □

Exercise 6.35. Prove the graph map $p \mapsto (p, F(p))$ is an immersion.

Hint. Differentiate it. □

Solution. Its differential is $v \mapsto (v, dF_p v)$, which is injective. □

Exercise 6.36. Show product-chart transitions are smooth.

Hint. Write them componentwise. □

Solution. They are $(x' \circ x^{-1}, y' \circ y^{-1})$, a product of smooth Euclidean maps. □

Exercise 6.37. Prove $\dim(M \times N) = \dim M + \dim N$.

Hint. Use product charts. □

Solution. A product chart lands in $\mathbb{R}^{m+n}$. □

Exercise 6.38. Show both product projections are submersions.

Hint. Use the tangent splitting. □

Solution. Their differentials are the canonical coordinate projections, hence surjective. □

Exercise 6.39. Prove the tangent splitting using product coordinates.

Hint. Write the coordinate basis. □

Solution. The first block of basis vectors comes from M , the second from N , yielding a direct sum. □

Exercise 6.40. Compute $d(F, G)_p$.

Hint. Compose with the two projections. □

Solution. Since $\pi_M \circ (F, G) = F$ and $\pi_N \circ (F, G) = G$, the tangent splitting gives $d(F, G) = (dF, dG)$. $\square$

Exercise 6.41. Prove the diagonal is an embedded submanifold.

Hint. Use $p \mapsto (p, p)$. $\square$

Solution. The diagonal map is an embedding; in product coordinates a linear change $(x, y) \mapsto (x, y - x)$ straightens it. $\square$

Exercise 6.42. Compute the tangent space of the diagonal.

Hint. Differentiate the diagonal map. $\square$

Solution. It is $\{(v, v) : v \in T_p M\}$. $\square$

Exercise 6.43. Show $M \times \{q\}$ is embedded and compute its tangent space.

Hint. Use a product chart. $\square$

Solution. It is a coordinate slice with tangent space $T_p M \oplus \{0\}$. $\square$

Exercise 6.44. For $T^2 = S^1 \times S^1$, compute its tangent dimension and splitting.

Hint. Use the product theorem. $\square$

Solution. It has dimension 2 and $T_{(p,q)} T^2 \cong T_p S^1 \oplus T_q S^1$. $\square$

Exercise 6.45. Check that two distinct coordinate axes in $\mathbb{R}^2$ meet transversely at the origin.

Hint. Add their tangent spaces. $\square$

Solution. The two one-dimensional tangent lines span $\mathbb{R}^2$. $\square$

Exercise 6.46. State the dimension expected for a transverse intersection $A^a \cap B^b \subset M^n$.

Hint. Use codimension addition. $\square$

Solution. The dimension is $a + b - n$, and the tangent space is $T_p A \cap T_p B$. $\square$

6.10 Solved problem dossiers

Problem 6.47 (Legacy submanifold problem: adapted coordinates). Prove that the coordinate-slice definition gives an intrinsic smooth structure on an embedded submanifold.

Solution. Restrict adapted charts to the slice. Ambient transition maps preserve the subset and restrict to smooth transitions; maximal completion gives the intrinsic structure. $\square$

Problem 6.48 (Legacy submanifold problem: tangent inclusion). Show the inclusion of an embedded submanifold identifies its tangent space with a linear subspace of the ambient tangent space.

Solution. In adapted coordinates di_p is the standard injection $\mathbb{R}^k \hookrightarrow \mathbb{R}^n$. $\square$

Problem 6.49 (Legacy regular-level problem). Prove that a regular level $F^{-1}(q)$ is an embedded submanifold with tangent space $\ker dF_p$.

Solution. The implicit-function normal form makes F a coordinate projection. Its level is a coordinate slice and its tangent space is exactly the kernel of the projection derivative. $\square$

Problem 6.50 (Legacy product-manifold problem). Construct the product smooth structure and prove its dimension.

Solution. Use product charts; their transitions are products of smooth transitions, and their target dimension is $m + n$. $\square$

Problem 6.51 (Legacy product tangent problem). Prove the canonical tangent splitting for a product.

Solution. Differentiate the two projections. In product coordinates the resulting map is the standard direct-sum identification. $\square$

Problem 6.52 (Legacy graph problem). Show the graph of a smooth map is an embedded submanifold.

Solution. The graph map is an embedding, or locally straighten the graph by $(x, y) \mapsto (x, y - F(x))$. $\square$

Problem 6.53 (Sphere as a regular level). Recover both the manifold structure and tangent space of a sphere from one constraint.

Solution. The value $r^2 > 0$ of $\|x\|^2$ is regular and the tangent kernel is $p^\perp$. $\square$

Problem 6.54 (Matrix-group level set). Show $SL_n(\mathbb{R})$ is a submanifold of $GL_n(\mathbb{R})$ and compute its tangent space.

Solution. Use the regular level $\det = 1$ and $d(\det)_A(H) = \det(A) \operatorname{tr}(A^{-1}H)$. The kernel gives the tangent space. $\square$

Problem 6.55 (Diagonal submanifold). Show the diagonal is embedded and identify its tangent space.

Solution. Differentiate $p \mapsto (p, p)$: the image is the diagonal subspace $\{(v, v)\}$. $\square$

Problem 6.56 (Product slice). Show $M \times \{q\}$ is canonically diffeomorphic to M and compute the inclusion differential.

Solution. The map $p \mapsto (p, q)$ is an embedding with differential $v \mapsto (v, 0)$. $\square$

Problem 6.57 (Transverse intersection). Derive the tangent-space formula for a transverse intersection.

Solution. Restrict a local defining submersion for one submanifold to the other; transversality gives surjectivity, and the regular-level theorem yields the intersection and tangent kernel. $\square$

Problem 6.58 (Embedding versus immersion). Explain precisely why immersion alone does not guarantee an embedded submanifold image.

Solution. Immersion controls only the differential. An embedding additionally requires global injectivity and the correct topology on the image; periodic immersed curves provide the simplest failure of injectivity. $\square$

6.11 Challenge problems

Problem 6.59 (Preimage theorem in full form). If $S \subset N$ is an embedded submanifold and $F : M \rightarrow N$ is transverse to S , show $F^{-1}(S)$ is an embedded submanifold and compute its tangent space.

Solution. Use adapted coordinates for S , take the normal-coordinate projection as a local defining submersion, and compose with F . Transversality makes the composite a submersion. Thus $F^{-1}(S)$ is regular and $T_p F^{-1}(S) = \{v : dF_p v \in T_{F(p)} S\}$. $\square$

Problem 6.60 (Fiber product preview). For transverse maps $F : M \rightarrow P$ and $G : N \rightarrow P$, show $M \times_P N = \{(m, n) : F(m) = G(n)\}$ is locally a submanifold.

Solution. Apply the preimage theorem to $F \times G : M \times N \rightarrow P \times P$ and the diagonal Δ_P . Transversality to the diagonal is the required condition. $\square$

Problem 6.61 (Codimension and conormal space). For a regular level $S = F^{-1}(q)$ with $F : M^n \rightarrow \mathbb{R}^k$, identify $(T_p S)^\circ$.

Solution. Since $T_p S = \ker dF_p$, VII/05 gives $(T_p S)^\circ = \text{im}(dF_p^*)$, spanned by the differentials of the component constraints. $\square$

Problem 6.62 (Products preserve embeddings). If $i : A \hookrightarrow M$ and $j : B \hookrightarrow N$ are embeddings, prove $i \times j : A \times B \hookrightarrow M \times N$ is an embedding.

Solution. The product map is an injective immersion because its differential is the direct sum of injective differentials, and the product of the two homeomorphisms onto their images is a homeomorphism onto the product image. $\square$

Problem 6.63 (Transversality dimension count). Explain algebraically why $\dim(T_p A \cap T_p B) = a + b - n$ under transversality.

Solution. For subspaces $U, V \subset E$, $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$. Setting $U + V = E$ gives the formula. $\square$

Bridge to VII/07

The spaces $T_p M$ now vary point by point and interact cleanly with products and embedded submanifolds. VII/07 assembles these fibers into vector bundles, beginning with the tangent and cotangent bundles and then abstracting local triviality, bundle charts, sections, and bundle morphisms.

Part II

Bundles and Forms

Chapter 7

Vector Bundles

A vector bundle is a smoothly varying family of vector spaces. Tangent and cotangent spaces are the motivating examples, but the abstraction is broader: local triviality lets one do linear algebra fiberwise while transition functions record the global twisting.

$$\pi : E \rightarrow M, \quad \pi^{-1}(U) \cong U \times \mathbb{R}^r, \quad (p, v)_\beta = (p, g_{\beta\alpha}(p)v)_\alpha.$$

7.1 Vector bundles and local triviality

Definition 7.1 (Smooth vector bundle). A smooth real vector bundle of rank r over a smooth manifold M is a smooth manifold E , a smooth surjection $\pi : E \rightarrow M$, and a vector-space structure on each fiber $E_p = \pi^{-1}(p)$, such that every $p \in M$ has an open neighborhood U and a diffeomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^r$$

with $\text{pr}_1 \circ \Phi = \pi$, and whose restriction to each fiber is linear.

Definition 7.2 (Bundle chart and local frame). A pair (U, Φ) as above is a bundle chart or local trivialization. Writing

$$\Phi(e) = (p, (a^1, \dots, a^r)),$$

the inverse images of the standard basis define smooth local sections $e_1, \dots, e_r$, called a local frame.

Proposition 7.3 (Dimension of the total space). *If M has dimension n and $E \rightarrow M$ has rank r , then E is a smooth manifold of dimension $n + r$.*

Proof. Every bundle chart identifies an open subset of E with an open subset of $M \times \mathbb{R}^r$, whose dimension is $n + r$. $\square$

7.2 Transition functions

Let (U_α, Φ_α) and (U_β, Φ_β) be bundle charts. On an overlap, fiberwise linearity forces

$$\Phi_\beta \Phi_\alpha^{-1}(p, v) = (p, g_{\beta\alpha}(p)v)$$

for a smooth map $g_{\beta\alpha} : U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{R})$.

Theorem 7.4 (Cocycle identities). *Transition functions satisfy*

$$g_{\alpha\alpha} = I, \quad g_{\alpha\beta} = g_{\beta\alpha}^{-1}, \quad g_{\gamma\beta}g_{\beta\alpha} = g_{\gamma\alpha}$$

on all relevant overlaps.

Proof. These identities are obtained by composing the corresponding changes of bundle coordinates. $\square$

Theorem 7.5 (Bundles from transition data). *Given an open cover $\{U_\alpha\}$ and smooth maps $g_{\beta\alpha} : U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{R})$ satisfying the cocycle identities, one obtains a rank- r vector bundle by gluing the products $U_\alpha \times \mathbb{R}^r$ using $(p, v)_\alpha \sim (p, g_{\beta\alpha}(p)v)_\beta$.*

7.3 Basic examples

Example 7.6 (Trivial bundle). For any manifold M , projection $M \times \mathbb{R}^r \rightarrow M$ is a rank- r vector bundle. A bundle is trivial if it is isomorphic to one of this form.

Example 7.7 (Tangent and cotangent bundles). The disjoint unions

$$TM = \coprod_{p \in M} T_p M, \quad T^*M = \coprod_{p \in M} T_p^* M$$

carry canonical smooth vector-bundle structures of rank $\dim M$. In a chart $x = (x^1, \dots, x^n)$, the coordinate vector fields and coordinate covectors provide local frames.

Example 7.8 (Möbius line bundle). Take $[0, 1] \times \mathbb{R}$ and identify $(0, t) \sim (1, -t)$. The quotient is a line bundle over S^1 , but it is not trivial. Its sign-changing transition function records the twist.

7.4 Sections

Definition 7.9 (Smooth section). A smooth section of $E \rightarrow M$ is a smooth map $s : M \rightarrow E$ satisfying $\pi \circ s = \text{id}_M$. The vector space of smooth sections is denoted $\Gamma(E)$.

Proposition 7.10 (Local coordinate description). *In a local frame $e_1, \dots, e_r$, every section has a unique expression*

$$s = \sum_{i=1}^r s^i e_i$$

with smooth coefficient functions s^i . On overlaps the column of coefficients transforms by the inverse frame-change matrix.

Proposition 7.11 (Global frame criterion). *A rank- r bundle is trivial if and only if it admits r smooth sections that form a basis of every fiber.*

Proof. A global frame defines a trivialization by $(p, a) \mapsto \sum_i a^i e_i(p)$. Conversely a trivialization pulls back the standard basis to a global frame. $\square$

7.5 Bundle maps and isomorphisms

Definition 7.12 (Vector-bundle morphism). For bundles $E \rightarrow M$ and $F \rightarrow N$, a vector-bundle morphism over $f : M \rightarrow N$ is a smooth map $\tilde{f} : E \rightarrow F$ such that $\pi_F \tilde{f} = f \pi_E$ and each fiber map $E_p \rightarrow F_{f(p)}$ is linear.

Example 7.13 (Differential as a bundle map). For smooth $f : M \rightarrow N$, the differential assembles into

$$df : TM \rightarrow TN, \quad v_p \mapsto df_p(v_p),$$

a vector-bundle morphism covering f .

7.6 Standard bundle constructions

Proposition 7.14 (Direct sum and dual). *For bundles $E, F \rightarrow M$, the fibers*

$$(E \oplus F)_p = E_p \oplus F_p, \quad (E^*)_p = E_p^*$$

define smooth vector bundles. In local frames their transition functions are respectively block diagonal $g_E \oplus g_F$ and $(g_E^{-1})^T$.

Proposition 7.15 (Tensor and Hom bundles). *Fiberwise tensor products and linear maps define bundles $E \otimes F$ and $\text{Hom}(E, F)$. Their local trivializations are obtained by applying the corresponding finite-dimensional linear-algebra constructions to bundle charts.*

7.7 Subbundles and quotient bundles

Definition 7.16 (Vector subbundle). A subset $S \subset E$ is a rank- k vector subbundle if each $S_p = S \cap E_p$ is a k -dimensional linear subspace and locally there exists a frame of E whose first k vectors frame S .

Proposition 7.17 (Kernel of constant-rank bundle map). *If $A : E \rightarrow F$ is a bundle map over id_M and $\text{rank } A_p$ is constant, then $\ker A = \coprod_p \ker A_p$ and $\text{im } A = \coprod_p \text{im } A_p$ are vector subbundles.*

7.8 Pullback bundles

Theorem 7.18 (Pullback bundle). *For $f : N \rightarrow M$ and a vector bundle $\pi : E \rightarrow M$, the set*

$$f^*E = \{(q, e) \in N \times E : f(q) = \pi(e)\}$$

*forms a vector bundle over N with fiber $(f^*E)_q \cong E_{f(q)}$. If $g_{\beta\alpha}$ are transition functions of E , then $g_{\beta\alpha} \circ f$ are transition functions of f^*E .*

Example 7.19 (Restriction). If $i : S \hookrightarrow M$ is an embedded submanifold, then i^*E is the restriction $E|_S$. In particular $TM|_S$ is an ambient tangent bundle over S , while TS is a generally smaller subbundle of it.

7.9 Exercises

Exercise 7.20. Show that every fiber of a rank- r vector bundle is isomorphic to $\mathbb{R}^r$.

Hint. Restrict a bundle chart to one fiber. □

Solution. A local trivialization sends E_p linearly and bijectively to $\{p\} \times \mathbb{R}^r$, hence to $\mathbb{R}^r$. □

Exercise 7.21. Prove that the bundle projection $\pi : E \rightarrow M$ is a submersion.

Hint. Write π in a bundle chart. □

Solution. Locally π is $(x, v) \mapsto x$, whose derivative is surjective. □

Exercise 7.22. Compute the dimension of the total space of a rank- r bundle over an n -manifold.

Hint. Use local triviality. □

Solution. Each bundle chart is modeled on $\mathbb{R}^n \times \mathbb{R}^r$, so $\dim E = n + r$. □

Exercise 7.23. Derive the cocycle identity for three bundle charts.

Hint. Compose two coordinate changes. □

Solution. The equality $\Phi_\gamma \Phi_\alpha^{-1} = (\Phi_\gamma \Phi_\beta^{-1})(\Phi_\beta \Phi_\alpha^{-1})$ gives $g_{\gamma\alpha} = g_{\gamma\beta} g_{\beta\alpha}$. □

Exercise 7.24. Show that a rank-zero vector bundle is canonically its base manifold.

Hint. Each fiber is the zero vector space. □

Solution. Local products are $U \times \{0\} \cong U$, and these identifications glue canonically. □

Exercise 7.25. Show that $M \times \mathbb{R}^r$ has the constant global frame.

Hint. Use the standard basis of $\mathbb{R}^r$. □

Solution. The sections $e_i(p) = (p, \mathbf{e}_i)$ are smooth and form a basis of each fiber. □

Exercise 7.26. Write the tangent-bundle transition matrix under a change of coordinates.

Hint. Differentiate the coordinate change. □

Solution. If $y = y(x)$, the tangent frame transforms by the Jacobian $(\partial y^a / \partial x^i)$. □

Exercise 7.27. Write the cotangent-bundle transition matrix.

Hint. Dualize the tangent transformation. □

Solution. Covectors transform by the inverse transpose of the tangent transition matrix. □

Exercise 7.28. Show that sections of the trivial line bundle are the same as smooth functions.

Hint. A section has the form $p \mapsto (p, f(p))$. □

Solution. Projection forces exactly that form, and smoothness is equivalent to smoothness of f . □

Exercise 7.29. Prove that the zero section of any vector bundle is smooth.

Hint. Use bundle coordinates. □

Solution. In a trivialization it is $p \mapsto (p, 0)$, hence smooth. □

Exercise 7.30. Show that pointwise sums and smooth-function multiples of sections are sections.

Hint. Check in a local frame. □

Solution. Local coefficient functions add and multiply smoothly, so $\Gamma(E)$ is a $C^\infty(M)$ -module. □

Exercise 7.31. Prove the global-frame criterion for triviality.

Hint. Construct a map from $M \times \mathbb{R}^r$. □

Solution. Send (p, a) to $\sum a^i e_i(p)$; fiberwise invertibility and smooth local matrices give a bundle isomorphism. □

Exercise 7.32. Show that $df : TM \rightarrow TN$ is a bundle morphism over f .

Hint. Use the chain-rule definition of df . □

Solution. It maps $T_p M$ linearly to $T_{f(p)} N$ and satisfies $\pi_{TN} \circ df = f \circ \pi_{TM}$. □

Exercise 7.33. Describe the transition functions of $E \oplus F$.

Hint. Use direct-sum frames. □

Solution. They are block-diagonal matrices $\text{diag}(g_E, g_F)$. □

Exercise 7.34. Describe the transition functions of the dual bundle.

Hint. Require evaluation to be invariant. □

Solution. If vectors transform by g , dual coordinates transform by $(g^{-1})^T$. □

Exercise 7.35. Show that $\text{Hom}(E, F) \cong E^* \otimes F$ fiberwise defines a bundle isomorphism.

Hint. Use the canonical finite-dimensional isomorphism. □

Solution. The standard map $\lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$ is natural under changes of frames, so it glues. □

Exercise 7.36. Give a local criterion for a family of subspaces $S_p \subset E_p$ to be a subbundle.

Hint. Look for a frame adapted to the family. □

Solution. It is enough that locally smooth sections $s_1, \dots, s_k$ span S_p and are linearly independent at every point. □

Exercise 7.37. Explain why a varying-dimensional kernel need not be a subbundle.

Hint. Rank is locally constant for a vector bundle. □

Solution. A subbundle has fixed rank on each connected trivializing neighborhood, so a dimension jump in $\ker A_p$ prevents local triviality. □

Exercise 7.38. Compute the pullback of a trivial bundle.

Hint. Use the defining fiber product. □

Solution. For $E = M \times \mathbb{R}^r$, $f^*E \cong N \times \mathbb{R}^r$. □

Exercise 7.39. Show $(g \circ f)^*E \cong f^*(g^*E)$.

Hint. Compare fibers. □

Solution. Both have fiber $E_{g(f(q))}$ over q , and the defining fiber-product identifications give a canonical bundle isomorphism. □

Exercise 7.40. Identify i^*TM for an inclusion $i : S \hookrightarrow M$.

Hint. Use the pullback definition. □

Solution. Its fiber at $p \in S$ is the ambient tangent space $T_p M$; thus it is $TM|_S$. □

Exercise 7.41. Show TS embeds as a subbundle of $TM|_S$ for an embedded submanifold.

Hint. Assemble the tangent inclusions from VII/06. □

Solution. The maps $di_p : T_p S \hookrightarrow T_p M$ vary smoothly in adapted coordinates and identify TS with a rank- $\dim S$ subbundle. □

Exercise 7.42. For the Möbius line bundle, explain why a nowhere-zero section would trivialize it.

Hint. Use the rank-one global-frame criterion. □

Solution. A nowhere-zero section is a basis of every one-dimensional fiber, hence a global frame and therefore a trivialization. □

Exercise 7.43. State what extra object VII/08 obtains by taking all ordered bases of the fibers.

Hint. Think of frames and the action of $GL(r)$. □

Solution. The set of ordered bases forms the frame bundle, a principal $GL(r, \mathbb{R})$ -bundle. □

7.10 Solved problem dossiers

Problem 7.44 (Legacy vector-bundle problem: local triviality). Starting from bundle charts, prove that the change of trivialization is encoded by a smooth $GL(r, \mathbb{R})$ -valued function.

Solution. The overlap map fixes the base point and is linear on each fiber, so it has the form $(p, v) \mapsto (p, g(p)v)$. Smoothness of the overlap and its inverse implies both g and g^{-1} vary smoothly, hence $g(p) \in GL(r)$. □

Problem 7.45 (Legacy vector-bundle problem: cocycles). Prove the three cocycle identities and explain why they are exactly the consistency conditions for gluing.

Solution. Identity and inverse changes give the first two identities. Equality of direct and two-step coordinate changes gives the third. These ensure the equivalence relation used in gluing is reflexive, symmetric, and transitive and that bundle charts agree smoothly. □

Problem 7.46 (Legacy tangent-bundle problem). Construct the smooth structure on TM from manifold charts and identify its transition functions.

Solution. A chart $x : U \rightarrow \mathbb{R}^n$ sends v_p to $(x(p), dx_p v_p)$. On overlaps the fiber coordinate changes by the Jacobian of $y \circ x^{-1}$, which is smooth and invertible. □

Problem 7.47 (Legacy cotangent-bundle problem). Construct T^*M and derive the inverse-transpose transition law.

Solution. Dual coordinate frames give local products. Preservation of the pairing $\alpha(v)$ forces the covector matrix to be the inverse transpose of the tangent matrix. □

Problem 7.48 (Legacy sections problem). Prove that a section is smooth exactly when its local coefficient functions are smooth.

Solution. Under a bundle chart, a section is $p \mapsto (p, s^1(p), \dots, s^r(p))$. Smoothness of this map is equivalent to smoothness of all coefficients. □

Problem 7.49 (Legacy triviality problem). Prove that a vector bundle is trivial exactly when it possesses a global frame.

Solution. A trivialization yields the standard global frame. Conversely, a global frame defines a fiberwise linear bijection from $M \times \mathbb{R}^r$ whose local matrix and inverse matrix are smooth. $\square$

Problem 7.50 (Möbius bundle dossier). Describe the Möbius line bundle using transition functions and isolate the obstruction visible in a loop around the base.

Solution. Cover S^1 by two arcs with disconnected overlap and choose transitions whose sign product records one reversal. Going once around the circle reverses a local basis; this cannot occur for a globally chosen nonzero frame. $\square$

Problem 7.51 (Bundle-map dossier). Show that a fiberwise linear smooth map over the identity has locally a matrix of smooth functions.

Solution. Choose local frames in source and target. Apply the map to each source frame vector and expand in the target frame; the resulting coefficient functions are smooth because the bundle map is smooth. $\square$

Problem 7.52 (Constant-rank kernel dossier). Prove that the kernel of a constant-rank bundle endomorphism is a vector subbundle.

Solution. In local frames the map is a smooth matrix of constant rank. The constant-rank normal form provides smooth local bases adapted to kernel and image, yielding a local frame for the kernel. $\square$

Problem 7.53 (Dual-bundle dossier). Explain intrinsically why dual transition functions are inverse transposes.

Solution. A change of frame is an isomorphism $g : V \rightarrow V$. The induced change on V^* is pullback by g^{-1} , represented by $(g^{-1})^T$. $\square$

Problem 7.54 (Pullback dossier). Construct the pullback bundle and verify local triviality.

Solution. Over $f^{-1}(U)$, where $E|_U \cong U \times \mathbb{R}^r$, map (q, e) to (q, v) using the fiber coordinate of e . This is a local trivialization, with transitions $g_{\beta\alpha} \circ f$. $\square$

Problem 7.55 (Tangent subbundle dossier). For an embedded submanifold $S \subset M$, prove $TS \subset TM|_S$ is a vector subbundle.

Solution. In an adapted chart, the first k coordinate vector fields span TS , while all n coordinate vector fields frame $TM|_S$. Thus the inclusion is locally a standard coordinate subbundle. $\square$

7.11 Challenge problems

Problem 7.56 (Bundles from cocycles). Give the full gluing construction from a $GL(r)$ -valued cocycle and prove the resulting quotient is a smooth vector bundle.

Solution. Take the disjoint union of $U_\alpha \times \mathbb{R}^r$ and quotient by the cocycle relation. The cocycle identities make this an equivalence relation. The natural images of the products provide compatible charts; on overlaps their transitions are exactly $(p, v) \mapsto (p, g_{\beta\alpha}(p)v)$, hence smooth. Fiberwise vector operations are well defined because the transition maps are linear. $\square$

Problem 7.57 (Determinant line bundle). For a rank- r bundle E , construct $\det E = \Lambda^r E$ and determine its transition functions.

Solution. In a local frame, the top exterior power has basis $e_1 \wedge \cdots \wedge e_r$. A frame change by g multiplies this basis by $\det g$, so $\det E$ is a line bundle with transition functions $\det g_{\beta\alpha}$. $\square$

Problem 7.58 (Short exact sequence of bundles). Suppose $A : E \rightarrow F$ has constant rank. Show locally that $0 \rightarrow \ker A \rightarrow E \rightarrow \operatorname{im} A \rightarrow 0$ is a short exact sequence of vector bundles.

Solution. Constant rank gives frames in which A is represented by a fixed block matrix $\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}$. In those frames the kernel and image are coordinate subbundles and the displayed sequence is fiberwise exact, hence a short exact sequence of bundles. $\square$

Problem 7.59 (Functoriality of pullback). Prove that pullback respects direct sums and duals: $f^*(E \oplus F) \cong f^*E \oplus f^*F$ and $f^*(E^*) \cong (f^*E)^*$.

Solution. Both statements are canonical fiberwise identifications. Local transition functions verify smoothness: pullback replaces each g by $g \circ f$, while direct sum and dual are algebraic operations commuting with composition by f . $\square$

Problem 7.60 (Frame-bundle preview). Let $E \rightarrow M$ have rank r . Show that changing a frame by a matrix in $GL(r)$ is free and transitive on each set of frames.

Solution. Given a frame $u : \mathbb{R}^r \rightarrow E_p$, right multiplication by g gives $u \circ g$. If $u \circ g = u$, injectivity gives $g = I$; given another frame v , the unique element $g = u^{-1}v$ satisfies $v = u \circ g$. This is the principal-bundle structure developed in VII/08. $\square$

Bridge to VII/08

A rank- r vector bundle can be studied either through its vectors or through all ordered bases of its fibers. VII/08 passes to the latter viewpoint: frames carry a free and transitive $GL(r, \mathbb{R})$ -action and lead naturally to principal bundles, transition functions, and associated bundles.

Chapter 8

Principal and Frame Bundles

Vector bundles encode linear fibers. Principal bundles isolate the symmetry behind their changes of frame. A principal G -bundle has fibers that look like the group G itself, but with no preferred identity element in an individual fiber.

$$P \times G \rightarrow P, \quad (u, g) \mapsto u \cdot g, \quad P|_U \cong U \times G.$$

8.1 Smooth group actions

Definition 8.1 (Right action). A smooth right action of a Lie group G on a manifold P is a smooth map $P \times G \rightarrow P$, $(u, g) \mapsto u \cdot g$, such that $u \cdot e = u$ and $(u \cdot g) \cdot h = u \cdot (gh)$.

Definition 8.2 (Free and transitive). An action is free if $u \cdot g = u$ implies $g = e$. It is transitive on a subset if any two points of that subset differ by the action of some group element.

8.2 Principal bundles

Definition 8.3 (Principal G -bundle). A principal G -bundle over M is a smooth manifold P , a smooth surjection $\pi : P \rightarrow M$, and a smooth right G -action preserving fibers, such that every $p \in M$ has a neighborhood U and a G -equivariant diffeomorphism

$$\Phi : \pi^{-1}(U) \rightarrow U \times G, \quad \Phi(u \cdot g) = (p, hg) \quad \text{if } \Phi(u) = (p, h).$$

Proposition 8.4 (Fiber torsor property). *The G -action on every fiber P_p is free and transitive. Hence for $u, v \in P_p$ there is a unique $g \in G$ with $v = u \cdot g$.*

Example 8.5 (Trivial principal bundle). The projection $M \times G \rightarrow M$ with $(p, h) \cdot g = (p, hg)$ is the trivial principal G -bundle.

8.3 Local sections and transition functions

Proposition 8.6 (Local section versus trivialization). *A local section $s : U \rightarrow P$ determines a trivialization by*

$$U \times G \rightarrow P|_U, \quad (p, g) \mapsto s(p) \cdot g,$$

and every principal-bundle trivialization determines a local section by taking the group identity.

Theorem 8.7 (Principal transition functions). *For local sections s_α, s_β , there is a unique smooth map*

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow G$$

such that $s_\beta = s_\alpha \cdot g_{\alpha\beta}$. These satisfy

$$g_{\alpha\alpha} = e, \quad g_{\beta\alpha} = g_{\alpha\beta}^{-1}, \quad g_{\alpha\gamma} = g_{\alpha\beta} g_{\beta\gamma}.$$

Warning 8.8 (Convention matters). Because we use a right group action, the order in the cocycle identity is tied to the convention $s_\beta = s_\alpha g_{\alpha\beta}$. Switching to left actions changes formulas. One should fix a convention before manipulating transition functions.

8.4 The frame bundle of a vector bundle

Definition 8.9 (Frame). If $E \rightarrow M$ has rank r , a frame at p is an ordered basis $(e_1, \dots, e_r)$ of E_p , equivalently a linear isomorphism

$$u : \mathbb{R}^r \rightarrow E_p.$$

Definition 8.10 (Frame bundle). The frame bundle $\text{Fr}(E)$ is the set of all frames in all fibers. It carries a right $GL(r, \mathbb{R})$ -action

$$u \cdot A = u \circ A,$$

and projection to the base point of the frame.

Theorem 8.11 (Frame bundle is principal). *For a rank- r vector bundle $E \rightarrow M$, $\text{Fr}(E) \rightarrow M$ is a principal $GL(r, \mathbb{R})$ -bundle.*

Proof. A bundle frame $e_1, \dots, e_r$ over U identifies every frame over $p \in U$ uniquely with a matrix $A \in GL(r)$. This gives an equivariant diffeomorphism $\text{Fr}(E)|_U \cong U \times GL(r)$. $\square$

Corollary 8.12 (Linear frame bundle of a manifold). *For an n -manifold M , $\text{Fr}(TM)$ is a principal $GL(n, \mathbb{R})$ -bundle, called the linear frame bundle of M .*

8.5 Principal bundle maps

Definition 8.13 (Principal-bundle morphism). A morphism of principal G -bundles over $f : M \rightarrow N$ is a smooth map $F : P \rightarrow Q$ satisfying $\pi_Q F = f \pi_P$ and $F(u \cdot g) = F(u) \cdot g$.

Proposition 8.14 (Equivariant maps are determined fiberwise). *If $F : P \rightarrow Q$ is equivariant and its value is known at one point of a fiber, its value on the entire fiber is forced by equivariance.*

8.6 Associated bundles

Let $\rho : G \rightarrow GL(V)$ be a finite-dimensional representation.

Definition 8.15 (Associated vector bundle). For a principal G -bundle $P \rightarrow M$, define

$$P \times_\rho V = (P \times V) / \sim, \quad (u \cdot g, v) \sim (u, \rho(g)v).$$

The projection $[u, v] \mapsto \pi(u)$ makes this a vector bundle over M .

Theorem 8.16 (Recovering a vector bundle from its frames). *If $E \rightarrow M$ has rank r and ρ is the standard representation of $GL(r)$ on $\mathbb{R}^r$, then*

$$\text{Fr}(E) \times_{GL(r)} \mathbb{R}^r \cong E, \quad [u, v] \mapsto u(v).$$

Proof. The map is well defined because $(uA, A^{-1}v)$ and (u, v) represent the same geometric vector under the consistent associated-bundle convention. Fiberwise it is a linear bijection, and local frame trivializations show smoothness. $\square$

Remark 8.17 (Convention for associated bundles). Equivalent conventions place g or g^{-1} in the relation. What matters is consistency with the chosen right action and representation convention.

8.7 Gauge changes without connections

Definition 8.18 (Gauge transformation in a trivialization). Changing a local section from s to $s' = s \cdot h$, where $h : U \rightarrow G$ is smooth, changes the local principal-bundle coordinates. This is a local gauge change.

Proposition 8.19 (Transition functions under gauge change). *If $s'_\alpha = s_\alpha h_\alpha$, then the transition functions change by*

$$g'_{\alpha\beta} = h_\alpha^{-1} g_{\alpha\beta} h_\beta.$$

Proof. Substitute the changed sections into $s'_\beta = s'_\alpha g'_{\alpha\beta}$ and use freeness of the right action. $\square$

8.8 Examples and geometric meaning

Example 8.20 (Frames on a trivial bundle). For $E = M \times \mathbb{R}^r$, the frame bundle is canonically $M \times GL(r, \mathbb{R})$.

Example 8.21 (Frame transitions from bundle transitions). If a vector bundle has transition matrices $g_{\beta\alpha}$, the induced frame-bundle transition functions are the same $GL(r)$ -valued maps, interpreted as changes of ordered basis.

Warning 8.22 (Connections come later). A principal bundle supplies fibers, symmetry, local sections, and transition functions. It does not yet choose horizontal directions or parallel transport. Those additional structures belong to VII/21 and later chapters.

8.9 Exercises

Exercise 8.23. Verify that right multiplication defines a smooth right action of G on itself.

Hint. Check identity and associativity. $\square$

Solution. The group multiplication map is smooth, $he = h$, and $(hg)k = h(gk)$. $\square$

Exercise 8.24. Show that the action on a principal-bundle fiber is free.

Hint. Use a local trivialization. $\square$

Solution. In $U \times G$, $(p, h)g = (p, h)$ implies $hg = h$, hence $g = e$. $\square$

Exercise 8.25. Show that the action on a principal-bundle fiber is transitive.

Hint. Again use $U \times G$. □

Solution. Given (p, h_1) and (p, h_2) , choose $g = h_1^{-1}h_2$. □

Exercise 8.26. Prove that a local section produces a principal-bundle trivialization.

Hint. Use $(p, g) \mapsto s(p)g$. □

Solution. Freeness and transitivity give a bijection on each fiber; local smoothness and the smooth division map in a principal chart give a diffeomorphism. □

Exercise 8.27. Recover a local section from a trivialization.

Hint. Insert the identity element. □

Solution. Define $s(p) = \Phi^{-1}(p, e)$; this is smooth and projects to p . □

Exercise 8.28. Derive the transition function relating two local sections.

Hint. Use the torsor property. □

Solution. For each p , there is a unique $g_{\alpha\beta}(p)$ with $s_\beta(p) = s_\alpha(p)g_{\alpha\beta}(p)$; local principal coordinates show smoothness. □

Exercise 8.29. Prove the principal cocycle identity.

Hint. Compare s_γ in two ways. □

Solution. From $s_\gamma = s_\beta g_{\beta\gamma} = s_\alpha g_{\alpha\beta} g_{\beta\gamma}$ and freeness, $g_{\alpha\gamma} = g_{\alpha\beta} g_{\beta\gamma}$. □

Exercise 8.30. Show that $M \times G \rightarrow M$ has the global section $p \mapsto (p, e)$.

Hint. Use the identity of G . □

Solution. The map is smooth and projects to p . □

Exercise 8.31. Show that a principal bundle with a global section is trivial.

Hint. Use the section-to-trivialization construction globally. □

Solution. The map $(p, g) \mapsto s(p)g$ is a global equivariant diffeomorphism $M \times G \rightarrow P$. □

Exercise 8.32. Show that a frame is equivalently a linear isomorphism $\mathbb{R}^r \rightarrow E_p$.

Hint. Send the standard basis to the ordered basis. □

Solution. Every ordered basis determines a unique linear isomorphism, and conversely the images of the standard basis form a frame. □

Exercise 8.33. Verify the right $GL(r)$ -action on frames is free.

Hint. Use injectivity of a frame map. □

Solution. If $uA = u$, then composing with u^{-1} gives $A = I$. □

Exercise 8.34. Verify the frame action is transitive on a fiber.

Hint. Compare two isomorphisms $u, v : \mathbb{R}^r \rightarrow E_p$. □

Solution. The unique matrix is $A = u^{-1}v$, so $v = uA$. □

Exercise 8.35. Compute $\dim \text{Fr}(E)$ for a rank- r bundle over an n -manifold.

Hint. Use local products with $GL(r)$. □

Solution. Since $\dim GL(r) = r^2$, the total dimension is $n + r^2$. □

Exercise 8.36. Show $\text{Fr}(M \times \mathbb{R}^r) \cong M \times GL(r)$.

Hint. Compare every frame to the standard frame. □

Solution. A frame is uniquely a standard frame followed by some $A \in GL(r)$, smoothly in p . □

Exercise 8.37. Explain why $\text{Fr}(TM)$ has structure group $GL(n)$.

Hint. A tangent frame is a basis of an n -dimensional vector space. □

Solution. The change from one ordered tangent basis to another is a unique element of $GL(n, \mathbb{R})$. □

Exercise 8.38. Check that a principal-bundle morphism preserves the group coordinate in local trivializations.

Hint. Use equivariance. □

Solution. Once the image of a chosen local section is known, equivariance forces $F(s(p)g) = F(s(p))g$. □

Exercise 8.39. Show the associated-bundle equivalence relation is compatible with the group action convention.

Hint. Check composition by two group elements. □

Solution. Applying g then h produces the same class as applying gh , precisely because ρ is a representation. □

Exercise 8.40. Compute the rank of $P \times_{\rho} V$.

Hint. Use a principal local trivialization. □

Solution. Over U , choosing a local section identifies the associated bundle with $U \times V$, so its rank is $\dim V$. □

Exercise 8.41. Show that the standard associated bundle of $\text{Fr}(E)$ has fiber E_p .

Hint. Evaluate a frame on a vector. □

Solution. The map $[u, v] \mapsto u(v)$ is fiberwise a well-defined linear bijection. □

Exercise 8.42. Derive the gauge-transformation law for principal transition functions.

Hint. Insert $s'_{\alpha} = s_{\alpha}h_{\alpha}$. □

Solution. Substitution into the chosen right-action convention gives $g'_{\alpha\beta} = h_{\alpha}^{-1}g_{\alpha\beta}h_{\beta}$. □

Exercise 8.43. For an abelian group G , simplify the gauge-transformation law.

Hint. Commute the factors. □

Solution. It becomes $g'_{\alpha\beta} = g_{\alpha\beta}h_{\alpha}^{-1}h_{\beta}$. □

Exercise 8.44. Show that a global frame of E is the same as a global section of $\text{Fr}(E)$.

Hint. A point of the frame bundle is itself a frame. □

Solution. A section selects smoothly one frame at every base point, exactly a global frame of E . □

Exercise 8.45. Use the previous exercise to restate the triviality criterion for vector bundles in principal-bundle language.

Hint. Apply the global-section criterion. □

Solution. E is trivial iff $\text{Fr}(E)$ has a global section, equivalently iff the frame principal bundle is trivial. □

Exercise 8.46. State what geometric datum has deliberately not been added in this chapter.

Hint. Read the final warning. □

Solution. No horizontal distribution, connection form, covariant derivative, or parallel transport has been chosen. □

8.10 Solved problem dossiers

Problem 8.47 (Legacy principal-bundle problem: torsors). Prove intrinsically that each principal-bundle fiber is a G -torsor.

Solution. A local principal trivialization identifies the fiber with G carrying right multiplication. Right multiplication is free and transitive, and these properties are invariant under equivariant diffeomorphism. □

Problem 8.48 (Legacy local-section problem). Prove the equivalence between local sections and local principal trivializations.

Solution. A section s gives $(p, g) \mapsto s(p)g$. Conversely a trivialization gives $s(p) = \Phi^{-1}(p, e)$. These constructions are inverse to one another. □

Problem 8.49 (Legacy transition-function problem). Derive transition functions from local sections and prove the cocycle condition.

Solution. Use the unique group element relating two section values in each fiber. Smoothness follows in a principal chart; comparing three sections gives the cocycle law. □

Problem 8.50 (Legacy frame-bundle problem). Construct the principal $GL(r)$ -bundle structure on $\text{Fr}(E)$.

Solution. A local frame identifies any frame uniquely with a matrix in $GL(r)$, giving $\text{Fr}(E)|_U \cong U \times GL(r)$. Composition on the right is exactly the principal action. □

Problem 8.51 (Tangent-frame dossier). Describe the frame bundle of an n -manifold directly in coordinates.

Solution. Over a coordinate chart, the coordinate tangent frame is a local section of $\text{Fr}(TM)$. Every other tangent frame is obtained by a unique $GL(n)$ matrix; chart changes induce Jacobian-valued transition functions. □

Problem 8.52 (Global-section dossier). Prove a principal G -bundle is trivial iff it has a global section.

Solution. A trivial bundle has the identity section. A global section gives the global equivariant trivialization $(p, g) \mapsto s(p)g$. $\square$

Problem 8.53 (Associated-bundle dossier). Construct the local trivializations of $P \times_{\rho} V$.

Solution. Choose a local section s . Every $u \in P_p$ is uniquely $s(p)g$; rewrite $[u, v]$ using the equivalence relation as a class represented by $[s(p), w]$. This identifies the fiber with V , smoothly over U . $\square$

Problem 8.54 (Recovering E from frames). Prove $\text{Fr}(E) \times_{GL(r)} \mathbb{R}^r \cong E$.

Solution. Evaluation $[u, v] \mapsto u(v)$ is compatible with the quotient relation, linear and bijective in every fiber, and in a local frame it is the identity trivialization. $\square$

Problem 8.55 (Gauge-change dossier). Show how changing local sections conjugates transition functions.

Solution. Write $s'_\alpha = s_\alpha h_\alpha$, impose $s'_\beta = s'_\alpha g'_{\alpha\beta}$, and cancel s_α using freeness to obtain $g'_{\alpha\beta} = h_\alpha^{-1} g_{\alpha\beta} h_\beta$. $\square$

Problem 8.56 (Equivariant-map dossier). Show an equivariant map between principal bundles is determined by its values on a local section.

Solution. Every point over U is uniquely $s(p)g$. Equivariance forces $F(s(p)g) = F(s(p))g$, so no additional choice remains. $\square$

Problem 8.57 (Vector/principal dictionary). Match vector-bundle frames, principal local sections, and vector-bundle local trivializations.

Solution. A vector local frame is a section of the frame bundle. Such a principal section trivializes the frame bundle and simultaneously identifies every vector in E_p with its coordinate column in $\mathbb{R}^r$. $\square$

Problem 8.58 (Why no preferred identity in a fiber). Explain why a principal fiber resembles G but is not canonically the group G .

Solution. Choosing one point $u \in P_p$ identifies $g \mapsto ug$ with G , but another choice translates this identification. Without a chosen section there is no canonical identity element in the fiber. $\square$

8.11 Challenge problems

Problem 8.59 (Reconstruct a principal bundle from a cocycle). Given smooth G -valued transition functions satisfying the cocycle law, construct a principal G -bundle.

Solution. Glue $U_\alpha \times G$ with the equivalence determined by the transition functions. The cocycle law makes the quotient consistent; right multiplication descends because the gluing maps are equivariant; the product pieces become local principal trivializations. $\square$

Problem 8.60 (Associated bundle functoriality). Let $T : V \rightarrow W$ intertwine two representations ρ and σ . Show it induces a vector-bundle map $P \times_{\rho} V \rightarrow P \times_{\sigma} W$.

Solution. Define $[u, v] \mapsto [u, Tv]$. If representatives are changed by g , equivariance $T\rho(g) = \sigma(g)T$ ensures the resulting class is unchanged. The map is locally $\text{id}_U \times T$, hence smooth. $\square$

Problem 8.61 (Frame bundle and bundle isomorphisms). Show a vector-bundle isomorphism $F : E \rightarrow E'$ over M induces a principal-bundle isomorphism $\text{Fr}(E) \rightarrow \text{Fr}(E')$.

Solution. Send a frame $u : \mathbb{R}^r \rightarrow E_p$ to $F_p \circ u$. This is smooth in local frames and commutes with the right $GL(r)$ -action: $F_p \circ (uA) = (F_p \circ u)A$. $\square$

Problem 8.62 (Triviality equivalence). Prove carefully that E is trivial iff its frame bundle is a trivial principal $GL(r)$ -bundle.

Solution. A global vector frame is exactly a global section of $\text{Fr}(E)$. VII/07 says a global frame trivializes E ; the principal global-section criterion says the same section trivializes $\text{Fr}(E)$. $\square$

Problem 8.63 (Why connections are additional structure). Using only local sections and transition functions, explain why no canonical horizontal complement to the vertical tangent spaces has yet appeared.

Solution. The principal-bundle axioms determine the vertical directions generated by the group action, but local gauge changes can mix any proposed complementary directions unless extra transformation-compatible data are specified. A connection is precisely such an additional choice, postponed to VII/21. $\square$

Bridge to VII/09

Principal bundles organize changes of frame. Differential forms, by contrast, are tensorial fields that can be pulled back along arbitrary smooth maps without choosing a frame. VII/09 develops alternating covariant tensors, wedge products, pullbacks, and the exterior derivative.

Chapter 9

Differential Forms

Differential forms are alternating covariant tensors that vary smoothly from point to point. They combine the cotangent bundle of VII/05 with the vector-bundle language of VII/07 and add two decisive operations: the wedge product and the exterior derivative.

$$\Omega^k(M) = \Gamma(\Lambda^k T^*M), \quad d : \Omega^k(M) \rightarrow \Omega^{k+1}(M), \quad d^2 = 0.$$

9.1 Alternating covariant tensors

Definition 9.1 (Alternating k -form at a point). A differential k -form at $p \in M$ is an alternating multilinear map

$$\omega_p : (T_p M)^k \rightarrow \mathbb{R}.$$

The vector space of such maps is $\Lambda^k T_p^* M$. By convention $\Lambda^0 T_p^* M = \mathbb{R}$.

Definition 9.2 (Smooth differential form). A smooth differential k -form is a smooth section of $\Lambda^k T^* M$. The space of these forms is denoted $\Omega^k(M)$, and $\Omega^0(M) = C^\infty(M)$.

Proposition 9.3 (Dimension). If $\dim M = n$, then

$$\dim \Lambda^k T_p^* M = \binom{n}{k},$$

and $\Lambda^k T_p^* M = 0$ for $k > n$.

9.2 Coordinate expressions

In local coordinates $x^1, \dots, x^n$, every k -form is uniquely

$$\omega = \sum_{i_1 < \dots < i_k} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

with smooth coefficient functions.

Example 9.4 (One-forms). A one-form is locally $\alpha = \sum_i a_i dx^i$. For a function f , its differential from VII/05 is the one-form

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i.$$

9.3 The wedge product

Definition 9.5 (Wedge product). For $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$, the wedge product $\alpha \wedge \beta \in \Omega^{k+\ell}(M)$ is the alternating product of the two covariant tensors.

Theorem 9.6 (Graded algebra identities). *For homogeneous forms,*

$$\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha, \quad (\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma).$$

The wedge product is bilinear over $\mathbb{R}$ and compatible with multiplication by smooth functions.

Corollary 9.7 (Odd forms square to zero). *If α has odd degree, then $\alpha \wedge \alpha = 0$.*

9.4 Pullback of forms

Definition 9.8 (Pullback). For a smooth map $F : M \rightarrow N$ and $\omega \in \Omega^k(N)$, define

$$(F^*\omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

Theorem 9.9 (Pullback identities). *Pullback is linear, preserves degree, and satisfies*

$$F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta, \quad (G \circ F)^* = F^* \circ G^*, \quad F^*(df) = d(f \circ F).$$

Example 9.10 (Polar coordinates). For $F(r, \theta) = (r \cos \theta, r \sin \theta)$,

$$F^*(dx) = \cos \theta dr - r \sin \theta d\theta, \quad F^*(dy) = \sin \theta dr + r \cos \theta d\theta,$$

so

$$F^*(dx \wedge dy) = r dr \wedge d\theta.$$

9.5 Exterior derivative

Theorem 9.11 (Exterior derivative). *There is a unique family of linear maps*

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

such that:

1. *for functions it is the ordinary differential;*
2. *$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \Omega^k(M)$;*
3. *$d^2 = 0$.*

In coordinates,

$$d\left(\sum_I a_I dx^I\right) = \sum_I da_I \wedge dx^I.$$

Proof. The coordinate formula defines d . Equality of mixed partial derivatives gives $d^2 = 0$, and a direct expansion proves the graded Leibniz rule. Naturality under coordinate changes shows the construction is intrinsic. $\square$

Example 9.12 (A one-form on $\mathbb{R}^2$). For $\alpha = P dx + Q dy$,

$$d\alpha = (Q_x - P_y) dx \wedge dy.$$

Example 9.13 (A two-form on $\mathbb{R}^3$). For

$$\beta = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy,$$

one obtains

$$d\beta = (P_x + Q_y + R_z) dx \wedge dy \wedge dz.$$

9.6 Naturality of the exterior derivative

Theorem 9.14 (Pullback commutes with d). *For every smooth $F : M \rightarrow N$,*

$$d(F^*\omega) = F^*(d\omega).$$

Proof. It holds first for functions by the chain rule, then for coordinate-generated forms by the graded Leibniz rule, and therefore locally for every form. $\square$

9.7 Closed and exact forms

Definition 9.15 (Closed and exact). A form ω is closed if $d\omega = 0$. It is exact if $\omega = d\eta$ for some form η of one lower degree.

Proposition 9.16 (Exact implies closed). *Every exact form is closed.*

Proof. If $\omega = d\eta$, then $d\omega = d^2\eta = 0$. $\square$

Theorem 9.17 (Poincaré lemma on star-shaped domains). *If $U \subset \mathbb{R}^n$ is star-shaped, then every closed k -form on U with $k \geq 1$ is exact.*

Remark 9.18. The theorem is local in spirit. Global closed forms need not be exact; the gap between the two notions becomes topological and leads later to de Rham cohomology.

9.8 Top-degree forms

Proposition 9.19 (Top-degree local form). *On an n -manifold, every n -form is locally*

$$\omega = f dx^1 \wedge \cdots \wedge dx^n.$$

Under a coordinate change $y = y(x)$, the coordinate volume element transforms by the Jacobian determinant.

Warning 9.20 (Integration is not yet defined). Top-degree forms are the correct objects to integrate on oriented manifolds, but orientation and integration are developed in VII/10. The present chapter uses only their algebraic and differential properties.

9.9 Exercises

Exercise 9.21. Show that a 0-form is exactly a smooth function.

Hint. Use $\Lambda^0 T_p^*M = \mathbb{R}$. $\square$

Solution. A smooth section of the trivial rank-one bundle $\Lambda^0 T^*M$ is precisely a smooth real-valued function. $\square$

Exercise 9.22. Compute $\dim \Lambda^2(\mathbb{R}^5)^*$.

Hint. Use the binomial coefficient. $\square$

Solution. The dimension is $\binom{5}{2} = 10$. $\square$

Exercise 9.23. Show that every k -form vanishes identically when $k > \dim M$.

Hint. Alternating tensors vanish on linearly dependent arguments. $\square$

Solution. Any k tangent vectors in an n -dimensional space with $k > n$ are linearly dependent, so every alternating k -linear map is zero. $\square$

Exercise 9.24. Show $dx^i \wedge dx^i = 0$.

Hint. Use graded commutativity for degree one. $\square$

Solution. $dx^i \wedge dx^i = -dx^i \wedge dx^i$, hence it vanishes over $\mathbb{R}$. $\square$

Exercise 9.25. Compute $dy \wedge dx$ in terms of $dx \wedge dy$.

Hint. Swap two one-forms. $\square$

Solution. $dy \wedge dx = -dx \wedge dy$. $\square$

Exercise 9.26. Expand $(a dx + b dy) \wedge (c dx + d dy)$.

Hint. Discard repeated factors. $\square$

Solution. The result is $(ad - bc) dx \wedge dy$. $\square$

Exercise 9.27. Show the wedge product of a 2-form and a 1-form commutes.

Hint. Evaluate $(-1)^{2 \cdot 1}$. $\square$

Solution. The sign is $+1$, so $\alpha \wedge \beta = \beta \wedge \alpha$. $\square$

Exercise 9.28. Show pullback preserves wedge products.

Hint. Apply both sides to tangent vectors. $\square$

Solution. Both sides evaluate by first applying dF to each tangent argument and then alternating the same tensor product, so they agree. $\square$

Exercise 9.29. Compute the pullback of dx under $F(t) = (t^2, t^3)$.

Hint. Differentiate the first coordinate. $\square$

Solution. $F^*(dx) = d(t^2) = 2t dt$. $\square$

Exercise 9.30. Compute the pullback of dy for the same map.

Hint. Differentiate the second coordinate. $\square$

Solution. $F^*(dy) = 3t^2 dt$. $\square$

Exercise 9.31. Explain why the pullback of every 2-form on $\mathbb{R}^2$ to a curve is zero.

Hint. A one-dimensional manifold has no nonzero two-forms. $\square$

Solution. The pullback is a two-form on the one-dimensional domain, hence vanishes. $\square$

Exercise 9.32. Verify $d(fg) = f dg + g df$.

Hint. Use the degree-zero Leibniz rule. $\square$

Solution. The graded Leibniz rule has sign $+1$ for degree zero and yields the ordinary product rule. $\square$

Exercise 9.33. Compute $d(P dx)$ on $\mathbb{R}^2$.

Hint. Use $dP \wedge dx$. □

Solution. Since $dP = P_x dx + P_y dy$, $d(Pdx) = -P_y dx \wedge dy$. □

Exercise 9.34. Compute $d(Q dy)$ on $\mathbb{R}^2$.

Hint. Use $dQ \wedge dy$. □

Solution. Only $Q_x dx \wedge dy$ survives, so $d(Qdy) = Q_x dx \wedge dy$. □

Exercise 9.35. Verify directly that $d(df) = 0$ in coordinates.

Hint. Use equality of mixed partials. □

Solution. The coefficient of $dx^i \wedge dx^j$ is $f_{x^i x^j} - f_{x^j x^i} = 0$. □

Exercise 9.36. Show every exact form is closed.

Hint. Apply d once more. □

Solution. If $\omega = d\eta$, then $d\omega = d^2\eta = 0$. □

Exercise 9.37. Give a closed one-form on $\mathbb{R}^2$ and exhibit a potential.

Hint. Take $x dx + y dy$. □

Solution. It equals $d(\frac{1}{2}(x^2 + y^2))$, hence is exact and therefore closed. □

Exercise 9.38. Determine when $Pdx + Qdy$ is closed.

Hint. Use the coordinate formula for $d\alpha$. □

Solution. It is closed exactly when $Q_x - P_y = 0$. □

Exercise 9.39. Show that pullback sends closed forms to closed forms.

Hint. Use naturality of d . □

Solution. If $d\omega = 0$, then $d(F^*\omega) = F^*(d\omega) = 0$. □

Exercise 9.40. Show that pullback sends exact forms to exact forms.

Hint. Pull back a primitive. □

Solution. If $\omega = d\eta$, then $F^*\omega = F^*d\eta = dF^*\eta$. □

Exercise 9.41. Compute $F^*(dx \wedge dy)$ for polar coordinates.

Hint. Use the pulled-back one-forms and wedge them. □

Solution. Expanding gives $r dr \wedge d\theta$. □

Exercise 9.42. Show that every top-degree form on an n -manifold is closed.

Hint. Its exterior derivative has degree $n + 1$. □

Solution. There are no nonzero $(n + 1)$ -forms, so $d\omega = 0$. □

Exercise 9.43. Explain why top-degree closedness does not imply global exactness.

Hint. Recall the warning after the Poincaré lemma. □

Solution. Closedness is local/differential, while exactness may be obstructed globally; de Rham cohomology measures this difference. $\square$

Exercise 9.44. State the next structural ingredient needed before integrating top-degree forms globally.

Hint. Look at VII/10. $\square$

Solution. One needs orientation, together with a compatible definition of integration on manifolds. $\square$

9.10 Solved problem dossiers

Problem 9.45 (Legacy differential-form problem: coordinate basis). Show that the wedge monomials $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ with increasing indices form a local frame of $\Lambda^k T^*M$.

Solution. At each point they are the standard basis of the exterior power of the cotangent space. Their number is $\binom{n}{k}$, and smooth dependence on the coordinate coframe makes them a local bundle frame. $\square$

Problem 9.46 (Legacy wedge-product problem). Derive graded commutativity from alternation.

Solution. Exchanging a block of k arguments with a block of ℓ arguments requires $k\ell$ transpositions, producing the sign $(-1)^{k\ell}$. $\square$

Problem 9.47 (Legacy pullback problem). Prove pullback is functorial.

Solution. For $F : M \rightarrow N$ and $G : N \rightarrow P$, the chain rule gives $d(G \circ F) = dG \circ dF$. Substituting this into the pointwise definition of pullback yields $(G \circ F)^* = F^*G^*$. $\square$

Problem 9.48 (Legacy exterior-derivative problem). Starting from the coordinate formula, prove the graded Leibniz rule.

Solution. Differentiate the coefficient products and use that moving a one-form da past a degree- k factor contributes the sign $(-1)^k$. The two groups of terms are exactly $d\alpha \wedge \beta$ and $(-1)^k \alpha \wedge d\beta$. $\square$

Problem 9.49 (Legacy $d^2 = 0$ problem). Prove $d^2 = 0$ in coordinates.

Solution. After applying d twice, coefficients involve symmetric second derivatives multiplied by antisymmetric wedge products. Pairing indices i, j cancels terms because mixed partials agree. $\square$

Problem 9.50 (Polar-coordinate dossier). Compute the pullback of the Euclidean area form under polar coordinates.

Solution. Pull back dx and dy , wedge the results, and cancel the $dr \wedge dr$ and $d\theta \wedge d\theta$ terms. The surviving coefficient is $r(\cos^2 \theta + \sin^2 \theta) = r$. $\square$

Problem 9.51 (Closed one-form dossier). For $\alpha = Pdx + Qdy$, derive the closedness condition and relate it locally to a potential.

Solution. The derivative is $(Q_x - P_y)dx \wedge dy$, so closedness is $Q_x = P_y$. On a star-shaped domain, the Poincaré lemma then gives $\alpha = df$ for some potential f . $\square$

Problem 9.52 (Three-dimensional two-form dossier). Derive the divergence-type formula for $d\beta$ when $\beta = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$.

Solution. Differentiate each coefficient. Only the term whose differential supplies the missing coordinate one-form survives in each summand; after reordering signs, the coefficient is $P_x + Q_y + R_z$. $\square$

Problem 9.53 (Naturality dossier). Prove $dF^* = F^*d$ for arbitrary forms from the function case.

Solution. Locally every form is a sum of functions times wedges of coordinate differentials. Pullback respects wedges, and both operators obey the graded Leibniz rule. The function case follows from the chain rule, so the identity extends to all local forms. $\square$

Problem 9.54 (Exact-versus-closed dossier). Explain the logical relation between exactness, closedness, and topology.

Solution. The identity $d^2 = 0$ makes exact forms closed on every manifold. The converse is true locally on contractible/star-shaped charts but may fail globally, so the quotient of closed forms by exact forms records global information. $\square$

Problem 9.55 (Top-form dossier). Show how the coefficient of a top-degree form changes under coordinates.

Solution. Wedge the transformed one-forms $dy^i = \sum_j (\partial y^i / \partial x^j) dx^j$. Multilinearity and alternation give the determinant of the Jacobian matrix. $\square$

Problem 9.56 (Restriction dossier). If $i : S \hookrightarrow M$ is an embedded submanifold, explain how ambient forms restrict to S .

Solution. Restriction is simply pullback $i^*\omega$. Pointwise it evaluates ω only on tangent vectors lying in $T_p S \subset T_p M$, and naturality gives $d(i^*\omega) = i^*(d\omega)$. $\square$

9.11 Challenge problems

Problem 9.57 (Coordinate independence of d). Prove that the coordinate formula for the exterior derivative defines an intrinsic operator.

Solution. On overlaps, pullback by the coordinate-change diffeomorphism commutes with the coordinate-defined derivative because the identity holds on functions and extends via wedges and the graded Leibniz rule. Thus the two coordinate formulas agree on overlaps and glue to a global operator. $\square$

Problem 9.58 (Poincaré homotopy operator). On a star-shaped domain, outline the homotopy operator proving that every closed positive-degree form is exact.

Solution. Use the radial homotopy $H_t(x) = tx$ and radial vector field $R = \sum x^i \partial_{x^i}$. Define $K\omega = \int_0^1 t^{k-1} \iota_R \omega(tx) dt$ for a k -form. The homotopy identity $dK + Kd = \text{id} - H_0^*$ implies $\omega = dK\omega$ when $k > 0$ and $d\omega = 0$. $\square$

Problem 9.59 (Exterior algebra under a linear map). For a linear map $A : V \rightarrow W$, prove that pullback $A^* : \Lambda^\bullet W^* \rightarrow \Lambda^\bullet V^*$ is a graded-algebra homomorphism.

Solution. Define $A^*\omega(v_1, \dots, v_k) = \omega(Av_1, \dots, Av_k)$. Multilinearity and alternation are preserved, and applying this definition to the alternation formula for wedges gives $A^*(\alpha \wedge \beta) = A^*\alpha \wedge A^*\beta$. $\square$

Problem 9.60 (A global closed non-exact form preview). On S^1 , explain why the angular one-form is locally $d\theta$ but cannot be globally the differential of a single-valued smooth angle function.

Solution. Local angle coordinates differ by additive constants, so their differentials glue to a global closed one-form. A global potential would provide a single-valued smooth angle whose change around the circle is zero, contradicting the nontrivial winding. A complete cohomological treatment comes later. $\square$

Problem 9.61 (Jacobian determinant and top forms). Prove directly that $dy^1 \wedge \cdots \wedge dy^n = \det(\partial y / \partial x) dx^1 \wedge \cdots \wedge dx^n$.

Solution. Expand each dy^i in the dx^j basis. Terms with repeated indices vanish. The remaining terms are indexed by permutations, and their wedge signs reproduce exactly the Leibniz formula for the determinant. $\square$

Bridge to VII/10

Top-degree forms already transform by Jacobian determinants, so the algebra is ready for integration. VII/10 introduces orientations, oriented atlases, partitions of unity where needed, and the coordinate-invariant integral of compactly supported top forms.

Chapter 10

Orientation and Integration

Orientation records a coherent choice of handedness on tangent spaces. Once such a choice is made, compactly supported top-degree differential forms can be integrated intrinsically. The two central principles are

orientation = coherent sign of frames, $\int_M \omega$ = coordinate integrals glued by a partition of unity.

10.1 Orientation of vector spaces

Definition 10.1 (Orientation of a vector space). Let V be an n -dimensional real vector space. Two ordered bases are equivalent when the determinant of the change-of-basis matrix is positive. An *orientation* of V is one of the two equivalence classes.

Proposition 10.2 (Top exterior power criterion). A nonzero element $\Omega \in \Lambda^n V^*$ determines an orientation by declaring a basis $(v_1, \dots, v_n)$ positive exactly when $\Omega(v_1, \dots, v_n) > 0$.

Proof. Under a basis change A , an alternating n -form is multiplied by $\det A$. Hence the sign changes precisely under orientation-reversing basis changes. $\square$

10.2 Oriented manifolds and oriented atlases

Definition 10.3 (Oriented atlas). An atlas on an n -manifold M is oriented if every transition map has positive Jacobian determinant wherever defined. An *orientation* of M is a maximal oriented atlas.

Theorem 10.4 (Tangent-space interpretation). An orientation of M is equivalent to a continuously varying choice of orientation of every tangent space $T_p M$.

Proof. An oriented chart transports the standard orientation of $\mathbb{R}^n$ to each tangent space. Positive transition Jacobians make the result chart-independent. Conversely, charts whose coordinate frames are positive form an oriented atlas. $\square$

Definition 10.5 (Orientation-preserving map). A local diffeomorphism $F : M \rightarrow N$ between oriented n -manifolds is orientation preserving at p when dF_p sends positive bases to positive bases; equivalently, its Jacobian in oriented charts has positive determinant.

Example 10.6 (Euclidean space). The standard ordered basis gives the standard orientation of $\mathbb{R}^n$. The coordinate form $dx^1 \wedge \dots \wedge dx^n$ is positive for this orientation.

10.3 Orientability and orientation forms

Definition 10.7 (Orientation form). A nowhere-vanishing smooth n -form on an n -manifold is called an orientation form or volume form in the purely differential-topological sense.

Theorem 10.8 (Orientability criterion). *An n -manifold is orientable if and only if it admits a nowhere-vanishing smooth n -form.*

Proof. A nowhere-zero top form chooses the positive bases pointwise. Conversely, choose positive coordinate top forms on an oriented atlas and glue them with a smooth partition of unity; positivity prevents cancellation to zero. $\square$

Proposition 10.9 (Connected manifolds have two orientations). *If M is connected and orientable, then it has exactly two orientations.*

Proof. Relative to one orientation, another assigns a sign ± 1 locally. This sign is locally constant and hence constant on a connected manifold. $\square$

Example 10.10 (Sphere). The sphere $S^n \subset \mathbb{R}^{n+1}$ is orientable. If $N(p) = p$ is the outward normal and Ω is the standard $(n+1)$ -form, then

$$\omega_p(v_1, \dots, v_n) = \Omega_p(N(p), v_1, \dots, v_n)$$

is a nowhere-vanishing n -form on S^n .

Warning 10.11 (Orientability is global). Every manifold is locally orientable. Nonorientability is the failure to make the local signs globally consistent; the Möbius band is the basic example.

10.4 Compact support and partitions of unity

Definition 10.12 (Compact support). For a differential form ω , its support is the closure of the set on which it is nonzero. We write $\Omega_c^k(M)$ for compactly supported smooth k -forms.

Theorem 10.13 (Partition of unity). *Every smooth manifold admits smooth partitions of unity subordinate to any open cover.*

Remark 10.14 (Why partitions of unity enter integration). A compactly supported top form need not lie in one chart. A partition of unity decomposes it into finitely many chart-supported pieces, each of which can be integrated by ordinary multivariable calculus.

10.5 Integration in an oriented chart

Definition 10.15 (Integral of a chart-supported top form). Let (U, x) be an oriented chart and let $\omega \in \Omega_c^n(M)$ have support contained in U . Write

$$(x^{-1})^*\omega = f(u) du^1 \wedge \dots \wedge du^n.$$

Define

$$\int_M \omega := \int_{x(U)} f(u) du.$$

Theorem 10.16 (Chart independence). *The preceding integral is independent of the oriented chart containing the support.*

Proof. For an orientation-preserving coordinate change $v = v(u)$, the coefficient of the pulled-back top form gains the factor $\det(Dv/Du)$, while the Euclidean change-of-variables theorem contributes exactly the same positive Jacobian factor. $\square$

10.6 Global integration

Definition 10.17 (Integral on an oriented manifold). Choose a partition of unity (ρ_i) subordinate to oriented coordinate neighborhoods. For $\omega \in \Omega_c^n(M)$ define

$$\int_M \omega := \sum_i \int_M \rho_i \omega.$$

Only finitely many terms meet $\text{supp } \omega$.

Theorem 10.18 (Well-definedness). *The global integral is independent of the subordinate atlas and partition of unity.*

Proof. Refine two decompositions simultaneously. On each overlap, chart independence applies; summing and using $\sum_i \rho_i = 1$ recovers the same integral. $\square$

Proposition 10.19 (Linearity and positivity). *Integration is linear. If $\omega = f\Omega$ where Ω is a positive orientation form and $f \geq 0$, then $\int_M \omega \geq 0$ whenever the integral is defined.*

Theorem 10.20 (Change of variables on manifolds). *If $F : M \rightarrow N$ is an orientation-preserving diffeomorphism of oriented n -manifolds and $\omega \in \Omega_c^n(N)$, then*

$$\int_M F^* \omega = \int_N \omega.$$

If F reverses orientation, the left-hand side equals $-\int_N \omega$.

Example 10.21 (Polar coordinates as pullback). For $F(r, \theta) = (r \cos \theta, r \sin \theta)$ on $r > 0$,

$$F^*(dx \wedge dy) = r dr \wedge d\theta.$$

Thus the familiar Jacobian r is already encoded by pullback of the Euclidean area form.

10.7 Integration on products

Proposition 10.22 (Product orientation). *If M^m and N^n are oriented, the ordered concatenation of positive bases of $T_p M$ and $T_q N$ defines the product orientation on $M \times N$.*

Proposition 10.23 (Separated product forms). *For compactly supported top forms $\alpha \in \Omega_c^m(M)$ and $\beta \in \Omega_c^n(N)$,*

$$\int_{M \times N} \pi_M^* \alpha \wedge \pi_N^* \beta = \left(\int_M \alpha \right) \left(\int_N \beta \right).$$

Proof. Reduce by partitions of unity to product charts and apply Fubini's theorem. $\square$

10.8 Exercises

Exercise 10.24. Show that positive-determinant change of basis is an equivalence relation on ordered bases.

Hint. Use multiplicativity of determinant. $\square$

Solution. Reflexivity uses determinant 1, symmetry uses $\det(A^{-1}) = (\det A)^{-1} > 0$, and transitivity uses $\det(AB) = \det A \det B > 0$. $\square$

Exercise 10.25. How many orientations does a nonzero finite-dimensional real vector space have?

Hint. There are two determinant signs. □

Solution. Exactly two: the two equivalence classes of ordered bases under positive determinant changes. □

Exercise 10.26. Show that a nonzero top form determines an orientation.

Hint. Track its value under a basis change. □

Solution. If $w_j = \sum_i A_{ij}v_i$, then $\Omega(w_1, \dots, w_n) = \det(A)\Omega(v_1, \dots, v_n)$, so positivity is preserved exactly by positive determinant changes. □

Exercise 10.27. Determine whether the map $(x, y) \mapsto (y, x)$ preserves the standard orientation of $\mathbb{R}^2$.

Hint. Compute its determinant. □

Solution. Its matrix has determinant -1 , so it reverses orientation. □

Exercise 10.28. Show that an oriented atlas induces an orientation on every tangent space.

Hint. Use coordinate frames. □

Solution. Declare the coordinate frame of any oriented chart positive. Transition derivatives have positive determinant, so the declaration is independent of the chosen chart. □

Exercise 10.29. Show that orientation preservation of a local diffeomorphism is coordinate independent.

Hint. Conjugate its Jacobian by oriented transition maps. □

Solution. Changing source and target oriented charts multiplies the determinant by positive transition determinants, so its sign does not change. □

Exercise 10.30. Show that $dx^1 \wedge \dots \wedge dx^n$ is positive for the standard orientation.

Hint. Evaluate on the standard basis. □

Solution. Its value on $(e_1, \dots, e_n)$ is $1 > 0$. □

Exercise 10.31. Explain why a nowhere-zero top form orients a manifold.

Hint. Use its sign on tangent bases. □

Solution. At each point declare bases on which the form is positive to be positive. Smooth nonvanishing prevents the sign convention from changing discontinuously. □

Exercise 10.32. Why does multiplying an orientation form by a positive function preserve orientation?

Hint. Check signs pointwise. □

Solution. If $f > 0$, then $(f\Omega)(v_1, \dots, v_n)$ has the same sign as $\Omega(v_1, \dots, v_n)$. □

Exercise 10.33. Show that negating an orientation form reverses orientation.

Hint. All signs reverse. □

Solution. $(-\Omega)(v_1, \dots, v_n) = -\Omega(v_1, \dots, v_n)$, so positive and negative bases are interchanged. □

Exercise 10.34. Show that a connected orientable manifold has exactly two orientations.

Hint. Compare any two orientations by a sign function. □

Solution. The relative sign is locally constant, hence constant by connectedness; only the original and its reverse occur. □

Exercise 10.35. Give the outward orientation of $S^{n-1} \subset \mathbb{R}^n$.

Hint. Insert the outward normal first. □

Solution. A basis $(v_1, \dots, v_{n-1})$ is positive when $(p, v_1, \dots, v_{n-1})$ is positive in $\mathbb{R}^n$. □

Exercise 10.36. Define the support of a differential form and explain compact support.

Hint. Take the closure of its nonzero locus. □

Solution. $\text{supp } \omega = \overline{\{p : \omega_p \neq 0\}}$; compact support means this closed set is compact. □

Exercise 10.37. Why is the partition-of-unity sum defining an integral finite for a compactly supported form?

Hint. Use local finiteness and compactness. □

Solution. A compact set meets only finitely many members of a locally finite family, so only finitely many $\rho_i \omega$ are nonzero. □

Exercise 10.38. Show that the chart-supported integral is unchanged under an oriented coordinate change.

Hint. Apply the Euclidean change-of-variables theorem. □

Solution. The top-form coefficient transforms by the positive Jacobian determinant, exactly matching the Euclidean substitution factor. □

Exercise 10.39. What happens to the integral under an orientation-reversing coordinate change?

Hint. The determinant has negative sign. □

Solution. The alternating top form contributes the signed determinant, while Euclidean measure uses its absolute value, producing an overall minus sign. □

Exercise 10.40. Prove linearity of integration.

Hint. Use linearity chartwise and in the finite partition sum. □

Solution. Each coordinate integral is linear, and a finite sum of linear maps is linear. □

Exercise 10.41. If $f \geq 0$ and Ω is positive, show $\int f\Omega \geq 0$ for compact support.

Hint. Use positive oriented charts. □

Solution. In every oriented chart the coefficient of $f\Omega$ is nonnegative, so each local Euclidean integral and hence their sum is nonnegative. □

Exercise 10.42. Prove the manifold change-of-variables formula for an orientation-preserving diffeomorphism.

Hint. Reduce to chart-supported pieces. □

Solution. Use a partition of unity on the target, pull it back to the source, apply Euclidean change of variables on each paired chart, then sum. □

Exercise 10.43. Compute $F^*(dx \wedge dy)$ for $F(r, \theta) = (r \cos \theta, r \sin \theta)$.

Hint. Differentiate x and y . □

Solution. $dx = \cos \theta dr - r \sin \theta d\theta$ and $dy = \sin \theta dr + r \cos \theta d\theta$, hence $dx \wedge dy = r dr \wedge d\theta$. □

Exercise 10.44. Show that the product of positive bases gives an orientation on $M \times N$.

Hint. Use product tangent splitting. □

Solution. Under $T_{(p,q)}(M \times N) \cong T_p M \oplus T_q N$, concatenate positive bases. Positive changes in each factor have positive product determinant. □

Exercise 10.45. What sign appears when the factors of an oriented product $M^m \times N^n$ are swapped?

Hint. Count transpositions. □

Solution. Moving m basis vectors past n basis vectors gives the sign $(-1)^{mn}$. □

Exercise 10.46. Verify the separated-product integral formula in Euclidean product charts.

Hint. Apply Fubini. □

Solution. The wedge product has coefficient $f(x)g(y)$ times the product coordinate volume form, so Fubini gives the product of the two integrals. □

Exercise 10.47. Explain why integration of top forms needs an orientation but integration of densities does not.

Hint. Top forms change sign under orientation reversal. □

Solution. A top form transforms by the signed determinant, so a global sign convention is needed. A density transforms by the absolute determinant and therefore carries no sign ambiguity. □

10.9 Solved problem dossiers

Problem 10.48 (Legacy orientation problem: positive atlases). Show that the positive-Jacobian condition on chart transitions is exactly what is needed to make coordinate-frame orientations consistent.

Solution. Coordinate frames on an overlap are related by the transition derivative. Their orientations agree exactly when its determinant is positive. Thus an oriented atlas is precisely a compatible system of tangent-space orientations. □

Problem 10.49 (Legacy orientation-form problem). Prove the equivalence between orientability and the existence of a nowhere-vanishing top form.

Solution. A nonzero top form selects positive tangent bases. Conversely, on an oriented atlas use the positive coordinate top forms and combine them with a partition of unity. Every summand is nonnegative relative to the chosen orientation and some summand is positive at each point, so the sum never vanishes. $\square$

Problem 10.50 (Legacy integration problem: chart independence). Prove that the integral of a compactly supported top form contained in one oriented chart is independent of that chart.

Solution. On an overlap write the form in both coordinate systems. The coefficients are related by the signed Jacobian determinant. Oriented transitions have positive determinant, so the Euclidean change-of-variables theorem shows the two integrals agree. $\square$

Problem 10.51 (Partition-of-unity globalization). Derive a global integral from chart-supported integrals and prove independence of the partition.

Solution. Decompose ω by a subordinate partition. Given two partitions, insert the common refinement $\rho_i\sigma_j$. Each refined term lies in a chart overlap, where chart independence holds. Summing first in either index gives the two original definitions. $\square$

Problem 10.52 (Change of variables). Prove $\int_M F^*\omega = \int_N \omega$ for an orientation-preserving diffeomorphism.

Solution. Choose a partition of unity subordinate to target charts. Its pullback is a partition on the source. Apply chartwise change of variables to each pulled-back summand and add. $\square$

Problem 10.53 (Outward orientation of the sphere). Construct an explicit orientation form on S^n from the ambient Euclidean form.

Solution. Let $N(p) = p$ and define $\omega = \iota_N(dx^1 \wedge \cdots \wedge dx^{n+1})|_{S^n}$. If $(v_1, \dots, v_n)$ spans $T_p S^n$, then $(N, v_1, \dots, v_n)$ is a basis of the ambient space, so ω never vanishes. $\square$

Problem 10.54 (The Möbius obstruction). Explain why the Möbius band cannot carry a global orientation.

Solution. Transport a local positive frame once around the central circle. The gluing reverses the transverse coordinate, returning the frame with opposite orientation. A globally consistent sign choice would therefore equal its own negative. $\square$

Problem 10.55 (Polar-coordinate Jacobian). Recover the polar area factor from differential forms.

Solution. Pull back $dx \wedge dy$ under $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$. The wedge computation gives $r dr \wedge d\theta$, so the Jacobian factor is intrinsic pullback data. $\square$

Problem 10.56 (Orientation of products). Prove the product orientation is natural and compute the sign of the flip map.

Solution. Concatenate positive bases of the factors. The flip moves m vectors past n vectors, hence changes orientation by $(-1)^{mn}$. $\square$

Problem 10.57 (Integration over a circle). Using the standard counterclockwise orientation, compute $\int_{S^1} (-y dx + x dy)$.

Solution. Parametrize by $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$. Then $\gamma^*(-y dx + x dy) = dt$, so the integral is 2π . $\square$

Problem 10.58 (Orientation-reversing symmetry). Suppose an oriented compact manifold admits an orientation-reversing diffeomorphism F and an invariant top form $F^*\omega = \omega$. Show $\int_M \omega = 0$.

Solution. Change of variables for an orientation-reversing diffeomorphism gives $\int_M F^*\omega = -\int_M \omega$. Invariance makes the left side equal to the original integral, so it must vanish. $\square$

Problem 10.59 (Top forms versus densities). Compare the transformation law of an n -form and a density and explain the role of orientation.

Solution. Top forms use $\det D\phi$ and therefore carry a sign. Densities use $|\det D\phi|$. Orientation chooses a coherent sign so top forms can be integrated globally as signed quantities. $\square$

10.10 Challenge problems

Problem 10.60 (Orientation double cover). Show that every connected manifold M has a canonical two-sheeted orientation cover whose points are pairs (p, o_p) with o_p an orientation of $T_p M$.

Solution. Use oriented local charts to trivialize the two choices over each coordinate neighborhood. Transition functions either preserve or swap the two sheets according to the sign of their Jacobian. This defines a smooth two-sheeted cover. A global orientation is exactly a continuous section of this cover. $\square$

Problem 10.61 (Orientability of real projective space). Determine for which n the manifold $\mathbb{RP}^n$ is orientable.

Solution. Use the antipodal quotient $S^n \rightarrow \mathbb{RP}^n$. The antipodal map on S^n has degree $(-1)^{n+1}$ and preserves the sphere orientation exactly when n is odd. Hence $\mathbb{RP}^n$ is orientable exactly for odd n . $\square$

Problem 10.62 (Integration and degree preview). Let $F : M \rightarrow N$ be an orientation-preserving d -sheeted covering of compact connected oriented n -manifolds. Show $\int_M F^*\omega = d \int_N \omega$.

Solution. Cover N by evenly covered coordinate neighborhoods and choose a subordinate partition of unity. Each local piece has d disjoint lifts, and change of variables gives the same integral on each lift. Summing yields the factor d . $\square$

Problem 10.63 (Compactly supported integral without global compactness). Explain why $\int_M \omega$ is still well defined when M is noncompact but ω has compact support.

Solution. Only a compact neighborhood of $\text{supp } \omega$ matters. A locally finite partition of unity intersects that compact set in finitely many terms, so the global definition reduces to a finite sum of ordinary chart integrals. $\square$

Problem 10.64 (Product orientation and Fubini sign). Let α be an m -form on M^m and β an n -form on N^n . Compare the integral of $\pi_M^*\alpha \wedge \pi_N^*\beta$ with the corresponding form after swapping factors.

Solution. The flip map exchanges the two factor blocks and changes orientation by $(-1)^{mn}$. Graded commutativity also gives $\beta \wedge \alpha = (-1)^{mn}\alpha \wedge \beta$. The two signs compensate exactly in the natural change-of-variables comparison. $\square$

Bridge to VII/11

Orientation makes signed integration possible. VII/11 identifies the precise compatibility between this integral and the exterior derivative: the integral of $d\omega$ over a manifold is the integral of ω over its oriented boundary.

Chapter 11

Stokes' Theorem

Stokes' theorem is the global statement that exterior differentiation and oriented boundary are dual operations:

$$\boxed{\int_M d\omega = \int_{\partial M} \omega.}$$

It contains the fundamental theorem of calculus, Green's theorem, the classical Stokes theorem in $\mathbb{R}^3$, and the divergence theorem as coordinate expressions of one geometric identity.

11.1 Manifolds with boundary

Definition 11.1 (Smooth manifold with boundary). An n -manifold with boundary is locally modeled on the half-space

$$\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}.$$

Points mapping to $x^n = 0$ form the boundary ∂M ; the remaining points form the interior M° .

Theorem 11.2 (Boundary is a manifold). *If M is a smooth n -manifold with boundary, then ∂M is a smooth $(n-1)$ -manifold without boundary.*

Definition 11.3 (Outward-normal-first orientation). Let M be oriented. A basis $(v_1, \dots, v_{n-1})$ of $T_p \partial M$ is positive when $(\nu, v_1, \dots, v_{n-1})$ is a positive basis of $T_p M$ for any outward-pointing transverse vector ν .

Example 11.4 (Interval boundary signs). For the standard orientation of $[a, b]$, the induced orientation on its 0-dimensional boundary is

$$\partial[a, b] = \{b\} - \{a\}.$$

11.2 Stokes' theorem

Theorem 11.5 (Stokes). *Let M be an oriented smooth n -manifold with boundary and let $\omega \in \Omega_c^{n-1}(M)$. Then*

$$\int_M d\omega = \int_{\partial M} \iota^* \omega,$$

where $\iota : \partial M \hookrightarrow M$ is inclusion and the boundary carries the outward-normal-first orientation.

Remark 11.6 (Boundaryless case). If $\partial M = \emptyset$, then every compactly supported exact top form has integral zero:

$$\int_M d\omega = 0.$$

11.3 Why the theorem is true

Proposition 11.7 (Local half-space calculation). *For a compactly supported $(n-1)$ -form on $\mathbb{H}^n$, Stokes reduces to the one-variable fundamental theorem of calculus in the normal coordinate.*

Proof. Write $\omega = \sum_i (-1)^{i-1} f_i dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n$. Then

$$d\omega = \left(\sum_i \partial_i f_i \right) dx^1 \wedge \cdots \wedge dx^n.$$

All tangential derivative integrals vanish by compact support. The x^n derivative contributes the boundary value at $x^n = 0$ with the sign fixed by the outward orientation. $\square$

Proof of Stokes' theorem. Choose a partition of unity subordinate to interior charts and boundary half-space charts. Apply the Euclidean compact-support calculation to each localized form. Interior pieces have no boundary contribution; boundary pieces contribute exactly the pullback to ∂M . Summing gives the theorem. $\square$

11.4 The fundamental theorem of calculus

Corollary 11.8 (Fundamental theorem). *For a smooth function f on $[a, b]$,*

$$\int_{[a,b]} df = f(b) - f(a).$$

11.5 Green's theorem

Corollary 11.9 (Green). *For a positively oriented planar region D with smooth boundary and smooth functions P, Q ,*

$$\oint_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

Proof. Apply Stokes to $\omega = P dx + Q dy$, for which

$$d\omega = (Q_x - P_y) dx \wedge dy.$$

$\square$

11.6 Classical Stokes in three dimensions

Corollary 11.10 (Curl theorem). *Let $S \subset \mathbb{R}^3$ be an oriented smooth surface with boundary and $F = (P, Q, R)$. For*

$$\omega = P dx + Q dy + R dz,$$

Stokes' theorem becomes

$$\int_{\partial S} F \cdot d\mathbf{r} = \int_S (\nabla \times F) \cdot \mathbf{n} dS.$$

11.7 Divergence theorem

Corollary 11.11 (Divergence theorem). *For a compact domain $M \subset \mathbb{R}^n$ with smooth boundary and a vector field $X = \sum_i X^i \partial_i$,*

$$\int_M \operatorname{div} X \, dV = \int_{\partial M} X \cdot \nu \, dS.$$

Proof. Let $dV = dx^1 \wedge \cdots \wedge dx^n$ and set $\omega = \iota_X dV$. Then

$$d\omega = (\operatorname{div} X) dV,$$

and the boundary pullback is the flux form. □

11.8 Consequences for exact and closed forms

Definition 11.12 (Closed and exact). A differential form α is closed if $d\alpha = 0$ and exact if $\alpha = d\beta$ for some form β . Since $d^2 = 0$, every exact form is closed.

Corollary 11.13 (Exact forms on cycles). *If C is a compact oriented k -manifold without boundary and $\alpha = d\beta$ is an exact k -form defined near C , then*

$$\int_C \alpha = 0.$$

Corollary 11.14 (Boundary of a boundary). *For an oriented manifold with corners treated facewise, codimension-two boundary contributions cancel in pairs, reflecting the algebraic identity $\partial^2 = 0$ and the differential identity $d^2 = 0$.*

11.9 Boundary orientation checks

Example 11.15 (Unit disk). The standard orientation $dx \wedge dy$ on the unit disk induces the counterclockwise orientation on S^1 .

Example 11.16 (Upper half-plane). For $\mathbb{H}^2 = \{(x, y) : y \geq 0\}$ with orientation (∂_x, ∂_y) , the outward vector along $y = 0$ is $-\partial_y$. Therefore the positive boundary tangent is ∂_x .

11.10 Exercises

Exercise 11.17. Identify the boundary and interior of $[0, 1]$ as a 1-manifold with boundary.

Hint. Use the half-line model. □

Solution. The boundary is $\{0, 1\}$ and the interior is $(0, 1)$. □

Exercise 11.18. Show that ∂M is locally modeled on $\mathbb{R}^{n-1}$.

Hint. Restrict a boundary chart to $x^n = 0$. □

Solution. A half-space chart restricts to a homeomorphism from the boundary slice onto an open subset of $\mathbb{R}^{n-1}$, and transition maps restrict smoothly. □

Exercise 11.19. For the standard orientation of $[a, b]$, compute the induced signs of the endpoints.

Hint. Use outward-normal-first. □

Solution. At b the outward vector agrees with the positive tangent, giving sign $+1$; at a it is negative, giving sign -1 . □

Exercise 11.20. Show that the standard orientation on the unit disk induces counterclockwise orientation on its boundary.

Hint. Test at $(1, 0)$. □

Solution. At $(1, 0)$ the outward normal is ∂_x ; the boundary vector ∂_y makes (∂_x, ∂_y) positive, so the induced direction is counterclockwise. □

Exercise 11.21. State Stokes' theorem for a compactly supported $(n - 1)$ -form.

Hint. Include the boundary pullback. □

Solution. $\int_M d\omega = \int_{\partial M} \iota^* \omega$ with the outward-normal-first boundary orientation. □

Exercise 11.22. Derive the fundamental theorem of calculus from Stokes.

Hint. Take a 0-form f on an interval. □

Solution. Stokes gives $\int_{[a,b]} df = \int_{\{b\} - \{a\}} f = f(b) - f(a)$. □

Exercise 11.23. Compute $d(P dx + Q dy)$.

Hint. Use $d(f\alpha) = df \wedge \alpha$. □

Solution. $d(P dx + Q dy) = (Q_x - P_y) dx \wedge dy$. □

Exercise 11.24. Recover Green's theorem from Stokes.

Hint. Use the preceding one-form. □

Solution. Substitute $d\omega = (Q_x - P_y)dx \wedge dy$ into Stokes on the oriented region. □

Exercise 11.25. Show that $d^2 f = 0$ for a smooth function.

Hint. Use equality of mixed partials. □

Solution. In coordinates the coefficient of $dx^i \wedge dx^j$ is $\partial_i \partial_j f - \partial_j \partial_i f = 0$. □

Exercise 11.26. Why is every exact form closed?

Hint. Apply d twice. □

Solution. If $\alpha = d\beta$, then $d\alpha = d^2\beta = 0$. □

Exercise 11.27. Show that an exact top form integrates to zero on a compact oriented manifold without boundary.

Hint. Apply Stokes. □

Solution. If $\alpha = d\beta$, then $\int_M \alpha = \int_{\partial M} \beta = 0$. □

Exercise 11.28. Compute $\oint_{S^1} (-y dx + x dy)$ by Stokes.

Hint. Differentiate the one-form over the disk. □

Solution. Its differential is $2 dx \wedge dy$, so the integral is 2 times the unit-disk area, namely 2π . $\square$

Exercise 11.29. For $F = (P, Q, R)$, relate $d(Pdx + Qdy + Rdz)$ to curl.

Hint. Expand the exterior derivative. $\square$

Solution. The coefficients of $dy \wedge dz$, $dz \wedge dx$, $dx \wedge dy$ are precisely the components of $\nabla \times F$. $\square$

Exercise 11.30. Use Stokes to show the circulation of a gradient around a closed curve is zero.

Hint. The corresponding one-form is exact. $\square$

Solution. If $F = \nabla f$, then $F \cdot d\mathbf{r} = df$, and the integral over a closed one-manifold is zero. $\square$

Exercise 11.31. For $X = \sum_i X^i \partial_i$, compute $d(\iota_X dV)$ in $\mathbb{R}^n$.

Hint. Differentiate each coefficient. $\square$

Solution. All cross terms vanish because of repeated differentials, leaving $(\sum_i \partial_i X^i) dV = (\operatorname{div} X) dV$. $\square$

Exercise 11.32. Recover the divergence theorem from Stokes.

Hint. Use $\omega = \iota_X dV$. $\square$

Solution. Stokes gives $\int_M (\operatorname{div} X) dV = \int_{\partial M} \iota^*(\iota_X dV)$, and the boundary form is the outward flux $X \cdot \nu dS$. $\square$

Exercise 11.33. Show that reversing the orientation of M reverses both sides of Stokes.

Hint. Track induced boundary orientation too. $\square$

Solution. The interior integral changes sign, and the boundary orientation induced by outward-normal-first also reverses, so the boundary integral changes sign as well. $\square$

Exercise 11.34. What happens to the induced boundary orientation if the outward vector is replaced by another outward transverse vector?

Hint. Connect the vectors through the outward half-space. $\square$

Solution. Any two outward transverse vectors differ by a positive normal component plus tangential terms; the change-of-basis determinant is positive, so the induced orientation is unchanged. $\square$

Exercise 11.35. Use Stokes to show there is no 1-form β on S^2 with $d\beta$ equal to the standard area form.

Hint. Integrate over S^2 . $\square$

Solution. If the area form were exact, its integral would be zero by Stokes, but its integral is 4π . $\square$

Exercise 11.36. For the annulus with standard orientation, determine the boundary directions.

Hint. Outward normals point opposite ways on the two circles. $\square$

Solution. The outer circle is counterclockwise; the inner circle is clockwise. □

Exercise 11.37. Why do interior partition-of-unity pieces contribute no boundary term in the proof of Stokes?

Hint. Their support avoids the boundary. □

Solution. Such a localized form is compactly supported in an open Euclidean chart, where integration of a total derivative vanishes by the ordinary fundamental theorem and compact support. □

Exercise 11.38. Explain why compact support is needed in the noncompact version of Stokes.

Hint. Prevent terms at infinity. □

Solution. Compact support ensures all coordinate integrations by parts have no contribution at infinity and makes the partition sum finite near the support. □

Exercise 11.39. Show that $\int_{\partial M} \iota^* d\eta = 0$ when ∂M is compact and has no boundary.

Hint. Apply Stokes on ∂M . □

Solution. $\int_{\partial M} d(\iota^* \eta) = \int_{\partial(\partial M)} \iota^* \eta = 0$. □

Exercise 11.40. Explain geometrically why $d^2 = 0$ mirrors $\partial^2 = 0$.

Hint. Pair Stokes twice. □

Solution. For suitable η , $\int_M d^2 \eta = \int_{\partial M} d\eta = \int_{\partial^2 M} \eta$. The algebraic zero on the left corresponds to cancellation of codimension-two boundary faces on the right. □

11.11 Solved problem dossiers

Problem 11.41 (Legacy Stokes problem: interval). Derive the fundamental theorem of calculus from the oriented-boundary convention.

Solution. The standard orientation on $[a, b]$ gives $\partial[a, b] = \{b\} - \{a\}$. Applying Stokes to the 0-form f gives $\int_a^b f'(t) dt = f(b) - f(a)$. □

Problem 11.42 (Legacy Stokes problem: planar Green theorem). Derive Green's theorem from the exterior derivative of $Pdx + Qdy$.

Solution. Compute $d(Pdx + Qdy) = (Q_x - P_y)dx \wedge dy$, then apply Stokes with the counterclockwise induced boundary orientation. □

Problem 11.43 (Legacy Stokes problem: boundary orientation). Prove the unit disk induces counterclockwise orientation on the unit circle.

Solution. At each point choose the outward radial vector first. The tangent obtained by rotating it counterclockwise by $\pi/2$ completes it to the standard positive basis of the plane. □

Problem 11.44 (Local half-space proof). Prove Stokes for a compactly supported $(n-1)$ -form on $\mathbb{H}^n$.

Solution. Expand into coordinate omissions. Tangential derivatives integrate to zero by compact support. The normal derivative integrates by the one-dimensional fundamental theorem to the boundary value, with the outward-normal-first sign. $\square$

Problem 11.45 (Partition-of-unity proof). Globalize the half-space calculation to an oriented manifold with boundary.

Solution. Choose a subordinate partition of unity. Apply the interior Euclidean and boundary half-space calculations to each localized form. Sum; the boundary restrictions of the partition still sum to one. $\square$

Problem 11.46 (Classical curl theorem). Translate Stokes for a one-form in $\mathbb{R}^3$ into the usual line-integral/curl formula.

Solution. Identify $Pdx + Qdy + Rdz$ with the vector field F . Its exterior derivative corresponds to $\nabla \times F$, while pullback to the boundary curve gives $F \cdot d\mathbf{r}$. $\square$

Problem 11.47 (Divergence theorem). Translate Stokes for $\iota_X dV$ into flux equals volume divergence.

Solution. The identity $d(\iota_X dV) = (\operatorname{div} X)dV$ supplies the interior term. Restricting $\iota_X dV$ to the oriented boundary evaluates the normal component of X times surface measure. $\square$

Problem 11.48 (Annulus signs). Determine the induced orientations on the two boundary circles of a standard oriented annulus.

Solution. Outward normal on the outer circle points radially outward, giving counterclockwise tangent. On the inner circle outward relative to the annulus points radially inward, forcing the positive tangent to be clockwise. $\square$

Problem 11.49 (Exact area-form obstruction). Show the standard area form on S^2 is not exact.

Solution. If $\omega = d\beta$, Stokes on boundaryless S^2 gives $\int_{S^2} \omega = 0$, contradicting the positive area 4π . $\square$

Problem 11.50 (Zero circulation of gradients). Show $\int_C df = 0$ for every closed oriented curve C .

Solution. The one-form df is exact. Applying Stokes to the closed one-manifold C gives zero; equivalently parameter endpoints cancel. $\square$

Problem 11.51 (Flux of radial field). Compute the flux of $X(x) = x$ across the unit sphere in $\mathbb{R}^n$ using divergence.

Solution. $\operatorname{div} X = n$. Thus the flux is $n \operatorname{Vol}(B^n)$, equal to the $(n-1)$ -area of S^{n-1} . $\square$

Problem 11.52 (Boundary of boundary cancellation). Explain the sign cancellation at corners of an oriented rectangle.

Solution. Each vertex is reached from two oriented edges with opposite endpoint signs. Hence the signed 0-chain $\partial^2 R$ vanishes, the geometric analogue of $d^2 = 0$. $\square$

11.12 Challenge problems

Problem 11.53 (Stokes and homotopy invariance preview). Let C_0 and C_1 be oriented closed curves bounding an oriented surface strip S . If α is closed, show $\int_{C_1} \alpha = \int_{C_0} \alpha$.

Solution. The oriented boundary is $C_1 - C_0$. Therefore $\int_{C_1} \alpha - \int_{C_0} \alpha = \int_{\partial S} \alpha = \int_S d\alpha = 0$. This is the basic mechanism behind homotopy invariance of periods. $\square$

Problem 11.54 (No compact manifold with one regular coordinate function). Suppose a compact connected oriented boundaryless manifold M admitted an $(n-1)$ -form η with $d\eta$ everywhere positive. Derive a contradiction.

Solution. Positivity gives $\int_M d\eta > 0$. But Stokes on a boundaryless compact manifold gives $\int_M d\eta = 0$. $\square$

Problem 11.55 (Gauss-law form). On $\mathbb{R}^n \setminus \{0\}$ construct the standard radial flux $(n-1)$ -form and explain why its integrals over concentric spheres agree.

Solution. Take $\omega = \iota_X dV / \|x\|^n$ with $X = \sum x^i \partial_i$. A direct computation gives $d\omega = 0$ away from the origin. Apply Stokes to the annulus between two spheres; their induced orientations yield equality of the two flux integrals. $\square$

Problem 11.56 (Exactness detected by an integral). Use Stokes to prove that the form $\alpha = (-y dx + x dy)/(x^2 + y^2)$ on $\mathbb{R}^2 \setminus \{0\}$ is closed but not exact.

Solution. A direct calculation gives $d\alpha = 0$. On the unit circle, $\alpha = d\theta$ along the parametrization and integrates to 2π . Every exact one-form has zero integral on a closed curve, so α cannot be exact. $\square$

Problem 11.57 (Boundary orientation under products). For oriented manifolds with boundary, derive the sign formula $\partial(M^m \times N) = \partial M \times N \cup (-1)^m M \times \partial N$.

Solution. Compare outward-normal-first frames with the product frame. For the second boundary piece, the outward normal from N must be moved past the m tangent vectors of M , producing the factor $(-1)^m$. $\square$

Bridge to VII/12

Stokes' theorem completes the first differential-form arc. VII/12 returns to one-dimensional geometry: regular parametrized curves, reparametrization, arc length, unit tangent, and line integrals, preparing the Frenet invariants of VII/13.

Part III

Curves and Surfaces

Chapter 12

Regular Curves

A parametrized curve is the simplest immersed manifold. Its geometry begins with a distinction that is easy to overlook: the image of a curve and its parametrization are not the same object. Regularity makes the velocity nonzero, while reparametrization isolates those quantities that belong to the geometric curve rather than to the chosen clock.

12.1 Parametrized and regular curves

Definition 12.1 (Parametrized curve). A smooth parametrized curve in a smooth manifold M is a smooth map

$$\gamma : I \rightarrow M$$

from an interval $I \subset \mathbb{R}$.

Definition 12.2 (Regular curve). A parametrized curve is regular when

$$\dot{\gamma}(t) = d\gamma_t \left(\frac{d}{dt} \right) \neq 0$$

for every $t \in I$. Equivalently, γ is an immersion.

Example 12.3 (Regular and singular planar curves). The circle $(\cos t, \sin t)$ is regular. The cusp (t^2, t^3) is not regular at $t = 0$ because its velocity vanishes there.

Proposition 12.4 (Local embeddedness). *Every regular curve is locally an embedded one-dimensional submanifold of M .*

Proof. A regular curve has rank one. The constant-rank theorem gives local coordinates in which it is $t \mapsto (t, 0, \dots, 0)$ after a local change of parameter. $\square$

12.2 Reparametrization

Definition 12.5 (Regular reparametrization). A reparametrization is a diffeomorphism $\phi : J \rightarrow I$. The same geometric curve is then represented by

$$\tilde{\gamma} = \gamma \circ \phi.$$

It is orientation preserving when $\phi' > 0$ and reversing when $\phi' < 0$.

Proposition 12.6 (Velocity transformation). *Under reparametrization,*

$$\dot{\tilde{\gamma}}(s) = \dot{\gamma}(\phi(s)) \phi'(s).$$

Hence regularity is invariant under reparametrization.

12.3 Curves in Euclidean space

Definition 12.7 (Speed). For a smooth curve $\gamma : I \rightarrow \mathbb{R}^n$, its speed is

$$v(t) = \|\dot{\gamma}(t)\|.$$

A regular curve has strictly positive speed.

Definition 12.8 (Length). For a smooth curve on $[a, b]$ define

$$L(\gamma) = \int_a^b \|\dot{\gamma}(t)\| dt.$$

For piecewise smooth curves, sum the lengths of the smooth pieces.

Theorem 12.9 (Length is reparametrization invariant). *If $\phi : [c, d] \rightarrow [a, b]$ is a diffeomorphism, then*

$$L(\gamma \circ \phi) = L(\gamma).$$

Proof. Use $\|\dot{\gamma}(\phi(s))\phi'(s)\| = \|\dot{\gamma}(\phi(s))\| |\phi'(s)|$ and the one-variable change-of-variables theorem. $\square$

12.4 Arc length

Definition 12.10 (Arc-length function). For a regular curve $\gamma : [a, b] \rightarrow \mathbb{R}^n$, define

$$s(t) = \int_a^t \|\dot{\gamma}(u)\| du.$$

Then $s'(t) = \|\dot{\gamma}(t)\| > 0$.

Theorem 12.11 (Unit-speed reparametrization). *Every regular Euclidean curve admits a local, and on an interval global, orientation-preserving reparametrization by arc length. In the new parameter s ,*

$$\left\| \frac{d\gamma}{ds} \right\| = 1.$$

Proof. Since $s'(t) > 0$, the inverse-function theorem gives an inverse $t = t(s)$. Then

$$\left\| \frac{d\gamma}{ds} \right\| = \|\dot{\gamma}(t(s))\| \frac{dt}{ds} = 1.$$

$\square$

Definition 12.12 (Unit tangent). For a regular Euclidean curve,

$$T(t) = \frac{\dot{\gamma}(t)}{\|\dot{\gamma}(t)\|}$$

is the unit tangent. Under orientation-preserving reparametrization it is unchanged; under orientation reversal it changes sign.

12.5 Simple and closed curves

Definition 12.13 (Simple curve). A parametrized curve is simple when it is injective, except that for a closed curve one permits the two endpoints of a compact parameter interval to have the same image.

Definition 12.14 (Closed curve). A smooth curve $\gamma : [a, b] \rightarrow M$ is smoothly closed when $\gamma(a) = \gamma(b)$ and the endpoint derivatives match under the identification needed to descend to a smooth map $S^1 \rightarrow M$.

Warning 12.15 (Same image, different traversal). Two curves may have the same image but traverse it with different speed, orientation, or multiplicity. Geometric invariants must therefore be checked for reparametrization invariance.

12.6 Standard examples

Example 12.16 (Line). For $\gamma(t) = p + tv$ with $v \neq 0$, speed is $\|v\|$ and length on $[a, b]$ is $(b - a)\|v\|$.

Example 12.17 (Circle). For $\gamma(t) = (R \cos t, R \sin t)$, speed is R and one full traversal has length $2\pi R$.

Example 12.18 (Helix). For $\gamma(t) = (R \cos t, R \sin t, ct)$,

$$\|\dot{\gamma}(t)\| = \sqrt{R^2 + c^2},$$

so length over $[a, b]$ is $(b - a)\sqrt{R^2 + c^2}$.

12.7 Curves on manifolds

Proposition 12.19 (Coordinate velocity). If $x = (x^1, \dots, x^n)$ is a chart and $\gamma(t)$ lies in its domain, then

$$\dot{\gamma}(t) = \sum_i \frac{d}{dt}(x^i \circ \gamma)(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}.$$

Proposition 12.20 (Velocity under smooth maps). For a smooth map $F : M \rightarrow N$,

$$\frac{d}{dt}(F \circ \gamma)(t) = dF_{\gamma(t)} \dot{\gamma}(t).$$

12.8 Piecewise regular curves and line integrals

Definition 12.21 (Piecewise regular). A continuous curve on $[a, b]$ is piecewise regular if there is a finite subdivision on whose subintervals it is smooth and regular, with one-sided derivatives at break points.

Definition 12.22 (Line integral of a one-form). For a piecewise smooth curve $\gamma : [a, b] \rightarrow M$ and a one-form α defined along its image,

$$\int_{\gamma} \alpha := \int_a^b \alpha_{\gamma(t)}(\dot{\gamma}(t)) dt.$$

This is invariant under orientation-preserving reparametrization and changes sign when orientation is reversed.

Proposition 12.23 (Exact one-forms along curves). *For a smooth function f ,*

$$\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)).$$

Proof. $df_{\gamma(t)}(\dot{\gamma}(t)) = \frac{d}{dt}(f \circ \gamma)(t)$, so integrate and apply the fundamental theorem of calculus. $\square$

Bridge to VII/13

Regularity and arc length provide the correct parameter-independent starting point for local curve geometry. VII/13 differentiates the unit tangent, constructs Frenet frames, and introduces curvature and torsion; those quantities are intentionally deferred until the parametrization issues have been settled here.

12.9 Exercises

Exercise 12.24. Determine whether $\gamma(t) = (t, t^2)$ is regular.

Hint. Differentiate. $\square$

Solution. $\dot{\gamma} = (1, 2t)$ never vanishes, so the curve is regular. $\square$

Exercise 12.25. Determine where $\gamma(t) = (t^2, t^3)$ fails to be regular.

Hint. Differentiate. $\square$

Solution. $\dot{\gamma} = (2t, 3t^2)$ vanishes exactly at $t = 0$. $\square$

Exercise 12.26. Show a regular curve is an immersion.

Hint. The domain is one-dimensional. $\square$

Solution. Its differential has rank one exactly when the velocity is nonzero, which is the definition of regularity. $\square$

Exercise 12.27. Explain why a regular curve is locally embedded.

Hint. Use the constant-rank theorem. $\square$

Solution. Rank one gives local coordinates in which the map is $t \mapsto (t, 0, \dots, 0)$, an embedding onto a coordinate line. $\square$

Exercise 12.28. For $\tilde{\gamma} = \gamma \circ \phi$, compute its velocity.

Hint. Apply the chain rule. $\square$

Solution. $\dot{\tilde{\gamma}}(s) = \dot{\gamma}(\phi(s))\phi'(s)$. $\square$

Exercise 12.29. Why does a diffeomorphic reparametrization preserve regularity?

Hint. Its derivative never vanishes. $\square$

Solution. Both factors in $\dot{\gamma}(\phi(s))\phi'(s)$ are nonzero for a regular curve and a parameter diffeomorphism. $\square$

Exercise 12.30. Compute the speed of $\gamma(t) = (3t, 4t)$.

Hint. Take the Euclidean norm. □

Solution. The speed is $\sqrt{3^2 + 4^2} = 5$. □

Exercise 12.31. Compute the length of $\gamma(t) = (3t, 4t)$ on $[0, 2]$.

Hint. Integrate its constant speed. □

Solution. $L = \int_0^2 5 \, dt = 10$. □

Exercise 12.32. Prove length invariance under orientation-preserving reparametrization.

Hint. Substitute $t = \phi(s)$. □

Solution. The transformed speed is $\|\dot{\gamma}(\phi(s))\|\phi'(s)$, and ordinary substitution yields the original length integral. □

Exercise 12.33. Why is the absolute value of ϕ' needed for orientation-reversing reparametrizations?

Hint. Length is nonnegative. □

Solution. The norm gives $\|\dot{\gamma}(\phi(s))\phi'(s)\| = \|\dot{\gamma}(\phi(s))\||\phi'(s)|$, so reversal does not make length negative. □

Exercise 12.34. Compute the speed of the radius- R circle $(R \cos t, R \sin t)$.

Hint. Differentiate. □

Solution. The velocity is $(-R \sin t, R \cos t)$, whose norm is R . □

Exercise 12.35. Compute one full circle length from its parametrization.

Hint. Integrate from 0 to 2π . □

Solution. $L = \int_0^{2\pi} R \, dt = 2\pi R$. □

Exercise 12.36. Compute the speed of the helix $(R \cos t, R \sin t, ct)$.

Hint. Differentiate and square components. □

Solution. The squared speed is $R^2 \sin^2 t + R^2 \cos^2 t + c^2 = R^2 + c^2$, so speed is $\sqrt{R^2 + c^2}$. □

Exercise 12.37. Show the arc-length function of a regular curve is strictly increasing.

Hint. Differentiate it. □

Solution. $s'(t) = \|\dot{\gamma}(t)\| > 0$, so s is strictly increasing. □

Exercise 12.38. Why can arc length be used as a parameter?

Hint. Apply the inverse-function theorem. □

Solution. Since $s'(t) > 0$, s is locally a diffeomorphism; on an interval it is globally one-to-one onto its image. □

Exercise 12.39. Show that an arc-length parametrized curve has unit speed.

Hint. Differentiate through the inverse $t(s)$. □

Solution. $d\gamma/ds = \dot{\gamma}(t(s))/s'(t(s))$, whose norm is 1. □

Exercise 12.40. How does the unit tangent transform under an orientation-preserving reparametrization?

Hint. Normalize the transformed velocity. □

Solution. The positive factor ϕ' cancels in normalization, so $\tilde{T}(s) = T(\phi(s))$. □

Exercise 12.41. How does the unit tangent transform under orientation reversal?

Hint. Now $\phi' < 0$. □

Solution. Normalization leaves the sign of ϕ' , so $\tilde{T}(s) = -T(\phi(s))$. □

Exercise 12.42. Give an example of two parametrizations with the same image but different multiplicity.

Hint. Use a circle. □

Solution. $(\cos t, \sin t)$ on $[0, 2\pi]$ traverses once, while $(\cos 2t, \sin 2t)$ on the same interval traverses twice. □

Exercise 12.43. State the coordinate formula for velocity on a manifold.

Hint. Differentiate coordinate functions. □

Solution. $\dot{\gamma} = \sum_i (d/dt)(x^i \circ \gamma) \partial/\partial x^i$. □

Exercise 12.44. Prove velocity is natural under smooth maps.

Hint. Use the chain rule for differentials. □

Solution. $d(F \circ \gamma)_t = dF_{\gamma(t)} \circ d\gamma_t$; applying to d/dt gives the formula. □

Exercise 12.45. Define a piecewise regular curve.

Hint. Use a finite subdivision. □

Solution. It is continuous and admits finitely many subintervals on which it is smooth and regular, with one-sided derivatives at the subdivision points. □

Exercise 12.46. Show the line integral of a one-form is invariant under orientation-preserving reparametrization.

Hint. Transform both velocity and dt . □

Solution. The factor ϕ' from the velocity combines with ds exactly as in one-variable substitution, giving the same integral. □

Exercise 12.47. Prove $\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a))$.

Hint. Use the chain rule. □

Solution. The integrand is $df(\dot{\gamma}) = d(f \circ \gamma)/dt$; integrate and apply the fundamental theorem. □

12.10 Solved problem dossiers

Problem 12.48 (Legacy regular-curve problem). Show that nonvanishing velocity is equivalent to the immersion condition for a curve.

Solution. Since $T_t I$ is one-dimensional, $d\gamma_t$ is injective exactly when $d\gamma_t(d/dt) = \dot{\gamma}(t)$ is nonzero. $\square$

Problem 12.49 (Legacy parametrization problem). Prove the reparametrization velocity formula and deduce invariance of regularity.

Solution. Apply the chain rule to $\gamma \circ \phi$. Because a parameter diffeomorphism has $\phi' \neq 0$, nonzero velocity is preserved. $\square$

Problem 12.50 (Legacy length problem). Prove Euclidean curve length is invariant under all diffeomorphic reparametrizations.

Solution. The transformed speed is the old speed times $|\phi'|$. The one-variable change-of-variables theorem, with orientation handled by the absolute value, recovers the same integral. $\square$

Problem 12.51 (Legacy arc-length problem). Construct the arc-length parameter for a regular curve and prove it gives unit speed.

Solution. Set $s(t) = \int_a^t \|\dot{\gamma}(u)\| du$. Since $s' > 0$, invert to $t(s)$ and differentiate $\gamma(t(s))$ to obtain norm one. $\square$

Problem 12.52 (Line and segment). Compute speed and length of $\gamma(t) = p + tv$ and explain its dependence on the parameter interval.

Solution. Velocity is the constant vector v , so speed is $\|v\|$ and length on $[a, b]$ is $(b - a)\|v\|$. Reparametrizing changes the clock but not the geometric segment length. $\square$

Problem 12.53 (Circle). Compute the length and unit tangent of a radius- R circle.

Solution. For $\gamma(t) = (R \cos t, R \sin t)$, speed is R , length over one turn is $2\pi R$, and $T(t) = (-\sin t, \cos t)$. $\square$

Problem 12.54 (Helix). Compute speed and length of a circular helix.

Solution. $\dot{\gamma} = (-R \sin t, R \cos t, c)$ has constant norm $\sqrt{R^2 + c^2}$, so length is parameter-span times that constant. $\square$

Problem 12.55 (Local embeddedness). Use the constant-rank theorem to show every regular curve is locally a coordinate axis.

Solution. Rank one allows source and target coordinates in which the map is $s \mapsto (s, 0, \dots, 0)$. Therefore the image near each point is an embedded one-dimensional submanifold. $\square$

Problem 12.56 (Same image, opposite orientation). Compare $\gamma(t) = (\cos t, \sin t)$ with $\tilde{\gamma}(t) = (\cos t, -\sin t)$.

Solution. They have the same circle image and equal speed, but traverse it in opposite orientations. Their unit tangents and line integrals of one-forms differ by sign after the corresponding orientation reversal. $\square$

Problem 12.57 (Coordinate velocity on a manifold). Derive the coordinate expression for $\dot{\gamma}$ from the derivation definition of tangent vectors.

Solution. Apply $\dot{\gamma}$ to a smooth function and use the coordinate chain rule. The coefficients multiplying the coordinate derivations are the derivatives of the coordinate functions along the curve. $\square$

Problem 12.58 (Line integral of an exact form). Show line integrals of df depend only on endpoints.

Solution. Pull df back to the interval: $\gamma^*df = d(f \circ \gamma)$. Integration gives $f(\gamma(b)) - f(\gamma(a))$. $\square$

Problem 12.59 (Piecewise regular concatenation). Show line integrals and lengths add under concatenation of piecewise regular curves with matching endpoints.

Solution. Choose a subdivision at the joining time. Both definitions are sums of ordinary integrals over the pieces, so additivity is immediate. $\square$

12.11 Challenge problems

Problem 12.60 (Arc length on a Riemannian manifold preview). Assume a Riemannian metric g is given. Show that $L_g(\gamma) = \int \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt$ is reparametrization invariant.

Solution. Under reparametrization, the velocity is multiplied by ϕ' . Bilinearity makes its squared norm multiply by $(\phi')^2$, so the speed multiplies by $|\phi'|$. One-variable substitution then proves invariance. $\square$

Problem 12.61 (Regular closed curve as an immersion of S^1). Show that a smoothly closed regular curve with matching endpoint jets descends to a smooth immersion $S^1 \rightarrow M$.

Solution. Identify S^1 with $[a, b]/(a \sim b)$. Matching derivatives ensure smoothness across the quotient point, and nonzero velocity gives rank one everywhere. $\square$

Problem 12.62 (Length versus multiplicity). If a regular closed curve is traversed k times by an orientation-preserving k -fold parameter map, compare its length with one traversal.

Solution. A k -fold traversal is not a diffeomorphic reparametrization of the once-traversed interval; it changes multiplicity. Splitting into k monotone pieces shows the total length is k times the one-traversal length. $\square$

Problem 12.63 (Line integrals detect orientation). Let $\alpha = -y dx + x dy$ on the unit circle. Compare its integral under counterclockwise and clockwise parametrizations.

Solution. Counterclockwise parametrization gives $\alpha(\dot{\gamma}) = 1$ and integral 2π . Reversing orientation multiplies the line integral by -1 , giving -2π . $\square$

Problem 12.64 (Why curvature belongs in the next chapter). Explain why $\|dT/dt\|$ is not itself a geometric invariant under arbitrary reparametrization, and identify the invariant combination that motivates arc length.

Solution. Under an orientation-preserving change $t = t(s)$, $dT/ds = (dT/dt)(dt/ds)$, so its norm scales with parameter speed. Differentiating with respect to arc length removes this factor. Thus $\|dT/ds\|$ is the natural next invariant, which is precisely the curvature introduced in VII/13. $\square$

Chapter 13

Frenet Frames, Curvature and Torsion

A regular curve carries more than a velocity. Once arc length is chosen, successive derivatives separate the bending of the curve from the twisting of its osculating plane. For a unit-speed space curve with nonzero curvature, the moving orthonormal frame

$$(T, N, B)$$

obeys the Frenet–Serret system

$$\boxed{T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N.}$$

This chapter develops these invariants carefully, including formulas that do not require an arc-length parameter.

13.1 Curvature of a unit-speed curve

Definition 13.1 (Curvature). Let $\gamma : I \rightarrow \mathbb{R}^3$ be unit speed. Its unit tangent is $T = \gamma'$, and its curvature is

$$\kappa(s) = \|T'(s)\| = \|\gamma''(s)\|.$$

At a point where $\kappa > 0$, the principal normal is

$$N = \frac{T'}{\kappa}.$$

Proposition 13.2 (Acceleration is normal for unit speed). *For a unit-speed curve, $T' \perp T$.*

Proof. Differentiate $\langle T, T \rangle = 1$ to obtain $2\langle T', T \rangle = 0$. □

Definition 13.3 (Binormal and Frenet frame). Where $\kappa > 0$, define

$$B = T \times N.$$

Then (T, N, B) is a positively oriented orthonormal frame, called the Frenet frame.

Warning 13.4 (Straight points). If $\kappa = 0$, the principal normal and binormal are not determined by the curve at that point. Curvature still makes sense, but the full Frenet frame does not.

13.2 Curvature in an arbitrary parameter

Theorem 13.5 (Parameter-independent curvature formula). *For a regular curve $\gamma(t) \subset \mathbb{R}^3$,*

$$\kappa(t) = \frac{\|\gamma'(t) \times \gamma''(t)\|}{\|\gamma'(t)\|^3}.$$

For a planar curve $\gamma = (x, y)$, this becomes

$$\kappa = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}}.$$

Proof. Write $v = ds/dt = \|\gamma'\|$ and $\gamma' = vT$. Then $\gamma'' = v'T + v^2\kappa N$, so $\gamma' \times \gamma'' = v^3\kappa B$. Taking norms gives the formula. $\square$

Example 13.6 (Circle). For $\gamma(t) = (R \cos t, R \sin t, 0)$, one finds

$$\kappa = \frac{1}{R}.$$

Thus curvature is the reciprocal of the radius for a circle.

13.3 Torsion and the Frenet–Serret equations

Definition 13.7 (Torsion). For a unit-speed curve with $\kappa > 0$, the torsion is

$$\tau = -\langle B', N \rangle.$$

Theorem 13.8 (Frenet–Serret equations). *Where $\kappa > 0$,*

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N.$$

Proof. The first equation is the definition of N . Differentiate the orthonormality relations among T, N, B . The derivative matrix of an orthonormal frame is skew-symmetric. Since $T' = \kappa N$ and $B' \perp B, T$, the remaining coefficients are forced to be those displayed, with $\tau = -\langle B', N \rangle$. $\square$

Theorem 13.9 (Torsion formula). *If $\gamma' \times \gamma'' \neq 0$, then for any regular parameter*

$$\tau = \frac{\det(\gamma', \gamma'', \gamma''')}{\|\gamma' \times \gamma''\|^2}.$$

Proof. The numerator is the scalar triple product measuring the component of γ''' in the binormal direction. Expanding derivatives through $\gamma' = vT$ and using the Frenet equations gives numerator $v^6\kappa^2\tau$, while the denominator is $v^6\kappa^2$. $\square$

13.4 Osculating geometry

Definition 13.10 (Osculating, normal and rectifying planes). At a point where $\kappa > 0$, the osculating plane is $\text{span}(T, N)$, the normal plane is $\text{span}(N, B)$, and the rectifying plane is $\text{span}(T, B)$.

Proposition 13.11 (Osculating circle). *At a point with $\kappa > 0$, the osculating circle has radius*

$$\rho = \frac{1}{\kappa}$$

and center $\gamma + \rho N$. It matches position, tangent direction, and curvature to second order.

13.5 Planar curves and torsion

Theorem 13.12 (Planarity criterion). *Let γ be connected, unit speed, and satisfy $\kappa > 0$. Then $\tau \equiv 0$ if and only if the image of γ lies in an affine plane.*

Proof. If $\tau = 0$, then $B' = 0$, so B is constant and $\langle \gamma, B \rangle$ is constant because $\gamma' = T \perp B$. Conversely, for a curve in a fixed plane its binormal is constant, so $B' = 0$ and $\tau = 0$. $\square$

13.6 The circular helix

Example 13.13 (Helix). For

$$\gamma(t) = (a \cos t, a \sin t, bt), \quad a > 0,$$

the speed is $\sqrt{a^2 + b^2}$, and

$$\kappa = \frac{a}{a^2 + b^2}, \quad \tau = \frac{b}{a^2 + b^2}.$$

Both invariants are constant. The ratio $\tau/\kappa = b/a$ records the slope of the helix.

Theorem 13.14 (General helix criterion). *A unit-speed curve with $\kappa > 0$ has tangent making a constant angle with a fixed direction if and only if τ/κ is constant.*

13.7 Rigid motions and the fundamental theorem

Proposition 13.15 (Euclidean invariance). *Orientation-preserving rigid motions preserve κ and τ . Orientation-reversing rigid motions preserve κ and reverse the sign convention for the oriented binormal, hence reverse the corresponding signed torsion convention.*

Theorem 13.16 (Fundamental theorem of space curves). *Let $\kappa : I \rightarrow (0, \infty)$ and $\tau : I \rightarrow \mathbb{R}$ be smooth. There exists a unit-speed curve with these curvature and torsion functions, unique up to an orientation-preserving Euclidean rigid motion.*

Proof idea. Solve the Frenet–Serret linear ODE for an oriented orthonormal frame with prescribed initial frame, then integrate $\gamma' = T$. ODE uniqueness shows that two solutions with the same initial position and frame coincide; arbitrary initial data differ by a rigid motion. $\square$

13.8 Exercises

Exercise 13.17. For a unit-speed curve prove $\gamma'' \perp \gamma'$.

Hint. Differentiate $\|\gamma'\|^2 = 1$. $\square$

Solution. The derivative gives $2\langle \gamma'', \gamma' \rangle = 0$. $\square$

Exercise 13.18. Compute the curvature of a straight line.

Hint. Its unit tangent is constant. $\square$

Solution. For $\gamma(s) = p + sT_0$, $T' = 0$, hence $\kappa = 0$. $\square$

Exercise 13.19. Compute the curvature of a circle of radius R .

Hint. Use the arbitrary-parameter formula. □

Solution. $\|\gamma' \times \gamma''\| = R^2$ and $\|\gamma'\|^3 = R^3$, so $\kappa = 1/R$. □

Exercise 13.20. Derive the planar determinant formula for curvature.

Hint. Embed the curve in the plane $z = 0$. □

Solution. The cross product has only third component $x'y'' - y'x''$, giving the stated formula. □

Exercise 13.21. Explain why N is undefined at a point with $\kappa = 0$.

Hint. Look at $N = T'/\kappa$. □

Solution. Both the normalization and a distinguished bending direction fail when $T' = 0$. □

Exercise 13.22. Prove $B \perp T, N$ and $\|B\| = 1$.

Hint. Use properties of the cross product. □

Solution. Since T, N are orthonormal, $T \times N$ is a unit vector perpendicular to both. □

Exercise 13.23. Show $B' \perp B$.

Hint. Differentiate $\langle B, B \rangle = 1$. □

Solution. One obtains $2\langle B', B \rangle = 0$. □

Exercise 13.24. Why is $B' \perp T$?

Hint. Differentiate $\langle B, T \rangle = 0$. □

Solution. $\langle B', T \rangle = -\langle B, T' \rangle = -\kappa \langle B, N \rangle = 0$. □

Exercise 13.25. Derive $B' = -\tau N$.

Hint. Use the previous two orthogonality relations. □

Solution. B' is parallel to N , and the definition $\tau = -\langle B', N \rangle$ gives the coefficient. □

Exercise 13.26. Derive the coefficient of T in N' .

Hint. Differentiate $\langle N, T \rangle = 0$. □

Solution. $\langle N', T \rangle = -\langle N, T' \rangle = -\kappa$. □

Exercise 13.27. Compute κ for $\gamma(t) = (t, t^2, 0)$.

Hint. Use the planar formula. □

Solution. $\kappa(t) = 2/(1 + 4t^2)^{3/2}$. □

Exercise 13.28. Show a planar curve with $\kappa > 0$ has zero torsion.

Hint. Its binormal is a fixed normal to the plane. □

Solution. The binormal is constant, hence $B' = 0$ and $\tau = 0$. □

Exercise 13.29. Compute curvature and torsion of $(a \cos t, a \sin t, bt)$.

Hint. Use the cross-product and determinant formulas. □

Solution. The formulas give $\kappa = a/(a^2 + b^2)$ and $\tau = b/(a^2 + b^2)$. □

Exercise 13.30. What geometric data does the osculating plane contain?

Hint. Recall its spanning frame vectors. □

Solution. It contains the tangent and principal normal directions, so it captures the second-order bending of the curve. □

Exercise 13.31. Find the osculating-circle radius in terms of curvature.

Hint. Use the circle model. □

Solution. The radius is $\rho = 1/\kappa$. □

Exercise 13.32. Prove curvature is unchanged by translation.

Hint. Derivatives of a translated curve are unchanged. □

Solution. Both γ' and γ'' are unchanged, hence so is the curvature formula. □

Exercise 13.33. Prove curvature is unchanged by an orthogonal linear map.

Hint. Orthogonal maps preserve norms and cross-product norms. □

Solution. The numerator and denominator in the curvature formula are preserved. □

Exercise 13.34. Explain why torsion is a third-order invariant.

Hint. Inspect its arbitrary-parameter formula. □

Solution. The determinant formula contains γ''' , so third derivatives are required. □

Exercise 13.35. Show that $\tau = 0$ implies constant binormal when $\kappa > 0$.

Hint. Use the third Frenet equation. □

Solution. $B' = -\tau N = 0$, hence B is constant on the interval. □

Exercise 13.36. If κ and τ are both constant with $\kappa > 0$, what model should one expect?

Hint. Compare with the circular helix. □

Solution. Up to a rigid motion the fundamental theorem identifies the curve with a circular helix; the case $\tau = 0$ is a circle. □

Exercise 13.37. Write the Frenet equations as a matrix ODE.

Hint. Use the ordered frame (T, N, B) . □

Solution. With frame columns, the coefficient matrix is skew-symmetric with entries κ and τ , expressing infinitesimal rotation. □

Exercise 13.38. Why does the Frenet coefficient matrix preserve orthonormality?

Hint. Check skew-symmetry. □

Solution. A solution F of $F' = FA$ with $A^T = -A$ satisfies $(F^T F)' = A^T F^T F + F^T F A = 0$. $\square$

Exercise 13.39. What is the torsion of a circle?

Hint. A circle lies in a plane. $\square$

Solution. Its binormal is constant, so $\tau = 0$. $\square$

Exercise 13.40. Explain why $\kappa > 0$ is assumed in the fundamental theorem stated here.

Hint. The Frenet normal must be defined. $\square$

Solution. Positive curvature guarantees a smooth distinguished normal and hence a nonsingular Frenet system. $\square$

13.9 Solved problem dossiers

Problem 13.41 (Legacy Frenet problem: construct the frame). Given a unit-speed curve with $\kappa > 0$, construct T, N, B and prove it is an oriented orthonormal frame.

Solution. Set $T = \gamma'$, $N = T'/\|T'\|$, and $B = T \times N$. Unit speed gives $T \perp T'$; the cross product then completes a positively oriented orthonormal frame. $\square$

Problem 13.42 (Legacy curvature problem). Derive curvature in a non-unit parameter.

Solution. Write $\gamma' = vT$ and differentiate. The cross product kills the tangential term and leaves $v^3 \kappa B$, yielding $\kappa = \|\gamma' \times \gamma''\|/\|\gamma'\|^3$. $\square$

Problem 13.43 (Legacy torsion problem). Derive the determinant formula for torsion.

Solution. Expand $\gamma', \gamma'', \gamma'''$ in the Frenet frame. The scalar triple product equals $v^6 \kappa^2 \tau$ and $\|\gamma' \times \gamma''\|^2 = v^6 \kappa^2$. $\square$

Problem 13.44 (Frenet equations from orthonormality). Recover the complete Frenet–Serret system from the derivatives of inner products.

Solution. The derivative matrix of an orthonormal frame is skew-symmetric. The known entry $T' = \kappa N$ and the definition of torsion determine all remaining entries. $\square$

Problem 13.45 (Planarity from torsion). Prove the zero-torsion criterion for a connected curve with $\kappa > 0$.

Solution. If $\tau = 0$, then B is constant and $\langle \gamma, B \rangle$ is constant, placing the curve in a plane. The converse follows from constant plane normal. $\square$

Problem 13.46 (Circular helix dossier). Compute the complete Frenet invariants of a circular helix.

Solution. Direct differentiation gives constant speed, then the cross-product and triple-product formulas yield $\kappa = a/(a^2 + b^2)$, $\tau = b/(a^2 + b^2)$. $\square$

Problem 13.47 (Osculating circle dossier). Explain why $\gamma + (1/\kappa)N$ is the center of the second-order approximating circle.

Solution. In arc length, Taylor expansion is $\gamma(s+h) = \gamma + hT + \frac{1}{2}h^2\kappa N + O(h^3)$, identical through second order to the circle with radius $1/\kappa$ and that center. $\square$

Problem 13.48 (Rigid-motion invariance). Prove curvature and signed torsion behavior under Euclidean rigid motions.

Solution. Translations disappear under differentiation. Orthogonal maps preserve norms and scalar triple products up to determinant; hence curvature is invariant and signed torsion changes by the orientation sign. $\square$

Problem 13.49 (General helix criterion). Prove that constant τ/κ is equivalent to a fixed direction making constant angle with T .

Solution. If $\tau/\kappa = c$, the vector $cT + B$ has derivative $c\kappa N - \tau N = 0$. Conversely, differentiating a constant-angle relation and resolving in the Frenet frame yields a constant ratio. $\square$

Problem 13.50 (Frame ODE dossier). Show the Frenet system evolves inside $SO(3)$.

Solution. Arrange T, N, B as an orthogonal matrix. Its logarithmic derivative is skew-symmetric, so the flow preserves determinant $+1$ and orthonormality. $\square$

Problem 13.51 (Constant invariants). Classify curves with positive constant curvature and constant torsion.

Solution. The circular helix has the prescribed constants; uniqueness in the fundamental theorem implies every such curve is congruent to it. $\square$

Problem 13.52 (Straight-point pathology). Explain what fails in Frenet theory at a zero-curvature point without claiming the curve itself is singular.

Solution. Regularity concerns $\gamma' \neq 0$, so the curve remains regular. What fails is normalization of T' , hence the distinguished principal normal and binormal. $\square$

13.10 Challenge problems

Problem 13.53 (Fundamental theorem by ODE). Give a detailed existence-and-uniqueness argument for prescribed smooth $\kappa > 0$ and τ .

Solution. Solve the skew-symmetric Frenet matrix ODE with an initial oriented orthonormal frame. Skew-symmetry preserves orthonormality. Integrate $\gamma' = T$. Any two choices of initial position and frame are related by one orientation-preserving rigid motion, and ODE uniqueness propagates that relation. $\square$

Problem 13.54 (Spherical-curve test). Assume γ is unit speed with $\kappa > 0$. Derive a condition involving κ and τ that can be used to test whether the curve lies on a sphere.

Solution. Write a candidate center in the normal plane as $C = \gamma + aN + bB$, differentiate, and impose $C' = 0$. The resulting system begins with $a = 1/\kappa$ and relates a' , b , and τ ; constancy of $a^2 + b^2$ then gives the radius. $\square$

Problem 13.55 (Curvature under scaling). If $\tilde{\gamma} = c\gamma$ with $c > 0$, determine how curvature and torsion scale.

Solution. Derivatives scale by c . Thus $\tilde{\kappa} = c^2/c^3 \kappa = \kappa/c$, and the torsion determinant scales by c^3 while its denominator scales by c^4 , giving $\tilde{\tau} = \tau/c$. $\square$

Problem 13.56 (Signed planar curvature). For an oriented regular planar curve define signed curvature and show it changes sign under orientation reversal.

Solution. For unit speed choose the positively rotated normal n and write $T' = k_s n$. Reversing the parameter reverses T but preserves the second derivative relation in a way that reverses the chosen oriented normal convention, hence k_s changes sign while $|k_s| = \kappa$ does not. $\square$

Problem 13.57 (Frenet data and local congruence). Two unit-speed curves have the same positive curvature and torsion on an interval. Prove they differ by one rigid motion on the entire interval.

Solution. Choose the unique orientation-preserving rigid motion matching one point and the two Frenet frames there. The transformed first curve and the second solve the same Frenet ODE with identical initial data; uniqueness makes the frames, and then the integrated positions, agree everywhere. $\square$

Bridge to VII/14

Curves now have a complete local moving-frame theory. VII/14 moves from one-dimensional images to regular surfaces in $\mathbb{R}^3$, where two independent tangent directions replace the single tangent vector and local parametrizations become the basic coordinate device.

Chapter 14

Regular Surfaces

A regular surface is a two-dimensional smooth manifold realized locally inside $\mathbb{R}^3$. The central local object is a surface patch

$$X(u, v) : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3$$

whose two coordinate derivatives are linearly independent. Equivalently,

$$X_u \times X_v \neq 0.$$

This rank condition produces tangent planes, local normals, coordinate curves, and the framework needed for the fundamental forms.

14.1 Regular surface patches

Definition 14.1 (Regular parametrized patch). A smooth map $X : U \rightarrow \mathbb{R}^3$, with $U \subset \mathbb{R}^2$ open, is a regular patch if $dX_{(u,v)}$ has rank 2 at every point and X is a homeomorphism onto its image.

Proposition 14.2 (Cross-product criterion). *A smooth parametrization has rank 2 exactly when*

$$X_u \times X_v \neq 0.$$

Proof. The derivative has columns X_u, X_v . It has rank 2 exactly when these vectors are linearly independent, equivalent in $\mathbb{R}^3$ to a nonzero cross product. $\square$

Definition 14.3 (Regular surface). A subset $S \subset \mathbb{R}^3$ is a regular surface if every point $p \in S$ lies in the image of a regular patch whose image is open in S with the subspace topology.

Warning 14.4 (Rank alone is not the whole patch definition). The rank condition is differential. A surface patch must also give the correct local topology; global self-overlap can prevent a parametrization from being a chart even if its derivative has rank 2.

14.2 Tangent planes

Definition 14.5 (Tangent plane). For $p = X(u_0, v_0)$, define

$$T_p S = \text{span}\{X_u(u_0, v_0), X_v(u_0, v_0)\}.$$

Theorem 14.6 (Curve characterization). *A vector $w \in \mathbb{R}^3$ lies in $T_p S$ if and only if there is a smooth curve α in S with $\alpha(0) = p$ and $\alpha'(0) = w$.*

Proof. If $\alpha(t) = X(u(t), v(t))$, then $\alpha' = u'X_u + v'X_v$. Conversely any combination $aX_u + bX_v$ is realized by the coordinate-domain line $(u_0 + at, v_0 + bt)$ for small t . $\square$

Definition 14.7 (Local unit normal). On a regular patch,

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|}$$

is a smooth unit normal field.

Proposition 14.8 (Tangent-plane equation). *At p with unit normal $N(p)$, the affine tangent plane is*

$$\{q \in \mathbb{R}^3 : \langle q - p, N(p) \rangle = 0\}.$$

14.3 Change of parameters

Theorem 14.9 (Reparametrization). *Let $\phi : \tilde{U} \rightarrow U$ be a diffeomorphism and $\tilde{X} = X \circ \phi$. Then $\tilde{X}$ is regular whenever X is regular, and the tangent plane is unchanged.*

Proof. The chain rule gives $d\tilde{X} = dX \circ d\phi$. Since $d\phi$ is invertible, both derivatives have the same rank and image. $\square$

Proposition 14.10 (Normal orientation under coordinates). *Under a change of parameters, the vector $X_u \times X_v$ is multiplied by the Jacobian determinant of the parameter change. Thus orientation-preserving changes preserve the chosen normal direction, while orientation-reversing changes reverse it.*

14.4 Graphs and implicit surfaces

Proposition 14.11 (Graphs are regular). *For smooth $f : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$,*

$$X(u, v) = (u, v, f(u, v))$$

is a regular patch. A normal vector is

$$(-f_u, -f_v, 1).$$

Theorem 14.12 (Implicit surface criterion). *Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ be smooth. If $\nabla F(p) \neq 0$ on $F^{-1}(c)$, then the level set is locally a regular surface and*

$$T_p S = \ker dF_p = \nabla F(p)^\perp.$$

Proof. This is the regular-level theorem of VII/06 in dimension $3 \rightarrow 1$. The implicit-function theorem also writes the level locally as a graph. $\square$

Example 14.13 (Sphere). The sphere S_R^2 is the regular level of $F(x, y, z) = x^2 + y^2 + z^2$ at R^2 . Hence

$$T_p S_R^2 = p^\perp.$$

Spherical coordinates provide useful patches away from their polar coordinate singularities.

Example 14.14 (Cylinder). The circular cylinder has patch

$$X(u, v) = (R \cos u, R \sin u, v),$$

with $X_u \times X_v \neq 0$ everywhere.

14.5 Coordinate curves and surface velocities

Definition 14.15 (Coordinate curves). For a patch $X(u, v)$, curves with $v = v_0$ and with $u = u_0$ are the coordinate curves. Their tangent vectors are X_u and X_v , respectively.

Proposition 14.16 (Velocity in surface coordinates). If $\alpha(t) = X(u(t), v(t))$, then

$$\alpha' = u'X_u + v'X_v.$$

Thus every surface velocity decomposes uniquely in the coordinate tangent basis.

14.6 Standard families of surfaces

Example 14.17 (Surface of revolution). If a profile $(r(v), z(v))$ with $r(v) > 0$ is revolved around the z -axis,

$$X(u, v) = (r(v) \cos u, r(v) \sin u, z(v)).$$

Regularity requires the profile to be regular and avoids points where the rotational coordinate degenerates.

Example 14.18 (Torus). For $R > r > 0$,

$$X(u, v) = ((R + r \cos v) \cos u, (R + r \cos v) \sin u, r \sin v)$$

is regular. Both parameters are periodic, so a global one-to-one rectangular patch is impossible, but finitely many patches cover the torus.

Example 14.19 (Helicoid). The map

$$X(u, v) = (u \cos v, u \sin v, av), \quad a \neq 0,$$

defines a regular helicoid. Its coordinate curves exhibit simultaneously radial and helical directions.

14.7 Overlap maps and intrinsic smoothness

Theorem 14.20 (Smooth overlap maps). If $X : U \rightarrow S$ and $Y : V \rightarrow S$ are overlapping regular surface patches, then

$$Y^{-1} \circ X$$

is smooth wherever defined.

Proof idea. Near an overlap point, choose a coordinate projection of $\mathbb{R}^3$ whose restriction has nonsingular derivative on the tangent plane. The inverse-function theorem makes that projection a local chart for both patches, and the transition is a composition of smooth local inverses. $\square$

Corollary 14.21 (Surface as a smooth manifold). Every regular surface in $\mathbb{R}^3$ carries a canonical smooth 2-manifold structure compatible with its regular patches.

14.8 Exercises

Exercise 14.22. Check that $X(u, v) = (u, v, u^2 + v^2)$ is regular.

Hint. Compute $X_u \times X_v$. □

Solution. $X_u = (1, 0, 2u)$, $X_v = (0, 1, 2v)$, so their cross product $(-2u, -2v, 1)$ never vanishes. □

Exercise 14.23. Find the tangent plane to $z = x^2 + y^2$ at $(1, 0, 1)$.

Hint. Use the graph normal. □

Solution. A normal is $(-2, 0, 1)$, so the plane is $-2(x - 1) + (z - 1) = 0$. □

Exercise 14.24. Prove the cross-product rank criterion.

Hint. Rank two means two independent columns. □

Solution. The columns are X_u, X_v ; in $\mathbb{R}^3$ they are independent exactly when their cross product is nonzero. □

Exercise 14.25. For the sphere show $T_p S^2 = p^\perp$.

Hint. Use the level-set function $F(x) = \|x\|^2$. □

Solution. $dF_p(v) = 2p \cdot v$, so the tangent space is its kernel $p^\perp$. □

Exercise 14.26. Check the cylinder patch is regular.

Hint. Compute its coordinate derivatives. □

Solution. $X_u = (-R \sin u, R \cos u, 0)$, $X_v = (0, 0, 1)$; their cross product has norm R . □

Exercise 14.27. Why do spherical coordinates fail to be regular at a pole?

Hint. Inspect $X_\theta \times X_\phi$. □

Solution. One coordinate vector collapses at a pole, so the cross product vanishes even though the sphere itself is regular there. □

Exercise 14.28. Prove graph patches are regular.

Hint. Look at the first two components of X_u, X_v . □

Solution. They contain the independent vectors $(1, 0)$ and $(0, 1)$, so the full vectors are independent. □

Exercise 14.29. Find a unit normal to a graph.

Hint. Normalize $(-f_u, -f_v, 1)$. □

Solution. $N = (-f_u, -f_v, 1)/\sqrt{1 + f_u^2 + f_v^2}$. □

Exercise 14.30. Prove the velocity formula $\alpha' = u'X_u + v'X_v$.

Hint. Apply the chain rule. □

Solution. Differentiate $X(u(t), v(t))$ componentwise to obtain the formula. □

Exercise 14.31. Why is the tangent basis decomposition unique?

Hint. Use regularity. □

Solution. X_u, X_v are linearly independent, so coordinates in their span are unique. □

Exercise 14.32. Show a regular change of parameters preserves the tangent plane.

Hint. Use $d(X \circ \phi) = dX d\phi$. □

Solution. Because $d\phi$ is invertible, the image of the composite derivative equals the image of dX . □

Exercise 14.33. What happens to the local normal under a parameter swap $(u, v) \mapsto (v, u)$?

Hint. The Jacobian determinant is -1 . □

Solution. The cross product reverses sign, so the chosen unit normal reverses. □

Exercise 14.34. Show $x^2 + y^2 - z = 0$ is a regular surface.

Hint. Check the gradient. □

Solution. $\nabla F = (2x, 2y, -1)$ never vanishes, so every level point is regular. □

Exercise 14.35. Where does $x^2 + y^2 - z^2 = 0$ fail the implicit regularity test?

Hint. Compute ∇F . □

Solution. The gradient $(2x, 2y, -2z)$ vanishes at the origin, the cone vertex. □

Exercise 14.36. Write the affine tangent-plane equation for an implicit surface.

Hint. Use the gradient as normal. □

Solution. At p , $\nabla F(p) \cdot (q - p) = 0$. □

Exercise 14.37. Check regularity of the helicoid patch.

Hint. Compute the cross product. □

Solution. $X_u = (\cos v, \sin v, 0)$, $X_v = (-u \sin v, u \cos v, a)$; the cross product has squared norm $a^2 + u^2 > 0$. □

Exercise 14.38. Why can a torus not be covered by one periodic rectangle as a one-to-one patch?

Hint. Opposite parameter edges represent the same points. □

Solution. Periodicity forces identifications, violating global injectivity on a full closed-period rectangle; overlapping open patches are used instead. □

Exercise 14.39. Describe the coordinate curves on a cylinder.

Hint. Hold one parameter fixed. □

Solution. Fixing height gives circles; fixing angle gives vertical lines. □

Exercise 14.40. For a surface of revolution compute X_u .

Hint. Differentiate in the angular coordinate. □

Solution. $X_u = (-r \sin u, r \cos u, 0)$. □

Exercise 14.41. State a regularity condition for a surface of revolution away from the axis.

Hint. Inspect $X_u \times X_v$. □

Solution. It is nonzero when $r > 0$ and the profile derivative (r', z') is nonzero. □

Exercise 14.42. Explain why a regular surface is locally a graph after a coordinate permutation.

Hint. Use a nonzero component of the normal. □

Solution. Rank two means some 2×2 minor of dX is nonsingular; projection to the corresponding coordinate plane is locally invertible, yielding a graph. □

Exercise 14.43. Show the tangent plane is independent of the chosen surface curve realizing a tangent vector.

Hint. Use the patch derivative image. □

Solution. Every curve velocity lies in $\text{im } dX$, and every vector in that image is realized by a parameter-domain curve, so the plane is intrinsic. □

Exercise 14.44. What is the dimension of a regular surface tangent space?

Hint. Use rank two. □

Solution. It is two-dimensional because it is the image of a rank-two derivative. □

Exercise 14.45. Explain why the cone vertex is not a regular-surface point.

Hint. Compare local topology or tangent behavior. □

Solution. No unique tangent plane exists there; equivalently the standard implicit gradient vanishes and the cone cannot be represented as a smooth graph near the vertex. □

14.9 Solved problem dossiers

Problem 14.46 (Legacy regular-patch problem). Prove the rank-two and cross-product criteria are equivalent.

Solution. The derivative has the two coordinate tangent vectors as columns; rank two is their linear independence, equivalent to nonzero cross product. □

Problem 14.47 (Legacy tangent-plane problem). Prove the tangent plane equals the set of velocities of curves on the surface.

Solution. Apply the chain rule in one direction and realize any coordinate linear combination by a straight parameter-domain curve in the other. □

Problem 14.48 (Legacy graph problem). Show a smooth graph is a regular surface and identify its normal.

Solution. Its coordinate tangent vectors are independent, and their cross product is $(-f_u, -f_v, 1)$. □

Problem 14.49 (Legacy implicit-surface problem). Recover a regular surface from a regular level set.

Solution. If $\nabla F \neq 0$, the regular-level theorem gives a two-dimensional embedded submanifold with tangent space $\ker dF = \nabla F^\perp$. $\square$

Problem 14.50 (Sphere atlas dossier). Explain why the sphere is regular despite singularities of spherical coordinates.

Solution. Regularity is a property of the surface, not one chosen parametrization. Near each point use an implicit-function graph or a coordinate projection patch with nonsingular derivative. $\square$

Problem 14.51 (Cylinder dossier). Compute tangent basis and normal for the circular cylinder.

Solution. The standard patch has $X_u = (-R \sin u, R \cos u, 0)$, $X_v = (0, 0, 1)$, and a radial normal $(\cos u, \sin u, 0)$. $\square$

Problem 14.52 (Change-of-parameters dossier). Prove tangent planes and regularity are invariant under a diffeomorphic parameter change.

Solution. The chain rule composes dX with an invertible 2×2 derivative, preserving rank and image. $\square$

Problem 14.53 (Surface-of-revolution dossier). Derive the regularity condition for a standard surface of revolution.

Solution. Compute $X_u \times X_v$; its norm is $r\sqrt{r'^2 + z'^2}$, so it is regular wherever $r > 0$ and the profile is regular. $\square$

Problem 14.54 (Torus dossier). Verify the standard torus parametrization is regular for $R > r > 0$.

Solution. The cross-product norm is $r(R + r \cos v)$, strictly positive because $R - r > 0$. $\square$

Problem 14.55 (Helicoid dossier). Verify regularity and identify tangent directions of the helicoid.

Solution. The cross product has norm $\sqrt{a^2 + u^2}$. The u -curves are horizontal radial lines and the v -curves are helices. $\square$

Problem 14.56 (Overlap-map dossier). Why are transition maps between regular surface patches smooth?

Solution. Choose an ambient coordinate projection that is nonsingular on the tangent plane. It supplies a common local Euclidean chart, reducing the transition to compositions of smooth maps and inverse-function-theorem inverses. $\square$

Problem 14.57 (Singular parametrization versus singular surface). Distinguish a bad parametrization from a genuinely singular surface point.

Solution. A bad parametrization can lose rank even when another patch is regular, as at spherical-coordinate poles. A genuinely singular point admits no regular patch at all, as at the cone vertex. $\square$

14.10 Challenge problems

Problem 14.58 (Local graph theorem for surfaces). Prove every regular surface is locally a graph over some coordinate plane.

Solution. At a regular point some 2×2 minor of the patch derivative is nonsingular. Project to those two ambient coordinates. The inverse-function theorem makes the projection composed with the patch a local diffeomorphism; solving for the remaining coordinate gives the graph representation. $\square$

Problem 14.59 (Equivalent tangent definitions). Show that the derivative-image, curve-velocity, and implicit-kernel definitions of the tangent plane agree whenever all are available.

Solution. The derivative-image and curve-velocity descriptions agree by the chain rule and curve realization. For a regular level $F = c$, differentiating $F \circ \alpha = c$ gives inclusion in $\ker dF$, and both spaces have dimension two, hence equality. $\square$

Problem 14.60 (Atlas from regular patches). Build explicitly the smooth-manifold atlas carried by a regular surface.

Solution. Use inverses of regular patches as charts on their images. The overlap theorem gives smooth transitions. The subspace topology is second countable and Hausdorff as a subspace of $\mathbb{R}^3$, so the resulting atlas defines a smooth two-manifold. $\square$

Problem 14.61 (Orientability preview). Explain how local normals on overlapping patches encode the obstruction to a global orientation.

Solution. Each patch supplies a unit normal. On a connected overlap two choices differ by sign. An orientation is a coherent choice with positive sign on all overlaps; failure to make these signs globally consistent is precisely the obstruction developed later in surface orientation theory. $\square$

Problem 14.62 (Regularity under ambient diffeomorphism). If $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a diffeomorphism and S is regular, prove $\Phi(S)$ is regular.

Solution. Compose every regular patch X with Φ . Its derivative is $d\Phi \circ dX$, rank two because $d\Phi$ is invertible, and the composition is a homeomorphism onto the corresponding image. $\square$

Bridge to VII/15

A regular patch provides a tangent basis X_u, X_v and a local normal N . VII/15 uses these to measure lengths, angles, areas, and normal bending through the first and second fundamental forms.

Chapter 15

First and Second Fundamental Forms

A surface patch supplies two kinds of local quadratic data. The first fundamental form is the Euclidean inner product restricted to the tangent plane; it measures intrinsic lengths and angles. The second fundamental form records the normal component of second derivatives and therefore measures how the surface bends inside $\mathbb{R}^3$. In coordinates,

$$I = E du^2 + 2F du dv + G dv^2, \quad II = e du^2 + 2f du dv + g dv^2.$$

15.1 The first fundamental form

Definition 15.1 (First fundamental form). For a regular surface $S \subset \mathbb{R}^3$, the first fundamental form at p is

$$I_p(v, w) = \langle v, w \rangle, \quad v, w \in T_p S.$$

It is a positive-definite symmetric bilinear form.

Definition 15.2 (Metric coefficients). For a patch $X(u, v)$, set

$$E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle.$$

Then for $w = aX_u + bX_v$,

$$I(w, w) = Ea^2 + 2Fab + Gb^2.$$

Proposition 15.3 (Positive definiteness). *For a regular patch,*

$$E > 0, \quad EG - F^2 > 0.$$

Proof. The matrix $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is the Gram matrix of the independent vectors X_u, X_v , hence positive definite. $\square$

15.2 Length, angle and area

Proposition 15.4 (Length of a surface curve). *If $\alpha(t) = X(u(t), v(t))$, then*

$$\|\alpha'\|^2 = E u'^2 + 2F u'v' + G v'^2,$$

so

$$L(\alpha) = \int \sqrt{E u'^2 + 2F u'v' + G v'^2} dt.$$

Proposition 15.5 (Angles). *For tangent vectors $a = a^i X_i$ and $b = b^j X_j$,*

$$\cos \theta = \frac{I(a, b)}{\sqrt{I(a, a)I(b, b)}}.$$

Coordinate curves are orthogonal exactly when $F = 0$.

Theorem 15.6 (Area element). *The Euclidean area element of a patch is*

$$dA = \|X_u \times X_v\| du dv = \sqrt{EG - F^2} du dv.$$

Proof. The square of the cross-product norm is the Gram determinant $\|X_u\|^2\|X_v\|^2 - \langle X_u, X_v \rangle^2 = EG - F^2$. $\square$

15.3 The second fundamental form

Definition 15.7 (Second fundamental form coefficients). Choose a local unit normal N . Define

$$e = \langle X_{uu}, N \rangle, \quad f = \langle X_{uv}, N \rangle, \quad g = \langle X_{vv}, N \rangle.$$

The second fundamental form is

$$II(aX_u + bX_v, aX_u + bX_v) = ea^2 + 2fab + gb^2.$$

Proposition 15.8 (Normal acceleration). *For a surface curve $\alpha(t) = X(u(t), v(t))$,*

$$\langle \alpha'', N \rangle = e u'^2 + 2f u'v' + g v'^2.$$

Thus II measures the normal component of acceleration determined by the tangent velocity.

Proof. Differentiate $\alpha' = u'X_u + v'X_v$. Terms involving $u''X_u + v''X_v$ are tangent and vanish after pairing with N ; the remaining second-derivative terms give the displayed quadratic form. $\square$

Warning 15.9 (Dependence on the chosen normal). Replacing N by $-N$ leaves I unchanged but replaces II by $-II$. Extrinsic signed bending therefore depends on a choice of local orientation.

15.4 Normal curvature

Definition 15.10 (Normal curvature). For a nonzero tangent direction $v \in T_p S$, define

$$k_n(v) = \frac{II(v, v)}{I(v, v)}.$$

For a unit-speed surface curve with tangent v , this equals the component of the curve curvature vector in the surface-normal direction.

Proposition 15.11 (Dependence only on direction). *For any nonzero scalar c ,*

$$k_n(cv) = k_n(v).$$

Proof. Both numerator and denominator are quadratic in v , so the factor c^2 cancels. $\square$

15.5 Coordinate change

Theorem 15.12 (Tensorial transformation). *Under a parameter change $(u, v) = (u(\xi, \eta), v(\xi, \eta))$, the coefficient matrices of I and II transform by*

$$\tilde{G} = J^T G J, \quad \tilde{B} = J^T B J,$$

provided the surface normal is transported consistently. Thus both forms define bilinear forms independent of the chosen coordinates.

15.6 Examples

Example 15.13 (Plane). For $X(u, v) = p + ua + vb$ with constant independent vectors a, b , all second derivatives vanish. Hence

$$II = 0.$$

The first form is the constant Gram form of a, b .

Example 15.14 (Sphere). On the sphere of radius R , choose the outward normal $N = X/R$. Differentiating $\langle X_i, N \rangle = 0$ gives

$$\langle X_{ij}, N \rangle = -\frac{1}{R} \langle X_i, X_j \rangle.$$

With the sign convention used here,

$$II = -\frac{1}{R} I.$$

Choosing the inward normal reverses the sign.

Example 15.15 (Cylinder). For $X(u, v) = (R \cos u, R \sin u, v)$ and outward normal $N = (\cos u, \sin u, 0)$,

$$E = R^2, \quad F = 0, \quad G = 1,$$

and

$$e = -R, \quad f = 0, \quad g = 0.$$

Thus circumferential directions bend normally while axial directions have zero normal curvature.

15.7 Local isometries and intrinsic data

Definition 15.16 (Local isometry of surfaces). A diffeomorphism $\Phi : S \rightarrow \tilde{S}$ is a local isometry if

$$I_{\tilde{S}}(d\Phi(v), d\Phi(w)) = I_S(v, w)$$

for tangent vectors in its domain.

Proposition 15.17 (Consequences of preserving I). *A local isometry preserves lengths of curves and angles between tangent vectors, and hence preserves the area density determined by I .*

Example 15.18 (Plane and cylinder). The map

$$(u, v) \longmapsto (R \cos(u/R), R \sin(u/R), v)$$

from a strip in the plane to the cylinder pulls back the cylinder's first form to

$$du^2 + dv^2.$$

Thus the cylinder is locally isometric to the plane even though their second fundamental forms differ.

Remark 15.19 (Intrinsic versus extrinsic). The first fundamental form depends only on metric geometry within the surface. The second fundamental form detects the embedding in $\mathbb{R}^3$. The plane–cylinder comparison is the basic example: same local first form, different bending.

15.8 Exercises

Exercise 15.20. Compute E, F, G for a graph $X(u, v) = (u, v, f(u, v))$.

Hint. Dot the two coordinate derivatives. □

Solution. $E = 1 + f_u^2$, $F = f_u f_v$, $G = 1 + f_v^2$. □

Exercise 15.21. Prove $EG - F^2 > 0$ for a regular patch.

Hint. Use the Gram determinant. □

Solution. $EG - F^2 = \|X_u \times X_v\|^2 > 0$. □

Exercise 15.22. When are coordinate curves orthogonal?

Hint. Inspect their tangent inner product. □

Solution. They are orthogonal exactly when $F = \langle X_u, X_v \rangle = 0$. □

Exercise 15.23. Derive the surface-curve speed formula.

Hint. Substitute $\alpha' = u'X_u + v'X_v$. □

Solution. Expanding its squared norm gives $Eu'^2 + 2Fu'v' + Gv'^2$. □

Exercise 15.24. Derive the area density from E, F, G .

Hint. Use the cross-product identity. □

Solution. $\|X_u \times X_v\|^2 = EG - F^2$, hence $dA = \sqrt{EG - F^2} du dv$. □

Exercise 15.25. Compute the first form of the cylinder.

Hint. Use the standard cylinder patch. □

Solution. $E = R^2$, $F = 0$, $G = 1$, so $I = R^2 du^2 + dv^2$. □

Exercise 15.26. Compute the first form of the plane patch $X(u, v) = (u, v, 0)$.

Hint. The coordinate tangent vectors are standard basis vectors. □

Solution. $E = G = 1$, $F = 0$, so $I = du^2 + dv^2$. □

Exercise 15.27. Why does reversing the surface normal reverse II ?

Hint. Each coefficient is paired with N . □

Solution. All of e, f, g change sign linearly, so the entire bilinear form changes sign. □

Exercise 15.28. Show a plane has zero second fundamental form.

Hint. Its patch is affine. □

Solution. All second derivatives of an affine patch vanish, hence $e = f = g = 0$. □

Exercise 15.29. Compute e, f, g for the standard cylinder with outward normal.

Hint. Differentiate twice. □

Solution. $X_{uu} = (-R \cos u, -R \sin u, 0)$, giving $e = -R$; mixed and vv derivatives vanish, so $f = g = 0$. □

Exercise 15.30. Find the normal curvature of the axial direction on a cylinder.

Hint. Use $v = X_v$. □

Solution. $II(X_v, X_v) = g = 0$, while $I(X_v, X_v) = 1$, so $k_n = 0$. □

Exercise 15.31. Find the normal curvature of the circumferential direction on the outward-oriented cylinder.

Hint. Use $v = X_u$. □

Solution. $k_n = e/E = -R/R^2 = -1/R$ with the stated sign convention. □

Exercise 15.32. Prove k_n depends only on a tangent line, not its scaling.

Hint. Use quadratic homogeneity. □

Solution. Both forms scale by c^2 , so their ratio is unchanged. □

Exercise 15.33. For the sphere with outward normal, compute normal curvature in any direction.

Hint. Use $II = -(1/R)I$. □

Solution. Every nonzero tangent direction has $k_n = -1/R$. □

Exercise 15.34. What does $F = 0$ mean geometrically?

Hint. It is the mixed coefficient of I . □

Solution. The coordinate tangent vectors, and hence the two coordinate curves at each point, meet orthogonally. □

Exercise 15.35. What does $E = G$ and $F = 0$ mean for coordinates?

Hint. Compare the two coordinate basis lengths and angle. □

Solution. The coordinate directions are orthogonal and have equal scale; such coordinates are conformal. □

Exercise 15.36. Show local isometries preserve curve speed.

Hint. Apply the defining identity to α' . □

Solution. $\|d\Phi(\alpha')\|^2 = I(d\Phi\alpha', d\Phi\alpha') = I(\alpha', \alpha')$. □

Exercise 15.37. Show local isometries preserve curve length.

Hint. Integrate preserved speed. □

Solution. Length is the integral of speed, so pointwise speed preservation gives equal lengths. □

Exercise 15.38. Show local isometries preserve tangent angles.

Hint. Use the cosine formula in terms of I . □

Solution. All numerator and denominator inner products are preserved, hence so is the angle. □

Exercise 15.39. Check the plane-to-cylinder map is locally isometric.

Hint. Differentiate with respect to u, v . □

Solution. The images of the coordinate vectors are orthonormal: the u -derivative has unit length, the v -derivative has unit length, and they are orthogonal. □

Exercise 15.40. Why are the plane and cylinder not congruent despite local isometry?

Hint. Compare second fundamental forms. □

Solution. The plane has $II = 0$, whereas the cylinder has nonzero circumferential second-form coefficient. □

Exercise 15.41. Show the normal component of surface-curve acceleration depends only on its velocity.

Hint. Pair the second derivative expansion with N . □

Solution. Tangential terms involving u'', v'' vanish against N , leaving only the quadratic expression in u', v' . □

Exercise 15.42. How do I and II matrices transform under coordinate change?

Hint. Use bilinearity and the Jacobian matrix J . □

Solution. Both transform by congruence: $\tilde{G} = J^T G J$ and $\tilde{B} = J^T B J$, with a consistent normal for II . □

Exercise 15.43. Explain why I is positive definite while II need not be.

Hint. Compare their definitions. □

Solution. $I(v, v) = \|v\|^2 > 0$ for $v \neq 0$. The second form is a signed normal-bending form and may be positive, negative, zero, or indefinite. □

15.9 Solved problem dossiers

Problem 15.44 (Legacy first-form problem). Derive the coefficients E, F, G and the quadratic line element.

Solution. Expand the Euclidean inner product of $dX = X_u du + X_v dv$ with itself to obtain $E du^2 + 2F du dv + G dv^2$. $\square$

Problem 15.45 (Legacy area problem). Prove $dA = \sqrt{EG - F^2} du dv$.

Solution. The area parallelogram is $\|X_u \times X_v\|$, whose square is the Gram determinant $EG - F^2$. $\square$

Problem 15.46 (Legacy second-form problem). Derive e, f, g from normal acceleration.

Solution. Differentiate a surface curve twice and pair with the normal. Terms multiplying u'', v'' are tangent and disappear, leaving the quadratic expression with X_{uu}, X_{uv}, X_{vv} . $\square$

Problem 15.47 (Graph fundamental forms). Compute both forms for a graph $z = f(u, v)$.

Solution. The first coefficients are $1 + f_u^2, f_u f_v, 1 + f_v^2$. With $W = \sqrt{1 + f_u^2 + f_v^2}$ and upward normal, the second coefficients are $f_{uu}/W, f_{uv}/W, f_{vv}/W$. $\square$

Problem 15.48 (Sphere dossier). Show the second form of the outward-oriented sphere is $-(1/R)I$ in the convention of this chapter.

Solution. Use $N = X/R$ and differentiate $\langle X_i, N \rangle = 0$: $\langle X_{ij}, N \rangle = -\langle X_i, X_j \rangle/R$. $\square$

Problem 15.49 (Cylinder dossier). Compute I, II , and normal curvature of the coordinate directions on the cylinder.

Solution. $I = R^2 du^2 + dv^2$, $II = -R du^2$. Thus circumferential normal curvature is $-1/R$ and axial normal curvature is zero. $\square$

Problem 15.50 (Normal-sign dossier). Prove exactly which quantities in this chapter change when N is reversed.

Solution. I, E, F, G , lengths, angles, and area are unchanged. The coefficients e, f, g, II , and signed normal curvature all reverse sign. $\square$

Problem 15.51 (Coordinate-invariance dossier). Prove the first and second forms are independent of the chosen patch.

Solution. Under a parameter transition tangent components transform by the Jacobian. Bilinearity gives congruence transformation of both coefficient matrices, representing the same geometric bilinear forms. $\square$

Problem 15.52 (Plane–cylinder local isometry). Prove a plane strip and cylinder patch have the same first fundamental form but different second forms.

Solution. Using arc-length coordinate $u = R\theta$, the cylinder first form becomes $du^2 + dv^2$, equal to the plane. The plane has $II = 0$, while the cylinder does not. $\square$

Problem 15.53 (Orthogonal coordinates dossier). Assume $F = 0$. Simplify length, angle, and area formulas.

Solution. Length density is $Eu'^2 + Gv'^2$, coordinate directions meet at right angles, and $dA = \sqrt{EG} du dv$. $\square$

Problem 15.54 (Conformal coordinates dossier). Assume $E = G = \lambda^2$ and $F = 0$. Explain angle preservation relative to the parameter plane.

Solution. The first form is $\lambda^2(du^2 + dv^2)$, a scalar multiple of the Euclidean metric, so all angles are unchanged although lengths are scaled by λ . $\square$

Problem 15.55 (Normal curvature dossier). Why does $II(v, v)/I(v, v)$ equal the surface-normal component of curvature for a unit-speed surface curve?

Solution. For unit speed, $I(v, v) = 1$ and the acceleration is the curvature vector. Its normal component is $\langle \alpha'', N \rangle = II(v, v)$. $\square$

15.10 Challenge problems

Problem 15.56 (Tensor transformation from scratch). Derive the coordinate-change formulas for both fundamental forms directly from $\tilde{X} = X \circ \phi$.

Solution. First derivatives satisfy $\tilde{X}_a = X_i J^i_a$, immediately giving $\tilde{G} = J^T G J$. Second derivatives contain an additional term $X_i \partial_a \partial_b \phi^i$, but this is tangent and vanishes when paired with N ; therefore $\tilde{B} = J^T B J$. $\square$

Problem 15.57 (Developability preview). Use the plane–cylinder example to explain how a surface can bend without changing its first fundamental form.

Solution. Rolling the plane into a cylinder preserves the metric coefficients after using arc-length angular coordinate, so lengths and angles are unchanged. The second form becomes nonzero in one direction, showing extrinsic bending without intrinsic stretching. $\square$

Problem 15.58 (Normal section principle). Let a plane through the surface normal and a tangent direction cut the surface in a regular curve. Show its signed curvature at the point equals the normal curvature of that tangent direction.

Solution. The section curve lies in a plane containing N , so its curvature vector at the point has no component orthogonal to that plane. The component along N is therefore its signed curvature, while the normal-acceleration formula identifies the same component with $II(v, v)/I(v, v)$. $\square$

Problem 15.59 (Recover area from the metric). Prove that the local area density is determined entirely by the first fundamental form, independent of the embedding coordinates.

Solution. In any basis the area-square of the tangent parallelogram is the determinant of its Gram matrix. Under coordinate change the determinant acquires $(\det J)^2$, exactly balanced by the coordinate area transformation, so $\sqrt{\det G} du dv$ is intrinsic. $\square$

Problem 15.60 (First versus second form data). Give a precise argument that equal first fundamental forms do not determine the embedding up to rigid motion.

Solution. The plane and cylinder provide explicit locally isometric patches with the same pulled-back first form. Their second forms differ, and a rigid motion would preserve both Euclidean first and second-order embedding data. Hence first-form equality alone does not imply congruence. $\square$

Bridge to VII/16

The second fundamental form records signed normal bending but has not yet been organized as a linear operator. VII/16 introduces the Gauss map and shape operator, turning the derivative of the unit normal into the linear-algebraic mechanism behind the second fundamental form.

Chapter 16

The Gauss Map and Shape Operator

For an oriented regular surface $S \subset \mathbb{R}^3$, the unit normal is not merely an auxiliary vector. Its variation records how the tangent plane turns. The two central objects are

$$N : S \rightarrow S^2, \quad S_p = -dN_p : T_p S \rightarrow T_p S.$$

The first is the Gauss map and the second is the shape operator (Weingarten map).

16.1 Local unit normals

Definition 16.1 (Local unit normal). For a regular parametrization $X : U \subset \mathbb{R}^2 \rightarrow S$, the associated unit normal is

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|}.$$

Changing the order of the coordinates reverses the sign.

Proposition 16.2 (Orthogonality). *At every point of a regular patch,*

$$\langle N, X_u \rangle = \langle N, X_v \rangle = 0, \quad \langle N, N \rangle = 1.$$

Consequently $dN_p(v) \perp N(p)$, so $dN_p(v) \in T_p S = T_{N(p)} S^2$.

Proof. The first two identities follow from the cross product. Differentiating $\langle N, N \rangle = 1$ in a tangent direction v gives $2\langle dN_p(v), N(p) \rangle = 0$. $\square$

16.2 Orientability and the global Gauss map

Definition 16.3 (Orientable surface). A regular surface is orientable if it admits a smooth global unit normal field $N : S \rightarrow S^2$.

Proposition 16.4 (Two choices on a connected surface). *If S is connected and orientable, any two global unit normals inducing the same orientation agree, while the opposite orientation is represented by $-N$.*

Warning 16.5 (The Gauss map requires an orientation). A nonorientable surface has no globally defined Gauss map into S^2 , although every coordinate patch has a local Gauss map.

16.3 Differential of the Gauss map

Definition 16.6 (Shape operator). For an oriented surface and $p \in S$, define

$$S_p = -dN_p : T_p S \rightarrow T_p S.$$

Theorem 16.7 (Weingarten self-adjointness). *The shape operator is self-adjoint with respect to the first fundamental form:*

$$\langle S_p u, v \rangle = \langle u, S_p v \rangle.$$

Proof. In a parametrization, differentiate $\langle N, X_u \rangle = \langle N, X_v \rangle = 0$. This gives

$$\langle N_u, X_v \rangle = \langle N_v, X_u \rangle$$

because mixed partials $X_{uv} = X_{vu}$. Since $S = -dN$, the matrix is symmetric with respect to the metric. $\square$

16.4 Coordinate matrix of the shape operator

Let

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} e & f \\ f & g \end{pmatrix},$$

where

$$E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle,$$

and

$$e = \langle X_{uu}, N \rangle, \quad f = \langle X_{uv}, N \rangle, \quad g = \langle X_{vv}, N \rangle.$$

Theorem 16.8 (Matrix relation). *In the basis (X_u, X_v) ,*

$$[S] = I^{-1} II.$$

Equivalently, for tangent vectors u, v ,

$$II(u, v) = I(Su, v).$$

Proof. From $\langle N, X_i \rangle = 0$, differentiation gives

$$-\langle N_i, X_j \rangle = \langle N, X_{ij} \rangle.$$

Thus the bilinear form associated with $-dN$ through the metric is exactly II . $\square$

16.5 Normal curvature

Definition 16.9 (Normal curvature). For a nonzero tangent vector $v \in T_p S$,

$$k_n(v) = \frac{II(v, v)}{I(v, v)} = \frac{\langle S_p v, v \rangle}{\langle v, v \rangle}.$$

Theorem 16.10 (Meusnier normal-curvature principle). *If a regular curve on the surface has unit tangent T at p and curvature vector κn_c , then the surface-normal component is*

$$k_n(T) = \langle \kappa n_c, N \rangle.$$

Thus normal curvature depends only on the tangent direction, not on the chosen surface curve realizing it.

16.6 Principal directions preview

Theorem 16.11 (Spectral theorem for the shape operator). *At every point, the self-adjoint map S_p admits an orthonormal eigenbasis of $T_p S$. Its eigenvalues are the principal curvatures and its eigenvectors are the principal directions.*

Remark 16.12. VII/17 develops the invariants built from these eigenvalues. Here the emphasis is the operator itself and its geometric origin as the derivative of the Gauss map.

16.7 Examples

Example 16.13 (Plane). For a plane, N is constant. Hence $dN = 0$ and $S = 0$.

Example 16.14 (Sphere). For the outward orientation of the sphere S_R^2 ,

$$N(p) = \frac{p}{R}, \quad dN_p(v) = \frac{v}{R}, \quad S_p = -\frac{1}{R} \text{id}.$$

With the inward orientation the signs reverse.

Example 16.15 (Cylinder). For $X(u, v) = (R \cos u, R \sin u, v)$, the normal is $N = (\cos u, \sin u, 0)$. The axial direction lies in $\ker S$, while the circular direction has nonzero normal bending.

16.8 Geometric meaning

The shape operator measures first-order rotation of tangent planes. A direction v lies in $\ker S_p$ precisely when the surface normal is stationary to first order along v . The determinant and trace of S_p will become the Gaussian and mean curvature in VII/17.

16.9 Exercises

Exercise 16.16. Show that $dN_p(v)$ is tangent to S^2 at $N(p)$.

Hint. Differentiate $\langle N, N \rangle = 1$. □

Solution. The derivative gives $2\langle dN_p(v), N(p) \rangle = 0$, which is exactly tangency to the unit sphere. □

Exercise 16.17. Verify that $N = (X_u \times X_v) / \|X_u \times X_v\|$ is orthogonal to the patch tangent plane.

Hint. Use the defining property of the cross product. □

Solution. The cross product is orthogonal to both X_u and X_v , which span the tangent plane. □

Exercise 16.18. Explain how reversing patch orientation changes the local normal.

Hint. Interchange u and v . □

Solution. Since $X_v \times X_u = -X_u \times X_v$, the unit normal changes from N to $-N$. □

Exercise 16.19. Show that if N is a global unit normal then $-N$ is another.

Hint. Check smoothness and norm. □

Solution. Negation preserves smoothness and unit length and reverses the chosen orientation. □

Exercise 16.20. Compute the shape operator of a plane.

Hint. Its normal is constant. □

Solution. Because $dN = 0$, one has $S = -dN = 0$. □

Exercise 16.21. Compute S for the outward oriented sphere of radius R .

Hint. Use $N(p) = p/R$. □

Solution. Then $dN_p(v) = v/R$, hence $S_p(v) = -v/R$. □

Exercise 16.22. How does the sphere shape operator change under reversal of orientation?

Hint. Replace N by $-N$. □

Solution. Then $S = -d(-N) = dN$, so all eigenvalues change sign. □

Exercise 16.23. For the standard cylinder, identify a direction in $\ker S$.

Hint. Move parallel to the axis. □

Solution. The normal is independent of the axial coordinate, so its derivative in the axial direction vanishes. □

Exercise 16.24. Prove the identity $II(u, v) = I(Su, v)$.

Hint. Differentiate $\langle N, X_i \rangle = 0$. □

Solution. The differentiated orthogonality equations yield $-\langle dN(u), v \rangle = \langle N, D_u v \rangle$, which is the second fundamental form. □

Exercise 16.25. Derive $[S] = I^{-1}II$.

Hint. Write the previous bilinear identity in matrix form. □

Solution. The identity is $II = I[S]$; multiplying by I^{-1} gives $[S] = I^{-1}II$. □

Exercise 16.26. Why is I invertible on a regular patch?

Hint. Use linear independence of X_u, X_v . □

Solution. Its Gram matrix is positive definite because the tangent vectors are independent. □

Exercise 16.27. Prove self-adjointness of S .

Hint. Use symmetry of II . □

Solution. Since $II(u, v) = II(v, u)$ and $II(u, v) = I(Su, v)$, one gets $I(Su, v) = I(u, Sv)$. □

Exercise 16.28. Show that eigenvectors belonging to distinct eigenvalues are orthogonal.

Hint. Apply self-adjointness. □

Solution. If $Su = \lambda u$, $Sv = \mu v$, then $(\lambda - \mu)\langle u, v \rangle = 0$. □

Exercise 16.29. Express normal curvature as a Rayleigh quotient.

Hint. Use $II(v, v) = I(Sv, v)$. □

Solution. Thus $k_n(v) = \langle Sv, v \rangle / \langle v, v \rangle$. □

Exercise 16.30. Show normal curvature is unchanged when v is rescaled.

Hint. Both numerator and denominator are quadratic. □

Solution. Replacing v by cv multiplies numerator and denominator by c^2 . □

Exercise 16.31. What happens to normal curvature when the surface orientation is reversed?

Hint. The shape operator changes sign. □

Solution. Therefore every normal curvature changes sign. □

Exercise 16.32. Show that $S_p = 0$ implies the Gauss map has zero differential at p .

Hint. Use the definition. □

Solution. Since $S_p = -dN_p$, the two statements are equivalent. □

Exercise 16.33. Interpret $\ker S_p$ geometrically.

Hint. Think about $dN_p(v)$. □

Solution. It consists of directions in which the unit normal is stationary to first order. □

Exercise 16.34. For a unit eigenvector e of S with eigenvalue k , compute $k_n(e)$.

Hint. Insert $Se = ke$. □

Solution. The quotient becomes $\langle ke, e \rangle / \langle e, e \rangle = k$. □

Exercise 16.35. Why does the spectral theorem apply to S_p ?

Hint. The tangent plane is a Euclidean inner-product space. □

Solution. It is finite-dimensional and S_p is self-adjoint. □

Exercise 16.36. Relate the second fundamental coefficients to derivatives of the normal.

Hint. Differentiate normal orthogonality. □

Solution. One gets $e = -\langle N_u, X_u \rangle$, $f = -\langle N_u, X_v \rangle = -\langle N_v, X_u \rangle$, $g = -\langle N_v, X_v \rangle$. □

Exercise 16.37. Show that f has the two equivalent expressions above.

Hint. Differentiate $\langle N, X_u \rangle$ in v and $\langle N, X_v \rangle$ in u . □

Solution. Equality follows from $X_{uv} = X_{vu}$. □

Exercise 16.38. Explain why the Gauss map is constant on a connected planar patch.

Hint. All tangent planes have the same normal line. □

Solution. A continuous choice of unit normal cannot jump between antipodal values on a connected patch, hence it is constant. □

Exercise 16.39. State the bridge from S_p to VII/17.

Hint. Use its two eigenvalues. □

Solution. VII/17 defines principal curvatures as the eigenvalues, Gaussian curvature as their product, and mean curvature as half their sum. □

16.10 Solved problem dossiers

Problem 16.40 (Legacy Gauss-map problem). Starting from a regular parametrization, construct the local Gauss map and prove its differential takes tangent vectors to the tangent plane.

Solution. Normalize $X_u \times X_v$. Differentiating $\langle N, N \rangle = 1$ shows $dN(v) \perp N$, hence $dN(v) \in T_p S$. $\square$

Problem 16.41 (Legacy shape-operator problem). Derive the shape operator from the Gauss map and explain the sign convention.

Solution. Define $S = -dN$. The minus sign makes the usual outward sphere carry negative eigenvalues under this convention; reversing orientation reverses S . $\square$

Problem 16.42 (Legacy Weingarten problem). Prove the shape operator is self-adjoint.

Solution. Differentiate the two equations $\langle N, X_u \rangle = 0 = \langle N, X_v \rangle$, use equality of mixed partials, and translate the symmetric coefficient matrix into the metric inner product. $\square$

Problem 16.43 (Coordinate-matrix problem). Prove $[S] = I^{-1}II$.

Solution. From $II(u, v) = I(Su, v)$, the matrices satisfy $II = I[S]$. Since I is positive definite, it is invertible. $\square$

Problem 16.44 (Sphere problem). Compute N, dN, S for the sphere of radius R .

Solution. For the outward normal, $N(p) = p/R$; thus $dN(v) = v/R$ and $S(v) = -v/R$. $\square$

Problem 16.45 (Cylinder problem). Compute the qualitative eigendirections of the circular cylinder.

Solution. The normal varies around circles but not along generators. Hence the axial direction has eigenvalue 0 and the circular direction has the nonzero eigenvalue. $\square$

Problem 16.46 (Normal-curvature problem). Prove that normal curvature depends only on a tangent direction.

Solution. The surface-normal component of the curve acceleration is $II(T, T)$, so for a unit tangent it equals $I(ST, T)$, determined by the tangent vector and the surface. $\square$

Problem 16.47 (Orientation problem). Track N, S, II , and normal curvature under orientation reversal.

Solution. Replacing N by $-N$ multiplies dN, S, II , and every normal curvature by -1 , while I is unchanged. $\square$

Problem 16.48 (Kernel problem). Characterize directions along which the normal is stationary to first order.

Solution. They are exactly $\ker dN_p = \ker S_p$. $\square$

Problem 16.49 (Principal-direction preview). Use the spectral theorem to produce an orthonormal basis adapted to the shape operator.

Solution. Self-adjointness gives an orthonormal eigenbasis of $T_p S$. $\square$

Problem 16.50 (Second-form reconstruction problem). Show I and S determine II .

Solution. The formula is $II(u, v) = I(Su, v)$. $\square$

Problem 16.51 (Gauss-map rank problem). Explain what rank 0, 1, 2 of dN_p means infinitesimally.

Solution. Rank records the number of independent tangent directions in which the normal changes; because $S = -dN$, it is the rank of the shape operator. $\square$

16.11 Challenge problems

Problem 16.52 (Weingarten equations). Derive explicit formulas for N_u and N_v in the basis X_u, X_v .

Solution. Write $N_u = aX_u + bX_v$ and pair with X_u, X_v . The resulting linear system is $I(a, b)^T = -(e, f)^T$. Similarly $I(c, d)^T = -(f, g)^T$ for $N_v = cX_u + dX_v$. $\square$

Problem 16.53 (Gauss-map local behavior). Suppose S_p is invertible. What does the inverse function theorem say about the Gauss map near p ?

Solution. Since $dN_p = -S_p$ is an isomorphism between two-dimensional tangent spaces, N is a local diffeomorphism from the surface to an open subset of S^2 . $\square$

Problem 16.54 (Umbilic operator criterion). Show that if every tangent direction has the same normal curvature at p , then $S_p = \lambda I$.

Solution. For a self-adjoint operator, equality of the Rayleigh quotient on every unit vector forces both eigenvalues to equal the same number λ ; hence $S_p = \lambda I$. $\square$

Problem 16.55 (Rank-one shape operator). Assume S_p has rank one. Describe its kernel and image geometrically.

Solution. Self-adjointness gives $T_p S = \ker S_p \oplus (\ker S_p)^\perp$. The image is the one-dimensional orthogonal complement of the kernel and is the unique direction in which the normal changes to first order. $\square$

Problem 16.56 (Coordinate invariance). Explain why $I^{-1}II$ transforms as an endomorphism under a change of surface coordinates.

Solution. Both forms undergo congruence transformations by the Jacobian. Their combination $I^{-1}II$ changes by similarity, exactly the transformation law of a linear operator, so the shape operator is intrinsic to the embedding and orientation rather than to the chosen coordinates. $\square$

Bridge to VII/17

VII/17 turns the spectral data of the shape operator into scalar curvature invariants: principal curvatures, Gaussian curvature, and mean curvature.

Chapter 17

Principal, Gaussian and Mean Curvature

The shape operator S_p is a self-adjoint endomorphism of the tangent plane. Its eigenvalues contain the fundamental local bending data:

$$k_1, k_2, \quad K = k_1 k_2 = \det S, \quad H = \frac{k_1 + k_2}{2} = \frac{1}{2} \operatorname{tr} S.$$

17.1 Principal curvatures and directions

Definition 17.1 (Principal curvatures). The eigenvalues k_1, k_2 of S_p are the principal curvatures. Corresponding unit eigenvectors are principal directions.

Theorem 17.2 (Euler formula). *If e_1, e_2 is an orthonormal principal basis and*

$$v = \cos \theta e_1 + \sin \theta e_2,$$

then the normal curvature is

$$k_n(v) = k_1 \cos^2 \theta + k_2 \sin^2 \theta.$$

Proof. Use $Sv = k_1 \cos \theta e_1 + k_2 \sin \theta e_2$ and take the inner product with v . □

Corollary 17.3 (Extremal property). *The principal curvatures are the minimum and maximum values of normal curvature among unit tangent directions.*

17.2 Gaussian and mean curvature

Definition 17.4 (Gaussian curvature).

$$K(p) = \det S_p = k_1 k_2.$$

Definition 17.5 (Mean curvature).

$$H(p) = \frac{1}{2} \operatorname{tr} S_p = \frac{k_1 + k_2}{2}.$$

Proposition 17.6 (Orientation dependence). *Reversing the orientation sends*

$$k_i \mapsto -k_i, \quad H \mapsto -H, \quad K \mapsto K.$$

Thus Gaussian curvature is independent of orientation.

17.3 Coordinate formulas

With

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} e & f \\ f & g \end{pmatrix},$$

and $S = I^{-1}II$, one obtains:

Theorem 17.7 (Curvature formulas).

$$K = \frac{eg - f^2}{EG - F^2},$$

and

$$H = \frac{Eg - 2Ff + Ge}{2(EG - F^2)}.$$

Proof. The first follows from $\det(I^{-1}II) = \det II / \det I$. The second is one half of the trace of the 2×2 matrix $I^{-1}II$. $\square$

17.4 Point classification

Definition 17.8 (Elliptic, hyperbolic, parabolic, planar). At a point p :

- $K > 0$: elliptic;
- $K < 0$: hyperbolic;
- $K = 0$ with $S \neq 0$: parabolic;
- $S = 0$: planar.

Proposition 17.9 (Signs of normal curvature). *At an elliptic point all nonzero normal curvatures have the same sign after choosing an orientation. At a hyperbolic point there are tangent directions with opposite signs of normal curvature.*

17.5 Asymptotic directions

Definition 17.10 (Asymptotic direction). A nonzero tangent direction v is asymptotic if

$$II(v, v) = 0,$$

equivalently $k_n(v) = 0$.

Proposition 17.11 (Existence). *At a hyperbolic point there are exactly two asymptotic lines in the tangent plane. At an elliptic point there are none.*

17.6 Umbilic points

Definition 17.12 (Umbilic). A point is umbilic if

$$S_p = \lambda \text{ id},$$

equivalently $k_1 = k_2$.

Proposition 17.13 (Umbilic curvature relation). *At an umbilic point,*

$$K = H^2.$$

In general,

$$H^2 - K = \frac{(k_1 - k_2)^2}{4} \geq 0.$$

17.7 Examples

Example 17.14 (Plane). A plane has $k_1 = k_2 = 0$, hence $K = H = 0$.

Example 17.15 (Sphere). For the outward normal under the convention $S = -dN$,

$$k_1 = k_2 = -\frac{1}{R}, \quad K = \frac{1}{R^2}, \quad H = -\frac{1}{R}.$$

Every point is umbilic.

Example 17.16 (Cylinder). A circular cylinder has one zero principal curvature and one nonzero principal curvature. Hence $K = 0$ and $H \neq 0$.

Example 17.17 (Saddle model). Near a saddle point such as the origin of $z = x^2 - y^2$, the principal curvatures have opposite signs, so $K < 0$.

17.8 Gauss-map interpretation

Theorem 17.18 (Jacobian of the Gauss map). *Up to the orientation convention for area forms, the Jacobian determinant of the Gauss map is*

$$\det dN_p = \det(-S_p) = K(p).$$

Thus Gaussian curvature measures infinitesimal area distortion of the Gauss map.

17.9 What is intrinsic and what is extrinsic?

The definition above uses the embedding in $\mathbb{R}^3$, but Gaussian curvature is much more rigid than mean curvature. Theorema Egregium, developed later in the Riemannian arc, shows that K is determined by the first fundamental form alone. Mean curvature remains extrinsic.

17.10 Exercises

Exercise 17.19. State the principal curvatures in terms of the shape operator.

Hint. Use eigenvalues. □

Solution. They are precisely the two eigenvalues of the self-adjoint map S_p . □

Exercise 17.20. Prove Euler's formula for normal curvature.

Hint. Use an orthonormal eigenbasis. □

Solution. Writing $v = \cos \theta e_1 + \sin \theta e_2$ and pairing Sv with v gives $k_1 \cos^2 \theta + k_2 \sin^2 \theta$. □

Exercise 17.21. Why are principal directions orthogonal when $k_1 \neq k_2$?

Hint. Use self-adjointness. □

Solution. Eigenvectors of a self-adjoint map corresponding to distinct eigenvalues are orthogonal. □

Exercise 17.22. Show $K = \det S$.

Hint. Use the eigenvalues. □

Solution. The determinant of a linear map equals the product of its eigenvalues, hence $K = k_1 k_2$. □

Exercise 17.23. Show $H = \frac{1}{2} \operatorname{tr} S$.

Hint. Use the eigenvalues. □

Solution. The trace is $k_1 + k_2$, so half the trace is the mean curvature. □

Exercise 17.24. Track K under orientation reversal.

Hint. Both eigenvalues change sign. □

Solution. Their product is unchanged, so K is orientation independent. □

Exercise 17.25. Track H under orientation reversal.

Hint. Both eigenvalues change sign. □

Solution. Their sum changes sign, so $H \mapsto -H$. □

Exercise 17.26. Derive $K = (eg - f^2)/(EG - F^2)$.

Hint. Use $\det(I^{-1}II)$. □

Solution. One has $K = \det II / \det I = (eg - f^2)/(EG - F^2)$. □

Exercise 17.27. Derive the coordinate formula for H .

Hint. Compute the trace of $I^{-1}II$. □

Solution. The trace equals $(Eg - 2Ff + Ge)/(EG - F^2)$, and division by 2 gives H . □

Exercise 17.28. Classify a point with $K < 0$.

Hint. Use the sign classification. □

Solution. It is hyperbolic. □

Exercise 17.29. Classify a point with $K > 0$.

Hint. Use the sign classification. □

Solution. It is elliptic. □

Exercise 17.30. Classify a point with $K = 0$ but $Seq0$.

Hint. One principal curvature vanishes. □

Solution. It is parabolic. □

Exercise 17.31. Explain why a plane is planar in the curvature classification.

Hint. Its shape operator vanishes. □

Solution. The normal is constant, so $S = 0$. □

Exercise 17.32. Show a sphere is umbilic.

Hint. Its shape operator is a scalar multiple of the identity. □

Solution. On a radius- R sphere, $S = -(1/R)I$ for outward orientation. □

Exercise 17.33. Prove $H^2 - K = (k_1 - k_2)^2/4$.

Hint. Expand both sides. □

Solution. From $H = (k_1 + k_2)/2$ and $K = k_1 k_2$, subtraction yields the stated square. □

Exercise 17.34. Deduce $H^2 \geq K$.

Hint. Use the previous identity. □

Solution. The right side is a square and is nonnegative. □

Exercise 17.35. When does equality $H^2 = K$ hold?

Hint. Set the square to zero. □

Solution. Exactly when $k_1 = k_2$, i.e. at umbilic points. □

Exercise 17.36. Define an asymptotic direction.

Hint. Use normal curvature. □

Solution. It is a nonzero tangent direction v with $II(v, v) = 0$, equivalently $k_n(v) = 0$. □

Exercise 17.37. Why do hyperbolic points have asymptotic directions?

Hint. Principal curvatures have opposite signs. □

Solution. Euler's formula varies continuously from one sign to the other and vanishes in two distinct lines. □

Exercise 17.38. Why do elliptic points have no asymptotic directions?

Hint. Principal curvatures have the same nonzero sign. □

Solution. Euler's formula is then a positive or negative weighted sum and cannot vanish. □

Exercise 17.39. Compute K for a cylinder.

Hint. One principal curvature is zero. □

Solution. Their product is zero. □

Exercise 17.40. Compute K, H for a plane.

Hint. Both principal curvatures vanish. □

Solution. Both invariants are zero. □

Exercise 17.41. Interpret K through the Gauss map.

Hint. Use $\det dN$. □

Solution. It is the infinitesimal signed area distortion factor of the Gauss map, up to the standard orientation convention. □

Exercise 17.42. State the bridge to ruled/developable surfaces.

Hint. Use $K = 0$. □

Solution. VII/18 studies ruled surfaces and characterizes developable behavior through vanishing Gaussian curvature and degeneracy of the Gauss map. □

17.11 Solved problem dossiers

Problem 17.43 (Legacy principal-curvature problem). Show that principal curvatures are the extremal normal curvatures.

Solution. Euler's formula expresses normal curvature as a convex combination of k_1, k_2 , so its extrema are the eigenvalues. $\square$

Problem 17.44 (Legacy Gaussian-curvature problem). Derive the determinant formula for K in coordinates.

Solution. Use $S = I^{-1}II$, then take determinants. $\square$

Problem 17.45 (Legacy mean-curvature problem). Derive the trace formula for H in coordinates.

Solution. Invert the 2×2 first-form matrix, multiply by II , and take half the trace. $\square$

Problem 17.46 (Orientation-invariance problem). Determine which standard curvature data survive orientation reversal.

Solution. Principal curvatures and H reverse sign; K , the unordered principal directions, and the point type determined by the sign of K do not. $\square$

Problem 17.47 (Umbilic problem). Characterize umbilic points using the equality $H^2 = K$.

Solution. The identity $H^2 - K = (k_1 - k_2)^2/4$ shows equality exactly when $k_1 = k_2$. $\square$

Problem 17.48 (Hyperbolic-point problem). Derive the existence of two asymptotic directions at a hyperbolic point.

Solution. In a principal basis solve $k_1 \cos^2 \theta + k_2 \sin^2 \theta = 0$; opposite signs yield two tangent lines. $\square$

Problem 17.49 (Sphere problem). Compute all principal, Gaussian, and mean curvature data for a sphere.

Solution. Both principal curvatures equal $-1/R$ for outward orientation, giving $K = 1/R^2$ and $H = -1/R$. $\square$

Problem 17.50 (Cylinder problem). Compute the curvature type of a circular cylinder.

Solution. One principal curvature is zero and the other nonzero, so $K = 0$, $H \neq 0$, and the points are parabolic. $\square$

Problem 17.51 (Plane problem). Explain why every plane point is umbilic and planar.

Solution. The zero operator is $0 \cdot I$, so the principal curvatures coincide at zero and $S = 0$. $\square$

Problem 17.52 (Gauss-map problem). Relate K to the differential of the Gauss map.

Solution. Because $dN = -S$ in dimension two, $\det dN = \det S = K$. $\square$

Problem 17.53 (Rayleigh-quotient problem). Recover k_1, k_2 from the normal-curvature function on the unit circle.

Solution. They are its minimum and maximum; their eigendirections are the directions at which these extrema occur. $\square$

Problem 17.54 (Intrinsic-versus-extrinsic problem). Explain the conceptual difference between K and H .

Solution. Both arise from the embedding through S , but later Gauss theory proves K is determined by the first fundamental form, whereas H changes under isometric bendings in general. $\square$

17.12 Challenge problems

Problem 17.55 (Asymptotic-angle formula). At a hyperbolic point derive the angle of an asymptotic direction relative to a principal direction.

Solution. Solve $k_1 \cos^2 \theta + k_2 \sin^2 \theta = 0$. Thus $\tan^2 \theta = -k_1/k_2$, which is positive exactly when $k_1 k_2 < 0$. $\square$

Problem 17.56 (Recover principal curvatures from H, K). Show that k_1, k_2 are roots of a quadratic determined by H, K .

Solution. The characteristic polynomial is $\lambda^2 - 2H\lambda + K$. Hence $k_{1,2} = H \pm \sqrt{H^2 - K}$. $\square$

Problem 17.57 (Gauss-map rank classification). Relate the rank of dN_p to K and the point type.

Solution. Since $dN = -S$, rank two means $K \neq 0$; rank one means $K = 0$ but $S \neq 0$, a parabolic point; rank zero means $S = 0$, a planar point. $\square$

Problem 17.58 (Normal-curvature reconstruction). Suppose the function $v \mapsto k_n(v)$ is known for every unit tangent vector. Show it determines S_p .

Solution. The quadratic form $q(v) = \langle Sv, v \rangle$ is known on the unit circle and hence by homogeneity on all vectors. Polarization recovers $\langle Su, v \rangle$, and the metric then determines Su for every u . $\square$

Problem 17.59 (Curvature discriminant). Interpret $H^2 - K$ geometrically.

Solution. It equals one quarter of the squared difference between principal curvatures. It measures anisotropy of normal bending and vanishes exactly at umbilics. $\square$

Bridge to VII/18

VII/18 applies the curvature machinery to ruled surfaces and isolates the developable case, where $K = 0$ and the Gauss map degenerates along rulings.

Chapter 18

Ruled and Developable Surfaces

A ruled surface is swept out by a one-parameter family of straight lines. This simple construction produces planes, cylinders, cones, hyperboloids, helicoids, and many classical engineering surfaces.

18.1 Ruled parametrizations

Definition 18.1 (Ruled surface). A parametrized surface is ruled if locally it can be written

$$X(u, v) = \alpha(u) + v\beta(u),$$

where $\beta(u) \neq 0$. For fixed u , the curve $v \mapsto X(u, v)$ is a straight line called a ruling.

Proposition 18.2 (Regularity criterion). *For*

$$X_u = \alpha'(u) + v\beta'(u), \quad X_v = \beta(u),$$

regularity is equivalent to

$$(\alpha'(u) + v\beta'(u)) \times \beta(u) \neq 0.$$

18.2 Standard examples

Example 18.3 (Plane). A plane is ruled in infinitely many ways.

Example 18.4 (Generalized cylinder). If β is constant,

$$X(u, v) = \alpha(u) + v\beta,$$

the surface is a generalized cylinder.

Example 18.5 (Cone). For a fixed vertex p ,

$$X(u, v) = p + v\beta(u)$$

is a cone away from singular points where the parametrization loses rank.

Example 18.6 (Helicoid).

$$X(u, v) = (v \cos u, v \sin u, cu)$$

is ruled: for each fixed u , varying v traces a horizontal line.

18.3 Developable surfaces

Definition 18.7 (Developable ruled surface). A regular ruled surface is developable if its tangent plane is constant along each ruling.

Theorem 18.8 (Triple-product criterion). *For*

$$X(u, v) = \alpha(u) + v\beta(u),$$

with β normalized to unit length, the ruled surface is developable precisely when

$$\det(\alpha'(u), \beta(u), \beta'(u)) = 0.$$

Proof. The tangent plane along a ruling is spanned by

$$\beta, \quad \alpha' + v\beta'.$$

Its normal direction is

$$(\alpha' + v\beta') \times \beta.$$

Constancy of the tangent plane as v varies is equivalent to the vectors α', β, β' being linearly dependent, which is the vanishing triple product. $\square$

Theorem 18.9 (Developable implies zero Gaussian curvature). *At every regular point of a developable ruled surface,*

$$K = 0.$$

Proof. Along a ruling the tangent plane, hence the Gauss map, is constant. Therefore the ruling direction lies in $\ker dN = \ker S$, so one principal curvature is zero and $K = \det S = 0$. $\square$

18.4 Gaussian curvature of a ruled surface

Proposition 18.10 (Sign restriction). *At regular nondevelopable points of a ruled surface in $\mathbb{R}^3$,*

$$K < 0.$$

Thus a ruled surface has $K \leq 0$ wherever regular.

Remark 18.11. The straight ruling has zero ambient curvature. Its normal curvature is therefore zero. If the surface is nondevelopable, this asymptotic direction is not a kernel direction of the whole shape operator, and the point is hyperbolic.

18.5 Striction and singular behavior

Definition 18.12 (Striction curve). For a noncylindrical ruled surface with unit ruling field β and $\beta' \neq 0$, a distinguished base curve can be chosen so that its derivative is orthogonal to β' . It is called a striction curve.

Remark 18.13. The striction curve organizes where neighboring rulings come closest. For tangent developables it is also closely tied to the singular edge of regression.

18.6 Tangent developables

Definition 18.14 (Tangent developable). For a regular space curve α ,

$$X(u, v) = \alpha(u) + v\alpha'(u)$$

is its tangent developable.

Proposition 18.15 (Singular base curve). *Along $v = 0$,*

$$X_u = \alpha', \quad X_v = \alpha',$$

so the parametrization is singular on the generating curve itself.

18.7 Cones and cylinders as developables

Proposition 18.16 (Cylinders). *Every generalized cylinder is developable.*

Proof. Here $\beta' = 0$, so the triple product criterion vanishes identically. □

Proposition 18.17 (Cones). *Every regular part of a cone is developable.*

Proof. For $X = p + v\beta(u)$, the vector α' in the ruled representation may be taken as zero; equivalently the tangent plane along a generator is spanned by β, β' , independent of v . □

18.8 The helicoid as a contrasting ruled surface

Proposition 18.18 (Helicoid is not developable). *For $c \neq 0$, the helicoid is ruled but not developable; at its regular points its Gaussian curvature is strictly negative.*

18.9 Geometric meaning

Developable surfaces are precisely the ruled surfaces whose tangent plane does not twist along a ruling. They are locally bendable from a plane without stretching in the classical smooth setting, and this is why cylinders and cones play a central role in fabrication, sheet materials, and geometric design.

18.10 Exercises

Exercise 18.19. Differentiate $X(u, v) = \alpha(u) + v\beta(u)$.

Hint. Differentiate in each parameter. □

Solution. One gets $X_u = \alpha' + v\beta'$ and $X_v = \beta$. □

Exercise 18.20. State the regularity condition for a ruled parametrization.

Hint. Use the cross product of the two partial derivatives. □

Solution. Regularity requires $(\alpha' + v\beta') \times \beta \neq 0$. □

Exercise 18.21. Show a plane is ruled.

Hint. Choose a line in the plane and a constant transverse direction. □

Solution. For example $X(u, v) = p + ua + vb$ with independent a, b is ruled in either parameter direction. □

Exercise 18.22. Show a generalized cylinder is ruled.

Hint. Take constant ruling direction. □

Solution. The expression $X = \alpha(u) + vb$ has straight rulings parallel to b . □

Exercise 18.23. Show a cone is ruled.

Hint. Fix the vertex. □

Solution. The curves $v \mapsto p + v\beta(u)$ are straight generators through the vertex. □

Exercise 18.24. Verify that the helicoid is ruled.

Hint. Fix u . □

Solution. Then $v \mapsto (v \cos u, v \sin u, cu)$ is an affine line. □

Exercise 18.25. Show a generalized cylinder is developable.

Hint. Use $\beta' = 0$. □

Solution. The triple product $\det(\alpha', \beta, \beta')$ vanishes identically. □

Exercise 18.26. Show a cone is developable away from its singular locus.

Hint. Check tangent planes along a generator. □

Solution. The tangent plane is spanned by β, β' , independent of the generator parameter. □

Exercise 18.27. Explain why tangent-plane constancy along a ruling forces dN to kill the ruling direction.

Hint. The normal is constant there. □

Solution. Differentiation of the constant Gauss map along the ruling gives $dN(X_v) = 0$. □

Exercise 18.28. Deduce $K = 0$ on a developable ruled surface.

Hint. Use the kernel of the shape operator. □

Solution. Since $S = -dN$ has a nonzero kernel vector, its determinant is zero. □

Exercise 18.29. Why is a ruling an asymptotic direction on any ruled surface?

Hint. The ruling is a straight line. □

Solution. Its ambient curvature is zero, so its normal curvature is zero. □

Exercise 18.30. What sign can K have on a regular ruled surface?

Hint. Use the asymptotic ruling direction. □

Solution. It cannot be positive; one has $K \leq 0$, with $K = 0$ in the developable case and $K < 0$ in the nondevelopable case. □

Exercise 18.31. State the triple-product criterion for developability.

Hint. Use α', β, β' . ☐

Solution. The criterion is $\det(\alpha', \beta, \beta') = 0$. ☐

Exercise 18.32. Interpret the vanishing triple product.

Hint. Use linear dependence. ☐

Solution. The three vectors α', β, β' are linearly dependent. ☐

Exercise 18.33. Compute the tangent vectors of a tangent developable.

Hint. Differentiate $X = \alpha + v\alpha'$. ☐

Solution. One gets $X_u = \alpha' + v\alpha''$ and $X_v = \alpha'$. ☐

Exercise 18.34. Locate the obvious singular set of a tangent developable.

Hint. Set $v = 0$. ☐

Solution. Then $X_u = X_v = \alpha'$, so rank drops along the base curve. ☐

Exercise 18.35. Why is a tangent developable developable away from singular points?

Hint. Use the triple product. ☐

Solution. Here $\beta = \alpha'$, so $\det(\alpha', \alpha', \alpha'') = 0$. ☐

Exercise 18.36. Explain the role of a striction curve.

Hint. Think of closest neighboring rulings. ☐

Solution. It provides a distinguished base curve organizing the family of rulings and satisfies a normalization such as orthogonality to β' . ☐

Exercise 18.37. Why is the helicoid not developable for nonzero pitch?

Hint. Its tangent plane rotates along rulings. ☐

Solution. The corresponding triple product does not vanish, so the developability criterion fails. ☐

Exercise 18.38. What is the Gaussian-curvature sign of a nondevelopable helicoid?

Hint. It is ruled and nondevelopable. ☐

Solution. It is strictly negative at regular points. ☐

Exercise 18.39. Give two classical developable surfaces.

Hint. Use standard examples. ☐

Solution. A circular cylinder and a cone are classical examples. ☐

Exercise 18.40. Give one ruled surface that is not developable.

Hint. Use the helicoid. ☐

Solution. The helicoid with nonzero pitch is ruled but not developable. ☐

Exercise 18.41. Relate developability to the rank of the Gauss map.

Hint. The ruling direction lies in the kernel. □

Solution. The differential of the Gauss map has rank at most one at regular developable points. □

Exercise 18.42. State the bridge to VII/19.

Hint. Move from $K = 0$ to $H = 0$. □

Solution. VII/19 studies minimal surfaces, where the mean curvature rather than Gaussian curvature is the central vanishing condition. □

18.11 Solved problem dossiers

Problem 18.43 (Legacy ruled-surface problem). Derive the basic tangent vectors and regularity condition for a ruled parametrization.

Solution. Differentiate $X = \alpha + v\beta$; regularity is exactly nonvanishing of $(\alpha' + v\beta') \times \beta$. □

Problem 18.44 (Legacy developability problem). Derive the scalar triple-product criterion.

Solution. Require the tangent plane along each fixed ruling to be independent of v . This is equivalent to linear dependence of α', β, β' . □

Problem 18.45 (Legacy curvature problem). Prove that a developable ruled surface has $K = 0$.

Solution. The Gauss map is constant along each ruling, so the shape operator has a nontrivial kernel and determinant zero. □

Problem 18.46 (Cylinder problem). Verify directly that generalized cylinders are developable.

Solution. The ruling field is constant, so $\beta' = 0$ and the triple product vanishes. □

Problem 18.47 (Cone problem). Verify directly that regular cone points are developable.

Solution. Along a generator, the tangent plane is spanned by β, β' and is independent of the radial parameter. □

Problem 18.48 (Tangent-developable problem). Show why the tangent developable of a space curve is developable yet singular along its generating curve.

Solution. The triple product vanishes because two vectors coincide, while at $v = 0$ the two parameter derivatives both equal α' . □

Problem 18.49 (Helicoid problem). Explain why the helicoid is ruled but nondevelopable.

Solution. It is affine in the ruling parameter v , but its tangent plane twists as v varies; equivalently the triple product criterion is nonzero. □

Problem 18.50 (Asymptotic-ruling problem). Show every ruling is an asymptotic direction.

Solution. Each ruling is a straight curve, so its curvature vector vanishes and hence its normal curvature is zero. □

Problem 18.51 (Sign-of- K problem). Prove $K \leq 0$ at regular points of a ruled surface.

Solution. The existence of a zero normal-curvature direction excludes elliptic points. Nondevelopable points are hyperbolic, while developable points have $K = 0$. $\square$

Problem 18.52 (Gauss-map rank problem). Describe the Gauss map on a developable surface.

Solution. It is constant along each ruling, so its differential has a ruling-direction kernel and rank at most one. $\square$

Problem 18.53 (Striction problem). Explain why changing the base curve by sliding along rulings can be used to impose a normalization.

Solution. Replacing α by $\alpha + s(u)\beta$ changes the base curve without changing the surface. Choosing s appropriately can enforce the standard striction orthogonality condition. $\square$

Problem 18.54 (Fabrication interpretation). Why are developables natural models for bending sheet material without stretching?

Solution. Vanishing Gaussian curvature and constant tangent-plane behavior along rulings are compatible with local isometric bending from planar sheets, unlike doubly curved surfaces. $\square$

18.12 Challenge problems

Problem 18.55 (Derive the triple-product criterion carefully). Starting from the normal $X_u \times X_v$, show tangent-plane constancy along v is equivalent to $\det(\alpha', \beta, \beta') = 0$.

Solution. The normal vector is $(\alpha' + v\beta') \times \beta = \alpha' \times \beta + v(\beta' \times \beta)$. Its direction is independent of v exactly when the two cross-product terms are linearly dependent, equivalent to α', β, β' being linearly dependent. $\square$

Problem 18.56 (Rank of the shape operator). Show that at a nonplanar regular point of a developable surface the shape operator has rank one.

Solution. Developability gives a nonzero kernel direction. If the point is nonplanar then $S \neq 0$. On a two-dimensional tangent plane, a nonzero endomorphism with nontrivial kernel has rank one. $\square$

Problem 18.57 (Edge of regression preview). For a tangent developable, explain geometrically why the base curve forms a singular edge.

Solution. Neighboring tangent lines envelope the original curve. Along the curve itself the two parameter directions collapse to the same tangent vector, so the surface parametrization loses rank precisely along this envelope. $\square$

Problem 18.58 (Nondevelopable ruled points). Show that a regular nondevelopable ruled point is hyperbolic.

Solution. The ruling gives a zero normal-curvature direction. If $K = 0$, the self-adjoint shape operator would have a kernel and the Gauss map would fail to vary along that direction in the developable way. At a genuinely nondevelopable regular ruled point the two principal curvatures must have opposite signs, hence $K < 0$. $\square$

Problem 18.59 (Local flattening principle). Explain, without proving the full theorem, why zero Gaussian curvature is the differential-geometric compatibility condition behind local flattening of a developable surface.

Solution. Gaussian curvature is intrinsic to the first fundamental form. A plane has $K = 0$, so any surface locally isometric to a plane must also have $K = 0$. Classical developables satisfy this condition and admit local unfoldings away from singularities. $\square$

Bridge to VII/19

VII/19 moves from the developable condition $K = 0$ to minimal surfaces, where the governing extrinsic condition is $H = 0$.

Chapter 19

Minimal Surfaces

Minimal surfaces are the critical points of area. For an oriented regular surface in $\mathbb{R}^3$, the local condition is strikingly simple:

$$H = 0.$$

The name “minimal” is historical: a critical surface need not globally minimize area, but its first variation vanishes for every compactly supported normal variation.

19.1 The mean-curvature condition

Definition 19.1 (Minimal surface). An oriented regular surface $S \subset \mathbb{R}^3$ is minimal if its mean curvature vanishes identically:

$$H \equiv 0.$$

This condition is independent of orientation because reversing the normal changes H to $-H$.

Proposition 19.2 (Principal-curvature characterization). *A surface is minimal exactly when*

$$k_1 + k_2 = 0.$$

At a nonplanar minimal point the principal curvatures have opposite signs, hence

$$K = k_1 k_2 = -k_1^2 < 0.$$

Corollary 19.3 (Gaussian-curvature sign). *Every minimal surface in $\mathbb{R}^3$ satisfies*

$$K \leq 0,$$

with equality at a point exactly when both principal curvatures vanish there.

19.2 Minimal graphs

Let a surface be given as a graph

$$X(x, y) = (x, y, u(x, y)).$$

Theorem 19.4 (Minimal-surface equation). *The graph of $u \in C^2(\Omega)$ is minimal exactly when*

$$(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0.$$

Equivalently,

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0.$$

Proof. For the upward unit normal

$$N = \frac{(-u_x, -u_y, 1)}{\sqrt{1 + |\nabla u|^2}},$$

the first and second fundamental forms give the displayed formula for $2H$. Setting $H = 0$ yields the equation. $\square$

Remark 19.5 (Linearization). For small slopes $|\nabla u| \ll 1$, the minimal-surface equation linearizes to

$$u_{xx} + u_{yy} = 0.$$

Thus harmonic functions are the infinitesimal model for minimal graphs.

19.3 Area functional and first variation

For a graph $u : \Omega \rightarrow \mathbb{R}$, its area is

$$\mathcal{A}[u] = \int_{\Omega} \sqrt{1 + |\nabla u|^2} \, dx \, dy.$$

Theorem 19.6 (Euler–Lagrange equation). *A C^2 graph is a critical point of $\mathcal{A}$ under compactly supported variations $u_t = u + t\varphi$ if and only if it satisfies the minimal-surface equation.*

Proof. Differentiate under the integral:

$$\left. \frac{d}{dt} \right|_0 \mathcal{A}[u + t\varphi] = \int_{\Omega} \frac{\nabla u \cdot \nabla \varphi}{\sqrt{1 + |\nabla u|^2}} \, dx \, dy.$$

Integration by parts, with φ compactly supported, gives

$$- \int_{\Omega} \varphi \operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) \, dx \, dy.$$

The fundamental lemma of the calculus of variations gives the result. $\square$

Theorem 19.7 (First variation in geometric form). *For a compactly supported normal variation with variational field $V = \phi N$,*

$$\left. \frac{d}{dt} \right|_0 \operatorname{Area}(S_t) = -2 \int_S H \phi \, dA$$

under the sign convention used in VII/17. Hence $H = 0$ is precisely the stationarity condition for area.

19.4 Conformal coordinates and harmonicity

Definition 19.8 (Conformal parametrization). A surface parametrization $X(u, v)$ is conformal if

$$E = G, \quad F = 0.$$

Theorem 19.9 (Harmonic coordinate criterion). *For a conformal regular parametrization,*

$$X_{uu} + X_{vv} = 2EH \, N.$$

Consequently the surface is minimal if and only if each coordinate function of X is harmonic:

$$X_{uu} + X_{vv} = 0.$$

19.5 Classical examples

Example 19.10 (Plane). A plane has $S = 0$, hence $H = 0$. It is the simplest minimal surface.

Example 19.11 (Catenoid). For $a > 0$,

$$X(u, v) = (a \cosh(v/a) \cos u, a \cosh(v/a) \sin u, v)$$

parametrizes a catenoid. Direct computation gives $H = 0$. It is the nonplanar minimal surface of revolution.

Example 19.12 (Helicoid). The helicoid

$$X(u, v) = (v \cos u, v \sin u, cu)$$

has $H = 0$. It is ruled but, unlike a developable surface, has negative Gaussian curvature away from degenerate limiting cases.

19.6 Minimal surfaces of revolution

Theorem 19.13 (Catenary profile). *A nonplanar minimal surface of revolution has a generating profile which, after translation and scaling, is a catenary*

$$r(z) = a \cosh\left(\frac{z - z_0}{a}\right).$$

Thus the corresponding surface is a catenoid.

19.7 Maximum-principle intuition

Proposition 19.14 (No strict interior graph extrema). *A nonconstant minimal graph cannot have a strict interior maximum or minimum.*

Remark 19.15. This reflects ellipticity of the minimal-surface equation. A full nonlinear maximum-principle treatment belongs to analysis; here the point is geometric: minimal graphs behave like nonlinear harmonic functions.

19.8 Area minimizing versus stationary

Warning 19.16 (Stationary is not automatically minimizing). The condition $H = 0$ says the first variation vanishes. Determining whether a minimal surface is stable or area minimizing requires second-variation information and global comparison arguments.

19.9 Boundary-value viewpoint

The Plateau problem asks for a surface of least area spanning a prescribed closed curve. Its solutions, when smooth and regular, satisfy $H = 0$ in the interior. This global problem motivates much of minimal-surface theory, but existence and regularity require tools beyond the present chapter.

19.10 Exercises

Exercise 19.17. Show that minimality is independent of the orientation choice.

Hint. Reverse N . □

Solution. Orientation reversal sends H to $-H$, so the equation $H = 0$ is unchanged. □

Exercise 19.18. Express minimality in terms of principal curvatures.

Hint. Use $H = (k_1 + k_2)/2$. □

Solution. Minimality is equivalent to $k_1 + k_2 = 0$. □

Exercise 19.19. Show that a nonplanar minimal point has $K < 0$.

Hint. Use $k_2 = -k_1$. □

Solution. Then $K = k_1 k_2 = -k_1^2 < 0$ unless $k_1 = 0$. □

Exercise 19.20. Verify that every plane is minimal.

Hint. Its normal is constant. □

Solution. The shape operator vanishes, so $H = 0$. □

Exercise 19.21. Write the upward unit normal to $z = u(x, y)$.

Hint. Take a cross product. □

Solution. It is $N = (-u_x, -u_y, 1)/\sqrt{1 + |\nabla u|^2}$. □

Exercise 19.22. State the minimal-surface equation for a graph.

Hint. Use the divergence form. □

Solution. The equation is $\operatorname{div}(\nabla u / \sqrt{1 + |\nabla u|^2}) = 0$. □

Exercise 19.23. Expand the graph equation into second derivatives.

Hint. Differentiate the divergence expression. □

Solution. It becomes $(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0$. □

Exercise 19.24. Linearize the minimal-surface equation at $u = 0$.

Hint. Discard quadratic slope terms. □

Solution. The linearized equation is $\Delta u = u_{xx} + u_{yy} = 0$. □

Exercise 19.25. Compute the graph area functional.

Hint. Use $\|X_x \times X_y\|$. □

Solution. One obtains $\mathcal{A}[u] = \int_{\Omega} \sqrt{1 + |\nabla u|^2} \, dx \, dy$. □

Exercise 19.26. Differentiate $\mathcal{A}[u + t\varphi]$ at $t = 0$.

Hint. Differentiate under the integral. □

Solution. The derivative is $\int_{\Omega} \nabla u \cdot \nabla \varphi / \sqrt{1 + |\nabla u|^2} \, dx \, dy$. □

Exercise 19.27. Recover the minimal-surface equation from the first variation.

Hint. Integrate by parts. □

Solution. Compact support removes the boundary term and the fundamental lemma gives the divergence equation. □

Exercise 19.28. Define a conformal parametrization.

Hint. Use the first fundamental coefficients. □

Solution. Conformality means $E = G$ and $F = 0$. □

Exercise 19.29. State the harmonic coordinate criterion for a conformal minimal immersion.

Hint. Use $X_{uu} + X_{vv}$. □

Solution. Minimality is equivalent to $X_{uu} + X_{vv} = 0$. □

Exercise 19.30. Why are the coordinate functions of a conformal minimal immersion harmonic?

Hint. Use the vector equation componentwise. □

Solution. The vector Laplacian vanishes, so each Euclidean coordinate function has zero scalar Laplacian. □

Exercise 19.31. Explain why a nonconstant minimal graph has no strict interior extremum.

Hint. Use elliptic maximum-principle intuition. □

Solution. The minimal-surface operator is elliptic; its solutions obey the same qualitative interior-extremum prohibition as harmonic functions. □

Exercise 19.32. Compute K at a minimal point in terms of k_1 .

Hint. Use $k_2 = -k_1$. □

Solution. One has $K = -k_1^2$. □

Exercise 19.33. Can a sphere be minimal?

Hint. Use its nonzero equal principal curvatures. □

Solution. No. Its mean curvature has magnitude $1/R$, so it never vanishes. □

Exercise 19.34. Can a circular cylinder be minimal?

Hint. Use its principal curvatures. □

Solution. No. One principal curvature is zero but the other is nonzero, so $H \neq 0$. □

Exercise 19.35. Name two nonplanar classical minimal surfaces.

Hint. Use the standard examples. □

Solution. The catenoid and helicoid are classical nonplanar minimal surfaces. □

Exercise 19.36. Why is the helicoid compatible with VII/18's ruled-surface sign restriction?

Hint. Compare H and K . □

Solution. The helicoid is ruled and nondevelopable, so $K < 0$; minimality imposes $H = 0$, which is fully compatible. □

Exercise 19.37. Distinguish minimal from developable.

Hint. Compare vanishing invariants. □

Solution. Minimal means $H = 0$, while developable means $K = 0$ in the regular ruled setting; these are different conditions. □

Exercise 19.38. State the geometric first-variation formula.

Hint. Use a normal field $V = \phi N$. □

Solution. The first variation is $-2 \int_S H \phi dA$ with the convention of this volume. □

Exercise 19.39. Why does $H = 0$ express stationarity rather than automatic minimization?

Hint. First variation only sees linear change. □

Solution. A vanishing first derivative does not control the sign of the second variation or global competitors. □

Exercise 19.40. State the bridge from minimal surfaces to Riemannian metrics.

Hint. Forget the ambient $\mathbb{R}^3$. □

Solution. VII/20 abstracts the first fundamental form into a Riemannian metric on an arbitrary smooth manifold. □

19.11 Solved problem dossiers

Problem 19.41 (Legacy minimal-surface problem). Prove the equivalence $H = 0 \iff k_1 + k_2 = 0$ and derive $K \leq 0$.

Solution. Use the definitions of H and K . If the sum vanishes then $k_2 = -k_1$, so $K = -k_1^2 \leq 0$. □

Problem 19.42 (Legacy minimal-graph problem). Derive the minimal-surface PDE for a graph.

Solution. Compute I , II , and H , or equivalently take the Euler–Lagrange equation of the graph area integrand $\sqrt{1 + |\nabla u|^2}$. □

Problem 19.43 (Area-variation problem). Show compactly supported critical graphs satisfy the divergence-form PDE.

Solution. Differentiate the area functional, integrate by parts, and apply the fundamental lemma. □

Problem 19.44 (Linearization problem). Explain why Laplace’s equation is the infinitesimal model of the minimal-surface equation.

Solution. At zero slope, the nonlinear coefficients reduce to $1, 0, 1$, leaving $u_{xx} + u_{yy} = 0$. □

Problem 19.45 (Conformal-coordinate problem). Prove a conformal immersion is minimal exactly when its coordinate functions are harmonic.

Solution. For $E = G, F = 0$, the normal component of $X_{uu} + X_{vv}$ is $2EHN$, while its tangent component vanishes; hence the vector Laplacian is zero precisely when $H = 0$. $\square$

Problem 19.46 (Plane problem). Show a plane is both minimal and globally area minimizing among graphs with the same affine boundary data.

Solution. The plane has $H = 0$. For the graph functional, convexity of $\sqrt{1 + |p|^2}$ gives the comparison with the affine solution. $\square$

Problem 19.47 (Catenoid problem). Explain why the catenary profile appears for a minimal surface of revolution.

Solution. Substituting a rotational ansatz into $H = 0$ yields an ODE with first integral $r/\sqrt{1 + r'^2} = a$, whose solutions are $r = a \cosh((z - z_0)/a)$. $\square$

Problem 19.48 (Helicoid problem). Verify the helicoid has $H = 0$.

Solution. Compute the first and second fundamental coefficients; the trace of $I^{-1}II$ vanishes. $\square$

Problem 19.49 (Sphere comparison problem). Why can no open piece of a round sphere be minimal?

Solution. Its two principal curvatures are equal and nonzero, so their mean cannot vanish. $\square$

Problem 19.50 (Stationary-versus-stable problem). Explain why $H = 0$ alone is insufficient for stability.

Solution. It annihilates the first variation only; stability requires nonnegativity of the second variation for admissible variations. $\square$

Problem 19.51 (Plateau problem preview). Explain why a smooth area-minimizing spanning surface must be minimal in its interior.

Solution. An interior compactly supported variation preserves the boundary. A minimizer has zero first variation for all such variations, hence $H = 0$. $\square$

Problem 19.52 (Intrinsic/extrinsic problem). Why is minimality extrinsic?

Solution. Mean curvature depends on how the surface sits in ambient space. Isometric surfaces can have different mean curvature. $\square$

19.12 Challenge problems

Problem 19.53 (Bernstein preview). State the classical Bernstein phenomenon for entire minimal graphs in low dimension and explain its geometric flavor.

Solution. In $\mathbb{R}^3$, every entire minimal graph over $\mathbb{R}^2$ is a plane. The theorem says the nonlinear minimal graph equation is globally rigid under the entire-graph hypothesis. $\square$

Problem 19.54 (First integral for revolution). Derive the first integral leading to the catenoid.

Solution. For a rotational area functional $2\pi \int r\sqrt{1+r'^2} dz$, the integrand has no explicit z -dependence. The Beltrami identity yields $r/\sqrt{1+r'^2} = a$, which integrates to the catenary. $\square$

Problem 19.55 (Conjugate-surface preview). Why do conformal harmonic parametrizations suggest a relation between the catenoid and helicoid?

Solution. Harmonic coordinate functions locally admit harmonic conjugates. Rotating the complex differential of a conformal minimal immersion can produce an associated family; the catenoid and helicoid occur as conjugate members. $\square$

Problem 19.56 (Minimal and developable). Show that a connected surface that is both minimal and developable must be planar wherever regular.

Solution. Minimal gives $k_2 = -k_1$; developable gives $k_1 k_2 = 0$. Thus both principal curvatures vanish. Hence $S = 0$, so the Gauss map is locally constant and the surface is planar on connected regular pieces. $\square$

Problem 19.57 (Scaling law). How do H and K change under dilation by a factor $\lambda > 0$, and why is minimality scale invariant?

Solution. Principal curvatures scale as $1/\lambda$, so $H \mapsto H/\lambda$ and $K \mapsto K/\lambda^2$. Therefore $H = 0$ is preserved by dilations. $\square$

Bridge to VII/20

The first fundamental form now becomes independent of any embedding: VII/20 develops Riemannian metrics on arbitrary smooth manifolds.

Part IV

Riemannian Geometry

Chapter 20

Riemannian Metrics

The first fundamental form of a surface is an inner product that varies smoothly from tangent plane to tangent plane. A Riemannian metric takes this as the definition on an arbitrary smooth manifold:

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}.$$

20.1 Definition and local coefficients

Definition 20.1 (Riemannian metric). A Riemannian metric on a smooth manifold M is a smooth symmetric positive definite $(0, 2)$ -tensor field g . Thus each g_p is an inner product on $T_p M$.

In coordinates $x^1, \dots, x^n$,

$$g = g_{ij} dx^i \otimes dx^j, \quad g_{ij} = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right).$$

Proposition 20.2 (Positive-definite matrix). *The matrix (g_{ij}) is symmetric and positive definite at every point, hence invertible. Its inverse is denoted (g^{ij}) .*

20.2 Lengths and angles

Definition 20.3 (Norm and angle). For $v, w \in T_p M$,

$$|v|_g = \sqrt{g(v, v)}.$$

For nonzero v, w ,

$$\cos \theta = \frac{g(v, w)}{|v|_g |w|_g}.$$

Definition 20.4 (Length of a curve). For a piecewise smooth curve $\gamma : [a, b] \rightarrow M$,

$$L_g(\gamma) = \int_a^b |\dot{\gamma}(t)|_g dt.$$

Proposition 20.5 (Reparametrization invariance). *Curve length is invariant under orientation-preserving C^1 reparametrization.*

20.3 Riemannian distance

Definition 20.6 (Riemannian distance). For points in the same connected component,

$$d_g(p, q) = \inf_{\gamma} L_g(\gamma),$$

where the infimum ranges over piecewise smooth curves from p to q .

Theorem 20.7 (Metric-space property). *On each connected component, d_g is a genuine metric and induces the original manifold topology.*

Remark 20.8. The existence and local minimizing behavior of geodesics will be developed after connections, in VII/23. The distance can be defined before solving the geodesic equation.

20.4 Pullback and induced metrics

Definition 20.9 (Pullback metric). If $F : M \rightarrow N$ is smooth and h is a Riemannian metric on N , define

$$(F^*h)_p(u, v) = h_{F(p)}(dF_p u, dF_p v).$$

If F is an immersion, F^*h is a Riemannian metric on M .

Example 20.10 (Induced metric on a surface). For an immersion $X : U \rightarrow \mathbb{R}^3$, the pullback of the Euclidean metric is

$$X^*\langle \cdot, \cdot \rangle = E du^2 + 2F du dv + G dv^2,$$

the first fundamental form from VII/15.

Definition 20.11 (Riemannian isometry). A diffeomorphism $F : (M, g) \rightarrow (N, h)$ is an isometry if

$$F^*h = g.$$

Proposition 20.12 (Isometries preserve length). *Riemannian isometries preserve tangent-vector norms, curve lengths, angles, and Riemannian distance.*

20.5 Product metrics

Definition 20.13 (Product metric). For Riemannian manifolds (M, g) and (N, h) , the product metric is

$$(g \oplus h)_{(p,q)}((u_1, v_1), (u_2, v_2)) = g_p(u_1, u_2) + h_q(v_1, v_2).$$

20.6 Musical isomorphisms

Definition 20.14 (Flat and sharp). A metric identifies tangent and cotangent spaces:

$$\flat : T_p M \rightarrow T_p^* M, \quad v^\flat = g(v, \cdot),$$

and its inverse is

$$\sharp : T_p^* M \rightarrow T_p M.$$

In coordinates,

$$(v^\flat)_i = g_{ij} v^j, \quad (\alpha^\sharp)^i = g^{ij} \alpha_j.$$

Definition 20.15 (Gradient). For $f \in C^\infty(M)$,

$$\text{grad } f = (df)^\sharp,$$

equivalently

$$g(\text{grad } f, X) = df(X)$$

for every vector field X .

Proposition 20.16 (Coordinate gradient).

$$\text{grad } f = g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial}{\partial x^i}.$$

20.7 Riemannian volume density

Definition 20.17 (Riemannian density). In a coordinate chart, the canonical positive density is

$$dV_g = \sqrt{\det(g_{ij})} |dx^1 \cdots dx^n|.$$

On an oriented Riemannian manifold this corresponds to the Riemannian volume form.

Proposition 20.18 (Coordinate invariance). *The factor $\sqrt{\det g}$ transforms with the inverse absolute Jacobian, so dV_g is globally well defined.*

20.8 Conformal metrics

Definition 20.19 (Conformal change). If $\varphi \in C^\infty(M)$, then

$$\tilde{g} = e^{2\varphi} g$$

is a conformally equivalent metric.

Proposition 20.20 (Effect on lengths and volume). *Under $\tilde{g} = e^{2\varphi} g$,*

$$|v|_{\tilde{g}} = e^\varphi |v|_g, \quad dV_{\tilde{g}} = e^{n\varphi} dV_g.$$

Angles are unchanged.

20.9 Existence of metrics

Theorem 20.21 (Existence). *Every smooth paracompact manifold admits a Riemannian metric.*

Proof. Choose a locally finite atlas, put the Euclidean inner product on each chart, and combine the resulting local metrics using a smooth partition of unity. Positive weighted sums of positive-definite bilinear forms remain positive definite. $\square$

20.10 Examples

Example 20.22 (Euclidean metric). On $\mathbb{R}^n$,

$$g = \sum_i dx^i \otimes dx^i.$$

Example 20.23 (Round sphere). The round metric on S^n is the metric induced by the Euclidean metric of $\mathbb{R}^{n+1}$.

Example 20.24 (Polar coordinates). On the Euclidean plane away from the origin,

$$g = dr^2 + r^2 d\theta^2.$$

The nonconstant coefficients reflect coordinates, not intrinsic curvature.

20.11 Exercises

Exercise 20.25. State the pointwise requirements on a Riemannian metric.

Hint. Think inner product. □

Solution. At each point it is bilinear, symmetric, and positive definite, and the coefficients vary smoothly. □

Exercise 20.26. Why is (g_{ij}) invertible?

Hint. Use positive definiteness. □

Solution. A positive-definite matrix has trivial kernel and hence is invertible. □

Exercise 20.27. Write the Euclidean metric in Cartesian coordinates.

Hint. Use the standard orthonormal basis. □

Solution. It is $g = \sum_i dx^i \otimes dx^i$. □

Exercise 20.28. Write the Euclidean plane metric in polar coordinates.

Hint. Differentiate $x = r \cos \theta, y = r \sin \theta$. □

Solution. One obtains $g = dr^2 + r^2 d\theta^2$. □

Exercise 20.29. Define the norm of a tangent vector.

Hint. Use $g(v, v)$. □

Solution. The norm is $|v|_g = \sqrt{g(v, v)}$. □

Exercise 20.30. Define the angle between two nonzero tangent vectors.

Hint. Use the inner-product formula. □

Solution. $\cos \theta = g(v, w) / (|v|_g |w|_g)$. □

Exercise 20.31. Define the length of a piecewise smooth curve.

Hint. Integrate speed. □

Solution. $L_g(\gamma) = \int |\dot{\gamma}|_g dt$. □

Exercise 20.32. Show curve length is invariant under increasing reparametrization.

Hint. Use the chain rule and substitution. □

Solution. The speed gains the factor ϕ' , which is exactly canceled by the change of variable. □

Exercise 20.33. Define Riemannian distance.

Hint. Take an infimum. □

Solution. It is the infimum of lengths of piecewise smooth curves joining the two points. □

Exercise 20.34. Why is $d_g(p, q) \leq L_g(\gamma)$ for any joining curve?

Hint. Use the definition of infimum. □

Solution. The distance is the infimum over a set containing $L_g(\gamma)$. □

Exercise 20.35. Define the pullback of a metric.

Hint. Insert differentials. □

Solution. $(F^*h)_p(u, v) = h_{F(p)}(dF_p u, dF_p v)$. □

Exercise 20.36. Why is F^*h positive definite when F is an immersion?

Hint. Use injectivity of dF_p . □

Solution. If $u \neq 0$, then $dF_p u \neq 0$, so $h(dF_p u, dF_p u) > 0$. □

Exercise 20.37. Recover the first fundamental form as a pullback metric.

Hint. Use a surface immersion into Euclidean space. □

Solution. The coefficients are $E = \langle X_u, X_u \rangle$, $F = \langle X_u, X_v \rangle$, and $G = \langle X_v, X_v \rangle$. □

Exercise 20.38. Define a Riemannian isometry.

Hint. Use pullback. □

Solution. A diffeomorphism F is an isometry when $F^*h = g$. □

Exercise 20.39. Show an isometry preserves tangent-vector norms.

Hint. Set the two arguments equal. □

Solution. $g(v, v) = h(dFv, dFv)$, hence the norms agree. □

Exercise 20.40. Define the product metric.

Hint. Use the tangent-space direct sum. □

Solution. It is $g \oplus h$, the sum of the two factor inner products. □

Exercise 20.41. Define $v^\flat$.

Hint. Pair with an arbitrary tangent vector. □

Solution. $v^\flat = g(v, \cdot)$. □

Exercise 20.42. Define $\alpha^\sharp$.

Hint. Invert the flat map. □

Solution. It is the unique vector satisfying $g(\alpha^\sharp, \cdot) = \alpha$. □

Exercise 20.43. Write the coordinate formula for the gradient.

Hint. Raise the index of df . □

Solution. $\text{grad } f = g^{ij}(\partial_j f)\partial_i$. □

Exercise 20.44. Why did VII/05 warn that gradients require extra structure?

Hint. Compare df and $\text{grad } f$. □

Solution. The covector df is intrinsic to the smooth structure; turning it into a vector uses the metric via $\sharp$. □

Exercise 20.45. State the Riemannian volume density.

Hint. Use the metric determinant. □

Solution. It is $\sqrt{\det(g_{ij})} |dx^1 \cdots dx^n|$. □

Exercise 20.46. How does a conformal change affect vector lengths?

Hint. Use $\tilde{g} = e^{2\varphi}g$. □

Solution. Lengths scale pointwise by e^φ . □

Exercise 20.47. How does a conformal change affect angles?

Hint. The same factor appears in numerator and denominator. □

Solution. Angles are unchanged. □

Exercise 20.48. State the bridge to connections.

Hint. Metrics compare vectors at one point only. □

Solution. VII/21 introduces connections, which differentiate vector fields and compare directions infinitesimally across nearby points. □

20.12 Solved problem dossiers

Problem 20.49 (Legacy metric problem). Show a smooth positive-definite matrix field g_{ij} defines a Riemannian metric in a coordinate chart.

Solution. Set $g(v, w) = g_{ij}v^iw^j$. Symmetry and positive definiteness are pointwise, while smoothness follows from the coefficient functions. □

Problem 20.50 (Coordinate-change problem). Derive the transformation law for metric coefficients.

Solution. If $x = x(y)$, substitute $\partial/\partial y^a = (\partial x^i/\partial y^a)\partial_i$. Then $\tilde{g}_{ab} = g_{ij}(\partial x^i/\partial y^a)(\partial x^j/\partial y^b)$. □

Problem 20.51 (Polar-coordinate problem). Derive $dr^2 + r^2 d\theta^2$ from the Cartesian Euclidean metric.

Solution. Substitute $dx = \cos \theta dr - r \sin \theta d\theta$ and $dy = \sin \theta dr + r \cos \theta d\theta$, then simplify. □

Problem 20.52 (Induced-metric problem). Show an immersion pulls an ambient metric back to a Riemannian metric.

Solution. Smoothness and symmetry are immediate. Positive definiteness follows from injectivity of the differential. $\square$

Problem 20.53 (Length-invariance problem). Prove curve length is parametrization invariant.

Solution. Under $t = \phi(s)$, $(\gamma \circ \phi)' = \dot{\gamma}(\phi)\phi'$; integrate the resulting speed and substitute variables. $\square$

Problem 20.54 (Distance problem). Explain why Riemannian distance satisfies the triangle inequality.

Solution. Concatenate a nearly minimizing curve from p to q with one from q to r , then take infima. $\square$

Problem 20.55 (Isometry problem). Show Riemannian isometries preserve distance.

Solution. They preserve every curve length, hence preserve the infimum of lengths over joining curves. $\square$

Problem 20.56 (Musical-isomorphism problem). Derive the coordinate formulas for $\flat$ and $\sharp$.

Solution. Lowering an index gives $\alpha_i = g_{ij}v^j$; multiplying by g^{ki} recovers $v^k = g^{ki}\alpha_i$. $\square$

Problem 20.57 (Gradient problem). Derive the coordinate expression for $\text{grad } f$.

Solution. Write $g_{ij}(\text{grad } f)^i = \partial_j f$ and multiply by g^{kj} . $\square$

Problem 20.58 (Volume-density problem). Explain why $\sqrt{\det g} |dx^1 \cdots dx^n|$ is coordinate invariant.

Solution. The metric matrix transforms by $J^T g J$, so its determinant gains $(\det J)^2$; the coordinate density contributes the inverse absolute Jacobian. $\square$

Problem 20.59 (Conformal problem). Compute the effect of $\tilde{g} = e^{2\varphi}g$ on volume.

Solution. In dimension n , the determinant scales by $e^{2n\varphi}$, so its square root scales by $e^{n\varphi}$. $\square$

Problem 20.60 (Existence problem). Construct a global Riemannian metric from local Euclidean metrics.

Solution. Choose a locally finite partition of unity subordinate to an atlas and form the positive weighted sum of the local metrics. $\square$

20.13 Challenge problems

Problem 20.61 (Topology from distance). Why does d_g induce the original manifold topology?

Solution. On a relatively compact coordinate ball, the eigenvalues of g_{ij} are bounded above and below by positive constants. Hence Riemannian length is locally comparable to Euclidean length, making the two local topologies identical. $\square$

Problem 20.62 (Metric completeness preview). Distinguish metric completeness from compactness and explain why completeness will matter for geodesics.

Solution. Metric completeness means every d_g -Cauchy sequence converges. Compact Riemannian manifolds are complete, but noncompact ones may also be complete. Later Hopf–Rinow theory relates metric completeness to extendability of geodesics. $\square$

Problem 20.63 (Conformal inverse metric). If $\tilde{g} = e^{2\varphi}g$, compute $\tilde{g}^{ij}$ and $\text{grad}_{\tilde{g}}f$.

Solution. The inverse metric scales by $e^{-2\varphi}$, hence $\text{grad}_{\tilde{g}}f = e^{-2\varphi} \text{grad}_g f$. $\square$

Problem 20.64 (Product distance bounds). For the product metric, compare $d_{g \oplus h}$ with the factor distances.

Solution. Any joining curve has length at least each projected length, so the product distance dominates both factor distances. A product of nearly minimizing constant-speed curves gives the sharper Euclidean-combination behavior. $\square$

Problem 20.65 (Metrics as bundle sections). Reinterpret a Riemannian metric in the language of VII/07.

Solution. A metric is a smooth section of $\text{Sym}^2 T^*M$ whose value lies in the open cone of positive-definite bilinear forms at every point. $\square$

Bridge to VII/21

A metric measures vectors at one point; VII/21 introduces connections to differentiate vector fields and compare directions infinitesimally across the manifold.

Chapter 21

Connections

A vector field assigns vectors in different tangent spaces, so ordinary subtraction at nearby points is meaningless. A connection supplies a derivative that respects the bundle structure:

$$\boxed{\nabla_X Y.}$$

This chapter develops general connections. VII/22 will single out the unique Levi–Civita connection associated with a Riemannian metric.

21.1 Affine connections on the tangent bundle

Definition 21.1 (Affine connection). An affine connection on TM is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M), \quad (X, Y) \mapsto \nabla_X Y,$$

such that for $f, g \in C^\infty(M)$,

$$\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z,$$

$$\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z,$$

and

$$\nabla_X(fY) = X(f)Y + f\nabla_X Y.$$

Remark 21.2 (Asymmetry of the two slots). A connection is $C^\infty(M)$ -linear in the direction field X , but only $\mathbb{R}$ -linear in the differentiated field Y ; the second slot obeys a Leibniz rule.

21.2 Local connection coefficients

In coordinates $x^1, \dots, x^n$, define

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

Definition 21.3 (Connection coefficients). The functions Γ_{ij}^k are the connection coefficients in the chosen coordinate frame.

Proposition 21.4 (Coordinate formula). *For*

$$X = X^i \partial_i, \quad Y = Y^j \partial_j,$$

$$\nabla_X Y = X^i \left(\partial_i Y^k + \Gamma_{ij}^k Y^j \right) \partial_k.$$

Warning 21.5 (Coefficients are not tensor components). The Γ_{ij}^k do not transform tensorially. Their transformation law contains second derivatives of the coordinate change. A connection itself is geometric even though its coefficient array is not a tensor.

21.3 Connections on vector bundles

Definition 21.6 (Vector-bundle connection). For a vector bundle $\pi : E \rightarrow M$, a connection is an $\mathbb{R}$ -linear map

$$\nabla : \Gamma(E) \rightarrow \Omega^1(M; E)$$

satisfying

$$\nabla(fs) = df \otimes s + f\nabla s.$$

Equivalently, for each vector field X ,

$$\nabla_X(fs) = X(f)s + f\nabla_X s.$$

If $e_1, \dots, e_r$ is a local frame, write

$$\nabla e_b = \omega^a_b \otimes e_a.$$

Definition 21.7 (Connection one-form matrix). The matrix

$$\omega = (\omega^a_b)$$

of one-forms is the connection form in the chosen local frame.

For $s = s^b e_b$,

$$\nabla s = (ds^a + \omega^a_b s^b) \otimes e_a.$$

21.4 Change of frame

Let $e' = eA$, where $A : U \rightarrow GL(r, \mathbb{R})$.

Theorem 21.8 (Gauge transformation law). *The connection-form matrices satisfy*

$$\omega' = A^{-1}\omega A + A^{-1}dA.$$

Proof. Differentiate $e' = eA$ using the Leibniz rule and rewrite the result in the new frame. $\square$

Remark 21.9. The inhomogeneous term $A^{-1}dA$ is the bundle analogue of the non-tensorial coordinate transformation law for connection coefficients.

21.5 Covariant derivative along a curve

Definition 21.10 (Covariant derivative along a curve). Let $\gamma : I \rightarrow M$ and let $V(t) \in E_{\gamma(t)}$ be a smooth vector field along γ . A connection induces

$$\frac{DV}{dt} = \nabla_{\dot{\gamma}} V.$$

In a local frame $V = V^a e_a$,

$$\frac{DV^a}{dt} = \frac{dV^a}{dt} + \omega^a_b(\dot{\gamma})V^b.$$

Definition 21.11 (Parallel field). A field V along γ is parallel if

$$\frac{DV}{dt} = 0.$$

Remark 21.12. Existence, uniqueness, and geometric consequences of parallel transport are developed systematically in VII/24.

21.6 Difference of two connections

Theorem 21.13 (Affine space of connections). *If ∇ and $\tilde{\nabla}$ are two connections on a vector bundle E , then*

$$A(X)s = \tilde{\nabla}_X s - \nabla_X s$$

is $C^\infty(M)$ -linear in both X and s . Hence

$$A \in \Omega^1(M; \text{End } E).$$

Conversely, adding any such A to a connection produces another connection.

21.7 Torsion of an affine connection

Definition 21.14 (Torsion). For a connection on TM ,

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

Proposition 21.15 (Torsion is tensorial). *T is $C^\infty(M)$ -linear in both arguments, hence a vector-valued two-form.*

Remark 21.16. VII/22 will impose the special condition $T = 0$, together with metric compatibility, to characterize the Levi–Civita connection.

21.8 Principal connections

Let $\pi : P \rightarrow M$ be a principal G -bundle with Lie algebra $\mathfrak{g}$.

Definition 21.17 (Horizontal distribution). A principal connection may be specified by a smooth G -equivariant choice of horizontal subspaces

$$T_p P = H_p P \oplus V_p P,$$

where $V_p P = \ker d\pi_p$ is the vertical space.

Definition 21.18 (Principal connection form). Equivalently, a principal connection is encoded by a $\mathfrak{g}$ -valued one-form

$$\omega \in \Omega^1(P; \mathfrak{g})$$

which reproduces fundamental vertical vector fields and is equivariant under the right G -action.

Proposition 21.19 (Horizontal kernel). *The horizontal distribution is*

$$H_p P = \ker \omega_p.$$

21.9 Induced connections

Proposition 21.20 (Associated bundles). *A principal connection on P induces connections on every associated vector bundle $P \times_\rho V$.*

Remark 21.21. This links the principal/frame-bundle language of VII/08 to covariant differentiation on tangent and tensor bundles.

21.10 Examples

Example 21.22 (Flat connection on $\mathbb{R}^n$). In standard Cartesian coordinates,

$$\nabla_{\partial_i} \partial_j = 0.$$

Thus

$$\nabla_X Y = X^i \partial_i Y^k \partial_k.$$

Example 21.23 (A connection on a trivial line bundle). On $M \times \mathbb{R}$, any one-form A defines

$$\nabla s = ds + A s.$$

Under a change of nonvanishing frame, A acquires the expected inhomogeneous gauge term.

21.11 Exercises

Exercise 21.24. State the three defining algebraic rules for an affine connection.

Hint. Compare the two slots. □

Solution. It is C^∞ -linear in the first slot, additive in the second, and obeys the Leibniz rule $\nabla_X(fY) = X(f)Y + f\nabla_X Y$. □

Exercise 21.25. Why is $\nabla_X Y$ not C^∞ -linear in Y ?

Hint. Use the Leibniz rule. □

Solution. The derivative term $X(f)Y$ prevents C^∞ -linearity. □

Exercise 21.26. Define the local coefficients Γ_{ij}^k .

Hint. Differentiate the coordinate frame. □

Solution. They are defined by $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$. □

Exercise 21.27. Derive the coordinate formula for $\nabla_X Y$.

Hint. Expand X, Y in the coordinate frame. □

Solution. Using the axioms gives $X^i(\partial_i Y^k + \Gamma_{ij}^k Y^j) \partial_k$. □

Exercise 21.28. Why are connection coefficients not tensor components?

Hint. Change coordinates. □

Solution. Their transformation law contains second derivatives of the coordinate change, unlike a tensorial law. □

Exercise 21.29. State the vector-bundle connection Leibniz rule.

Hint. Use a section s and function f . □

Solution. $\nabla(fs) = df \otimes s + f\nabla s$. □

Exercise 21.30. Write a connection in a local frame e_a .

Hint. Differentiate the frame vectors. □

Solution. $\nabla e_b = \omega^a{}_b \otimes e_a$. □

Exercise 21.31. Compute $\nabla(s^b e_b)$.

Hint. Apply Leibniz. □

Solution. $\nabla s = (ds^a + \omega^a{}_b s^b) \otimes e_a$. □

Exercise 21.32. State the gauge transformation law for connection forms.

Hint. Use $e' = eA$. □

Solution. $\omega' = A^{-1}\omega A + A^{-1}dA$. □

Exercise 21.33. Identify the inhomogeneous part of the gauge law.

Hint. Look for the term independent of ω . □

Solution. It is $A^{-1}dA$. □

Exercise 21.34. Define covariant differentiation along a curve.

Hint. Insert the curve velocity. □

Solution. $DV/dt = \nabla_{\dot{\gamma}} V$. □

Exercise 21.35. Write the local ODE for a parallel field.

Hint. Set $DV/dt = 0$. □

Solution. $\dot{V}^a + \omega^a{}_b(\dot{\gamma})V^b = 0$. □

Exercise 21.36. Define a parallel field along a curve.

Hint. Set its covariant derivative to zero. □

Solution. A field is parallel when $DV/dt = 0$. □

Exercise 21.37. Show the difference of two connections is C^∞ -linear in the section.

Hint. Subtract the two Leibniz rules. □

Solution. The derivative terms $X(f)s$ cancel, leaving $f(\tilde{\nabla}_X s - \nabla_X s)$. □

Exercise 21.38. Where does the difference of two bundle connections live?

Hint. Use linearity in X and s . □

Solution. It is an $\text{End}(E)$ -valued one-form. □

Exercise 21.39. Define torsion.

Hint. Compare covariant derivatives and the Lie bracket. □

Solution. $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$. □

Exercise 21.40. Why is torsion tensorial?

Hint. Expand $T(fX, Y)$ and $T(X, fY)$. □

Solution. Leibniz terms cancel against those from the Lie bracket, leaving C^∞ -linearity. □

Exercise 21.41. What does torsion-free mean?

Hint. Set the torsion tensor to zero. □

Solution. It means $\nabla_X Y - \nabla_Y X = [X, Y]$ for all vector fields. □

Exercise 21.42. Define the vertical space of a principal bundle.

Hint. Use $d\pi$. □

Solution. $V_p P = \ker d\pi_p$. □

Exercise 21.43. Define a horizontal distribution for a principal connection.

Hint. Split the tangent space. □

Solution. It is an equivariant choice $T_p P = H_p P \oplus V_p P$. □

Exercise 21.44. State the connection-form description on a principal bundle.

Hint. Use a Lie-algebra-valued one-form. □

Solution. A principal connection is encoded by $\omega \in \Omega^1(P; \mathfrak{g})$ with reproduction and equivariance properties. □

Exercise 21.45. How is the horizontal space recovered from the principal connection form?

Hint. Take the kernel. □

Solution. $H_p P = \ker \omega_p$. □

Exercise 21.46. Give the standard flat connection on $\mathbb{R}^n$.

Hint. Use Cartesian coordinates. □

Solution. Set all $\Gamma_{ij}^k = 0$. □

Exercise 21.47. State the bridge to VII/22.

Hint. Add metric structure and a normalization. □

Solution. VII/22 proves that a Riemannian metric determines a unique connection that is metric compatible and torsion free. □

21.12 Solved problem dossiers

Problem 21.48 (Legacy connection problem). Verify the coordinate formula for an affine connection from its coefficients.

Solution. Expand $X = X^i \partial_i$, $Y = Y^j \partial_j$, use C^∞ -linearity in X and the Leibniz rule in Y , then collect coefficients. □

Problem 21.49 (Non-tensorial coefficient problem). Explain why vanishing connection coefficients in one coordinate system need not imply vanishing in another.

Solution. The transformation law contains second derivatives of the coordinate change. Even the flat Cartesian connection acquires nonzero coefficients in curvilinear coordinates. □

Problem 21.50 (Bundle-connection problem). Derive the local formula $\nabla s = ds + \omega s$.

Solution. Write $s = s^b e_b$, apply the Leibniz rule, and substitute $\nabla e_b = \omega^a{}_b e_a$. $\square$

Problem 21.51 (Gauge-law problem). Derive $\omega' = A^{-1}\omega A + A^{-1}dA$.

Solution. Differentiate the new frame $e' = eA$, rewrite the result in the new frame, and compare coefficients. $\square$

Problem 21.52 (Along-curve problem). Derive the linear ODE governing parallel sections.

Solution. Substitute $V = V^a e_a$ into $DV/dt = 0$. The equation is $\dot{V}^a + \omega^a{}_b(\dot{\gamma})V^b = 0$. $\square$

Problem 21.53 (Difference-tensor problem). Prove the space of connections is affine over $\Omega^1(\text{End } E)$.

Solution. The difference of two connections is tensorial; conversely adding an endomorphism-valued one-form preserves the connection Leibniz rule. $\square$

Problem 21.54 (Torsion problem). Prove torsion is a tensor.

Solution. Expand with multiplied arguments and use $[fX, Y] = f[X, Y] - Y(f)X$; derivative terms cancel. $\square$

Problem 21.55 (Coordinate torsion problem). Compute torsion coefficients in a coordinate frame.

Solution. Because $[\partial_i, \partial_j] = 0$, $T^k{}_{ij} = \Gamma^k_{ij} - \Gamma^k_{ji}$. $\square$

Problem 21.56 (Flat-space problem). Show the standard connection on $\mathbb{R}^n$ differentiates vector components ordinarily.

Solution. With $\Gamma^k_{ij} = 0$, the coordinate formula reduces to $\nabla_X Y = X^i \partial_i Y^k \partial_k$. $\square$

Problem 21.57 (Principal-horizontal problem). Explain the equivalence between a horizontal distribution and a principal connection form.

Solution. The form projects a tangent vector onto its vertical Lie-algebra component. Its kernel is horizontal; conversely an equivariant horizontal splitting defines that projection. $\square$

Problem 21.58 (Associated-bundle problem). Explain how a principal connection induces covariant differentiation on an associated vector bundle.

Solution. Lift directions horizontally to the principal bundle, differentiate equivariant representative functions along those lifts, and descend the result through the representation. $\square$

Problem 21.59 (Levi-Civita boundary problem). Which facts should be deferred to VII/22?

Solution. The formula determined by g , uniqueness, metric compatibility, torsion-free characterization, and the explicit Christoffel-symbol formula belong to VII/22. $\square$

21.13 Challenge problems

Problem 21.60 (Coordinate transformation law). Derive the full transformation law for affine connection coefficients under $x = x(y)$.

Solution. Apply $\nabla_{\partial/\partial y^a}(\partial/\partial y^b)$, express both y -basis fields through the x -basis, and use the Leibniz rule. The resulting law consists of the tensor-like conjugation term plus a term containing second derivatives $\partial^2 x/\partial y^a \partial y^b$. $\square$

Problem 21.61 (Connection from a projection). Let $E \subset M \times \mathbb{R}^N$ be a vector subbundle with orthogonal projection P_x . Show $\nabla_X s = P(ds(X))$ defines a connection.

Solution. Linearity in X is immediate. For fs , $d(fs)(X) = X(f)s + f ds(X)$; since $s(x) \in E_x$, $P(s) = s$, yielding the required Leibniz rule. $\square$

Problem 21.62 (Gauge covariance). Show the covariant derivative of section components transforms homogeneously although ω transforms inhomogeneously.

Solution. Under $s' = A^{-1}s$ and $\omega' = A^{-1}\omega A + A^{-1}dA$, direct substitution shows $ds' + \omega' s' = A^{-1}(ds + \omega s)$. The derivative of A^{-1} cancels the inhomogeneous term. $\square$

Problem 21.63 (Principal connection and local gauge potential). Given a local section $\sigma : U \rightarrow P$, define the local gauge potential and describe its change under another section.

Solution. Set $A = \sigma^*\omega$. If $\sigma' = \sigma g$ for $g : U \rightarrow G$, then $A' = \text{Ad}_{g^{-1}} A + g^{-1}dg$, the principal-bundle version of the frame gauge law. $\square$

Problem 21.64 (Why a metric does not itself differentiate). Explain conceptually why VII/20's metric is insufficient to define $\nabla_X Y$ until an additional characterization is imposed.

Solution. A metric compares vectors only inside one tangent space. Infinitely many connections can coexist with the same smooth manifold and metric. VII/22 adds metric compatibility and zero torsion, which uniquely select the Levi-Civita connection. $\square$

Bridge to VII/22

VII/22 combines the general connection formalism with a Riemannian metric and proves the existence and uniqueness of the Levi-Civita connection.

Chapter 22

The Levi–Civita Connection

A Riemannian metric measures lengths and angles inside each tangent space. To differentiate vector fields one also needs a connection. The fundamental theorem of Riemannian geometry says that the metric selects a unique connection once two natural requirements are imposed:

$$\boxed{\nabla g = 0, \quad T = 0.}$$

This distinguished connection is the Levi–Civita connection.

22.1 Metric compatibility

Definition 22.1 (Metric-compatible connection). Let (M, g) be Riemannian. A connection ∇ on TM is metric compatible if

$$X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

for all vector fields X, Y, Z .

Proposition 22.2 (Preservation of norms). *For a vector field V along a curve γ ,*

$$\frac{d}{dt} g(V, V) = 2g\left(\frac{DV}{dt}, V\right).$$

In particular, a parallel vector field has constant length.

Proof. Apply metric compatibility with $X = \dot{\gamma}$ and $Y = Z = V$. □

22.2 Torsion-free connections

Recall from VII/21 that

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

Definition 22.3 (Torsion free). A connection is torsion free if $T = 0$, equivalently

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

In a coordinate frame, where $[\partial_i, \partial_j] = 0$, torsion freeness is equivalent to

$$\Gamma_{ij}^k = \Gamma_{ji}^k.$$

22.3 The fundamental theorem

Theorem 22.4 (Fundamental theorem of Riemannian geometry). *Every Riemannian manifold (M, g) admits a unique affine connection ∇ which is both metric compatible and torsion free.*

The key formula is obtained by cycling metric compatibility and using zero torsion.

Theorem 22.5 (Koszul formula). *The Levi-Civita connection satisfies*

$$2g(\nabla_X Y, Z) = Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]).$$

Conversely this identity defines the Levi-Civita connection.

Proof. Write metric compatibility for the cyclic triples (X, Y, Z) , (Y, Z, X) , and (Z, X, Y) . Add the first two equations and subtract the third. Replace expressions such as $\nabla_X Y - \nabla_Y X$ by $[X, Y]$. This gives the formula.

For uniqueness, the right side depends only on g, X, Y, Z , so it determines $g(\nabla_X Y, Z)$ for every Z , hence determines $\nabla_X Y$. Conversely, the formula can be checked to define a connection satisfying both required properties. $\square$

22.4 Christoffel symbols from the metric

Let

$$g_{ij} = g(\partial_i, \partial_j), \quad (g^{ij}) = (g_{ij})^{-1}.$$

Theorem 22.6 (Christoffel formula). *For the Levi-Civita connection,*

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

Proof. Apply the Koszul formula to the commuting coordinate vector fields $\partial_i, \partial_j, \partial_\ell$:

$$2g_{\ell k} \Gamma_{ij}^k = \partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}.$$

Multiply by $g^{m\ell}$. $\square$

Corollary 22.7 (Symmetry).

$$\Gamma_{ij}^k = \Gamma_{ji}^k.$$

22.5 Euclidean space

Example 22.8 (Cartesian coordinates). For the Euclidean metric

$$g = \sum_i (dx^i)^2,$$

all g_{ij} are constant, so every Christoffel symbol vanishes:

$$\Gamma_{ij}^k = 0.$$

Warning 22.9 (Vanishing coefficients are coordinate dependent). The Euclidean connection has nonzero Christoffel symbols in curvilinear coordinates. A connection is geometric; its coefficient array is not a tensor.

Example 22.10 (Polar coordinates). For

$$g = dr^2 + r^2 d\theta^2,$$

the nonzero Christoffel symbols are

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}.$$

22.6 Conformal metrics

Example 22.11 (Conformal Euclidean metric). On a domain in $\mathbb{R}^n$, let

$$g_{ij} = e^{2\varphi} \delta_{ij}.$$

Then

$$\Gamma_{ij}^k = \delta_j^k \partial_i \varphi + \delta_i^k \partial_j \varphi - \delta_{ij} \partial^k \varphi.$$

22.7 Tangential projection for submanifolds

Theorem 22.12 (Induced Levi-Civita connection). *Let $M \subset \mathbb{R}^N$ be an embedded submanifold with induced metric. For tangent vector fields X, Y ,*

$$\nabla_X Y = (D_X Y)^\top,$$

the tangential projection of the ambient derivative.

Proof. The projected derivative is a connection. Euclidean differentiation preserves the ambient inner product, so its tangential part is metric compatible. Since the ambient derivative has zero torsion and the Lie bracket of tangent fields is tangent, the projected connection is torsion free. Uniqueness identifies it with the Levi-Civita connection. $\square$

For a surface in $\mathbb{R}^3$, the complementary normal part gives the Gauss formula

$$D_X Y = \nabla_X Y + II(X, Y)N.$$

22.8 Covariant Hessian

Definition 22.13 (Hessian). For $f \in C^\infty(M)$,

$$\text{Hess } f(X, Y) = g(\nabla_X \text{grad } f, Y) = X(Yf) - (\nabla_X Y)f.$$

Proposition 22.14 (Symmetry of the Hessian). *For the Levi-Civita connection,*

$$\text{Hess } f(X, Y) = \text{Hess } f(Y, X).$$

Proof. Subtract the two expressions:

$$X(Yf) - Y(Xf) - (\nabla_X Y - \nabla_Y X)f.$$

Torsion freeness turns the second parenthesis into $[X, Y]$, and the result vanishes. $\square$

22.9 Why uniqueness matters

The Levi–Civita connection removes the arbitrary choice present in VII/21. Once g is fixed, covariant differentiation, geodesics, parallel transport, and the Riemann curvature tensor become canonical Riemannian constructions.

22.10 Exercises

Exercise 22.15. State metric compatibility.

Hint. Differentiate an inner product. □

Solution. It is $Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$. □

Exercise 22.16. State torsion freeness.

Hint. Use the Lie bracket. □

Solution. It is $\nabla_X Y - \nabla_Y X = [X, Y]$. □

Exercise 22.17. What characterizes the Levi–Civita connection?

Hint. Combine two conditions. □

Solution. It is the unique connection that is metric compatible and torsion free. □

Exercise 22.18. Show a parallel field has constant norm for a metric-compatible connection.

Hint. Differentiate $g(V, V)$. □

Solution. Metric compatibility gives $\frac{d}{dt}g(V, V) = 2g(DV/dt, V) = 0$. □

Exercise 22.19. Show two parallel fields have constant inner product.

Hint. Differentiate $g(V, W)$. □

Solution. Both covariant derivative terms vanish, so the inner product is constant. □

Exercise 22.20. Write the Koszul formula.

Hint. Cycle metric compatibility. □

Solution. The formula expresses $2g(\nabla_X Y, Z)$ as three metric derivatives plus three Lie-bracket terms. □

Exercise 22.21. Why does the Koszul formula prove uniqueness?

Hint. Vary Z . □

Solution. It determines the inner product of $\nabla_X Y$ with every Z , hence determines the vector. □

Exercise 22.22. Why do coordinate Lie-bracket terms vanish in the Koszul formula?

Hint. Coordinate vector fields commute. □

Solution. $[\partial_i, \partial_j] = 0$. □

Exercise 22.23. State the Christoffel formula for the Levi–Civita connection.

Hint. Use $g^{k\ell}$. □

Solution. $\Gamma_{ij}^k = \frac{1}{2}g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$. □

Exercise 22.24. Show the Levi–Civita Christoffel symbols are symmetric in i, j .

Hint. Inspect the formula. □

Solution. The right side is unchanged when i and j are interchanged. □

Exercise 22.25. Compute the Christoffel symbols of Euclidean Cartesian coordinates.

Hint. The metric coefficients are constant. □

Solution. All Christoffel symbols vanish. □

Exercise 22.26. Why can Euclidean Christoffel symbols be nonzero in polar coordinates?

Hint. Connection coefficients are not tensor components. □

Solution. The coordinate frame itself varies, producing nonzero coefficients even for the flat Euclidean connection. □

Exercise 22.27. Compute $\Gamma_{\theta\theta}^r$ for $dr^2 + r^2 d\theta^2$.

Hint. Differentiate $g_{\theta\theta} = r^2$. □

Solution. $\Gamma_{\theta\theta}^r = -\frac{1}{2}\partial_r(r^2) = -r$. □

Exercise 22.28. Compute $\Gamma_{r\theta}^\theta$ in polar coordinates.

Hint. Use $g^{\theta\theta} = 1/r^2$. □

Solution. $\Gamma_{r\theta}^\theta = \frac{1}{2}r^{-2}\partial_r(r^2) = 1/r$. □

Exercise 22.29. Show the projected ambient derivative defines a connection on an embedded submanifold.

Hint. Project after applying D_X . □

Solution. Linearity in the direction and the Leibniz rule survive orthogonal projection onto the tangent bundle. □

Exercise 22.30. Why is the projected connection metric compatible?

Hint. Use the Euclidean product rule. □

Solution. The normal parts are orthogonal to tangent vectors, leaving exactly the intrinsic metric-compatibility identity. □

Exercise 22.31. Why is the projected connection torsion free?

Hint. Compare ambient derivatives. □

Solution. $(D_X Y - D_Y X)^\top = [X, Y]$, since the bracket is tangent. □

Exercise 22.32. State the Gauss formula for a surface.

Hint. Split ambient derivative into tangent and normal parts. □

Solution. $D_X Y = \nabla_X Y + II(X, Y)N$. □

Exercise 22.33. Define the Hessian of a function.

Hint. Differentiate the gradient covariantly. □

Solution. $\text{Hess } f(X, Y) = g(\nabla_X \text{grad } f, Y)$. □

Exercise 22.34. Give the equivalent Hessian formula without the gradient.

Hint. Expand using metric compatibility. □

Solution. $\text{Hess } f(X, Y) = X(Yf) - (\nabla_X Y)f$. □

Exercise 22.35. Why is the Levi–Civita Hessian symmetric?

Hint. Subtract the two orders. □

Solution. The difference equals $[X, Y]f - (\nabla_X Y - \nabla_Y X)f = 0$ by torsion freeness. □

Exercise 22.36. For constant metric coefficients, what does the Christoffel formula give?

Hint. All first derivatives vanish. □

Solution. It gives $\Gamma_{ij}^k = 0$ in that coordinate system. □

Exercise 22.37. What data of VII/21 become canonical after choosing g ?

Hint. Use uniqueness. □

Solution. The general connection is replaced by the unique Levi–Civita connection, making its covariant derivative canonical. □

Exercise 22.38. State the bridge to geodesics.

Hint. Use covariant acceleration. □

Solution. VII/23 defines geodesics by the Levi–Civita equation $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$. □

22.11 Solved problem dossiers

Problem 22.39 (Fundamental-theorem problem). Derive the Koszul formula from metric compatibility and torsion freeness.

Solution. Write the three cyclic metric-compatibility identities, add the first two and subtract the third, then replace antisymmetric covariant-derivative differences with Lie brackets. □

Problem 22.40 (Uniqueness problem). Prove at most one metric-compatible torsion-free connection can exist.

Solution. Both must satisfy the same Koszul formula, which determines $g(\nabla_X Y, Z)$ for every Z . Nondegeneracy of g gives equality. □

Problem 22.41 (Existence problem). Explain why the Koszul formula defines a connection.

Solution. For fixed X, Y , its right side is C^∞ -linear in Z , hence represents a unique vector. Checking the connection axioms, metric compatibility, and zero torsion is direct. □

Problem 22.42 (Christoffel problem). Derive the coordinate Christoffel formula from Koszul.

Solution. Set $X = \partial_i, Y = \partial_j, Z = \partial_\ell$, use commuting coordinate fields, then raise the ℓ index with $g^{k\ell}$. $\square$

Problem 22.43 (Polar-coordinate problem). Compute all nonzero Euclidean Christoffel symbols in polar coordinates.

Solution. For $g_{rr} = 1, g_{\theta\theta} = r^2$, only the derivative $\partial_r g_{\theta\theta} = 2r$ is nonzero, yielding $\Gamma_{\theta\theta}^r = -r$ and $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = 1/r$. $\square$

Problem 22.44 (Conformal-metric problem). Derive the Christoffel symbols for $g = e^{2\varphi}\delta$.

Solution. Substitute $g_{ij} = e^{2\varphi}\delta_{ij}$ and $g^{ij} = e^{-2\varphi}\delta^{ij}$ into the metric formula and cancel the common factor. $\square$

Problem 22.45 (Submanifold problem). Show tangential projection of the Euclidean derivative is the induced Levi–Civita connection.

Solution. Verify connection axioms, metric compatibility, and zero torsion; uniqueness then completes the argument. $\square$

Problem 22.46 (Gauss-formula problem). Recover the second fundamental form from the ambient/intrinsic derivative split.

Solution. The normal component of $D_X Y$ is $\langle D_X Y, N \rangle N = II(X, Y)N$, leaving the Levi–Civita tangential component. $\square$

Problem 22.47 (Hessian-symmetry problem). Prove the Hessian is symmetric precisely because the Levi–Civita torsion vanishes.

Solution. Compute $\text{Hess } f(X, Y) - \text{Hess } f(Y, X)$; the result is $[X, Y]f - (\nabla_X Y - \nabla_Y X)f = -T(X, Y)f$. $\square$

Problem 22.48 (Metric-compatibility problem). Show parallel transport for the Levi–Civita connection must preserve inner products.

Solution. Along a curve, differentiate $g(V, W)$. Metric compatibility expresses the derivative as the sum of pairings with DV/dt and DW/dt , both zero for parallel fields. $\square$

Problem 22.49 (Normal-coordinate preview). Why is it plausible that Levi–Civita Christoffel symbols can be made zero at one point?

Solution. The connection coefficients are not tensors and can be canceled at a chosen point by a suitable coordinate choice. VII/23 constructs such coordinates via the exponential map. $\square$

Problem 22.50 (Connection-versus-curvature problem). Why does zero Christoffel symbol at one point not imply zero curvature?

Solution. Curvature depends on first derivatives and quadratic combinations of connection coefficients. Coordinates can kill the coefficients at one point but cannot generally kill their derivatives there. $\square$

22.12 Challenge problems

Problem 22.51 (Difference of metric connections). Suppose $\nabla, \tilde{\nabla}$ are both metric compatible. Describe the algebraic constraint on $A(X)Y = \tilde{\nabla}_X Y - \nabla_X Y$.

Solution. Metric compatibility for both connections implies $g(A(X)Y, Z) + g(Y, A(X)Z) = 0$. Thus for each X , the endomorphism $A(X)$ is skew-adjoint. $\square$

Problem 22.52 (Uniqueness via difference tensor). Use the previous fact plus zero torsion to prove uniqueness without invoking the full Koszul formula.

Solution. Torsion freeness gives $A(X)Y = A(Y)X$, while metric compatibility gives skew-adjointness in the last two slots. Cycling the three arguments forces the trilinear form $g(A(X)Y, Z)$ to equal its own negative, hence vanish. $\square$

Problem 22.53 (Conformal connection difference). Compare the Levi-Civita connections of g and $\tilde{g} = e^{2\varphi}g$.

Solution. The difference is $\tilde{\nabla}_X Y - \nabla_X Y = X(\varphi)Y + Y(\varphi)X - g(X, Y)\text{grad}_g \varphi$. This follows from the Koszul formula or the conformal Christoffel formula. $\square$

Problem 22.54 (Normal coordinates preview). Assume coordinates satisfy $g_{ij}(p) = \delta_{ij}$ and $\partial_k g_{ij}(p) = 0$. What are the Levi-Civita coefficients at p ?

Solution. The Christoffel formula is built from first derivatives of g_{ij} , so every $\Gamma_{ij}^k(p)$ vanishes. $\square$

Problem 22.55 (Variational bridge). Explain why metric compatibility will force a geodesic to have constant speed.

Solution. For a geodesic, $D\dot{\gamma}/dt = 0$. Metric compatibility then gives $\frac{d}{dt}g(\dot{\gamma}, \dot{\gamma}) = 2g(D\dot{\gamma}/dt, \dot{\gamma}) = 0$. $\square$

Bridge to VII/23

With the Levi-Civita connection fixed canonically by the metric, VII/23 studies its autoparallel curves: geodesics, the exponential map, normal coordinates, and variational length geometry.

Chapter 23

Geodesics

Straight lines in Euclidean space have zero acceleration. On a Riemannian manifold the intrinsic replacement is zero covariant acceleration:

$$\boxed{\nabla_{\dot{\gamma}} \dot{\gamma} = 0.}$$

Geodesics are simultaneously the autoparallel curves of the Levi-Civita connection and the local critical curves of the energy functional.

23.1 Definition

Definition 23.1 (Geodesic). A smooth curve $\gamma : I \rightarrow M$ is an affinely parametrized geodesic if

$$\frac{D\dot{\gamma}}{dt} = \nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

Proposition 23.2 (Constant speed). *Every Levi-Civita geodesic has constant speed.*

Proof. Metric compatibility gives

$$\frac{d}{dt} |\dot{\gamma}|^2 = 2g\left(\frac{D\dot{\gamma}}{dt}, \dot{\gamma}\right) = 0.$$

□

Warning 23.3 (Affine parameters). A nonlinear reparametrization of a geodesic generally does not satisfy the geodesic equation. Affine changes $t \mapsto at + b$ preserve the equation.

23.2 Coordinate geodesic equation

Write $\gamma(t) = (x^1(t), \dots, x^n(t))$.

Theorem 23.4 (Geodesic ODE). *The geodesic equation is*

$$\ddot{x}^k + \Gamma_{ij}^k(x(t)) \dot{x}^i \dot{x}^j = 0, \quad k = 1, \dots, n.$$

Proof. Expand

$$\nabla_{\dot{\gamma}} \dot{\gamma} = \left(\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j \right) \partial_k.$$

□

23.3 Existence and uniqueness

Theorem 23.5 (Initial-value problem). *For every $p \in M$ and $v \in T_p M$, there exists a unique maximal geodesic γ_v satisfying*

$$\gamma_v(0) = p, \quad \dot{\gamma}_v(0) = v.$$

It depends smoothly on the initial data wherever defined.

Proof. The coordinate geodesic equation is a smooth second-order ODE. Rewrite it as a first-order system on TM and apply standard existence, uniqueness, and smooth dependence for ODEs. $\square$

23.4 The exponential map

Definition 23.6 (Exponential map). For $p \in M$, define

$$\exp_p(v) = \gamma_v(1)$$

whenever the geodesic with initial velocity v exists up to time 1.

Equivalently,

$$\exp_p(tv) = \gamma_v(t)$$

whenever both sides are defined.

Proposition 23.7 (Derivative at the origin).

$$d(\exp_p)_0 = \text{id}_{T_p M}.$$

Proof. For $w \in T_p M$, the curve $s \mapsto \exp_p(sw)$ is the geodesic with initial velocity w . Its derivative at $s = 0$ is w . $\square$

Corollary 23.8 (Normal neighborhood). *The inverse function theorem implies that $\exp_p$ is a diffeomorphism from a neighborhood of $0 \in T_p M$ onto a neighborhood of $p \in M$.*

23.5 Normal coordinates

Definition 23.9 (Normal coordinates). Choose a basis of $T_p M$ and use $\exp_p$ to transfer linear coordinates near 0 to coordinates near p . These are normal coordinates centered at p .

Theorem 23.10 (Metric normalization). *For an orthonormal basis used in normal coordinates,*

$$g_{ij}(p) = \delta_{ij}, \quad \Gamma_{ij}^k(p) = 0, \quad \partial_\ell g_{ij}(p) = 0.$$

Remark 23.11. These identities hold at the center point, not throughout the coordinate neighborhood. Curvature appears in the higher-order deviations from Euclidean behavior.

23.6 Energy and length

Definition 23.12 (Energy). For a smooth curve $\gamma : [a, b] \rightarrow M$,

$$E(\gamma) = \frac{1}{2} \int_a^b |\dot{\gamma}|^2 dt.$$

Proposition 23.13 (Energy–length inequality).

$$L(\gamma)^2 \leq 2(b-a)E(\gamma),$$

with equality exactly when the speed is constant.

Proof. Apply Cauchy–Schwarz to

$$L(\gamma) = \int_a^b |\dot{\gamma}| dt.$$

□

Theorem 23.14 (First variation of energy). *Let $\gamma_s(t)$ be a variation with fixed endpoints and variational field $V = \partial_s \gamma_s|_{s=0}$. Then*

$$\left. \frac{d}{ds} \right|_0 E(\gamma_s) = - \int_a^b g\left(V, \frac{D\dot{\gamma}}{dt}\right) dt.$$

Consequently the fixed-endpoint critical curves of energy are precisely the geodesics.

23.7 Local minimizing property

Theorem 23.15 (Local minimization). *Every sufficiently short geodesic segment minimizes length among nearby curves with the same endpoints.*

Warning 23.16 (Not globally minimizing forever). A geodesic may cease to minimize after traveling far enough. Global minimizing behavior is governed by cut loci, conjugate points, and completeness, which belong to later Riemannian geometry.

23.8 Examples

Example 23.17 (Euclidean space). Since all Cartesian Christoffel symbols vanish,

$$\ddot{x}^k = 0.$$

Thus Euclidean geodesics are affine lines.

Example 23.18 (Round sphere). The geodesics of the round sphere are great circles, parametrized with constant speed.

Proof. For a unit sphere embedded in Euclidean space, the intrinsic covariant derivative is tangential projection. A great circle has ambient acceleration normal to the sphere, so its tangential acceleration vanishes. □

Example 23.19 (Cylinder). A circular cylinder is locally isometric to the Euclidean plane. Its geodesics therefore correspond under unrolling to straight lines: circles, generators, and helices appear as special cases.

Example 23.20 (Euclidean polar coordinates). Using

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{r\theta}^\theta = \frac{1}{r},$$

the Euclidean geodesic equations become

$$\ddot{r} - r\dot{\theta}^2 = 0, \quad \ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} = 0.$$

These equations still describe straight lines, merely in polar coordinates.

23.9 Geodesic completeness preview

Definition 23.21 (Geodesically complete). A Riemannian manifold is geodesically complete if every geodesic can be extended for all $t \in \mathbb{R}$.

Remark 23.22. The Hopf–Rinow theorem later links geodesic completeness with completeness of the Riemannian distance and existence of globally minimizing geodesics between points.

23.10 Exercises

Exercise 23.23. Define an affinely parametrized geodesic.

Hint. Use covariant acceleration. □

Solution. It satisfies $D\dot{\gamma}/dt = 0$. □

Exercise 23.24. Prove a geodesic has constant speed.

Hint. Differentiate $g(\dot{\gamma}, \dot{\gamma})$. □

Solution. Metric compatibility gives derivative $2g(D\dot{\gamma}/dt, \dot{\gamma}) = 0$. □

Exercise 23.25. Which reparametrizations preserve the geodesic equation?

Hint. Check the chain rule. □

Solution. Affine changes $t \mapsto at + b$ preserve affine geodesic parametrization. □

Exercise 23.26. Write the coordinate geodesic equation.

Hint. Expand the covariant derivative. □

Solution. $\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0$. □

Exercise 23.27. Why is the geodesic equation a second-order ODE?

Hint. It contains $\ddot{x}^k$. □

Solution. The unknowns are coordinate functions and their second derivatives appear explicitly. □

Exercise 23.28. What initial data determine a unique local geodesic?

Hint. Use ODE theory. □

Solution. A point p and tangent vector $v \in T_p M$. □

Exercise 23.29. Define $\exp_p(v)$.

Hint. Use the geodesic at time 1. □

Solution. $\exp_p(v) = \gamma_v(1)$, wherever defined. □

Exercise 23.30. Show $\exp_p(tv) = \gamma_v(t)$.

Hint. Use uniqueness and rescaling. □

Solution. Both sides are geodesics with the same initial point and appropriate rescaled initial velocity. □

Exercise 23.31. Compute $d(\exp_p)_0$.

Hint. Differentiate $s \mapsto \exp_p(sw)$. □

Solution. It is the identity on T_pM . □

Exercise 23.32. Why does $\exp_p$ give local coordinates near p ?

Hint. Use the inverse function theorem. □

Solution. Its derivative at 0 is an isomorphism, so it is a local diffeomorphism. □

Exercise 23.33. Define normal coordinates.

Hint. Transfer linear tangent coordinates through $\exp_p$. □

Solution. They are coordinates obtained from an orthonormal basis of T_pM and the exponential map. □

Exercise 23.34. What are $g_{ij}(p)$ in orthonormal normal coordinates?

Hint. The tangent basis at the center is orthonormal. □

Solution. $g_{ij}(p) = \delta_{ij}$. □

Exercise 23.35. What are $\Gamma_{ij}^k(p)$ in normal coordinates?

Hint. Radial coordinate lines are geodesics. □

Solution. All Christoffel symbols vanish at the center. □

Exercise 23.36. What happens to $\partial_\ell g_{ij}(p)$ in normal coordinates?

Hint. Use the Christoffel formula plus metric compatibility. □

Solution. All first derivatives vanish at the center. □

Exercise 23.37. Define the energy of a curve.

Hint. Integrate squared speed. □

Solution. $E(\gamma) = \frac{1}{2} \int |\dot{\gamma}|^2 dt$. □

Exercise 23.38. State the energy–length inequality.

Hint. Apply Cauchy–Schwarz. □

Solution. $L^2 \leq 2(b-a)E$. □

Exercise 23.39. When does equality in the energy–length inequality hold?

Hint. Use equality in Cauchy–Schwarz. □

Solution. Exactly when the speed is constant. □

Exercise 23.40. State the fixed-endpoint first variation of energy.

Hint. Use the variational field V . □

Solution. It is $-\int g(V, D\dot{\gamma}/dt) dt$. □

Exercise 23.41. Why are geodesics critical points of energy?

Hint. Insert the geodesic equation. □

Solution. The integrand vanishes because $D\dot{\gamma}/dt = 0$. □

Exercise 23.42. Why does every energy-critical fixed-endpoint curve satisfy the geodesic equation?

Hint. Use arbitrary compactly supported variational fields. □

Solution. The fundamental lemma forces $D\dot{\gamma}/dt = 0$. □

Exercise 23.43. Describe Euclidean geodesics.

Hint. Cartesian Christoffel symbols vanish. □

Solution. They are affine straight lines. □

Exercise 23.44. Describe round-sphere geodesics.

Hint. Use tangential acceleration. □

Solution. They are great circles with affine constant-speed parametrization. □

Exercise 23.45. Describe cylinder geodesics via unrolling.

Hint. Use the local isometry to the plane. □

Solution. They become straight lines in the developed plane, producing generators, circles, and helices. □

Exercise 23.46. State the bridge to parallel transport.

Hint. Look at the tangent field of a geodesic. □

Solution. A geodesic is characterized by its tangent vector being parallel along itself; VII/24 develops parallel transport for arbitrary vectors along arbitrary curves. □

23.11 Solved problem dossiers

Problem 23.47 (Geodesic-ODE problem). Derive the coordinate geodesic equation.

Solution. Write $\dot{\gamma} = \dot{x}^i \partial_i$, covariantly differentiate, and use $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$. □

Problem 23.48 (Initial-value problem). Explain geodesic existence, uniqueness, and smooth dependence from standard ODE theory.

Solution. Introduce velocities $v^i = \dot{x}^i$ and rewrite the geodesic equation as a first-order smooth vector field on TM . □

Problem 23.49 (Exponential-map problem). Prove $d(\exp_p)_0 = \text{id}$.

Solution. For $w \in T_p M$, differentiate the radial curve $s \mapsto \exp_p(sw) = \gamma_w(s)$ at zero. □

Problem 23.50 (Normal-coordinate problem). Explain why Christoffel symbols vanish at the center of normal coordinates.

Solution. Every radial line $t \mapsto tv$ in the tangent space maps to a geodesic. Substitution into the geodesic equation for arbitrary v forces the symmetric coefficients at the center to vanish. $\square$

Problem 23.51 (Energy problem). Derive $L^2 \leq 2(b-a)E$.

Solution. Apply Cauchy–Schwarz to $1 \cdot |\dot{\gamma}|$ on $[a, b]$. $\square$

Problem 23.52 (First-variation problem). Derive the fixed-endpoint first variation of energy.

Solution. Differentiate $g(\partial_t \gamma_s, \partial_t \gamma_s)$, exchange covariant derivatives using torsion freeness, integrate by parts, and use the fixed endpoints. $\square$

Problem 23.53 (Euclidean problem). Solve the Euclidean geodesic equation.

Solution. All $\Gamma_{ij}^k = 0$, so $\ddot{x}^k = 0$, giving $x^k(t) = a^k t + b^k$. $\square$

Problem 23.54 (Polar-coordinate problem). Derive the Euclidean geodesic equations in polar coordinates.

Solution. Insert the nonzero polar Christoffel symbols into the general equation. $\square$

Problem 23.55 (Sphere problem). Show a unit-speed great circle is a geodesic on the round sphere.

Solution. Its ambient acceleration points radially and therefore has zero tangential projection, which equals its intrinsic covariant acceleration. $\square$

Problem 23.56 (Cylinder problem). Classify cylinder geodesics by unrolling.

Solution. The cylinder is locally isometric to a strip in the plane. Straight planar lines descend to constant-slope helices, with horizontal and vertical limiting cases. $\square$

Problem 23.57 (Local-minimization problem). Why is a sufficiently short radial geodesic in a normal neighborhood minimizing?

Solution. Use Gauss-lemma geometry: radial motion contributes directly to length, while angular motion can only increase it. Hence the radial geodesic realizes the local distance. $\square$

Problem 23.58 (Completeness problem). Distinguish local geodesic existence from geodesic completeness.

Solution. ODE theory always gives a local solution. Completeness requires every maximal solution to extend for all real parameter values. $\square$

23.12 Challenge problems

Problem 23.59 (Affine reparametrization). Show explicitly why $t \mapsto at + b$ preserves the geodesic equation and a general nonlinear change does not.

Solution. Under $\tilde{\gamma}(s) = \gamma(\phi(s))$, covariant acceleration becomes $(\phi')^2 D\dot{\gamma}/dt + \phi''\dot{\gamma}$. For a geodesic the first term vanishes, leaving $\phi''\dot{\gamma}$, so affine ϕ is required. $\square$

Problem 23.60 (Geodesic spray). Recast the geodesic equation as a vector field on TM .

Solution. In coordinates (x^i, v^i) , the geodesic spray is $G = v^i \partial_{x^i} - \Gamma_{ij}^k v^i v^j \partial_{v^k}$. Its integral curves project to geodesics. $\square$

Problem 23.61 (Normal-coordinate metric expansion preview). Why can curvature first appear only at second order in the metric coefficients in normal coordinates?

Solution. At the center $g_{ij} = \delta_{ij}$ and all first derivatives vanish. Therefore the first nontrivial Taylor data are second derivatives, which are constrained by the Riemann curvature tensor. $\square$

Problem 23.62 (Geodesic on an embedded surface). Show that a unit-speed curve on a surface is geodesic exactly when its ambient acceleration is normal to the surface.

Solution. The intrinsic acceleration is the tangential projection of the ambient acceleration by the induced Levi–Civita connection. It vanishes exactly when ambient acceleration has no tangent component. $\square$

Problem 23.63 (Constant-speed energy equivalence). For fixed endpoints and fixed parameter interval, explain why minimizing energy and minimizing length agree among constant-speed curves.

Solution. For constant speed equality holds in $L^2 \leq 2(b-a)E$, giving $E = L^2/[2(b-a)]$. Thus the two functionals are monotone functions of one another on that class. $\square$

Bridge to VII/24

The tangent of a geodesic is parallel along itself. VII/24 generalizes this idea to arbitrary vectors along arbitrary curves and develops the resulting transport maps and horizontal lifts.

Chapter 24

Parallel Transport

A connection tells us how to differentiate a vector field along a curve. Parallel transport solves the inverse problem: start with one vector and extend it along the curve so that its covariant derivative vanishes.

$$\boxed{\frac{DV}{dt} = 0.}$$

For the Levi-Civita connection, this extension preserves lengths and angles.

24.1 Parallel vector fields

Definition 24.1 (Parallel field). Let $\gamma : I \rightarrow M$ be smooth. A vector field $V(t) \in T_{\gamma(t)}M$ is parallel along γ if

$$\frac{DV}{dt} = \nabla_{\dot{\gamma}} V = 0.$$

In coordinates,

$$V = V^k \partial_k,$$

so the equation becomes

$$\dot{V}^k + \Gamma_{ij}^k(\gamma(t)) \dot{x}^i V^j = 0.$$

24.2 Existence and uniqueness

Theorem 24.2 (Parallel initial-value problem). *Given a smooth curve $\gamma : [a, b] \rightarrow M$, a time $t_0 \in [a, b]$, and a vector $v \in T_{\gamma(t_0)}M$, there exists a unique parallel field V along γ satisfying*

$$V(t_0) = v.$$

Proof. The coordinate equation is a linear first-order ODE for the component vector $V(t)$. Standard linear ODE theory gives existence and uniqueness. $\square$

24.3 The transport map

Definition 24.3 (Parallel transport). The parallel transport map along γ from a to b is

$$P_\gamma : T_{\gamma(a)}M \rightarrow T_{\gamma(b)}M, \quad P_\gamma(v) = V(b),$$

where V is the unique parallel field with $V(a) = v$.

Proposition 24.4 (Linearity and invertibility). *P_γ is a linear isomorphism.*

Proof. Linearity follows from linearity of the parallel ODE. Reversing the curve solves the inverse problem, so

$$P_{\gamma^{-1}} = P_{\gamma}^{-1}.$$

□

Proposition 24.5 (Concatenation). *If γ_1 ends where γ_2 begins, then*

$$P_{\gamma_2 * \gamma_1} = P_{\gamma_2} \circ P_{\gamma_1}.$$

24.4 Levi–Civita transport is orthogonal

Theorem 24.6 (Metric preservation). *For the Levi–Civita connection, if V, W are parallel along γ , then*

$$g(V(t), W(t))$$

is constant. Consequently

$$P_{\gamma} : T_{\gamma(a)}M \rightarrow T_{\gamma(b)}M$$

is an inner-product isometry.

Proof. Metric compatibility gives

$$\frac{d}{dt}g(V, W) = g(DV/dt, W) + g(V, DW/dt) = 0.$$

□

Corollary 24.7 (Orthonormal frames remain orthonormal). *Parallel transport of an orthonormal basis along a curve produces an orthonormal frame along that curve.*

24.5 Geodesics revisited

Proposition 24.8 (Geodesic tangent). *A curve is an affinely parametrized geodesic exactly when its tangent vector is parallel along itself:*

$$P_{\gamma| [a, t]}(\dot{\gamma}(a)) = \dot{\gamma}(t).$$

24.6 Transport in a local frame

Let $e_1, \dots, e_n$ be a local frame with connection one-form matrix $\omega = (\omega^a_b)$. If $V = V^a e_a$, then parallelism is

$$\dot{V}^a + \omega^a_b(\dot{\gamma})V^b = 0.$$

Proposition 24.9 (Fundamental matrix). *If $U(t)$ is the matrix transporting frame components, then*

$$\dot{U}(t) = -A(t)U(t), \quad A^a_b(t) = \omega^a_b(\dot{\gamma}(t)), \quad U(a) = I.$$

Thus P_{γ} is represented by the endpoint value $U(b)$.

Remark 24.10 (Path ordering). If the matrices $A(t)$ fail to commute at different times, one cannot replace the solution by an ordinary exponential of $-\int A dt$. The correct object is the path-ordered exponential, or equivalently the fundamental solution of the ODE.

24.7 Euclidean examples

Example 24.11 (Cartesian Euclidean transport). In $\mathbb{R}^n$ with Cartesian coordinates, $\Gamma_{ij}^k = 0$. Parallel fields have constant components, so parallel transport is ordinary translation of vectors.

Example 24.12 (Polar frame warning). In Euclidean polar coordinates, a vector with constant (r, θ) -coordinate components is generally not parallel because the coordinate frame varies. Parallelism must be tested with the Christoffel terms.

24.8 Parallel transport on a sphere

Consider the round sphere. Along a great circle, the tangent vector of the great circle is parallel. A second parallel vector can be chosen orthogonal to it within the tangent plane, producing a parallel orthonormal frame along the arc.

Example 24.13 (Meridian). Along a meridian on the unit sphere, the unit tangent to the meridian is parallel, as is the east–west tangent vector obtained by the corresponding Levi–Civita parallel ODE along that meridian.

Warning 24.14 (Transport depends on the path). On a curved manifold, parallel transport between the same endpoints can depend on the chosen path. The systematic study of transport around closed loops is holonomy, the subject of VII/25.

24.9 Covectors and tensors

A connection on TM induces connections on T^*M and all tensor bundles.

Definition 24.15 (Parallel covector). A covector field $\alpha(t) \in T_{\gamma(t)}^*M$ is parallel if

$$\frac{D\alpha}{dt} = 0.$$

Equivalently, for every parallel vector field V ,

$$\alpha(V)$$

is constant.

Proposition 24.16 (Metric duals). *For the Levi–Civita connection, V is parallel if and only if $V^\flat$ is parallel.*

24.10 Principal-bundle horizontal lifts

Let $P \rightarrow M$ be a principal bundle with a principal connection.

Definition 24.17 (Horizontal lift of a curve). A curve $\tilde{\gamma}(t) \in P$ above $\gamma(t)$ is horizontal if

$$\pi(\tilde{\gamma}(t)) = \gamma(t), \quad \dot{\tilde{\gamma}}(t) \in H_{\tilde{\gamma}(t)}P.$$

Theorem 24.18 (Horizontal lift). *Given γ and an initial point $p_0 \in P_{\gamma(a)}$, there is a unique horizontal lift $\tilde{\gamma}$ with $\tilde{\gamma}(a) = p_0$.*

Remark 24.19. For the frame bundle, horizontal lifting of a frame is precisely parallel transport of that frame. This unifies the vector-bundle and principal-bundle pictures developed in VII/08 and VII/21.

24.11 Reparametrization

Proposition 24.20 (Orientation-preserving reparametrization). *Parallel transport depends on the oriented image of the curve, not on its regular orientation-preserving parametrization.*

Proof. Under $t = \phi(s)$, the covariant derivative along the reparametrized curve gains the scalar factor $\phi'(s)$, so the zero equation remains zero. $\square$

24.12 Exercises

Exercise 24.21. Define a parallel vector field along a curve.

Hint. Set covariant derivative to zero. $\square$

Solution. It satisfies $DV/dt = 0$. $\square$

Exercise 24.22. Write the coordinate parallel-transport ODE.

Hint. Expand $V = V^k \partial_k$. $\square$

Solution. $\dot{V}^k + \Gamma_{ij}^k \dot{x}^i V^j = 0$. $\square$

Exercise 24.23. Why does a unique parallel field exist for prescribed initial data?

Hint. Use linear ODE theory. $\square$

Solution. The component equation is a smooth linear first-order system. $\square$

Exercise 24.24. Define the parallel transport map P_γ .

Hint. Use the endpoint of the unique parallel extension. $\square$

Solution. $P_\gamma(v) = V(b)$ for the parallel field with $V(a) = v$. $\square$

Exercise 24.25. Why is P_γ linear?

Hint. The ODE is linear in V . $\square$

Solution. Linear combinations of parallel solutions are parallel with the corresponding initial data. $\square$

Exercise 24.26. Why is P_γ invertible?

Hint. Reverse the curve. $\square$

Solution. Transport along the reversed curve is the inverse map. $\square$

Exercise 24.27. State the concatenation law.

Hint. Transport in two stages. $\square$

Solution. $P_{\gamma_2 * \gamma_1} = P_{\gamma_2} \circ P_{\gamma_1}$. $\square$

Exercise 24.28. Show Levi-Civita transport preserves vector lengths.

Hint. Differentiate $g(V, V)$. $\square$

Solution. Parallelism and metric compatibility make the derivative zero. $\square$

Exercise 24.29. Show Levi–Civita transport preserves angles.

Hint. Preserve all pairwise inner products. □

Solution. Inner products and norms are constant, hence the angle formula is unchanged. □

Exercise 24.30. What happens to an orthonormal basis under Levi–Civita transport?

Hint. Use metric preservation. □

Solution. It remains an orthonormal basis along the curve. □

Exercise 24.31. How is a geodesic characterized by parallel transport?

Hint. Look at its tangent field. □

Solution. Its tangent vector is parallel along the curve. □

Exercise 24.32. Write parallel transport in a local frame.

Hint. Use connection one-forms. □

Solution. $\dot{V}^a + \omega^a_b(\dot{\gamma})V^b = 0$. □

Exercise 24.33. Write the fundamental-matrix equation.

Hint. Collect the linear system. □

Solution. $\dot{U} = -A(t)U$ with $A^a_b = \omega^a_b(\dot{\gamma})$ and $U(a) = I$. □

Exercise 24.34. When can the transport matrix be written as an ordinary exponential?

Hint. Require commutation in time. □

Solution. If $A(t_1)$ and $A(t_2)$ commute for all times, then $U(b) = \exp(-\int_a^b A(t) dt)$. □

Exercise 24.35. Describe Euclidean Cartesian parallel transport.

Hint. All Christoffel symbols vanish. □

Solution. Vector components stay constant. □

Exercise 24.36. Why are constant polar-coordinate components not necessarily parallel in the Euclidean plane?

Hint. The polar frame varies. □

Solution. Nonzero Christoffel symbols contribute to the covariant derivative. □

Exercise 24.37. Along a geodesic, is the tangent field parallel?

Hint. Use the geodesic equation. □

Solution. Yes: $D\dot{\gamma}/dt = 0$. □

Exercise 24.38. Define a parallel covector.

Hint. Use the induced dual connection. □

Solution. It satisfies $D\alpha/dt = 0$. □

Exercise 24.39. How can parallel covectors be characterized using parallel vectors?

Hint. Pair them. ☐

Solution. $\alpha(V)$ is constant for every parallel V . ☐

Exercise 24.40. Why does $\flat$ carry parallel vectors to parallel covectors for Levi-Civita?

Hint. Use $\nabla g = 0$. ☐

Solution. Metric compatibility says covariant differentiation commutes with lowering indices. ☐

Exercise 24.41. Define a horizontal lift in a principal bundle.

Hint. Lift the curve with horizontal velocity. ☐

Solution. It projects to the base curve and has velocity in the horizontal distribution. ☐

Exercise 24.42. What is the frame-bundle meaning of parallel transport?

Hint. Lift a frame horizontally. ☐

Solution. A horizontal frame lift is exactly a frame whose vectors are parallel along the base curve. ☐

Exercise 24.43. Does parallel transport depend on speed of parametrization?

Hint. Use orientation-preserving reparametrization. ☐

Solution. No; regular orientation-preserving reparametrization leaves the zero covariant-derivative equation unchanged. ☐

Exercise 24.44. State the bridge to holonomy.

Hint. Close the curve. ☐

Solution. VII/25 studies the transformations obtained by parallel transport around loops and the groups they generate. ☐

24.13 Solved problem dossiers

Problem 24.45 (Parallel-ODE problem). Derive the coordinate ODE for a parallel vector field.

Solution. Substitute $V = V^j \partial_j$ and $\dot{\gamma} = \dot{x}^i \partial_i$ into $\nabla_{\dot{\gamma}} V = 0$ and apply the connection Leibniz rule. ☐

Problem 24.46 (Existence-uniqueness problem). Prove the parallel transport map is well defined.

Solution. The coordinate equation is a linear ODE with smooth coefficients, so each initial vector has a unique solution along the curve. ☐

Problem 24.47 (Composition problem). Prove the concatenation law for parallel transport.

Solution. Transport the initial vector along the first curve, then use that endpoint as initial data on the second. Uniqueness identifies the piecewise field with transport along the concatenation. $\square$

Problem 24.48 (Inverse problem). Prove transport along the reversed path is the inverse.

Solution. Reverse a parallel field in parameter. The reversed field remains parallel, so transporting out and back returns the initial vector. $\square$

Problem 24.49 (Metric-preservation problem). Prove Levi–Civita parallel transport is an isometry.

Solution. For two parallel fields V, W , differentiate $g(V, W)$ and use metric compatibility. $\square$

Problem 24.50 (Geodesic problem). Relate the geodesic equation to parallel transport.

Solution. A curve is geodesic precisely when its tangent field solves the parallel ODE along the same curve. $\square$

Problem 24.51 (Fundamental-matrix problem). Represent parallel transport in a moving frame.

Solution. The connection matrix evaluated on $\dot{\gamma}$ gives $A(t)$, and the transport matrix solves $\dot{U} = -AU$, $U(a) = I$. $\square$

Problem 24.52 (Euclidean polar problem). Explain how a constant Euclidean vector acquires changing polar components while remaining parallel.

Solution. The vector itself is fixed in Cartesian space, but the polar coordinate basis rotates and rescales. Its changing components exactly cancel the Christoffel terms in the parallel ODE. $\square$

Problem 24.53 (Sphere problem). Construct a parallel orthonormal frame along a great-circle arc.

Solution. The unit tangent is parallel because the arc is geodesic. Choose an orthogonal unit tangent vector at the start and parallel transport it; metric preservation keeps the frame orthonormal. $\square$

Problem 24.54 (Dual-transport problem). Describe transport on covectors.

Solution. Define P_γ^* contragrediently so pairings with transported vectors remain constant; this agrees with parallel transport for the induced dual connection. $\square$

Problem 24.55 (Horizontal-lift problem). Explain how a principal connection produces transport.

Solution. Lift the base curve horizontally from a chosen point in the initial fiber. The endpoint defines an equivariant map between fibers, and associated-bundle representations turn it into vector transport. $\square$

Problem 24.56 (Path-dependence problem). Why can two paths with the same endpoints produce different transport maps?

Solution. The connection prescribes infinitesimal comparison, but curvature can make the accumulated comparison path dependent. Closed-loop discrepancies are organized by holonomy. $\square$

24.14 Challenge problems

Problem 24.57 (Path-ordered exponential). Explain why noncommuting connection matrices require path ordering.

Solution. For $\dot{U} = -A(t)U$, successive infinitesimal factors occur in chronological order. If $A(t_1)A(t_2) \neq A(t_2)A(t_1)$, rearranging them changes the product, so the solution is a time/path-ordered exponential rather than $\exp(-\int A)$. $\square$

Problem 24.58 (Gauge covariance of transport). How does the matrix of parallel transport change under a frame change $e' = eA$?

Solution. If endpoint frames change by $A(a)$ and $A(b)$, the transport matrix changes by $U' = A(b)^{-1}UA(a)$. Thus the underlying linear map is frame independent. $\square$

Problem 24.59 (Flat connection and path independence). On a simply connected domain with a globally trivial flat connection that admits a parallel frame, why is transport path independent?

Solution. In a global parallel frame the connection matrix vanishes, so parallel components are constant. The transport map then depends only on the endpoint frame identification, not on the chosen path. $\square$

Problem 24.60 (Parallel tensors). Show that tensor products and contractions of parallel tensor fields remain parallel.

Solution. The induced connection obeys a Leibniz rule on tensor products and commutes with contractions. If each factor has zero covariant derivative, so does the tensor product, and contraction preserves zero. $\square$

Problem 24.61 (Holonomy preview). Why does transport around loops based at p naturally form a subgroup of the orthogonal group $O(T_pM, g_p)$?

Solution. Each loop gives a Levi-Civita transport isometry $T_pM \rightarrow T_pM$. Concatenation gives composition, reversal gives inverses, and the constant loop gives the identity. VII/25 studies this group in detail. $\square$

Bridge to VII/25

Parallel transport around closed loops need not return a vector unchanged. VII/25 organizes these loop transports into holonomy groups and relates them to curvature and global geometry.

Chapter 25

Holonomy

The previous four chapters form a single geometric mechanism:

connection $\longrightarrow$ Levi–Civita connection $\longrightarrow$ geodesics $\longrightarrow$ parallel transport $\longrightarrow$ holonomy.

A connection tells us how to differentiate vector fields. The Levi–Civita connection makes that rule canonical for a Riemannian metric. Geodesics are curves whose tangent vector transports parallel to itself. Parallel transport compares vectors along an arbitrary path. Holonomy records what happens when the path closes into a loop.

25.1 Transport around loops

Fix a Riemannian manifold (M, g) and a base point $p \in M$.

Definition 25.1 (Holonomy transformation). If $\gamma : [0, 1] \rightarrow M$ is a piecewise smooth loop with $\gamma(0) = \gamma(1) = p$, its Levi–Civita parallel transport is an orthogonal linear map

$$P_\gamma : T_p M \rightarrow T_p M.$$

This map is called the holonomy transformation of γ .

Definition 25.2 (Holonomy group). The holonomy group at p is

$$\text{Hol}_p(M, g) = \{P_\gamma : \gamma \text{ is a piecewise smooth loop based at } p\}.$$

Theorem 25.3 (Group property). $\text{Hol}_p(M, g)$ is a subgroup of the orthogonal group

$$O(T_p M, g_p).$$

Proof. The constant loop gives the identity. Concatenation of loops corresponds to composition of transport maps, and reversing a loop gives the inverse transport. Levi–Civita transport preserves the inner product, so every element is orthogonal. $\square$

25.2 Why holonomy is genuinely geometric

Remark 25.4 (Path dependence becomes structure). Parallel transport between fixed endpoints may depend on the path. Holonomy does not regard that dependence as a defect: it packages all loop-dependent discrepancies into a group.

Example 25.5 (Euclidean space). In $\mathbb{R}^n$ with its standard metric, Cartesian parallel transport keeps components constant. Hence every loop has

$$P_\gamma = \text{id}, \quad \text{Hol}_p(\mathbb{R}^n) = \{I\}.$$

25.3 Base-point dependence

Let p, q lie in the same connected component and let σ be a path from p to q .

Theorem 25.6 (Base-point conjugacy). *Parallel transport along σ conjugates the two holonomy groups:*

$$\text{Hol}_q = P_\sigma \text{Hol}_p P_\sigma^{-1}.$$

Proof. Given a loop γ based at p , form the loop at q

$$\sigma^{-1} * \gamma * \sigma$$

with the convention that paths are traversed from right to left in the displayed composition. The concatenation law for parallel transport gives the conjugation formula. $\square$

Remark 25.7. Thus the holonomy group is canonically attached to a base point only up to conjugacy. Its abstract isomorphism type is constant on a connected component.

25.4 Restricted holonomy

Definition 25.8 (Restricted holonomy). The restricted holonomy group

$$\text{Hol}_p^0(M, g)$$

is generated by holonomy transformations of loops based at p that are homotopic to the constant loop through based loops.

Proposition 25.9 (Normal subgroup). Hol_p^0 is a normal subgroup of Hol_p .

Proof. Conjugating a contractible loop by any based loop again gives a contractible loop after transporting the contraction along the conjugating path. $\square$

Remark 25.10 (Local versus global information). Restricted holonomy is controlled by local differential geometry. The quotient

$$\text{Hol}_p / \text{Hol}_p^0$$

can also contain information arising from the global topology of the manifold.

25.5 Flat connections and topology

Proposition 25.11 (Parallel frame criterion). *If a connected open set U admits a global orthonormal frame $E_1, \dots, E_n$ satisfying*

$$\nabla E_i = 0,$$

then parallel transport in U is path independent and every loop in U has trivial holonomy.

Proof. Write any vector as $V = v^i E_i$. Along a curve,

$$\frac{DV}{dt} = \dot{v}^i E_i.$$

Thus parallel fields are exactly those with constant coefficients in the parallel frame. $\square$

Warning 25.12 (Flat does not automatically mean globally trivial). Vanishing curvature is a local statement. On a nonsimply connected manifold a flat connection may still have nontrivial full holonomy because loops can carry global monodromy.

25.6 A two-dimensional picture

On an oriented Riemannian surface, every tangent plane is an oriented Euclidean two-plane. Hence an orientation-preserving holonomy transformation is a rotation.

Proposition 25.13 (Surface holonomy angle). *For an oriented Riemannian surface, the restricted holonomy group is contained in*

$$SO(2).$$

Thus sufficiently small contractible loops act by rotations of the tangent plane.

25.7 Sphere: transport detects curvature

Example 25.14 (Spherical triangle). On the unit sphere, parallel transport around a positively oriented geodesic triangle returns a tangent vector rotated by the spherical excess

$$\alpha + \beta + \gamma - \pi.$$

For the unit sphere this equals the area of the triangle.

Remark 25.15 (Gauss–Bonnet preview). The equality between holonomy angle and enclosed curvature is a first glimpse of Gauss–Bonnet. The present chapter uses it as geometric motivation; the systematic curvature tensor comes first in VII/26.

25.8 Small loops and infinitesimal holonomy

Holonomy around a very small loop is close to the identity. Its leading nontrivial deviation is governed by curvature.

Theorem 25.16 (Infinitesimal holonomy preview). *Let $X, Y \in T_p M$, and consider a sufficiently small coordinate parallelogram with oriented area ε^2 tangent to X, Y . Parallel transport around its boundary has the schematic expansion*

$$P_{\partial \square_\varepsilon} = I + \varepsilon^2 R(X, Y) + O(\varepsilon^3),$$

up to the sign convention used in defining the curvature operator.

Remark 25.17. VII/26 will define $R(X, Y)$ precisely and derive the algebraic identities of the Riemann curvature tensor. The present statement explains conceptually why curvature is the infinitesimal generator of holonomy.

25.9 Holonomy of a connection on a vector bundle

The construction is not limited to the tangent bundle.

Definition 25.18 (Bundle holonomy). Let $E \rightarrow M$ be a vector bundle with connection. For a loop γ based at p , parallel transport gives

$$P_\gamma : E_p \rightarrow E_p.$$

The subgroup of $GL(E_p)$ generated by all such maps is the holonomy group of the connection on E .

If the connection preserves a fiber metric, its holonomy lies in the corresponding orthogonal or unitary group.

25.10 Principal-bundle holonomy

Let $P \rightarrow M$ be a principal G -bundle with connection.

Definition 25.19 (Principal holonomy). Choose $u \in P_p$. Horizontally lift a based loop γ beginning at u . Its endpoint lies in the same fiber and therefore has the form

$$u \cdot h_\gamma$$

for a unique $h_\gamma \in G$. The elements h_γ form the principal holonomy group based at u .

Proposition 25.20 (Change of point in the fiber). *Replacing u by $u \cdot g$ conjugates the principal holonomy subgroup by $g^{-1}(\cdot)g$.*

25.11 Holonomy and invariant tensors

Theorem 25.21 (Parallel tensors give holonomy invariants). *If a tensor field T is parallel,*

$$\nabla T = 0,$$

then its value T_p is fixed by the natural action of Hol_p .

Proof. Parallel transport carries T_p to T_q along every path. For a loop based at p , the endpoint is again p , so the induced holonomy action must fix T_p . $\square$

Remark 25.22 (Reverse direction). On suitable connected domains, holonomy-invariant fiber data can be extended by parallel transport to parallel tensor fields. This principle is one of the main reasons holonomy is so powerful.

25.12 Riemannian holonomy and special geometry

For an n -dimensional Riemannian manifold,

$$\text{Hol}_p \subseteq O(n),$$

and on an oriented manifold the restricted holonomy lies in $SO(n)$. A proper reduction of holonomy means that parallel transport preserves additional algebraic structure beyond the metric.

Example 25.23 (Parallel complex structure preview). If a Riemannian manifold carries a parallel orthogonal complex structure J , then holonomy preserves J , so the holonomy group is contained in a unitary group.

Remark 25.24. This observation leads toward Kähler, Calabi–Yau, G_2 , and Spin(7) geometry. Those theories lie beyond the present volume, but holonomy is the organizing language behind all of them.

25.13 The five-chapter arc

The conceptual chain can now be read backwards as well as forwards:

holonomy measures accumulated parallel transport,
parallel transport integrates a connection along curves,
geodesics parallel-transport their own tangent vectors,
Levi–Civita makes the connection canonical from the metric,
general connections provide the differentiation framework.

The next chapter identifies the infinitesimal obstruction responsible for nontrivial local holonomy: the Riemann curvature tensor.

25.14 Exercises

Exercise 25.25. Define the holonomy transformation of a based loop.

Hint. Use parallel transport from the base point back to itself. □

Solution. For a loop γ based at p , the map $P_\gamma : T_p M \rightarrow T_p M$ is its holonomy transformation. □

Exercise 25.26. Define $\text{Hol}_p(M, g)$.

Hint. Collect all loop transports. □

Solution. It is the set of all P_γ for piecewise smooth loops based at p . □

Exercise 25.27. Why is Levi-Civita holonomy contained in an orthogonal group?

Hint. Use metric preservation of parallel transport. □

Solution. Each P_γ preserves g_p , so $\text{Hol}_p \subset O(T_p M, g_p)$. □

Exercise 25.28. What loop gives the identity holonomy element?

Hint. Use the constant loop. □

Solution. Parallel transport along the constant loop does nothing, hence gives the identity. □

Exercise 25.29. How does loop concatenation act on holonomy transformations?

Hint. Use the concatenation law for transport. □

Solution. Transport around the concatenation is the composition of the two loop transport maps. □

Exercise 25.30. What is the inverse of P_γ ?

Hint. Reverse the loop. □

Solution. $P_{\gamma^{-1}} = P_\gamma^{-1}$. □

Exercise 25.31. Compute the holonomy group of Euclidean space.

Hint. Use Cartesian parallel transport. □

Solution. Every loop transports vectors identically, so the holonomy group is trivial. □

Exercise 25.32. State base-point conjugacy of holonomy groups.

Hint. Transport between the base points. □

Solution. If $\sigma : p \rightarrow q$, then $\text{Hol}_q = P_\sigma \text{Hol}_p P_\sigma^{-1}$. □

Exercise 25.33. Define restricted holonomy.

Hint. Use contractible based loops. □

Solution. Hol_p^0 is generated by transport around based loops homotopic to the constant loop. $\square$

Exercise 25.34. Why is restricted holonomy normal in full holonomy?

Hint. Conjugate a contractible loop. $\square$

Solution. Conjugation by any based loop preserves contractibility, so the corresponding subgroup is normal. $\square$

Exercise 25.35. What information can full holonomy contain beyond restricted holonomy?

Hint. Think about noncontractible loops. $\square$

Solution. Full holonomy can contain global monodromy arising from the topology of the manifold. $\square$

Exercise 25.36. Give a sufficient condition for trivial holonomy on an open set.

Hint. Use a global parallel frame. $\square$

Solution. If the open set admits a global frame with $\nabla E_i = 0$, parallel components are constant and all loop transports are identity. $\square$

Exercise 25.37. Why does flatness not by itself force globally trivial holonomy?

Hint. Separate local geometry from topology. $\square$

Solution. A flat connection on a nonsimply connected space may have nontrivial monodromy around noncontractible loops. $\square$

Exercise 25.38. On an oriented surface, in which group does restricted holonomy lie?

Hint. Use oriented orthogonal transformations of a two-plane. $\square$

Solution. It lies in $SO(2)$, so its elements are rotations. $\square$

Exercise 25.39. What is the holonomy angle around a unit-sphere geodesic triangle?

Hint. Use spherical excess. $\square$

Solution. It is $\alpha + \beta + \gamma - \pi$, equal to the triangle's area. $\square$

Exercise 25.40. What does small-loop holonomy measure infinitesimally?

Hint. Use the VII/26 preview. $\square$

Solution. Its leading deviation from the identity is governed by the curvature operator $R(X, Y)$. $\square$

Exercise 25.41. Define holonomy for a general vector bundle connection.

Hint. Transport around loops acts on one fiber. $\square$

Solution. The holonomy group is the subgroup of $GL(E_p)$ generated by loop transport maps $E_p \rightarrow E_p$. $\square$

Exercise 25.42. Where does holonomy lie for a metric-preserving vector bundle connection?

Hint. Use preservation of the fiber inner product. $\square$

Solution. It lies in the orthogonal group of the fiber metric, or in a unitary group in the Hermitian case. $\square$

Exercise 25.43. Define principal-bundle holonomy.

Hint. Use horizontal lifts of loops. $\square$

Solution. A horizontal lift starting at $u \in P_p$ ends at uh_γ ; the resulting $h_\gamma \in G$ form the holonomy group. $\square$

Exercise 25.44. How does principal holonomy change when u is replaced by ug ?

Hint. Use equivariance. $\square$

Solution. The subgroup is conjugated by $g^{-1}(\cdot)g$. $\square$

Exercise 25.45. If T is parallel, what does holonomy do to T_p ?

Hint. Transport T around a loop. $\square$

Solution. It fixes T_p . $\square$

Exercise 25.46. Why can reduced holonomy signal extra geometry?

Hint. Look for preserved tensors. $\square$

Solution. A proper subgroup may preserve tensors or structures not forced by the metric alone, such as a complex structure. $\square$

Exercise 25.47. Which five chapters form the connection-to-holonomy arc?

Hint. Read the construction forward. $\square$

Solution. VII/21 connections, VII/22 Levi–Civita, VII/23 geodesics, VII/24 parallel transport, and VII/25 holonomy. $\square$

Exercise 25.48. State the bridge to VII/26.

Hint. Ask what generates infinitesimal loop transport. $\square$

Solution. VII/26 defines the Riemann curvature tensor, the infinitesimal obstruction to path-independent parallel transport. $\square$

25.15 Solved problem dossiers

Problem 25.49 (Holonomy-group problem). Prove the set of Levi–Civita loop transports at p forms a subgroup of $O(T_pM)$.

Solution. The constant loop gives identity, concatenation gives composition, reversal gives inverses, and metric compatibility makes every transport orthogonal. $\square$

Problem 25.50 (Base-point problem). Derive the conjugacy relation between holonomy groups at two points.

Solution. Join the base points by a path σ , conjugate loops using σ , and apply the parallel-transport concatenation rule. $\square$

Problem 25.51 (Restricted-holonomy problem). Show contractible-loop transports form a normal subgroup after generation.

Solution. Conjugating a contractible loop by any based loop yields another contractible loop, so the generated subgroup is normal. $\square$

Problem 25.52 (Flat-frame problem). Show a global parallel frame implies path-independent transport.

Solution. Every vector has constant coefficients in the parallel frame along a parallel extension; the endpoint depends only on those coefficients, not on the path. $\square$

Problem 25.53 (Sphere-triangle problem). Explain geometrically why transport around a spherical triangle produces a rotation.

Solution. Transport preserves the oriented metric, so the result lies in $SO(2)$. Following the vector through the three geodesic edges yields a net turning equal to the spherical excess. $\square$

Problem 25.54 (Small-loop problem). Why is holonomy around a shrinking loop close to the identity?

Solution. The parallel ODE integrates a smooth connection over a path whose size goes to zero. First-order edge effects cancel around a closed infinitesimal parallelogram, leaving curvature at second order. $\square$

Problem 25.55 (Bundle-holonomy problem). Generalize tangent-bundle holonomy to a connection on $E \rightarrow M$.

Solution. Loop transport is a linear automorphism of E_p ; collect all such automorphisms into a subgroup of $GL(E_p)$. $\square$

Problem 25.56 (Principal-holonomy problem). Relate horizontal lifts in a principal bundle to holonomy elements of the structure group.

Solution. The horizontal lift of a based loop ends in the same fiber as it starts, so the endpoint differs by a unique right action $h_\gamma \in G$. $\square$

Problem 25.57 (Invariant-tensor problem). Prove a parallel tensor at p is fixed by holonomy.

Solution. Parallel transport carries the tensor consistently along every path. A loop returns to the same fiber, so its induced action must leave the initial tensor unchanged. $\square$

Problem 25.58 (Full-versus-restricted problem). Explain conceptually why restricted holonomy is local while full holonomy can be global.

Solution. Contractible loops can be shrunk into local coordinate neighborhoods and probe curvature. Noncontractible loops can also detect topological monodromy. $\square$

Problem 25.59 (Euclidean-versus-sphere problem). Contrast loop transport in $\mathbb{R}^2$ and on S^2 .

Solution. Euclidean transport is path independent and all loop holonomy is trivial. On the sphere, contractible loops can rotate vectors because curvature is nonzero. $\square$

Problem 25.60 (Arc-synthesis problem). Explain the logical chain VII/21 through VII/25 in one paragraph.

Solution. A connection defines covariant differentiation; a Riemannian metric selects the Levi-Civita connection; its autoparallel curves are geodesics; integrating the connection along curves gives parallel transport; closing those curves produces holonomy. $\square$

25.16 Challenge problems

Problem 25.61 (Topology without local curvature). Describe how a flat connection can have trivial restricted holonomy but nontrivial full holonomy.

Solution. Local flatness makes transport around contractible loops trivial, so restricted holonomy vanishes. On a nonsimply connected base, a representation of the fundamental group can still assign nontrivial transport to noncontractible loops, producing global monodromy. $\square$

Problem 25.62 (Holonomy representation preview). When a flat connection has path-independent transport up to homotopy, what algebraic object does loop transport define?

Solution. It defines a group homomorphism from the fundamental group $\pi_1(M, p)$ to $GL(E_p)$ (or the relevant structure group), called the monodromy or holonomy representation. $\square$

Problem 25.63 (Infinitesimal rectangle). Explain why the first nontrivial term in a small rectangular loop is second order in its side lengths.

Solution. Transport along one edge is first order in the connection. Opposite edges cancel those first-order contributions when the loop closes. The surviving commutator/derivative contribution is proportional to enclosed area and is encoded by curvature. $\square$

Problem 25.64 (Holonomy reduction). Suppose a nonzero vector field V is globally parallel on a connected Riemannian manifold. What does this imply about holonomy?

Solution. For every loop based at p , parallel transport must return V_p to itself. Hence Hol_p lies in the stabilizer of V_p , a proper subgroup of $O(T_p M)$ when $V_p \neq 0$. $\square$

Problem 25.65 (From holonomy to curvature). Why is VII/26 the inevitable next chapter after holonomy?

Solution. Holonomy records finite accumulated failure of path-independent transport. Shrinking loops converts that finite discrepancy into an infinitesimal bilinear operator in the spanning directions; that operator is the Riemann curvature tensor. $\square$

Bridge to VII/26

Holonomy detects the accumulated effect of curvature around loops. VII/26 now defines the Riemann curvature tensor directly from covariant derivatives and shows how its algebraic identities control the infinitesimal geometry behind this transport phenomenon.

Chapter 26

The Riemann Curvature Tensor

Holonomy records the finite failure of parallel transport to be path independent. Curvature is the infinitesimal version of that failure. For the Levi–Civita connection we use the convention

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

With this convention the round sphere has positive sectional curvature.

26.1 Definition and tensoriality

Definition 26.1 (Curvature operator). For vector fields X, Y, Z ,

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Theorem 26.2 (Tensoriality). *The map $R(X, Y)Z$ is $C^\infty(M)$ -linear in each of X, Y, Z . Hence*

$$R \in \Gamma(\Lambda^2 T^* M \otimes \text{End}(TM)).$$

Proof. Linearity in X and Y follows after the derivative terms created by multiplication by a function cancel against the Lie-bracket term. Linearity in Z follows because the second-derivative terms in a function cancel between the two commutators. Therefore the value at a point depends only on the tangent vectors at that point. $\square$

Proposition 26.3 (Skew symmetry in the first two entries).

$$R(X, Y) = -R(Y, X).$$

26.2 The $(0, 4)$ curvature tensor

Definition 26.4 (Riemann tensor). Define

$$\text{Rm}(X, Y, Z, W) = g(R(X, Y)Z, W).$$

For the Levi–Civita connection this tensor satisfies strong algebraic symmetries.

Theorem 26.5 (Riemann symmetries). *For all tangent vectors,*

$$\text{Rm}(X, Y, Z, W) = -\text{Rm}(Y, X, Z, W),$$

$$\text{Rm}(X, Y, Z, W) = -\text{Rm}(X, Y, W, Z),$$

and

$$\text{Rm}(X, Y, Z, W) = \text{Rm}(Z, W, X, Y).$$

Theorem 26.6 (First Bianchi identity).

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0.$$

Equivalently,

$$\text{Rm}(X, Y, Z, W) + \text{Rm}(Y, Z, X, W) + \text{Rm}(Z, X, Y, W) = 0.$$

Proof. Expand the definition of R , sum cyclically, and use torsion freeness $\nabla_X Y - \nabla_Y X = [X, Y]$ together with the Jacobi identity for Lie brackets. $\square$

26.3 Coordinate components

Let

$$R(\partial_i, \partial_j)\partial_k = R^\ell_{kij}\partial_\ell.$$

Theorem 26.7 (Coordinate formula). *In a coordinate chart,*

$$R^\ell_{kij} = \partial_i \Gamma^\ell_{jk} - \partial_j \Gamma^\ell_{ik} + \Gamma^m_{jk} \Gamma^\ell_{im} - \Gamma^m_{ik} \Gamma^\ell_{jm}.$$

Lowering the first index gives

$$R_{\ell kij} = g_{\ell m} R^m_{kij}.$$

Warning 26.8 (Index conventions). Different books permute the positions of i, j, k, ℓ and may define R with the opposite overall sign. Always check the operator convention before comparing component formulas or the sign of sectional curvature.

26.4 Normal coordinates

At a point p , choose normal coordinates:

$$g_{ij}(p) = \delta_{ij}, \quad \Gamma^k_{ij}(p) = 0.$$

Proposition 26.9 (Curvature survives normal coordinates). *At the center of normal coordinates,*

$$R^\ell_{kij}(p) = \partial_i \Gamma^\ell_{jk}(p) - \partial_j \Gamma^\ell_{ik}(p).$$

Thus one can kill the connection coefficients at a point, but not generally their first derivatives.

26.5 Sectional curvature

Definition 26.10 (Sectional curvature). Let $\sigma \subset T_p M$ be a two-plane spanned by linearly independent u, v . Its sectional curvature is

$$K(\sigma) = \frac{g(R(u, v)v, u)}{g(u, u)g(v, v) - g(u, v)^2}.$$

For an orthonormal pair u, v ,

$$K(\sigma) = g(R(u, v)v, u).$$

Proposition 26.11 (Well-definedness). *$K(\sigma)$ depends only on the two-plane σ , not on the chosen basis.*

Remark 26.12. In dimension two there is only one tangent two-plane at each point. Its sectional curvature is exactly the Gaussian curvature of the Riemannian surface.

26.6 Constant sectional curvature

Definition 26.13 (Space form curvature). A Riemannian manifold has constant sectional curvature c if every tangent two-plane has sectional curvature c .

Theorem 26.14 (Curvature tensor of a space form). *With the convention of this chapter, constant sectional curvature c is equivalent to*

$$R(X, Y)Z = c(g(Y, Z)X - g(X, Z)Y).$$

Equivalently,

$$\text{Rm}(X, Y, Z, W) = c(g(Y, Z)g(X, W) - g(X, Z)g(Y, W)).$$

Example 26.15 (Euclidean space). For $\mathbb{R}^n$ with its Euclidean metric,

$$R = 0, \quad K = 0.$$

Example 26.16 (Round sphere). The round sphere of radius a has

$$K = \frac{1}{a^2}.$$

Hence

$$R(X, Y)Z = \frac{1}{a^2}(g(Y, Z)X - g(X, Z)Y).$$

Example 26.17 (Hyperbolic model preview). Hyperbolic space has constant negative sectional curvature. Concrete models are developed in VII/31–VII/35.

26.7 Curvature and commutators of covariant derivatives

The defining equation may be read as

$$[\nabla_X, \nabla_Y]Z = \nabla_{[X, Y]}Z + R(X, Y)Z.$$

Thus curvature is exactly the correction term preventing covariant derivatives from commuting as though the connection were globally flat.

26.8 Second Bianchi identity

Theorem 26.18 (Second Bianchi identity). *For the Levi–Civita connection,*

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Remark 26.19. Contracting this differential identity produces the contracted Bianchi identity, which becomes central after Ricci and scalar curvature are introduced in VII/27.

26.9 Curvature and holonomy

For a small loop spanning tangent directions X, Y , the leading departure of parallel transport from the identity is governed by $R(X, Y)$. Thus VII/25 and VII/26 are two scales of the same phenomenon:

$$\text{finite loop transport} \longleftrightarrow \text{infinitesimal curvature}.$$

26.10 Exercises

Exercise 26.20. State the curvature-operator convention used in this volume.

Hint. Write the commutator and bracket correction. □

Solution. $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$. □

Exercise 26.21. Show $R(X, Y) = -R(Y, X)$.

Hint. Interchange X, Y . □

Solution. Every term changes sign, including the Lie bracket because $[Y, X] = -[X, Y]$. □

Exercise 26.22. Why must the Lie-bracket term appear in the definition?

Hint. Covariant derivatives along noncommuting fields have a built-in commutator. □

Solution. Subtracting $\nabla_{[X, Y]}$ makes the expression tensorial in the direction fields. □

Exercise 26.23. What bundle contains R ?

Hint. Use two covariant direction slots and one endomorphism. □

Solution. $R \in \Gamma(\Lambda^2 T^* M \otimes \text{End}(TM))$. □

Exercise 26.24. Define $\text{Rm}(X, Y, Z, W)$.

Hint. Lower the output index with the metric. □

Solution. $\text{Rm}(X, Y, Z, W) = g(R(X, Y)Z, W)$. □

Exercise 26.25. State skew symmetry in the last two entries.

Hint. Use metric compatibility. □

Solution. $\text{Rm}(X, Y, Z, W) = -\text{Rm}(X, Y, W, Z)$. □

Exercise 26.26. State pair symmetry.

Hint. Exchange the ordered pairs. □

Solution. $\text{Rm}(X, Y, Z, W) = \text{Rm}(Z, W, X, Y)$. □

Exercise 26.27. State the first Bianchi identity.

Hint. Take the cyclic sum in X, Y, Z . □

Solution. $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$. □

Exercise 26.28. Write R^ℓ_{kij} in terms of Christoffel symbols.

Hint. Differentiate Γ and add quadratic terms. □

Solution. $R^\ell_{kij} = \partial_i \Gamma^\ell_{jk} - \partial_j \Gamma^\ell_{ik} + \Gamma^m_{jk} \Gamma^\ell_{im} - \Gamma^m_{ik} \Gamma^\ell_{jm}$. □

Exercise 26.29. What simplifies at the center of normal coordinates?

Hint. Set $\Gamma(p) = 0$. □

Solution. The quadratic Christoffel terms vanish, leaving first derivatives of Γ . □

Exercise 26.30. Can curvature be made zero by choosing normal coordinates?

Hint. Only the connection coefficients vanish. □

Solution. No. Curvature is tensorial and survives coordinate changes unless it is geometrically zero. □

Exercise 26.31. Define sectional curvature.

Hint. Normalize $g(R(u, v)v, u)$ by the Gram determinant. □

Solution. $K(\sigma) = g(R(u, v)v, u)/(g(u, u)g(v, v) - g(u, v)^2)$. □

Exercise 26.32. What is $K(\sigma)$ for an orthonormal basis u, v ?

Hint. The denominator is 1. □

Solution. $K(\sigma) = g(R(u, v)v, u)$. □

Exercise 26.33. How many sectional curvatures exist at a point of a surface?

Hint. There is only one tangent two-plane. □

Solution. Exactly one, and it equals the Gaussian curvature. □

Exercise 26.34. Compute R in Euclidean space.

Hint. Use zero Cartesian Christoffel symbols. □

Solution. All curvature components vanish. □

Exercise 26.35. What is the sectional curvature of a round sphere of radius a ?

Hint. Use the space-form formula. □

Solution. It is $1/a^2$. □

Exercise 26.36. State the space-form curvature operator.

Hint. Use constant c . □

Solution. $R(X, Y)Z = c(g(Y, Z)X - g(X, Z)Y)$. □

Exercise 26.37. What sign of sectional curvature does hyperbolic space have?

Hint. Recall the model-space convention. □

Solution. It has constant negative sectional curvature. □

Exercise 26.38. Interpret R as a commutator correction.

Hint. Rearrange the definition. □

Solution. $[\nabla_X, \nabla_Y]Z = \nabla_{[X, Y]}Z + R(X, Y)Z$. □

Exercise 26.39. State the second Bianchi identity.

Hint. Take a cyclic covariant derivative of R . □

Solution. $(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0$. □

Exercise 26.40. Why is curvature related to small-loop holonomy?

Hint. Shrink a transport loop. □

Solution. The first nontrivial area-order deviation of loop transport from identity is generated by $R(X, Y)$. □

Exercise 26.41. What is the geometric meaning of $R = 0$?

Hint. Think infinitesimal path independence. □

Solution. The connection is locally flat: infinitesimal loop transport has no curvature obstruction. □

Exercise 26.42. Why do authors sometimes disagree on signs?

Hint. Check the definition of R . □

Solution. Some define the negative operator or permute component indices, reversing sectional-curvature signs. □

Exercise 26.43. State the bridge to Ricci curvature.

Hint. Contract the Riemann tensor. □

Solution. VII/27 traces the Riemann tensor to obtain Ricci curvature and then traces again to obtain scalar curvature. □

26.11 Solved problem dossiers

Problem 26.44 (Tensoriality problem). Prove $R(X, Y)Z$ is C^∞ -linear in all three vector fields.

Solution. Expand after multiplying each entry by a function. Leibniz terms cancel between the covariant-derivative commutator and the Lie-bracket correction. □

Problem 26.45 (First-Bianchi problem). Derive the first Bianchi identity for a torsion-free connection.

Solution. Expand the cyclic sum and use $\nabla_X Y - \nabla_Y X = [X, Y]$ together with the Jacobi identity. □

Problem 26.46 (Component problem). Derive the coordinate formula for R^ℓ_{kij} .

Solution. Apply the definition to coordinate vector fields, use $[\partial_i, \partial_j] = 0$, and expand two covariant derivatives of ∂_k . □

Problem 26.47 (Normal-coordinate problem). Explain what normal coordinates reveal about curvature.

Solution. At the center $\Gamma = 0$, so curvature is built directly from first derivatives of connection coefficients, showing why curvature is second-order metric information. □

Problem 26.48 (Sectional-curvature problem). Show the sectional-curvature quotient is invariant under changing the basis of the same plane.

Solution. Both numerator and denominator transform by the square of the determinant of the 2×2 basis-change matrix; the quotient is unchanged. □

Problem 26.49 (Surface problem). Show sectional curvature specializes to Gaussian curvature in dimension two.

Solution. The unique tangent two-plane has orthonormal basis e_1, e_2 , and $K = g(R(e_1, e_2)e_2, e_1)$, the intrinsic Gaussian curvature. $\square$

Problem 26.50 (Space-form problem). Verify the constant-curvature tensor has sectional curvature c .

Solution. For orthonormal u, v , the formula gives $R(u, v)v = cu$, so $g(R(u, v)v, u) = c$. $\square$

Problem 26.51 (Sphere problem). Determine the Riemann tensor of a round sphere of radius a .

Solution. The sphere is a space form with $c = 1/a^2$, so substitute this value in the space-form formula. $\square$

Problem 26.52 (Flatness problem). Why does $R = 0$ imply local path independence on a simply connected sufficiently small domain?

Solution. A flat connection satisfies the integrability condition for a local parallel frame. In such a frame the connection form vanishes and transport depends only on endpoints. $\square$

Problem 26.53 (Holonomy problem). Relate the sign convention for small-loop transport to the sign convention for R .

Solution. The area-order term may be written $I \pm \varepsilon^2 R(X, Y)$ depending on loop orientation and curvature convention. Fixing both removes the ambiguity. $\square$

Problem 26.54 (Second-Bianchi problem). Explain why the second Bianchi identity is a differential analogue of Jacobi.

Solution. Curvature is a commutator of covariant derivatives corrected by Lie brackets; cyclically differentiating and applying commutator/Jacobi relations yields the identity. $\square$

Problem 26.55 (Dimension-count problem). How many independent components can an algebraic Riemann tensor have in dimension n ?

Solution. Using the Riemann symmetries and Bianchi identity gives $n^2(n^2 - 1)/12$ independent components. $\square$

26.12 Challenge problems

Problem 26.56 (Independent-component count). Derive $n^2(n^2 - 1)/12$ for the dimension of the space of algebraic curvature tensors.

Solution. View Rm as a symmetric bilinear form on $\Lambda^2 V$, then impose the first Bianchi identity. Starting with $m(m + 1)/2$, $m = n(n - 1)/2$, subtract the $\binom{n}{4}$ Bianchi constraints and simplify. $\square$

Problem 26.57 (Curvature from sectional curvature). Explain why knowledge of all sectional curvatures determines the full Riemann tensor.

Solution. The values $g(R(u, v)v, u)$ determine the curvature quadratic form on decomposable bivectors. Polarization, together with the Riemann symmetries and Bianchi identity, recovers the full algebraic tensor. $\square$

Problem 26.58 (Normal-coordinate expansion preview). What kind of term should appear at second order in the Taylor expansion of g_{ij} in normal coordinates?

Solution. Since $g_{ij}(p) = \delta_{ij}$ and first derivatives vanish, the quadratic term must be a universal contraction of the Riemann tensor with $x^k x^\ell$; its precise coefficient depends on the component convention. $\square$

Problem 26.59 (Jacobi-field preview). Why should $R(J, \dot{\gamma})\dot{\gamma}$ appear in the equation for infinitesimal separation of nearby geodesics?

Solution. Commuting covariant derivatives of a two-parameter family of geodesics produces curvature. The variation field therefore satisfies a second-order equation whose restoring/focusing term is the curvature operator. $\square$

Problem 26.60 (Curvature operator on bivectors). Describe how Rm can be viewed as a self-adjoint operator on $\Lambda^2 T_p M$.

Solution. The pair symmetry makes the bilinear form on bivectors symmetric. Using the metric-induced inner product on $\Lambda^2 T_p M$, Riesz representation yields a self-adjoint curvature operator. $\square$

Bridge to VII/27

VII/27 contracts the Riemann tensor once to obtain Ricci curvature and once more to obtain scalar curvature, revealing directional and total curvature averages.

Chapter 27

Ricci and Scalar Curvature

The Riemann tensor contains all local curvature information, but it has many components. Two canonical contractions summarize important averages of sectional curvature:

$$\boxed{\text{Ric} \quad \text{and} \quad \text{Scal}.}$$

27.1 Ricci curvature

Definition 27.1 (Ricci tensor). For $X, Y \in T_p M$, define

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y).$$

If $e_1, \dots, e_n$ is an orthonormal basis,

$$\text{Ric}(X, Y) = \sum_{i=1}^n g(R(e_i, X)Y, e_i).$$

Proposition 27.2 (Symmetry). *For the Levi-Civita connection,*

$$\text{Ric}(X, Y) = \text{Ric}(Y, X).$$

Proof. Use the pair symmetry and first Bianchi identity of the Riemann tensor, then sum over an orthonormal basis. $\square$

27.2 Ricci curvature as an average

Let v be a unit tangent vector and complete it to an orthonormal basis

$$v, e_2, \dots, e_n.$$

Proposition 27.3 (Sum of sectional curvatures).

$$\text{Ric}(v, v) = \sum_{i=2}^n K(\text{span}\{v, e_i\}).$$

Thus Ricci curvature in direction v is the sum of sectional curvatures of all orthogonal two-planes containing v .

27.3 Scalar curvature

Definition 27.4 (Scalar curvature). The scalar curvature is the metric trace of the Ricci tensor:

$$\text{Scal} = \text{tr}_g \text{Ric}.$$

For an orthonormal basis,

$$\text{Scal} = \sum_{i=1}^n \text{Ric}(e_i, e_i).$$

Proposition 27.5 (Sum over sectional curvatures). *For an orthonormal basis $e_1, \dots, e_n$,*

$$\text{Scal} = 2 \sum_{1 \leq i < j \leq n} K(e_i, e_j).$$

27.4 Coordinate contractions

With curvature components consistent with VII/26,

$$R^\ell_{kij},$$

the Ricci components are obtained by contraction:

$$\text{Ric}_{kj} = R^i_{kij}.$$

The scalar curvature is

$$\text{Scal} = g^{kj} \text{Ric}_{kj}.$$

Warning 27.6 (Component conventions). If a source uses the opposite sign for the Riemann tensor, both Ricci and scalar curvature change sign. Geometric comparisons must therefore begin with the underlying Riemann convention.

27.5 Constant sectional curvature

Theorem 27.7 (Space-form contractions). *If an n -dimensional manifold has constant sectional curvature c , then*

$$\text{Ric} = (n-1)c g,$$

and

$$\text{Scal} = n(n-1)c.$$

Proof. For a unit vector v , there are $n-1$ orthogonal two-planes $\text{span}\{v, e_i\}$, each of curvature c . Hence $\text{Ric}(v, v) = (n-1)c$. Polarization gives the Ricci formula, and one more trace gives the scalar formula. $\square$

Example 27.8 (Round sphere). For the round sphere $S^n(a)$,

$$\text{Ric} = \frac{n-1}{a^2} g, \quad \text{Scal} = \frac{n(n-1)}{a^2}.$$

Example 27.9 (Euclidean space). For Euclidean space,

$$\text{Ric} = 0, \quad \text{Scal} = 0.$$

27.6 Einstein metrics

Definition 27.10 (Einstein metric). A Riemannian metric is Einstein if

$$\text{Ric} = \lambda g$$

for some function λ .

Proposition 27.11 (Constancy in dimension at least three). *On a connected Einstein manifold of dimension $n \geq 3$, the factor λ is constant.*

Proof. The contracted Bianchi identity gives

$$\text{div Ric} = \frac{1}{2} d \text{Scal}.$$

If $\text{Ric} = \lambda g$, then

$$d\lambda = \frac{n}{2} d\lambda.$$

For $n \neq 2$, this forces $d\lambda = 0$. □

27.7 The Einstein tensor

Definition 27.12 (Einstein tensor). Define

$$G = \text{Ric} - \frac{1}{2} \text{Scal } g.$$

Theorem 27.13 (Divergence-free identity). *The contracted second Bianchi identity implies*

$$\text{div } G = 0.$$

Remark 27.14. This identity is purely Riemannian and differential-geometric. Its Lorentzian role in gravitation belongs later, after indefinite metrics are introduced in VII/29–VII/30.

27.8 Low dimensions

Proposition 27.15 (Dimension two). *On a Riemannian surface,*

$$\text{Ric} = K g, \quad \text{Scal} = 2K.$$

Proposition 27.16 (Dimension three preview). *In dimension three the Riemann tensor is completely determined by the Ricci tensor and scalar curvature. Equivalently, the Weyl tensor introduced in VII/28 vanishes identically.*

27.9 Geometric meaning

Sectional curvature resolves curvature plane by plane. Ricci curvature averages those planes through one direction. Scalar curvature averages over all orthogonal two-planes at a point. These successive contractions trade directional detail for compact geometric information.

27.10 Exercises

Exercise 27.17. Define the Ricci tensor by tracing the Riemann operator.

Hint. Trace $Z \mapsto R(Z, X)Y$. □

Solution. $\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y)$. □

Exercise 27.18. Write Ricci curvature in an orthonormal basis.

Hint. Expand the trace. □

Solution. $\text{Ric}(X, Y) = \sum_i g(R(e_i, X)Y, e_i)$. □

Exercise 27.19. Is the Levi-Civita Ricci tensor symmetric?

Hint. Use Riemann symmetries. □

Solution. Yes, $\text{Ric}(X, Y) = \text{Ric}(Y, X)$. □

Exercise 27.20. For unit v , interpret $\text{Ric}(v, v)$.

Hint. Complete v to an orthonormal basis. □

Solution. It is the sum of sectional curvatures of the $n - 1$ orthogonal two-planes containing v . □

Exercise 27.21. Define scalar curvature.

Hint. Trace Ricci with the metric. □

Solution. $\text{Scal} = \text{tr}_g \text{Ric}$. □

Exercise 27.22. Write scalar curvature in an orthonormal basis.

Hint. Take the Ricci trace. □

Solution. $\text{Scal} = \sum_i \text{Ric}(e_i, e_i)$. □

Exercise 27.23. Express scalar curvature as a sum of sectional curvatures.

Hint. Each plane appears twice in the double trace. □

Solution. $\text{Scal} = 2 \sum_{i < j} K(e_i, e_j)$. □

Exercise 27.24. State the coordinate contraction for Ricci.

Hint. Contract one upper and one lower Riemann index. □

Solution. With the convention of VII/26, $\text{Ric}_{kj} = R^i_{kij}$. □

Exercise 27.25. State the coordinate contraction for scalar curvature.

Hint. Trace Ricci with g^{-1} . □

Solution. $\text{Scal} = g^{ij} \text{Ric}_{ij}$. □

Exercise 27.26. What happens to Ricci under an overall reversal of the Riemann sign convention?

Hint. Ricci is a linear contraction. □

Solution. It also reverses sign. □

Exercise 27.27. Compute Ricci for constant sectional curvature c .

Hint. Sum $n - 1$ equal sectional curvatures. □

Solution. $\text{Ric} = (n - 1)cg$. □

Exercise 27.28. Compute scalar curvature for constant sectional curvature c .

Hint. Trace the Ricci formula. □

Solution. $\text{Scal} = n(n - 1)c$. □

Exercise 27.29. Compute Ricci on a round sphere of radius a .

Hint. Use $c = 1/a^2$. □

Solution. $\text{Ric} = ((n - 1)/a^2)g$. □

Exercise 27.30. Compute scalar curvature on a round sphere of radius a .

Hint. Trace the previous result. □

Solution. $\text{Scal} = n(n - 1)/a^2$. □

Exercise 27.31. Compute Ricci and scalar curvature in Euclidean space.

Hint. Riemann curvature vanishes. □

Solution. Both contractions vanish. □

Exercise 27.32. Define an Einstein metric.

Hint. Require Ricci proportional to the metric. □

Solution. $\text{Ric} = \lambda g$. □

Exercise 27.33. For an Einstein metric, what is scalar curvature?

Hint. Take the trace. □

Solution. $\text{Scal} = n\lambda$. □

Exercise 27.34. Why is λ constant for connected Einstein manifolds with $n \geq 3$?

Hint. Use contracted Bianchi. □

Solution. The identity gives $(1 - n/2)d\lambda = 0$, so $d\lambda = 0$. □

Exercise 27.35. Define the Einstein tensor.

Hint. Combine Ricci and scalar curvature. □

Solution. $G = \text{Ric} - \frac{1}{2}\text{Scal } g$. □

Exercise 27.36. What differential identity does G satisfy?

Hint. Use contracted Bianchi. □

Solution. $\text{div } G = 0$. □

Exercise 27.37. On a surface, relate Ricci and Gaussian curvature.

Hint. There is one sectional curvature. □

Solution. $\text{Ric} = Kg$. □

Exercise 27.38. On a surface, relate scalar and Gaussian curvature.

Hint. $\text{Trace Ric} = Kg$. □

Solution. $\text{Scal} = 2K$. □

Exercise 27.39. What special fact holds in dimension three?

Hint. Preview Weyl curvature. □

Solution. The Weyl tensor vanishes identically, so Riemann curvature is determined by Ricci and scalar curvature. □

Exercise 27.40. State the bridge to Weyl curvature.

Hint. Remove all traces from Riemann. □

Solution. VII/28 isolates the trace-free part of the Riemann tensor, called the Weyl tensor. □

27.11 Solved problem dossiers

Problem 27.41 (Ricci-contraction problem). Derive the orthonormal-basis expression for the Ricci tensor from the trace definition.

Solution. The trace of $Z \mapsto R(Z, X)Y$ is the sum of its diagonal matrix entries $g(R(e_i, X)Y, e_i)$. □

Problem 27.42 (Symmetry problem). Prove the Ricci tensor is symmetric.

Solution. Use the Riemann pair symmetry and first Bianchi identity to rewrite the two traces and show their difference vanishes. □

Problem 27.43 (Sectional-average problem). Prove $\text{Ric}(v, v) = \sum_{i=2}^n K(v, e_i)$ for unit v .

Solution. Choose an orthonormal basis beginning with v . The $i = 1$ term vanishes by skew symmetry; the remaining terms are exactly sectional curvatures. □

Problem 27.44 (Scalar-average problem). Derive $\text{Scal} = 2 \sum_{i < j} K(e_i, e_j)$.

Solution. Expand the Ricci trace as a double sum. Terms with $i = j$ vanish, and every unordered pair occurs twice. □

Problem 27.45 (Space-form problem). Derive Ricci and scalar curvature for constant sectional curvature.

Solution. Apply the sectional-average formula and then take the metric trace. □

Problem 27.46 (Sphere problem). Compute the contractions for $S^n(a)$.

Solution. Insert $c = 1/a^2$ into the space-form formulas. □

Problem 27.47 (Surface problem). Show $\text{Ric} = Kg$ and $\text{Scal} = 2K$ in dimension two.

Solution. For any unit v , the single orthogonal plane has curvature K , hence $\text{Ric}(v, v) = K$; polarization gives the tensor identity. $\square$

Problem 27.48 (Einstein problem). Show a constant-sectional-curvature metric is Einstein.

Solution. Space forms satisfy $\text{Ric} = (n - 1)c g$, which is exactly the Einstein condition. $\square$

Problem 27.49 (Contracted-Bianchi problem). State the contracted Bianchi identity and derive divergence freeness of the Einstein tensor.

Solution. The identity is $\text{div Ric} = \frac{1}{2}d \text{Scal}$. Since $\nabla g = 0$, subtracting $\frac{1}{2} \text{Scal} g$ yields zero divergence. $\square$

Problem 27.50 (Einstein-factor problem). Prove the proportionality factor of an Einstein metric is constant when $n \geq 3$.

Solution. Substitute $\text{Ric} = \lambda g$ and $\text{Scal} = n\lambda$ into contracted Bianchi to obtain $(1 - n/2)d\lambda = 0$. $\square$

Problem 27.51 (Dimension-three problem). Explain why Ricci contains all Riemann curvature information in dimension three.

Solution. The trace-free Weyl component is identically zero in dimension three, leaving only the algebraic part constructed from Ricci, scalar curvature, and the metric. $\square$

Problem 27.52 (Hierarchy problem). Compare sectional, Ricci, and scalar curvature.

Solution. Sectional curvature is plane-resolved; Ricci sums planes containing a chosen direction; scalar curvature traces once more and averages all orthogonal two-planes. $\square$

27.12 Challenge problems

Problem 27.53 (Scaling law). If $\tilde{g} = c^2 g$ for constant $c > 0$, determine the scaling of sectional, Ricci-as-a-(0, 2)-tensor, and scalar curvature.

Solution. The Levi-Civita connection is unchanged. Sectional curvature scales as c^{-2} ; the (0, 2) Ricci tensor is unchanged under constant scaling; scalar curvature scales as c^{-2} because the inverse metric contributes c^{-2} . $\square$

Problem 27.54 (Product manifolds). Describe Ricci curvature of a Riemannian product $M \times N$.

Solution. The Levi-Civita connection splits and mixed sectional curvatures vanish. Ricci is the direct sum $\text{Ric}_{M \times N} = \text{Ric}_M \oplus \text{Ric}_N$, and scalar curvatures add. $\square$

Problem 27.55 (Einstein product criterion). When is a product of two Einstein manifolds Einstein under the unscaled product metric?

Solution. If $\text{Ric}_M = \lambda_M g_M$ and $\text{Ric}_N = \lambda_N g_N$, the product is Einstein exactly when $\lambda_M = \lambda_N$. $\square$

Problem 27.56 (Ricci versus sectional sign). Does positive Ricci curvature force every sectional curvature to be positive?

Solution. No in dimensions at least three. Ricci is only a sum of sectional curvatures through a direction, so some sectional curvatures can be negative while the total sum remains positive. $\square$

Problem 27.57 (Scalar curvature locality). Explain why scalar curvature is a much coarser invariant than the full Riemann tensor.

Solution. It is only a double contraction. Many distinct algebraic curvature tensors have the same scalar trace; Weyl curvature in particular is completely invisible to scalar curvature. $\square$

Bridge to VII/28

VII/28 subtracts precisely these Ricci and scalar traces from the Riemann tensor, leaving the trace-free Weyl curvature.

Chapter 28

Weyl Curvature

Ricci and scalar curvature are traces of the Riemann tensor. What remains after all trace information is removed is the Weyl tensor. In dimensions $n \geq 4$ it captures the conformally invariant part of Riemannian curvature.

28.1 Kulkarni–Nomizu product

Definition 28.1 (Kulkarni–Nomizu product). For symmetric $(0, 2)$ -tensors h, k , define

$$(h \mathring{\wedge} k)(X, Y, Z, W) = h(X, Z)k(Y, W) + h(Y, W)k(X, Z) \\ - h(X, W)k(Y, Z) - h(Y, Z)k(X, W).$$

This product produces a $(0, 4)$ -tensor with the algebraic symmetries of a curvature tensor.

28.2 Schouten tensor

Definition 28.2 (Schouten tensor). For dimension $n \geq 3$, define

$$P = \frac{1}{n-2} \left(\text{Ric} - \frac{\text{Scal}}{2(n-1)} g \right).$$

28.3 Weyl decomposition

Definition 28.3 (Weyl tensor). For $n \geq 3$, the Weyl tensor is

$$W = \text{Rm} - g \mathring{\wedge} P.$$

Equivalently,

$$W = \text{Rm} - \frac{1}{n-2} g \mathring{\wedge} \text{Ric} + \frac{\text{Scal}}{2(n-1)(n-2)} g \mathring{\wedge} g.$$

Theorem 28.4 (Trace-free property). *Every metric contraction of the Weyl tensor vanishes.*

Remark 28.5. Thus the Riemann tensor splits into a trace-free part W and a part determined entirely by Ric , Scal , and g .

28.4 Low dimensions

Theorem 28.6 (Dimensions two and three). *In dimension two the Riemann tensor is determined by Gaussian curvature. In dimension three,*

$$W \equiv 0,$$

so the Riemann tensor is completely determined by Ricci curvature.

Warning 28.7 (Weyl begins in dimension four). The Weyl tensor carries independent local curvature information only in dimensions $n \geq 4$.

28.5 Constant curvature and Einstein metrics

Proposition 28.8 (Space forms are conformally flat). *Every constant-sectional-curvature metric has*

$$W = 0.$$

Proof. For constant sectional curvature c ,

$$\text{Rm} = \frac{c}{2} g \mathring{\wedge} g$$

with the Kulkarni–Nomizu convention above. Substituting $\text{Ric} = (n-1)cg$ and $\text{Scal} = n(n-1)c$ into the Weyl decomposition cancels all terms. $\square$

Proposition 28.9 (Einstein simplification). *If*

$$\text{Ric} = \lambda g,$$

then

$$\text{Rm} = W + \frac{\lambda}{2(n-1)} g \mathring{\wedge} g.$$

Thus on an Einstein manifold, Weyl curvature measures exactly the difference from constant sectional curvature at the algebraic level.

28.6 Conformal change

Let

$$\tilde{g} = e^{2\varphi} g.$$

Theorem 28.10 (Conformal covariance). *For $n \geq 4$, the Weyl tensor is conformally invariant as a $(1,3)$ -tensor. As a fully covariant $(0,4)$ -tensor,*

$$\widetilde{W} = e^{2\varphi} W.$$

Consequently,

$$W = 0$$

is a conformally invariant condition.

Remark 28.11. Riemann, Ricci, and scalar curvature all acquire derivative terms under a general conformal change. The specific trace subtraction in W cancels those non-invariant pieces.

28.7 Local conformal flatness

Definition 28.12 (Locally conformally flat). A Riemannian metric is locally conformally flat if every point has a neighborhood on which

$$g = e^{2\varphi} g_{\text{flat}}$$

in suitable local coordinates.

Theorem 28.13 (Weyl criterion in dimension at least four). *For $n \geq 4$,*

$$W = 0$$

if and only if the metric is locally conformally flat.

Warning 28.14 (Dimension three). In dimension three W vanishes for every metric, so it cannot detect local conformal flatness. The relevant obstruction is the Cotton tensor.

28.8 Examples

Example 28.15 (Euclidean space). Euclidean space has $\text{Rm} = 0$, hence $W = 0$.

Example 28.16 (Round sphere). The round sphere has constant sectional curvature, so $W = 0$. Stereographic projection also exhibits its metric as conformal to the Euclidean metric.

Example 28.17 (Dimension three). Every three-dimensional Riemannian metric has $W = 0$, even though it may have nonzero Riemann, Ricci, and scalar curvature.

28.9 Weyl curvature as the trace-free remainder

At a point in dimension $n \geq 4$, the decomposition

$$\text{Rm} = W + g \mathring{\wedge} P$$

separates curvature into:

$$\text{trace-free conformal curvature} \quad + \quad \text{Ricci-controlled curvature}.$$

This decomposition is the natural endpoint of the Riemannian curvature contraction arc begun in VII/26.

28.10 Transition beyond positive-definite metrics

Everything so far has used positive-definite Riemannian metrics. The algebra of connections and curvature extends much further. VII/29 replaces positive definiteness by a nondegenerate bilinear form of arbitrary signature and develops indefinite metrics.

28.11 Exercises

Exercise 28.18. Define the Kulkarni–Nomizu product of symmetric tensors h, k .

Hint. Use four terms with alternating signs. □

Solution. $(h \overset{\circ}{\wedge} k)(X, Y, Z, W) = h(X, Z)k(Y, W) + h(Y, W)k(X, Z) - h(X, W)k(Y, Z) - h(Y, Z)k(X, W)$. $\square$

Exercise 28.19. What algebraic symmetries does $h \overset{\circ}{\wedge} k$ have?

Hint. Compare with Riemann curvature symmetries. $\square$

Solution. It is skew in each pair, symmetric under pair exchange, and satisfies the first Bianchi identity. $\square$

Exercise 28.20. Define the Schouten tensor.

Hint. Normalize Ricci and scalar traces. $\square$

Solution. $P = \frac{1}{n-2}(\text{Ric} - \frac{\text{Scal}}{2(n-1)}g)$. $\square$

Exercise 28.21. Define the Weyl tensor using P .

Hint. Subtract the trace-generated part. $\square$

Solution. $W = \text{Rm} - g \overset{\circ}{\wedge} P$. $\square$

Exercise 28.22. Write the Weyl decomposition directly using Ricci and scalar curvature.

Hint. Substitute the Schouten tensor. $\square$

Solution. $W = \text{Rm} - \frac{1}{n-2}g \overset{\circ}{\wedge} \text{Ric} + \frac{\text{Scal}}{2(n-1)(n-2)}g \overset{\circ}{\wedge} g$. $\square$

Exercise 28.23. What is the defining trace property of W ?

Hint. Contract any metric pair. $\square$

Solution. Every metric contraction of W vanishes. $\square$

Exercise 28.24. In which dimensions can Weyl carry independent information?

Hint. Check the low-dimensional theorem. $\square$

Solution. Only in dimensions $n \geq 4$. $\square$

Exercise 28.25. What is W in dimension three?

Hint. Use the algebraic decomposition. $\square$

Solution. It vanishes identically for every Riemannian metric. $\square$

Exercise 28.26. Does $W = 0$ imply flatness in dimension three?

Hint. Distinguish Weyl from full Riemann curvature. $\square$

Solution. No. Weyl always vanishes in dimension three while Ricci curvature may be nonzero. $\square$

Exercise 28.27. What is W for constant sectional curvature?

Hint. Substitute the space-form contractions. $\square$

Solution. It vanishes. $\square$

Exercise 28.28. What is W for Euclidean space?

Hint. Riemann curvature is zero. ☐

Solution. It vanishes. ☐

Exercise 28.29. What is W for a round sphere?

Hint. Round spheres are space forms. ☐

Solution. It vanishes. ☐

Exercise 28.30. State the Einstein simplification of Riemann curvature.

Hint. Set Ricci proportional to the metric. ☐

Solution. $\text{Rm} = W + \frac{\lambda}{2(n-1)} g \mathring{\wedge} g$. ☐

Exercise 28.31. For an Einstein metric, what does $W = 0$ imply algebraically?

Hint. Only the metric-wedge-metric part remains. ☐

Solution. The curvature tensor is that of constant sectional curvature $c = \lambda/(n-1)$. ☐

Exercise 28.32. How does the $(0, 4)$ Weyl tensor transform under $\tilde{g} = e^{2\varphi}g$?

Hint. Use conformal covariance. ☐

Solution. $\tilde{W} = e^{2\varphi}W$. ☐

Exercise 28.33. How does Weyl transform as a $(1, 3)$ -tensor?

Hint. Raise one index with the transformed metric. ☐

Solution. It is unchanged under conformal rescaling. ☐

Exercise 28.34. Is the condition $W = 0$ conformally invariant?

Hint. Scaling by a nowhere-zero factor preserves zero. ☐

Solution. Yes. ☐

Exercise 28.35. Define local conformal flatness.

Hint. Compare locally to a scaled Euclidean metric. ☐

Solution. Locally $g = e^{2\varphi}g_{\text{flat}}$ in suitable coordinates. ☐

Exercise 28.36. For $n \geq 4$, what characterizes local conformal flatness?

Hint. Use the Weyl criterion. ☐

Solution. It is equivalent to $W = 0$. ☐

Exercise 28.37. Why does that criterion fail in dimension three?

Hint. Weyl vanishes identically. ☐

Solution. Because W cannot distinguish conformally flat from non-conformally-flat three-metrics; the Cotton tensor is needed. ☐

Exercise 28.38. What does the decomposition $\text{Rm} = W + g \mathring{\wedge} P$ separate?

Hint. Think traces versus trace-free data. □

Solution. It separates conformal trace-free curvature from the curvature determined by Ricci and scalar traces. □

Exercise 28.39. Can two metrics have the same Ricci tensor but different Weyl tensors in dimension four or higher?

Hint. Ricci does not determine all of Riemann. □

Solution. Yes. Weyl contains independent trace-free curvature information. □

Exercise 28.40. What is special about dimension two in this hierarchy?

Hint. Riemann is determined by one scalar. □

Solution. The Gaussian curvature determines the full Riemann tensor; there is no independent Weyl component. □

Exercise 28.41. State the bridge to VII/29.

Hint. Change the metric signature. □

Solution. VII/29 extends metric geometry from positive-definite Riemannian metrics to nondegenerate indefinite metrics. □

28.12 Solved problem dossiers

Problem 28.42 (Decomposition problem). Derive the Weyl decomposition from the requirement that W have the Riemann symmetries and zero traces.

Solution. Make the ansatz $W = \text{Rm} + a g \overset{\circ}{\wedge} \text{Ric} + b \text{Scal } g \overset{\circ}{\wedge} g$, contract, and solve for $a = -1/(n-2)$ and $b = 1/[2(n-1)(n-2)]$. □

Problem 28.43 (Trace-free problem). Verify one Ricci-type contraction of the Weyl tensor vanishes.

Solution. Contract the decomposition with g^{ik} , use the contractions of $g \overset{\circ}{\wedge} \text{Ric}$ and $g \overset{\circ}{\wedge} g$, and observe exact cancellation. □

Problem 28.44 (Dimension-three problem). Explain algebraically why $W \equiv 0$ in dimension three.

Solution. The dimension of the space of algebraic Riemann tensors equals the dimension carried by the Ricci tensor in dimension three; the trace-free remainder has zero dimension. □

Problem 28.45 (Space-form problem). Show constant-sectional-curvature tensors have zero Weyl part.

Solution. Insert $\text{Ric} = (n-1)cg$, $\text{Scal} = n(n-1)c$, and $\text{Rm} = (c/2)g \overset{\circ}{\wedge} g$ into the decomposition. □

Problem 28.46 (Einstein problem). Derive $\text{Rm} = W + \frac{\lambda}{2(n-1)}g \overset{\circ}{\wedge} g$ for an Einstein metric.

Solution. Substitute $\text{Ric} = \lambda g$ and $\text{Scal} = n\lambda$ into the general decomposition and simplify. □

Problem 28.47 (Conformal problem). Explain why Weyl is the natural conformal curvature tensor.

Solution. Under $g \mapsto e^{2\varphi}g$, derivative terms entering Riemann curvature are all contained in the Schouten part $g \mathring{\wedge} P$; the trace-free remainder is conformally invariant as a $(1, 3)$ -tensor. $\square$

Problem 28.48 (Sphere problem). Give two reasons the round sphere has vanishing Weyl tensor.

Solution. It is a constant-curvature space form, and stereographic coordinates show it is locally conformal to Euclidean space; either property forces $W = 0$. $\square$

Problem 28.49 (Local-flatness problem). Why does $W = 0$ mean local conformal flatness only for $n \geq 4$?

Solution. In dimension at least four Weyl is the complete local conformal obstruction. In dimension three it vanishes identically and the Cotton tensor supplies the missing differential obstruction. $\square$

Problem 28.50 (Ricci-information problem). Explain what information Ricci loses that Weyl retains.

Solution. Ricci is a contraction and records averages of sectional curvature; Weyl captures trace-free anisotropic curvature patterns invisible to every Ricci contraction. $\square$

Problem 28.51 (Scaling problem). Under a constant rescaling $\tilde{g} = c^2g$, how does the fully covariant Weyl tensor scale?

Solution. Constant rescaling is a conformal change with $e^{2\varphi} = c^2$, so $\tilde{W} = c^2W$ as a $(0, 4)$ -tensor. $\square$

Problem 28.52 (Low-dimensional hierarchy problem). Summarize curvature data in dimensions two, three, and at least four.

Solution. In dimension two one scalar K determines Riemann; in dimension three Ricci determines Riemann; from dimension four onward the independent Weyl tensor appears. $\square$

Problem 28.53 (Arc-synthesis problem). Explain the curvature arc VII/26–VII/28.

Solution. VII/26 defines full Riemann curvature, VII/27 extracts Ricci and scalar traces, and VII/28 isolates the remaining trace-free conformal curvature. $\square$

28.13 Challenge problems

Problem 28.54 (Dimension count for Weyl). Derive the number of independent Weyl components in dimension $n \geq 4$.

Solution. The algebraic Riemann tensor has $n^2(n^2 - 1)/12$ components. The symmetric Ricci tensor has $n(n + 1)/2$ independent components and the decomposition is direct, so Weyl has $(n + 2)(n + 1)n(n - 3)/12$ components. $\square$

Problem 28.55 (Einstein plus Weyl zero). Show that an Einstein metric with $W = 0$ has constant sectional curvature when $n \geq 3$.

Solution. The Einstein decomposition becomes $\text{Rm} = \frac{\lambda}{2(n-1)}g \mathring{\wedge} g$. Since λ is constant for $n \geq 3$, this is the space-form tensor with $c = \lambda/(n - 1)$. $\square$

Problem 28.56 (Conformal-flat sphere). Use stereographic projection to explain why the round sphere is conformally flat.

Solution. In stereographic coordinates the round metric has the form $g = \Omega(x)^2 g_{\text{Eucl}}$ for an explicit positive factor Ω . Hence the sphere is locally conformal to Euclidean space and must have $W = 0$. $\square$

Problem 28.57 (Weyl versus scalar curvature). Can $W \neq 0$ while scalar curvature is zero?

Solution. Yes. Scalar curvature is only a double trace, while Weyl is trace free. A metric can have vanishing scalar curvature yet retain nonzero trace-free Riemann curvature. $\square$

Problem 28.58 (Curvature decomposition as representation theory preview). Why is the decomposition into Weyl, trace-free Ricci, and scalar parts natural under orthogonal coordinate changes?

Solution. Each piece is invariant under the orthogonal group action and corresponds to a distinct algebraic trace type. The decomposition separates irreducible-style components of the curvature tensor under the metric symmetry group. $\square$

Bridge to VII/29

VII/29 keeps the connection-and-curvature machinery but relaxes positive definiteness, introducing nondegenerate metrics of arbitrary signature.

Part V

Lorentzian Geometry

Chapter 29

Indefinite Metrics

Riemannian geometry assumes positive definiteness. Many geometric constructions require only nondegeneracy. Replacing an inner product by a nondegenerate symmetric bilinear form of arbitrary signature leads to pseudo-Riemannian geometry.

29.1 Pseudo-Riemannian metrics

Definition 29.1 (Pseudo-Riemannian metric). A pseudo-Riemannian metric on a smooth manifold M is a smooth symmetric $(0, 2)$ -tensor g such that each bilinear form

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

is nondegenerate.

Nondegeneracy means

$$g_p(v, w) = 0 \quad \text{for all } w \quad \implies \quad v = 0.$$

Warning 29.2 (Nonzero vectors may have zero norm-square). Unlike the Riemannian case, nondegeneracy does not imply $g(v, v) > 0$ for $v \neq 0$. A nonzero vector may satisfy

$$g(v, v) = 0.$$

Such vectors are called null.

29.2 Signature and Sylvester's law

Theorem 29.3 (Sylvester's law of inertia). *For every nondegenerate symmetric bilinear form on a finite-dimensional real vector space, there exists a basis in which the matrix is diagonal with entries*

$$\underbrace{+1, \dots, +1}_p, \underbrace{-1, \dots, -1}_q.$$

The pair (p, q) is independent of the diagonalizing basis.

Definition 29.4 (Signature). The ordered pair (p, q) is the signature, with $p + q = n$. Some authors instead call $p - q$ the signature; throughout this volume the pair (p, q) is used.

Proposition 29.5 (Signature is locally constant). *For a smooth nondegenerate metric, the inertia pair (p, q) is locally constant. Hence on each connected component it is constant.*

29.3 Lorentzian metrics

Definition 29.6 (Lorentzian metric). A Lorentzian metric is a pseudo-Riemannian metric of index one: in dimension n its signature is either

$$(n-1, 1)$$

or, under the opposite sign convention, $(1, n-1)$. We use the mostly-plus convention $(n-1, 1)$.

Example 29.7 (Minkowski metric). On $\mathbb{R}^n$ with coordinates

$$(t, x^1, \dots, x^{n-1}),$$

the flat Lorentzian metric is

$$\eta = -dt^2 + \sum_{i=1}^{n-1} (dx^i)^2.$$

29.4 Causal types

Let $v \neq 0$ in a Lorentzian tangent space with mostly-plus convention.

Definition 29.8 (Timelike, null, spacelike). The vector v is

$$\begin{array}{ll} \text{timelike} & \text{if } g(v, v) < 0, \\ \text{null} & \text{if } g(v, v) = 0, \\ \text{spacelike} & \text{if } g(v, v) > 0. \end{array}$$

Remark 29.9. These inequalities reverse if one reverses the overall sign convention for the metric. The causal trichotomy itself is invariant.

Proposition 29.10 (Orthogonal complement of a timelike vector). *If u is timelike in a Lorentzian vector space, then the restriction of g to*

$$u^\perp$$

is positive definite.

29.5 The null cone

Definition 29.11 (Null cone). At p , the null cone is

$$\mathcal{N}_p = \{v \in T_p M \setminus \{0\} : g(v, v) = 0\}.$$

In Minkowski space,

$$-(v^0)^2 + (v^1)^2 + \dots + (v^{n-1})^2 = 0.$$

The null cone separates timelike from spacelike directions.

29.6 Musical maps still exist

Proposition 29.12 (Flat and sharp). *Nondegeneracy of g still gives an isomorphism*

$$\flat : T_p M \rightarrow T_p^* M, \quad v^\flat = g(v, \cdot),$$

with inverse $\sharp$.

Remark 29.13. Positive definiteness is unnecessary for raising and lowering indices. What is essential is nondegeneracy.

29.7 Volume density

In coordinates,

$$(g_{ij})$$

is invertible but may have negative determinant.

Definition 29.14 (Pseudo-Riemannian density). The canonical metric density is

$$dV_g = \sqrt{|\det(g_{ij})|} |dx^1 \cdots dx^n|.$$

On an oriented pseudo-Riemannian manifold this determines a signed volume form after an orientation is chosen.

29.8 Levi–Civita connection survives unchanged

Theorem 29.15 (Pseudo-Riemannian fundamental theorem). *Every pseudo-Riemannian metric has a unique torsion-free metric-compatible connection.*

Proof. The Koszul formula from VII/22 requires only nondegeneracy of g , not positive definiteness. It therefore determines a unique vector $\nabla_X Y$ from its pairings with all Z , and the standard verification remains valid. $\square$

Consequently the Christoffel formula remains

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

29.9 Geodesics and curvature

The definitions

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0, \quad R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

carry over verbatim. Ricci, scalar, and Weyl curvature also remain meaningful.

Warning 29.16 (What does not carry over verbatim). The positive-definite length and distance theory changes radically. Null vectors have zero norm-square, timelike vectors have negative norm-square, and the Riemannian distance construction is no longer the correct basic global invariant.

29.10 Orthogonal groups of signature

For the standard form

$$I_{p,q} = \text{diag}(1, \dots, 1, -1, \dots, -1),$$

define

$$O(p, q) = \{A \in GL(n, \mathbb{R}) : A^T I_{p,q} A = I_{p,q}\}.$$

Proposition 29.17 (Pseudo-orthogonal symmetry). *Changes of pseudo-orthonormal frame take values in $O(p, q)$. In the Lorentzian case the relevant group is $O(n-1, 1)$.*

29.11 A two-dimensional model

Example 29.18 (Lorentzian plane). On $\mathbb{R}^2$,

$$g = -dt^2 + dx^2.$$

The null lines are

$$x = \pm t.$$

They divide the tangent plane into timelike and spacelike sectors.

29.12 Exercises

Exercise 29.19. Define a pseudo-Riemannian metric.

Hint. Replace positive definiteness by nondegeneracy. □

Solution. It is a smooth symmetric nondegenerate $(0, 2)$ -tensor. □

Exercise 29.20. State nondegeneracy of a bilinear form.

Hint. Use the radical. □

Solution. If $g(v, w) = 0$ for every w , then $v = 0$. □

Exercise 29.21. Can a nonzero vector satisfy $g(v, v) = 0$ in an indefinite metric?

Hint. Recall null vectors. □

Solution. Yes; such a vector is null. □

Exercise 29.22. State Sylvester's law of inertia.

Hint. Diagonalize over the reals. □

Solution. A nondegenerate symmetric form has a basis with p plus ones and q minus ones, and (p, q) is basis independent. □

Exercise 29.23. Define the signature convention used here.

Hint. Use the inertia pair. □

Solution. The signature is the ordered pair (p, q) of positive and negative squares. □

Exercise 29.24. Why is signature constant on a connected manifold?

Hint. Eigenvalues cannot cross zero without degeneracy. □

Solution. The inertia indices are locally constant for a smooth nondegenerate form, hence constant on connected components. □

Exercise 29.25. Define a Lorentzian metric.

Hint. Use index one. □

Solution. It is a pseudo-Riemannian metric of signature $(n-1, 1)$ in the mostly-plus convention. □

Exercise 29.26. Write the Minkowski metric.

Hint. One time coordinate has negative sign. ☐

Solution. $\eta = -dt^2 + \sum_i (dx^i)^2$. ☐

Exercise 29.27. Classify a vector with $g(v, v) < 0$ in the mostly-plus convention.

Hint. Use the causal trichotomy. ☐

Solution. It is timelike. ☐

Exercise 29.28. Classify a nonzero vector with $g(v, v) = 0$.

Hint. Use the null cone. ☐

Solution. It is null. ☐

Exercise 29.29. Classify a vector with $g(v, v) > 0$.

Hint. Use the mostly-plus convention. ☐

Solution. It is spacelike. ☐

Exercise 29.30. Write the null-cone equation in Minkowski space.

Hint. Set the quadratic form equal to zero. ☐

Solution. $-(v^0)^2 + \sum_{i=1}^{n-1} (v^i)^2 = 0$. ☐

Exercise 29.31. What is special about the orthogonal complement of a timelike vector?

Hint. Use Lorentzian index one. ☐

Solution. The metric restricted to it is positive definite. ☐

Exercise 29.32. Do musical maps exist for an indefinite metric?

Hint. Ask whether g is invertible. ☐

Solution. Yes; nondegeneracy gives $\flat$ and $\sharp$ isomorphisms. ☐

Exercise 29.33. Which hypothesis is needed for raising and lowering indices?

Hint. Not positivity. ☐

Solution. Nondegeneracy is sufficient. ☐

Exercise 29.34. State the pseudo-Riemannian volume density.

Hint. Use the absolute determinant. ☐

Solution. $dV_g = \sqrt{|\det g|} |dx^1 \cdots dx^n|$. ☐

Exercise 29.35. Does the Levi–Civita theorem remain valid?

Hint. Check the Koszul formula. ☐

Solution. Yes; every pseudo-Riemannian metric has a unique metric-compatible torsion-free connection. ☐

Exercise 29.36. Does the Christoffel formula change?

Hint. Only invertibility is needed. □

Solution. No; the same formula using g^{ij} holds. □

Exercise 29.37. Do geodesics remain defined by zero covariant acceleration?

Hint. Use the same connection. □

Solution. Yes, $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$. □

Exercise 29.38. Does the Riemann curvature definition change?

Hint. It depends only on the connection. □

Solution. No; the same curvature operator applies. □

Exercise 29.39. Why is ordinary Riemannian distance problematic in indefinite signature?

Hint. Norm-square can be zero or negative. □

Solution. There is no positive norm on all tangent vectors, so the Riemannian length-infimum construction no longer supplies the same metric-space structure. □

Exercise 29.40. Define $O(p, q)$.

Hint. Preserve the standard signature matrix. □

Solution. $O(p, q) = \{A : A^T I_{p,q} A = I_{p,q}\}$. □

Exercise 29.41. What is the null set in the Lorentzian plane $-dt^2 + dx^2$?

Hint. Solve $dx^2 - dt^2 = 0$. □

Solution. The null directions are $x = \pm t$. □

Exercise 29.42. State the bridge to VII/30.

Hint. Compare positive-definite and index-one geometry. □

Solution. VII/30 systematically contrasts Riemannian and Lorentzian lengths, causal classes, geodesics, orthogonality, and global behavior. □

29.13 Solved problem dossiers

Problem 29.43 (Signature problem). Show two diagonalizations of a real nondegenerate symmetric form have the same numbers of positive and negative squares.

Solution. This is Sylvester's law of inertia. One argument identifies the maximal dimension of a positive-definite subspace and of a negative-definite subspace; these dimensions are basis independent. □

Problem 29.44 (Null-vector problem). Give a nonzero null vector for the Lorentzian plane.

Solution. For $g = -dt^2 + dx^2$, the vector $(1, 1)$ satisfies $-1 + 1 = 0$. □

Problem 29.45 (Timelike complement problem). Prove $u^\perp$ is positive definite when u is timelike.

Solution. Normalize u to one negative basis vector and extend it to a pseudo-orthonormal basis. The remaining basis vectors all carry positive squares. $\square$

Problem 29.46 (Signature-continuity problem). Explain why signature cannot change without degeneracy.

Solution. Eigenvalues of a symmetric matrix vary continuously. Changing the number of positive or negative eigenvalues requires an eigenvalue to pass through zero, making the form degenerate. $\square$

Problem 29.47 (Musical problem). Prove $v \mapsto g(v, \cdot)$ is an isomorphism for an indefinite nondegenerate form.

Solution. Its kernel is exactly the radical of g , which is zero. In finite dimensions injectivity implies bijectivity. $\square$

Problem 29.48 (Volume problem). Why does the volume density use $|\det g|$?

Solution. An indefinite metric matrix may have negative determinant. Taking the absolute value before the square root produces a real positive density with the correct Jacobian transformation law. $\square$

Problem 29.49 (Levi-Civita problem). Why does the Koszul proof work without positive definiteness?

Solution. The proof uses symmetry, nondegeneracy, torsion freeness, and metric compatibility. Nondegeneracy is enough to recover a vector from all of its metric pairings. $\square$

Problem 29.50 (Minkowski-geodesic problem). Determine geodesics of flat Minkowski space in Cartesian coordinates.

Solution. The metric coefficients are constant, so all Christoffel symbols vanish and the geodesic equations are $\ddot{x}^\mu = 0$; geodesics are affine lines. $\square$

Problem 29.51 (Causal-line problem). Classify an affine line in Minkowski space by its constant tangent vector.

Solution. If the tangent has negative, zero, or positive norm-square, the line is respectively timelike, null, or spacelike. $\square$

Problem 29.52 (Orthogonal-group problem). Show pseudo-orthonormal changes of frame form $O(p, q)$.

Solution. If the metric matrix is $I_{p,q}$ in both frames and the frame-change matrix is A , then $A^T I_{p,q} A = I_{p,q}$. $\square$

Problem 29.53 (Curvature problem). Why do Riemann, Ricci, scalar, and Weyl curvature extend to pseudo-Riemannian metrics?

Solution. The Levi-Civita connection exists, and all these tensors are obtained algebraically from the connection, the nondegenerate metric, and its inverse. $\square$

Problem 29.54 (Distance-failure problem). Give a concrete reason a null curve cannot be treated by the usual positive length norm.

Solution. A nonzero null tangent has $g(\dot{\gamma}, \dot{\gamma}) = 0$, so the positive-definite formula $\sqrt{g(\dot{\gamma}, \dot{\gamma})}$ assigns zero despite nonzero motion. $\square$

29.14 Challenge problems

Problem 29.55 (Witt decomposition preview). Explain how null directions lead to a basis adapted to pairs of null vectors.

Solution. In an indefinite space one can choose null vectors ℓ, n with $g(\ell, n) \neq 0$, rescale to $g(\ell, n) = -1$, and complement their span by a nondegenerate orthogonal subspace. This is the basic idea behind a Witt decomposition. $\square$

Problem 29.56 (Reverse Cauchy-Schwarz). For future-directed timelike vectors in Minkowski space, what replaces the Euclidean Cauchy-Schwarz inequality?

Solution. With normalized time orientation one obtains a reversed inequality $-g(u, v) \geq \sqrt{-g(u, u)}\sqrt{-g(v, v)}$, with equality when the vectors are positively proportional. $\square$

Problem 29.57 (Null orthogonal complement). Why is the orthogonal complement of a null vector degenerate?

Solution. If ℓ is null, then $\ell \in \ell^\perp$. Moreover $g(\ell, w) = 0$ for every $w \in \ell^\perp$, so ℓ lies in the radical of the restricted form. $\square$

Problem 29.58 (Constant conformal scaling). How does multiplying a pseudo-Riemannian metric by a positive constant affect its signature and Levi-Civita connection?

Solution. The signature is unchanged because all signs are preserved. The constant factor cancels between g_{ij} and g^{ij} in the Christoffel formula, so the Levi-Civita connection is unchanged. $\square$

Problem 29.59 (Hyperboloid bridge). Why will an indefinite ambient metric become useful again in hyperbolic geometry?

Solution. The hyperboloid model realizes hyperbolic space as one sheet of a quadratic hypersurface in Minkowski space; the ambient Lorentzian form restricts to a positive-definite metric on each tangent space of the hyperboloid. $\square$

Bridge to VII/30

VII/30 keeps the same pseudo-Riemannian formalism but specializes to Lorentzian signature and compares it point by point with positive-definite Riemannian geometry.

Chapter 30

Riemannian versus Lorentzian Geometry

Riemannian and Lorentzian geometry share the same differential machinery: metrics, Levi-Civita connections, geodesics, curvature, and parallel transport. What changes is the algebra of the metric, and with it the global geometry of length, orthogonality, and causality.

30.1 One formalism, two signatures

A Riemannian metric has signature

$$(n, 0),$$

while a Lorentzian metric in our convention has signature

$$(n - 1, 1).$$

Proposition 30.1 (Shared local formulas). *The Koszul formula, Christoffel symbols, geodesic equation, parallel-transport equation, Riemann tensor, Ricci tensor, scalar curvature, and Weyl tensor have the same coordinate formulas in both signatures.*

Remark 30.2. The formulas agree because they require nondegeneracy. Their geometric interpretations differ because the sign of $g(v, v)$ now matters.

30.2 Norm versus causal type

In Riemannian geometry,

$$g(v, v) > 0 \quad (v \neq 0).$$

In Lorentzian geometry a nonzero vector is timelike, null, or spacelike.

Warning 30.3 (There is no single positive norm from g). The expression $\sqrt{g(v, v)}$ is not a real positive norm on all Lorentzian vectors. Timelike vectors have negative norm-square and null vectors have zero norm-square.

30.3 Proper time

For a timelike curve γ , define its proper-time length by

$$\tau(\gamma) = \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$$

in the mostly-plus convention.

Definition 30.4 (Proper time). The quantity

$$d\tau = \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$$

is the infinitesimal proper time along a timelike curve.

Remark 30.5. Riemannian geodesics locally minimize length. Timelike Lorentzian geodesics, under suitable local conditions and fixed endpoints, locally maximize proper time. This reversal is one of the basic signature effects.

30.4 Null curves

Definition 30.6 (Null curve). A regular curve is null if

$$g(\dot{\gamma}, \dot{\gamma}) = 0$$

everywhere.

Null curves have zero proper time but are geometrically nontrivial. Null geodesics therefore cannot be understood as shortest curves in the Riemannian sense.

30.5 Time orientation

At each point of a Lorentzian manifold, timelike vectors form two connected cones.

Definition 30.7 (Time orientation). A time orientation is a continuous choice of one of the two timelike cones as future-directed.

Proposition 30.8 (Criterion). *A Lorentzian manifold is time-orientable if and only if it admits a continuous nowhere-zero timelike vector field.*

30.6 Causal curves

Definition 30.9 (Causal curve). A future-directed causal curve is a curve whose tangent is everywhere future-directed timelike or null.

Causal curves introduce a partial ordering flavor absent from Riemannian geometry: one can ask whether q lies in the causal or chronological future of p .

Definition 30.10 (Chronological and causal future). Write

$$q \in I^+(p)$$

if there exists a future-directed timelike curve from p to q , and

$$q \in J^+(p)$$

if there exists a future-directed causal curve from p to q .

30.7 Minkowski intervals

For two points in Minkowski space, let

$$\Delta x = (\Delta t, \Delta \mathbf{x}).$$

Then

$$\eta(\Delta x, \Delta x) = -(\Delta t)^2 + |\Delta \mathbf{x}|^2.$$

Proposition 30.11 (Interval classification). *The separation is timelike, null, or spacelike according as*

$$\eta(\Delta x, \Delta x)$$

is negative, zero, or positive.

30.8 Geodesics

Proposition 30.12 (Causal type is constant along a geodesic). *For an affinely parametrized Lorentzian geodesic,*

$$g(\dot{\gamma}, \dot{\gamma})$$

is constant. Therefore a nonconstant geodesic is everywhere timelike, everywhere null, or everywhere spacelike.

Proof. Metric compatibility and

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0$$

give

$$\frac{d}{dt} g(\dot{\gamma}, \dot{\gamma}) = 0.$$

□

30.9 Orthogonality

In Riemannian geometry,

$$v^\perp$$

is always a complementary positive-definite hyperplane for $v \neq 0$. Lorentzian orthogonality depends on causal type.

Proposition 30.13 (Lorentzian orthogonal complements). *If v is timelike, $v^\perp$ is spacelike and positive definite. If v is null, then $v \in v^\perp$, so the restriction to $v^\perp$ is degenerate.*

30.10 Distance versus time separation

Riemannian distance is symmetric:

$$d(p, q) = d(q, p) \geq 0.$$

Lorentzian geometry instead uses causal relations and, on suitable spacetimes, a time-separation function based on the supremum of proper times of future-directed causal curves.

Warning 30.14 (Not a metric-space distance). Lorentzian time separation is generally not symmetric and obeys a reverse triangle-type behavior along causally ordered points. It is not a metric in the ordinary metric-space sense.

30.11 Curvature interpretation

Positive, zero, and negative sectional curvature remain meaningful for nondegenerate two-planes, but a Lorentzian tangent space contains planes of different signatures. Curvature inequalities therefore require more care than in the Riemannian setting.

30.12 Why return to Riemannian hyperbolic geometry?

The next chapters study hyperbolic geometry, which is Riemannian and positive-definite. Yet the Lorentzian material is immediately useful: the hyperboloid model realizes the hyperbolic plane inside three-dimensional Minkowski space.

30.13 Exercises

Exercise 30.15. What signature distinguishes Riemannian from Lorentzian geometry?

Hint. Compare $(n, 0)$ with index one. □

Solution. Riemannian signature is $(n, 0)$; Lorentzian signature is $(n - 1, 1)$ in our convention. □

Exercise 30.16. Name three constructions with identical formal definitions in both signatures.

Hint. Use the Levi–Civita machinery. □

Solution. For example: the Levi–Civita connection, geodesic equation, and Riemann curvature tensor. □

Exercise 30.17. Why is $\sqrt{g(v, v)}$ not a Lorentzian norm?

Hint. Consider timelike and null vectors. □

Solution. It is imaginary for timelike vectors and zero for nonzero null vectors. □

Exercise 30.18. Define proper time along a timelike curve.

Hint. Insert a minus sign under the square root. □

Solution. $\tau = \int \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$. □

Exercise 30.19. What is the proper time of a null curve?

Hint. Its tangent has zero norm-square. □

Solution. It is zero. □

Exercise 30.20. Define a null curve.

Hint. Classify its tangent. □

Solution. It satisfies $g(\dot{\gamma}, \dot{\gamma}) = 0$ everywhere. □

Exercise 30.21. What is a time orientation?

Hint. Choose one timelike cone continuously. □

Solution. It is a continuous choice of one timelike cone as future-directed. □

Exercise 30.22. Give a criterion for time orientability.

Hint. Use a timelike vector field. □

Solution. A Lorentzian manifold is time-orientable iff it admits a continuous nowhere-zero timelike vector field. □

Exercise 30.23. Define a future-directed causal curve.

Hint. Allow timelike or null tangents. □

Solution. Its tangent is everywhere future-directed and nonspacelike. □

Exercise 30.24. Define $I^+(p)$.

Hint. Use timelike curves. □

Solution. It is the set of points reachable from p by future-directed timelike curves. □

Exercise 30.25. Define $J^+(p)$.

Hint. Allow causal curves. □

Solution. It is the set of points reachable from p by future-directed causal curves. □

Exercise 30.26. Classify a Minkowski separation with negative interval-square.

Hint. Use mostly-plus convention. □

Solution. It is timelike. □

Exercise 30.27. Classify zero interval-square.

Hint. Use the light cone. □

Solution. It is null. □

Exercise 30.28. Classify positive interval-square.

Hint. Use causal type. □

Solution. It is spacelike. □

Exercise 30.29. Why does causal type remain constant along an affine geodesic?

Hint. Differentiate $g(\dot{\gamma}, \dot{\gamma})$. □

Solution. Metric compatibility and zero covariant acceleration make it constant. □

Exercise 30.30. What is special about $v^\perp$ for timelike v ?

Hint. Index one leaves no negative direction. □

Solution. The restriction of the metric to $v^\perp$ is positive definite. □

Exercise 30.31. What is special about $v^\perp$ for null v ?

Hint. Notice $v \perp v$. □

Solution. It contains v in its radical, so the restricted metric is degenerate. □

Exercise 30.32. Is Lorentzian time separation symmetric?

Hint. Future and past differ. ☐

Solution. No, in general it is directed and not symmetric. ☐

Exercise 30.33. Does Lorentzian time separation satisfy the ordinary triangle inequality?

Hint. Think proper-time maximization. ☐

Solution. No; for causally ordered points it has a reverse triangle flavor. ☐

Exercise 30.34. What extremal behavior distinguishes short Riemannian and timelike Lorentzian geodesics?

Hint. Compare length and proper time. ☐

Solution. Riemannian geodesics locally minimize length, while timelike Lorentzian geodesics locally maximize proper time under suitable conditions. ☐

Exercise 30.35. Do null geodesics minimize ordinary distance?

Hint. There is no positive-definite distance from the Lorentzian metric. ☐

Solution. No; null geodesics are characterized by the connection and causal structure, not by Riemannian minimization. ☐

Exercise 30.36. Can sectional curvature be discussed in Lorentzian geometry?

Hint. Restrict to nondegenerate two-planes. ☐

Solution. Yes, but the signature of the plane matters and curvature inequalities require care. ☐

Exercise 30.37. Which ambient metric is used for the hyperboloid model?

Hint. Recall Minkowski space. ☐

Solution. A Lorentzian metric of signature $(2, 1)$ on $\mathbb{R}^3$. ☐

Exercise 30.38. State the bridge to VII/31.

Hint. Move from Lorentzian ambient space to a Riemannian submanifold. ☐

Solution. VII/31 introduces hyperbolic plane models, including the hyperboloid in Minkowski space and its disk and half-plane images. ☐

30.14 Solved problem dossiers

Problem 30.39 (Shared-formulas problem). Explain why the Levi-Civita formulas are signature independent.

Solution. The Koszul and Christoffel formulas use symmetry and nondegeneracy of the metric; they do not require positivity. ☐

Problem 30.40 (Proper-time problem). Compute proper time along an inertial Minkowski line $\gamma(t) = (t, vt)$ with $|v| < 1$.

Solution. Here $g(\dot{\gamma}, \dot{\gamma}) = -(1 - |v|^2)$, so $\tau = \sqrt{1 - |v|^2} \Delta t$. $\square$

Problem 30.41 (Null-line problem). Show $\gamma(t) = (t, t)$ in $1 + 1$ Minkowski space is null.

Solution. Its tangent is $(1, 1)$, whose norm-square is $-1 + 1 = 0$. $\square$

Problem 30.42 (Time-orientation problem). Show Minkowski space is time orientable.

Solution. The constant vector field ∂_t is everywhere timelike. Choose the cone containing ∂_t as future-directed. $\square$

Problem 30.43 (Causal-future problem). Describe $I^+(0)$ and $J^+(0)$ in $1 + 1$ Minkowski space.

Solution. With future $t > 0$, $I^+(0) = \{t > |x|\}$ and $J^+(0) = \{t \geq |x|, t \geq 0\}$. $\square$

Problem 30.44 (Geodesic-type problem). Prove causal type cannot change along a nonconstant affine geodesic.

Solution. The squared tangent norm is constant; its sign therefore cannot change. $\square$

Problem 30.45 (Orthogonality problem). Contrast the orthogonal complement of timelike and null vectors.

Solution. Timelike complement is nondegenerate positive definite. A null vector belongs to its own orthogonal complement, making that restriction degenerate. $\square$

Problem 30.46 (Riemannian-distance problem). Why does infimizing Lorentzian proper time not reproduce a useful analogue of Riemannian distance?

Solution. Timelike proper time is maximized rather than minimized by local geodesics, while null curves have zero proper time. The causal order is an essential part of the geometry. $\square$

Problem 30.47 (Reverse-triangle problem). Explain heuristically why Lorentzian time separation has a reverse triangle inequality.

Solution. A maximizing proper-time curve from p to r competes with the concatenation of maximizing segments $p \rightarrow q$ and $q \rightarrow r$; the supremum from p to r is at least their sum. $\square$

Problem 30.48 (Curvature problem). Which curvature tensors from VII/26–VII/28 remain available Lorentzianly?

Solution. All of them: Riemann, Ricci, scalar, and Weyl tensors are defined using the same Levi-Civita formalism. $\square$

Problem 30.49 (Hyperboloid tangent problem). Why is the tangent metric on the hyperboloid positive definite although the ambient metric is Lorentzian?

Solution. The normal to the upper hyperboloid is timelike. Its Lorentz-orthogonal complement is therefore spacelike and positive definite. $\square$

Problem 30.50 (Comparison problem). Summarize the principal structural difference between Riemannian and Lorentzian geometry.

Solution. The differential formulas are largely shared, but positive definiteness is replaced by causal cone structure, changing length extremals, orthogonality, global separation, and admissible curves. $\square$

30.15 Challenge problems

Problem 30.51 (Twin-paradox geometry). Use proper-time geometry to explain why a straight timelike Minkowski geodesic has greater proper time than a nearby broken timelike path with the same endpoints.

Solution. For inertial segments the reverse Cauchy–Schwarz inequality implies proper-time concavity. Summing the broken segments gives less total proper time than the direct geodesic, illustrating local/global maximization in flat spacetime. $\square$

Problem 30.52 (Null reparametrization). Why can a null geodesic have zero proper time while still possessing a meaningful affine parameter?

Solution. Proper time uses $\sqrt{-g(\dot{\gamma}, \dot{\gamma})}$, which vanishes for null tangents. Affine parameter instead comes from the connection equation $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$ and remains nontrivial. $\square$

Problem 30.53 (Conformal causal invariance). What happens to causal type under a positive conformal rescaling $\tilde{g} = e^{2\varphi}g$?

Solution. The squared norm is multiplied by the positive factor $e^{2\varphi}$, so its sign is unchanged. Timelike, null, and spacelike cones are therefore conformally invariant. $\square$

Problem 30.54 (Riemannian versus Lorentzian spheres). Why does the ordinary sphere inherit a Riemannian metric from Euclidean space while the hyperboloid inherits one from Lorentzian space?

Solution. In both cases the tangent space is the orthogonal complement of the normal. The Euclidean normal has positive norm and leaves a positive tangent form; the hyperboloid normal is timelike, whose Lorentz-orthogonal complement is likewise positive definite. $\square$

Problem 30.55 (Boundary of causal future). In flat Minkowski space, why is the boundary of $I^+(p)$ generated by null geodesics?

Solution. Timelike points satisfy a strict cone inequality; the boundary occurs at equality, where the separation is null. Straight null rays are affine geodesics and generate this cone boundary. $\square$

Bridge to VII/31

VII/31 returns to positive-definite geometry and uses the Lorentzian hyperboloid as one of several equivalent models of the hyperbolic plane.

Part VI

Hyperbolic Geometry

Chapter 31

Hyperbolic Plane Models

Hyperbolic geometry is the model geometry of constant negative curvature. There is not a single preferred coordinate picture. Instead several equivalent models make different structures visible:

$$\boxed{\text{hyperboloid} \longleftrightarrow \text{disk} \longleftrightarrow \text{upper half-plane} \longleftrightarrow \text{Klein disk}.}$$

This chapter develops the spaces and their explicit identifications. VII/32 focuses on the Poincaré metric itself.

31.1 The hyperboloid model

Let $\mathbb{R}^{2,1}$ carry the Lorentzian bilinear form

$$\langle x, y \rangle_L = x_1y_1 + x_2y_2 - x_3y_3.$$

Definition 31.1 (Hyperboloid model). The hyperboloid model is

$$\mathbb{H}_{\text{hyp}}^2 = \left\{ x \in \mathbb{R}^{2,1} : \langle x, x \rangle_L = -1, x_3 > 0 \right\}.$$

Proposition 31.2 (Positive tangent metric). For $x \in \mathbb{H}_{\text{hyp}}^2$,

$$T_x \mathbb{H}_{\text{hyp}}^2 = \{ v : \langle v, x \rangle_L = 0 \}.$$

The Lorentzian form restricted to this tangent plane is positive definite.

Proof. Differentiating $\langle x, x \rangle_L = -1$ gives the tangent equation. Since x is timelike, its Lorentz-orthogonal complement is spacelike and positive definite by VII/29. $\square$

31.2 Geodesics in the hyperboloid picture

Proposition 31.3 (Plane sections). Hyperbolic geodesics in the hyperboloid model are intersections of $\mathbb{H}_{\text{hyp}}^2$ with two-dimensional linear subspaces through the origin on which the Lorentzian form has signature $(1, 1)$.

This is the hyperbolic analogue of great circles on the sphere.

31.3 The Poincaré disk as a set

Definition 31.4 (Disk model). The Poincaré disk is the open unit disk

$$\mathbb{D} = \{ z \in \mathbb{C} : |z| < 1 \}.$$

The Euclidean boundary circle

$$S^1 = \{|z| = 1\}$$

does not belong to the hyperbolic plane. It represents ideal boundary points.

Definition 31.5 (Ideal boundary). The ideal boundary of the disk model is

$$\partial_\infty \mathbb{H}^2 \cong S^1.$$

31.4 Hyperboloid to disk

Theorem 31.6 (Stereographic-type projection). *The map*

$$\Phi(x_1, x_2, x_3) = \frac{x_1 + ix_2}{x_3 + 1}$$

sends $\mathbb{H}_{\text{hyp}}^2$ diffeomorphically onto $\mathbb{D}$. Its inverse is

$$\Phi^{-1}(z) = \left(\frac{2\Re z}{1 - |z|^2}, \frac{2\Im z}{1 - |z|^2}, \frac{1 + |z|^2}{1 - |z|^2} \right).$$

Proof. Substituting the inverse formula gives Lorentz norm -1 and third coordinate positive. Direct substitution in the two compositions gives the identity. $\square$

31.5 The upper half-plane

Definition 31.7 (Upper half-plane). The upper half-plane model is

$$\mathbb{H} = \{z = x + iy \in \mathbb{C} : y > 0\}.$$

Its ideal boundary is

$$\mathbb{R} \cup \{\infty\}.$$

31.6 Cayley transform

Theorem 31.8 (Disk–half-plane equivalence). *The Cayley transform*

$$C(z) = i \frac{1 + z}{1 - z}$$

maps $\mathbb{D}$ biholomorphically onto $\mathbb{H}$. Its inverse is

$$C^{-1}(w) = \frac{w - i}{w + i}.$$

Proposition 31.9 (Boundary correspondence). *The Cayley transform extends to the ideal boundaries in the Möbius sense:*

$$S^1 \longleftrightarrow \mathbb{R} \cup \{\infty\}.$$

In particular,

$$1 \mapsto \infty, \quad -1 \mapsto 0, \quad i \mapsto -1.$$

31.7 Geodesics in conformal models

Theorem 31.10 (Disk geodesic shapes). *In the Poincaré disk, geodesics are Euclidean diameters and circular arcs meeting the boundary circle orthogonally.*

Theorem 31.11 (Half-plane geodesic shapes). *In the upper half-plane, geodesics are vertical lines and semicircles meeting the real axis orthogonally.*

Remark 31.12. VII/32 derives these facts from the Poincaré metric. Here they provide the geometric dictionary among models.

31.8 The Klein disk

Definition 31.13 (Klein model). The Klein disk uses the same underlying open unit disk, but with a different projective realization of hyperbolic geometry.

Theorem 31.14 (Central projection). *The map*

$$K(x_1, x_2, x_3) = \left(\frac{x_1}{x_3}, \frac{x_2}{x_3} \right)$$

sends the hyperboloid onto the open Euclidean unit disk. Its inverse is

$$K^{-1}(u) = \frac{(u_1, u_2, 1)}{\sqrt{1 - |u|^2}}.$$

Proposition 31.15 (Klein geodesics). *Hyperbolic geodesics appear in the Klein model as Euclidean chords of the disk.*

Warning 31.16 (Klein is not conformal). The Klein model makes geodesics straight, but it does not preserve Euclidean angles. The Poincaré disk is conformal; the Klein disk is projective.

31.9 Disk–Klein conversion

For $z \in \mathbb{D}$, identify z with its vector in $\mathbb{R}^2$.

Proposition 31.17 (Radial conversion). *The Poincaré-disk coordinate z and Klein coordinate u for the same hyperboloid point satisfy*

$$u = \frac{2z}{1 + |z|^2}.$$

Conversely,

$$z = \frac{u}{1 + \sqrt{1 - |u|^2}}.$$

31.10 Ideal points and asymptotic geodesics

Definition 31.18 (Ideal endpoint). A hyperbolic geodesic has two endpoints on

$$\partial_\infty \mathbb{H}^2.$$

These endpoints are not points of $\mathbb{H}^2$, but encode the asymptotic direction of the geodesic.

Two distinct geodesics may share one ideal endpoint; they are then asymptotic.

31.11 Model comparison

Proposition 31.19 (What each model makes easy). *The hyperboloid model makes Lorentz symmetry and curvature algebra transparent. The Poincaré disk makes angles and ideal-boundary symmetry visual. The upper half-plane makes real Möbius transformations natural. The Klein disk makes geodesics straight.*

31.12 Negative curvature preview

All these models describe the same simply connected complete surface of constant sectional curvature -1 , once equipped with their corresponding hyperbolic metrics.

Warning 31.20 (Euclidean pictures are not Euclidean geometry). The disk and half-plane are subsets of the Euclidean plane only as coordinate models. Euclidean lengths, areas, and straightness are generally not the hyperbolic ones.

31.13 Exercises

Exercise 31.21. Define the hyperboloid model of $\mathbb{H}^2$.

Hint. Use Lorentz norm -1 and the upper sheet. □

Solution. $\{x : \langle x, x \rangle_L = -1, x_3 > 0\}$. □

Exercise 31.22. Describe the tangent space at x on the hyperboloid.

Hint. Differentiate the quadratic constraint. □

Solution. $T_x \mathbb{H}^2 = \{v : \langle v, x \rangle_L = 0\}$. □

Exercise 31.23. Why is the induced tangent metric positive definite?

Hint. The normal x is timelike. □

Solution. The Lorentz-orthogonal complement of a timelike vector is positive definite. □

Exercise 31.24. How are hyperboloid geodesics obtained?

Hint. Intersect with planes through the origin. □

Solution. They are intersections with suitable two-dimensional linear subspaces through the origin. □

Exercise 31.25. Define the Poincaré disk as a set.

Hint. Use the unit disk. □

Solution. $\mathbb{D} = \{z : |z| < 1\}$. □

Exercise 31.26. What is the ideal boundary of the disk model?

Hint. Use the unit circle. □

Solution. $\partial_\infty \mathbb{H}^2 \cong S^1$. □

Exercise 31.27. Write the map from hyperboloid to disk.

Hint. Divide by $x_3 + 1$. □

Solution. $\Phi(x) = (x_1 + ix_2)/(x_3 + 1)$. □

Exercise 31.28. Write the inverse disk-to-hyperboloid map.

Hint. Use denominator $1 - |z|^2$. □

Solution. $\Phi^{-1}(z) = (2\Re z, 2\Im z, 1 + |z|^2)/(1 - |z|^2)$. □

Exercise 31.29. Define the upper half-plane.

Hint. Require positive imaginary part. □

Solution. $\mathbb{H} = \{x + iy : y > 0\}$. □

Exercise 31.30. What is its ideal boundary?

Hint. Compactify the real line by infinity. □

Solution. $\mathbb{R} \cup \{\infty\}$. □

Exercise 31.31. State the Cayley transform from disk to half-plane.

Hint. Use $i(1 + z)/(1 - z)$. □

Solution. $C(z) = i(1 + z)/(1 - z)$. □

Exercise 31.32. State the inverse Cayley transform.

Hint. Solve for z . □

Solution. $C^{-1}(w) = (w - i)/(w + i)$. □

Exercise 31.33. Where does $z = 1$ go under the boundary extension?

Hint. The denominator vanishes. □

Solution. It maps to ∞ . □

Exercise 31.34. What are Poincaré-disk geodesics as Euclidean curves?

Hint. Meet the boundary orthogonally. □

Solution. They are diameters and circular arcs orthogonal to S^1 . □

Exercise 31.35. What are half-plane geodesics as Euclidean curves?

Hint. Meet the real axis orthogonally. □

Solution. They are vertical lines and semicircles orthogonal to the real axis. □

Exercise 31.36. Define the Klein disk model.

Hint. Use central projective coordinates on the open disk. □

Solution. It is the open unit disk with the projective hyperbolic realization obtained from the hyperboloid. □

Exercise 31.37. Write the hyperboloid-to-Klein map.

Hint. Divide spatial coordinates by x_3 . ☐

Solution. $K(x) = (x_1/x_3, x_2/x_3)$. ☐

Exercise 31.38. What do hyperbolic geodesics look like in the Klein disk?

Hint. Central projection sends planes to lines. ☐

Solution. They are Euclidean chords. ☐

Exercise 31.39. Is the Klein model conformal?

Hint. Compare its angle behavior. ☐

Solution. No. ☐

Exercise 31.40. Which disk model is conformal?

Hint. Recall the Poincaré disk. ☐

Solution. The Poincaré disk model. ☐

Exercise 31.41. Write the Poincaré-disk to Klein radial conversion.

Hint. Use $2z/(1 + |z|^2)$. ☐

Solution. $u = 2z/(1 + |z|^2)$. ☐

Exercise 31.42. What is an ideal endpoint of a geodesic?

Hint. It lies on the boundary at infinity. ☐

Solution. It is one of the two boundary-at-infinity points approached by the geodesic. ☐

Exercise 31.43. Why are Euclidean lengths in the disk not hyperbolic lengths?

Hint. The disk is only a coordinate picture. ☐

Solution. The hyperbolic metric is non-Euclidean and blows up toward the ideal boundary. ☐

Exercise 31.44. State the bridge to VII/32.

Hint. Move from model dictionaries to metric calculation. ☐

Solution. VII/32 derives and analyzes the Poincaré metric on the half-plane and disk, including distance, geodesics, area, and curvature. ☐

31.14 Solved problem dossiers

Problem 31.45 (Hyperboloid tangent problem). Derive the tangent-space equation for the hyperboloid.

Solution. For a curve $x(t)$ on the hyperboloid, differentiate $\langle x(t), x(t) \rangle_L = -1$ to obtain $2\langle x, \dot{x} \rangle_L = 0$. $\square$

Problem 31.46 (Positive-metric problem). Prove the hyperboloid's induced metric is Riemannian.

Solution. The position vector x is timelike. Its Lorentz-orthogonal complement is a positive-definite two-plane, and this complement is exactly the tangent space. $\square$

Problem 31.47 (Disk-map problem). Verify the stated inverse Φ^{-1} lands on the hyperboloid.

Solution. Compute the Lorentz norm of $(2x, 2y, 1 + r^2)/(1 - r^2)$; the numerator becomes $4r^2 - (1 + r^2)^2 = -(1 - r^2)^2$. $\square$

Problem 31.48 (Cayley problem). Show $C(z) = i(1 + z)/(1 - z)$ maps $\mathbb{D}$ to the upper half-plane.

Solution. A direct calculation gives $\text{Im } C(z) = (1 - |z|^2)/|1 - z|^2 > 0$. $\square$

Problem 31.49 (Boundary problem). Explain the ideal-boundary correspondence between S^1 and $\mathbb{R} \cup \{\infty\}$.

Solution. The Möbius Cayley transform extends to the Riemann sphere, carrying the unit circle to the extended real line. $\square$

Problem 31.50 (Disk-geodesic problem). How does the Cayley transform carry disk geodesics to half-plane geodesics?

Solution. Möbius maps send generalized circles to generalized circles and preserve angles; therefore circles orthogonal to S^1 map to circles or lines orthogonal to the real axis. $\square$

Problem 31.51 (Klein-map problem). Verify central projection maps the hyperboloid inside the unit disk.

Solution. From $x_1^2 + x_2^2 - x_3^2 = -1$, one gets $(x_1^2 + x_2^2)/x_3^2 = 1 - 1/x_3^2 < 1$. $\square$

Problem 31.52 (Klein-geodesic problem). Explain why hyperboloid geodesics become chords in the Klein disk.

Solution. A hyperboloid geodesic lies in a plane through the origin. Central projection sends that plane's intersection with $x_3 = 1$ to a Euclidean line. $\square$

Problem 31.53 (Disk-Klein problem). Derive $u = 2z/(1 + |z|^2)$.

Solution. Substitute the disk inverse $\Phi^{-1}(z)$ into $K(x) = (x_1/x_3, x_2/x_3)$. $\square$

Problem 31.54 (Ideal-endpoint problem). Why are boundary points not points of the hyperbolic plane?

Solution. In disk and half-plane coordinates the hyperbolic metric sends the boundary to infinite distance; the boundary records limiting directions rather than interior points. $\square$

Problem 31.55 (Model-choice problem). Which model is best suited to straight-line convexity arguments?

Solution. The Klein model, because its geodesics are Euclidean chords, though it sacrifices conformality. $\square$

Problem 31.56 (Model-synthesis problem). Explain why the four pictures describe one geometry rather than four unrelated geometries.

Solution. Explicit diffeomorphisms identify the underlying spaces, and when each is equipped with the transported hyperbolic metric these maps are isometries. Different models emphasize different features of the same constant-curvature surface. $\square$

31.15 Challenge problems

Problem 31.57 (Hyperboloid distance preview). Guess the invariant quantity that will determine hyperbolic distance between x, y on the hyperboloid.

Solution. The Lorentz product is preserved by $O^+(2, 1)$; for unit timelike points one has $-\langle x, y \rangle_L \geq 1$, and the distance is characterized by $\cosh d(x, y) = -\langle x, y \rangle_L$. $\square$

Problem 31.58 (Geodesic endpoints). Given two distinct ideal boundary points in the disk, why is there a unique hyperbolic geodesic joining them?

Solution. There is a unique generalized Euclidean circle orthogonal to the unit circle through the two boundary points, with a diameter as the limiting straight-line case. $\square$

Problem 31.59 (Cross-model vertical line). Under the inverse Cayley map, what kind of disk geodesic corresponds to a vertical half-plane geodesic?

Solution. A vertical line is a generalized circle through ∞ orthogonal to the real axis. Möbius inversion carries it to a generalized circle orthogonal to S^1 , hence a disk geodesic. $\square$

Problem 31.60 (Klein boundary behavior). Why do Klein and Poincaré disk coordinates share the same ideal circle although interior points differ radially?

Solution. The radial map $u = 2z/(1 + |z|^2)$ tends to $u = z$ as $|z| \rightarrow 1$, so it fixes every boundary direction in the limit. $\square$

Problem 31.61 (Lorentz group bridge). Why should Lorentz transformations preserving the upper hyperboloid become hyperbolic isometries?

Solution. They preserve the ambient Lorentz form and the hyperboloid equation, hence preserve the induced tangent metric. Later chapters identify these symmetries with Möbius transformations through the model maps. $\square$

Bridge to VII/32

VII/32 puts the analytic metric structure onto the disk and half-plane models, deriving the Poincaré line element, distance formula, geodesics, area element, and curvature.

Chapter 32

The Poincaré Metric

VII/31 introduced the upper half-plane and disk as coordinate models of the hyperbolic plane. The geometry becomes hyperbolic only after we equip these domains with the Poincaré metric.

32.1 Upper half-plane metric

Definition 32.1 (Poincaré metric on the half-plane). On

$$\mathbb{H} = \{z = x + iy : y > 0\},$$

define

$$g_{\mathbb{H}} = \frac{dx^2 + dy^2}{y^2}.$$

Equivalently,

$$ds_{\mathbb{H}} = \frac{|dz|}{\operatorname{Im} z}.$$

Proposition 32.2 (Conformality). *The Poincaré metric is conformal to the Euclidean metric, with conformal factor y^{-2} . Hence hyperbolic angles agree with Euclidean angles.*

32.2 Lengths and area

For a smooth curve $\gamma(t) = x(t) + iy(t)$,

$$L_{\mathbb{H}}(\gamma) = \int \frac{\sqrt{\dot{x}^2 + \dot{y}^2}}{y} dt.$$

Definition 32.3 (Hyperbolic area). The Riemannian area form is

$$dA_{\mathbb{H}} = \frac{dx dy}{y^2}.$$

Remark 32.4. The factor y^{-1} in length and y^{-2} in area makes the Euclidean boundary $y = 0$ infinitely far away in the hyperbolic metric.

32.3 Vertical geodesics

Consider the vertical curve

$$\gamma(t) = x_0 + iy(t).$$

Proposition 32.5 (Vertical distance). *For $0 < y_1 < y_2$,*

$$d_{\mathbb{H}}(x_0 + iy_1, x_0 + iy_2) = \log \frac{y_2}{y_1}.$$

Proof. The vertical segment has length

$$\int_{y_1}^{y_2} \frac{dy}{y} = \log \frac{y_2}{y_1}.$$

The geodesic equations or isometric normalization developed below show that this segment is minimizing. $\square$

32.4 Christoffel symbols and geodesic equations

For

$$g_{xx} = g_{yy} = y^{-2}, \quad g_{xy} = 0,$$

the nonzero Christoffel symbols are

$$\Gamma_{xy}^x = \Gamma_{yx}^x = -\frac{1}{y}, \quad \Gamma_{xx}^y = \frac{1}{y}, \quad \Gamma_{yy}^y = -\frac{1}{y}.$$

Proposition 32.6 (Geodesic equations). *An affinely parametrized geodesic satisfies*

$$\ddot{x} - \frac{2}{y}\dot{x}\dot{y} = 0,$$

$$\ddot{y} + \frac{1}{y}(\dot{x}^2 - \dot{y}^2) = 0.$$

32.5 Geodesic shapes

Theorem 32.7 (Half-plane geodesics). *The unparametrized geodesics of $(\mathbb{H}, g_{\mathbb{H}})$ are precisely*

vertical lines and Euclidean semicircles orthogonal to $\mathbb{R}$.

Remark 32.8. This proves the model statement previewed in VII/31. Orthogonality is Euclidean because the metric is conformal.

32.6 Hyperbolic distance formula

Theorem 32.9 (Distance in the upper half-plane). *For $z, w \in \mathbb{H}$,*

$$\cosh d_{\mathbb{H}}(z, w) = 1 + \frac{|z - w|^2}{2 \operatorname{Im} z \operatorname{Im} w}.$$

Equivalently,

$$\sinh \frac{d_{\mathbb{H}}(z, w)}{2} = \frac{|z - w|}{2\sqrt{\operatorname{Im} z \operatorname{Im} w}}.$$

Example 32.10 (Points on one vertical line). For $z = iy_1$ and $w = iy_2$,

$$d_{\mathbb{H}}(z, w) = \left| \log \frac{y_2}{y_1} \right|.$$

32.7 The disk metric

Let

$$\mathbb{D} = \{z : |z| < 1\}.$$

Definition 32.11 (Poincaré disk metric). The disk metric of curvature -1 is

$$g_{\mathbb{D}} = \frac{4|dz|^2}{(1 - |z|^2)^2}.$$

Thus

$$ds_{\mathbb{D}} = \frac{2|dz|}{1 - |z|^2}.$$

Proposition 32.12 (Cayley isometry). *The Cayley transform*

$$C(z) = i \frac{1+z}{1-z}$$

satisfies

$$C^* g_{\mathbb{H}} = g_{\mathbb{D}}.$$

Hence it is an isometry between the two Poincaré models.

32.8 Disk distance

Theorem 32.13 (Distance in the disk). *For $z, w \in \mathbb{D}$,*

$$\tanh \frac{d_{\mathbb{D}}(z, w)}{2} = \left| \frac{z - w}{1 - \bar{w}z} \right|.$$

In particular,

$$d_{\mathbb{D}}(0, r) = 2 \operatorname{artanh} r = \log \frac{1+r}{1-r}.$$

32.9 Infinite distance to the ideal boundary

Proposition 32.14 (Completeness at the boundary). *As $r \rightarrow 1^-$,*

$$d_{\mathbb{D}}(0, r) \rightarrow \infty.$$

Likewise, approaching the real axis in the half-plane requires infinite hyperbolic length.

Remark 32.15. The ideal boundary is therefore a genuine boundary at infinity, not a missing finite-distance edge of the surface.

32.10 Gaussian curvature

For a conformal metric

$$g = e^{2u}(dx^2 + dy^2)$$

on a surface,

$$K = -e^{-2u} \Delta u.$$

Theorem 32.16 (Constant curvature). *The Poincaré half-plane and disk metrics both have*

$$K = -1.$$

Proof. For the half-plane,

$$u = -\log y.$$

Hence

$$\Delta u = \frac{1}{y^2}, \quad e^{-2u} = y^2,$$

so

$$K = -y^2 \frac{1}{y^2} = -1.$$

The disk result follows from the Cayley isometry. $\square$

32.11 Hyperbolic circles

Proposition 32.17 (Euclidean appearance of hyperbolic circles). *Metric circles in the Poincaré disk are Euclidean circles, although their Euclidean center generally differs from their hyperbolic center.*

Example 32.18 (Circle centered at the origin). The hyperbolic circle of radius R centered at 0 is the Euclidean circle

$$|z| = \tanh \frac{R}{2}.$$

32.12 Area growth

A hyperbolic disk of radius R has area

$$A(R) = 2\pi(\cosh R - 1) = 4\pi \sinh^2 \frac{R}{2}.$$

Remark 32.19. Exponential area growth is one of the clearest quantitative signatures of negative curvature and sharply contrasts with the Euclidean formula πR^2 .

32.13 Exercises

Exercise 32.20. State the half-plane Poincaré metric.

Hint. Divide Euclidean metric by y^2 . $\square$

Solution. $g = (dx^2 + dy^2)/y^2$. $\square$

Exercise 32.21. Why is it conformal to Euclidean geometry?

Hint. Identify a scalar factor. $\square$

Solution. It is y^{-2} times the Euclidean metric. $\square$

Exercise 32.22. Write the hyperbolic length of a curve $x(t) + iy(t)$.

Hint. Take the metric norm of the tangent. $\square$

Solution. $L = \int \sqrt{\dot{x}^2 + \dot{y}^2} dt/y$. $\square$

Exercise 32.23. Write the hyperbolic area element.

Hint. Take $\sqrt{\det g}$. $\square$

Solution. $dA = dx dy/y^2$. □

Exercise 32.24. Compute vertical distance between heights y_1, y_2 .

Hint. Integrate dy/y . □

Solution. $|\log(y_2/y_1)|$. □

Exercise 32.25. List the nonzero half-plane Christoffel symbols.

Hint. Differentiate y^{-2} . □

Solution. $\Gamma_{xy}^x = \Gamma_{yx}^x = -1/y$, $\Gamma_{xx}^y = 1/y$, $\Gamma_{yy}^y = -1/y$. □

Exercise 32.26. State the half-plane geodesic shapes.

Hint. Use orthogonality to the boundary. □

Solution. Vertical lines and semicircles orthogonal to the real axis. □

Exercise 32.27. State the half-plane distance formula.

Hint. Use the $\cosh d$ form. □

Solution. $\cosh d = 1 + |z - w|^2/(2\Im z \Im w)$. □

Exercise 32.28. Give the equivalent $\sinh(d/2)$ formula.

Hint. Use $\cosh d - 1 = 2 \sinh^2(d/2)$. □

Solution. $\sinh(d/2) = |z - w|/(2\sqrt{\Im z \Im w})$. □

Exercise 32.29. State the Poincaré disk metric.

Hint. Use the factor $4/(1 - |z|^2)^2$. □

Solution. $g_{\mathbb{D}} = 4|dz|^2/(1 - |z|^2)^2$. □

Exercise 32.30. Why is the Cayley transform an isometry?

Hint. Pull back the half-plane metric. □

Solution. Direct differentiation and the formula for $\Im C(z)$ give $C^*g_{\mathbb{H}} = g_{\mathbb{D}}$. □

Exercise 32.31. State the disk distance formula.

Hint. Use the Möbius invariant ratio. □

Solution. $\tanh(d/2) = |(z - w)/(1 - \bar{w}z)|$. □

Exercise 32.32. Compute $d_{\mathbb{D}}(0, r)$.

Hint. Set $w = 0$. □

Solution. $2 \operatorname{artanh} r = \log((1 + r)/(1 - r))$. □

Exercise 32.33. What happens to $d(0, r)$ as $r \rightarrow 1$?

Hint. Inspect the logarithm. □

Solution. It diverges to infinity. □

Exercise 32.34. Why is the ideal boundary infinitely far away?

Hint. The conformal factor blows up. ☐

Solution. Any path approaching the boundary accumulates infinite hyperbolic length. ☐

Exercise 32.35. State the curvature formula for $g = e^{2u}(dx^2 + dy^2)$.

Hint. Use the Euclidean Laplacian. ☐

Solution. $K = -e^{-2u}\Delta u$. ☐

Exercise 32.36. Compute u for the half-plane Poincaré metric.

Hint. Set $e^{2u} = y^{-2}$. ☐

Solution. $u = -\log y$. ☐

Exercise 32.37. Compute its Gaussian curvature.

Hint. Use $\Delta(-\log y) = 1/y^2$. ☐

Solution. $K = -1$. ☐

Exercise 32.38. What is the curvature of the disk metric?

Hint. Use the Cayley isometry. ☐

Solution. It is also -1 . ☐

Exercise 32.39. What is the Euclidean radius of a hyperbolic circle of radius R centered at 0?

Hint. Invert $d(0, r)$. ☐

Solution. $r = \tanh(R/2)$. ☐

Exercise 32.40. State the area of a hyperbolic radius- R disk.

Hint. Recall constant curvature -1 . ☐

Solution. $A(R) = 2\pi(\cosh R - 1)$. ☐

Exercise 32.41. Compare large-radius area growth with Euclidean area.

Hint. Use $\cosh R \sim e^R/2$. ☐

Solution. Hyperbolic area grows exponentially, whereas Euclidean area grows quadratically. ☐

Exercise 32.42. Why do Euclidean and hyperbolic angles agree in Poincaré models?

Hint. Conformal rescaling preserves angles. ☐

Solution. The metric differs from Euclidean by a positive scalar factor. ☐

Exercise 32.43. State the bridge to VII/33.

Hint. Ask which transformations preserve these formulas. ☐

Solution. VII/33 identifies Möbius transformations in $PSL(2, \mathbb{R})$ as the natural orientation-preserving isometries of the half-plane. ☐

32.14 Solved problem dossiers

Problem 32.44 (Vertical-distance problem). Derive the logarithmic distance formula along a vertical geodesic.

Solution. Restrict the line element to $dx = 0$, obtaining $ds = dy/y$, then integrate. $\square$

Problem 32.45 (Christoffel problem). Derive the nonzero Christoffel symbols of $g = y^{-2}(dx^2 + dy^2)$.

Solution. Use $g^{ij} = y^2\delta^{ij}$ and the Levi-Civita formula; only y -derivatives are nonzero. $\square$

Problem 32.46 (Geodesic-equation problem). Derive the coordinate geodesic equations.

Solution. Insert the four nonzero Christoffel symbols into $\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0$. $\square$

Problem 32.47 (Curvature problem). Compute K directly from the conformal factor.

Solution. Write $u = -\log y$, compute $\Delta u = 1/y^2$, and substitute in $K = -e^{-2u}\Delta u$. $\square$

Problem 32.48 (Cayley-isometry problem). Verify $C^*g_{\mathbb{H}} = g_{\mathbb{D}}$.

Solution. Use $C'(z) = 2i/(1-z)^2$ and $\Im C(z) = (1-|z|^2)/|1-z|^2$; all $|1-z|$ factors cancel. $\square$

Problem 32.49 (Disk-radial problem). Derive $d_{\mathbb{D}}(0, r)$.

Solution. Along a radius, integrate $2 dr/(1-r^2)$ from 0 to r . $\square$

Problem 32.50 (Boundary problem). Prove radial approach to the unit circle has infinite length.

Solution. The integral $\int^1 2 dr/(1-r^2)$ diverges logarithmically. $\square$

Problem 32.51 (Distance-identity problem). Show the two half-plane distance formulas are equivalent.

Solution. Use $\cosh d = 1 + 2 \sinh^2(d/2)$. $\square$

Problem 32.52 (Circle problem). Find the Euclidean circle representing a hyperbolic circle centered at 0.

Solution. Set $d(0, z) = R$, so $|z| = \tanh(R/2)$. $\square$

Problem 32.53 (Area problem). Compute the area of a disk centered at the origin using disk coordinates.

Solution. Integrate $4r dr d\theta/(1-r^2)^2$ up to $r = \tanh(R/2)$ and simplify. $\square$

Problem 32.54 (Completeness problem). Explain why the Poincaré disk is complete.

Solution. A curve escaping every compact set must approach $|z| = 1$, but the boundary is at infinite metric distance; Cauchy sequences therefore cannot escape through it. $\square$

Problem 32.55 (Conformal-angle problem). Prove Euclidean orthogonality to the model boundary is meaningful hyperbolically.

Solution. A positive conformal factor multiplies all inner products by the same scalar at a point, leaving angle cosines unchanged. $\square$

32.15 Challenge problems

Problem 32.56 (Distance from invariance). Outline how one can derive the full half-plane distance formula by mapping one point to i .

Solution. Use a real Möbius isometry to send the geodesic through z, w to a vertical line, compute logarithmic distance there, and rewrite the result in an invariant algebraic form. $\square$

Problem 32.57 (Hyperbolic circumference). Compute the circumference of a hyperbolic circle of radius R .

Solution. Differentiate the disk-area formula or integrate the metric along $|z| = \tanh(R/2)$; the result is $2\pi \sinh R$. $\square$

Problem 32.58 (Small-radius asymptotics). Show hyperbolic disk area agrees with Euclidean area to leading order.

Solution. Expand $\cosh R = 1 + R^2/2 + O(R^4)$, giving $A(R) = \pi R^2 + O(R^4)$. $\square$

Problem 32.59 (Curvature scaling). If the metric is multiplied by a^2 , what is the new curvature?

Solution. Constant metric scaling multiplies sectional curvature by a^{-2} , so $a^2 g_{\mathbb{H}}$ has curvature $-1/a^2$. $\square$

Problem 32.60 (Ideal-boundary metric effect). Why can Euclidean points very near the ideal boundary be hyperbolically far apart even when their Euclidean separation is tiny?

Solution. The line element magnifies Euclidean displacement by $1/y$ in the half-plane or $2/(1 - |z|^2)$ in the disk, which diverges near the boundary. $\square$

Bridge to VII/33

VII/33 identifies the Möbius transformations whose derivative and imaginary-part factors cancel exactly in the Poincaré line element, producing the group $PSL(2, \mathbb{R})$ of orientation-preserving isometries.

Chapter 33

Möbius Transformations and $PSL(2, \mathbb{R})$

The Poincaré metric is not merely a formula; it has a large symmetry group. Real Möbius transformations act transitively on the upper half-plane and preserve its hyperbolic metric.

33.1 Fractional linear transformations

Definition 33.1 (Möbius transformation). For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc \neq 0,$$

define

$$A \cdot z = \frac{az + b}{cz + d}$$

on the Riemann sphere where the expression is defined.

Multiplying A by a nonzero scalar does not change the transformation.

33.2 The real determinant-one group

Definition 33.2 (Special linear group).

$$SL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) : \det A = 1\}.$$

The matrices A and $-A$ define the same Möbius map.

Definition 33.3 (Projective special linear group).

$$PSL(2, \mathbb{R}) = SL(2, \mathbb{R}) / \{\pm I\}.$$

33.3 Preservation of the upper half-plane

Theorem 33.4 (Imaginary-part transformation). *For*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{R}),$$

$$\operatorname{Im}(A \cdot z) = \frac{\operatorname{Im} z}{|cz + d|^2}.$$

Therefore

$$A(\mathbb{H}) = \mathbb{H}.$$

Proof. Subtract the complex conjugate:

$$\frac{az + b}{cz + d} - \frac{a\bar{z} + b}{c\bar{z} + d} = \frac{(ad - bc)(z - \bar{z})}{|cz + d|^2}.$$

Since $ad - bc = 1$, taking imaginary parts gives the formula. $\square$

33.4 Derivative

Proposition 33.5 (Möbius derivative). *For $A \in SL(2, \mathbb{R})$,*

$$(A \cdot z)' = \frac{1}{(cz + d)^2}.$$

Hence

$$|d(A \cdot z)| = \frac{|dz|}{|cz + d|^2}.$$

33.5 Poincaré invariance

Theorem 33.6 (Metric invariance). *Every element of $PSL(2, \mathbb{R})$ is an orientation-preserving isometry of $(\mathbb{H}, g_{\mathbb{H}})$:*

$$A^* g_{\mathbb{H}} = g_{\mathbb{H}}.$$

Proof. Using the derivative and imaginary-part formulas,

$$\frac{|d(Az)|}{\operatorname{Im}(Az)} = \frac{|dz|/|cz + d|^2}{\operatorname{Im} z / |cz + d|^2} = \frac{|dz|}{\operatorname{Im} z}.$$

$\square$

33.6 Generators

Several elementary matrices produce familiar transformations:

$$T_b(z) = z + b, \quad D_\lambda(z) = \lambda z, \quad S(z) = -\frac{1}{z},$$

where $b \in \mathbb{R}$ and $\lambda > 0$.

Proposition 33.7 (Elementary isometries). *Translations T_b , positive dilations D_λ , and inversion S are hyperbolic isometries of $\mathbb{H}$.*

33.7 Transitivity

Theorem 33.8 (Transitive action). *The action of $PSL(2, \mathbb{R})$ on $\mathbb{H}$ is transitive.*

Proof. For $z = x + iy$, the map

$$w \mapsto yw + x$$

is represented projectively by an element of $PSL(2, \mathbb{R})$ and sends i to z . $\square$

33.8 Stabilizer of i

Theorem 33.9 (Isotropy subgroup). *The stabilizer of i consists of matrices*

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

modulo $\pm I$. It is isomorphic to $SO(2)$.

Thus

$$\mathbb{H} \cong PSL(2, \mathbb{R})/SO(2)$$

as a homogeneous Riemannian space.

33.9 Boundary action

The same fractional linear formula acts on

$$\partial_\infty \mathbb{H} = \mathbb{R} \cup \{\infty\}.$$

Proposition 33.10 (Boundary extension). *Every $A \in PSL(2, \mathbb{R})$ extends continuously to a homeomorphism of the extended real line.*

The boundary action will be central for classifying isometries in VII/34.

33.10 Generalized circles

Theorem 33.11 (Circle preservation). *Möbius transformations send generalized circles—Euclidean circles or straight lines on the Riemann sphere—to generalized circles.*

Since real Möbius maps preserve the real axis and conformal angles, they send half-plane geodesics to half-plane geodesics.

33.11 Cross ratios

Definition 33.12 (Cross ratio). For four distinct points,

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

Theorem 33.13 (Cross-ratio invariance). *For every Möbius transformation M ,*

$$[Mz_1, Mz_2; Mz_3, Mz_4] = [z_1, z_2; z_3, z_4].$$

Cross ratios provide an efficient bridge between boundary geometry and hyperbolic distance.

33.12 Disk form of the group

Conjugating $PSL(2, \mathbb{R})$ by the Cayley transform produces the orientation-preserving isometry group of the Poincaré disk.

Proposition 33.14 (Disk automorphism form). *Every orientation-preserving disk isometry has the form*

$$z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z}, \quad |a| < 1.$$

33.13 Orientation reversal

Complex conjugation

$$z \mapsto -\bar{z}$$

preserves the upper half-plane and the Poincaré metric but reverses orientation.

Remark 33.15. Thus $PSL(2, \mathbb{R})$ is the orientation-preserving isometry group. The full isometry group is obtained by adjoining one orientation-reversing isometry.

33.14 Exercises

Exercise 33.16. Define a Möbius transformation.

Hint. Use a fractional linear formula. □

Solution. $z \mapsto (az + b)/(cz + d)$ with $ad - bc \neq 0$. □

Exercise 33.17. Why do scalar multiples of a matrix define the same Möbius map?

Hint. Cancel the common scalar. □

Solution. Both numerator and denominator are multiplied by the same nonzero factor. □

Exercise 33.18. Define $SL(2, \mathbb{R})$.

Hint. Require determinant one. □

Solution. It is the group of real 2×2 matrices with determinant 1. □

Exercise 33.19. Define $PSL(2, \mathbb{R})$.

Hint. Identify A and $-A$. □

Solution. $SL(2, \mathbb{R})/\{\pm I\}$. □

Exercise 33.20. State the imaginary-part transformation formula.

Hint. Use $|cz + d|^2$. □

Solution. $\Im(Az) = \Im z / |cz + d|^2$. □

Exercise 33.21. Why does $PSL(2, \mathbb{R})$ preserve $\mathbb{H}$?

Hint. Positive imaginary parts stay positive. □

Solution. The denominator factor is positive, so $\Im z > 0$ implies $\Im(Az) > 0$. □

Exercise 33.22. Compute the derivative of $A \cdot z$ when $\det A = 1$.

Hint. Differentiate the quotient. □

Solution. $(Az)' = 1/(cz + d)^2$. □

Exercise 33.23. Why is A a Poincaré isometry?

Hint. Compare derivative scaling with imaginary-part scaling. □

Solution. The same factor $|cz + d|^{-2}$ appears in numerator and denominator of $ds = |dz|/\Im z$, so it cancels. □

Exercise 33.24. Give the Möbius form of a real translation.

Hint. Use $c = 0$.

□

Solution. $T_b(z) = z + b$.

□

Exercise 33.25. Give a positive dilation.

Hint. Projectively normalize determinant.

□

Solution. $D_\lambda(z) = \lambda z$, $\lambda > 0$.

□

Exercise 33.26. Give the inversion generator.

Hint. Use the matrix $\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}$.

□

Solution. $S(z) = -1/z$.

□

Exercise 33.27. Is the action on $\mathbb{H}$ transitive?

Hint. Send i to $x + iy$.

□

Solution. Yes.

□

Exercise 33.28. What is the stabilizer of i ?

Hint. Think rotations in tangent space.

□

Solution. It is isomorphic to $SO(2)$.

□

Exercise 33.29. Write $\mathbb{H}$ as a homogeneous space.

Hint. Quotient by the stabilizer.

□

Solution. $\mathbb{H} \cong PSL(2, \mathbb{R})/SO(2)$.

□

Exercise 33.30. Where does the boundary action live?

Hint. Use the extended real line.

□

Solution. On $\mathbb{R} \cup \{\infty\}$.

□

Exercise 33.31. What does Möbius geometry do to generalized circles?

Hint. Recall the circle theorem.

□

Solution. It sends generalized circles to generalized circles.

□

Exercise 33.32. Why are hyperbolic geodesics mapped to geodesics?

Hint. Preserve circles, boundary, and orthogonality.

□

Solution. Half-plane geodesics are generalized circles orthogonal to the real boundary, and Möbius maps preserve all three properties.

□

Exercise 33.33. Define the cross ratio.

Hint. Use four differences.

□

Solution. $[z_1, z_2; z_3, z_4] = (z_1 - z_3)(z_2 - z_4)/((z_1 - z_4)(z_2 - z_3))$.

□

Exercise 33.34. Is cross ratio Möbius invariant?

Hint. Use cancellation of denominator factors. □

Solution. Yes. □

Exercise 33.35. Give the general orientation-preserving disk isometry.

Hint. Use one disk translation and a rotation. □

Solution. $z \mapsto e^{i\theta}(z - a)/(1 - \bar{a}z)$. □

Exercise 33.36. What does conjugating by the Cayley transform do?

Hint. Transfer the action from half-plane to disk. □

Solution. It identifies $PSL(2, \mathbb{R})$ with the orientation-preserving Poincaré-disk isometry group. □

Exercise 33.37. Give an orientation-reversing half-plane isometry.

Hint. Reflect across the imaginary axis. □

Solution. $z \mapsto -\bar{z}$. □

Exercise 33.38. Is $PSL(2, \mathbb{R})$ the full isometry group?

Hint. Account for orientation reversal. □

Solution. No; it is the orientation-preserving subgroup of index two. □

Exercise 33.39. State the bridge to VII/34.

Hint. Study individual Möbius elements. □

Solution. VII/34 classifies hyperbolic isometries by fixed points, trace, axes, and translation lengths. □

33.15 Solved problem dossiers

Problem 33.40 (Imaginary-part problem). Derive $\Im(Az) = \Im z/|cz + d|^2$.

Solution. Subtract the conjugate fractional linear expressions and use $ad - bc = 1$. □

Problem 33.41 (Derivative problem). Derive $A'(z) = 1/(cz + d)^2$.

Solution. Differentiate the quotient: the numerator simplifies to $ad - bc = 1$. □

Problem 33.42 (Metric-invariance problem). Prove the half-plane metric is invariant under $PSL(2, \mathbb{R})$.

Solution. Substitute the derivative and imaginary-part formulas into $|dw|/\Im w$; the same denominator cancels. □

Problem 33.43 (Transitivity problem). Construct an element sending i to $x + iy$.

Solution. The affine map $z \mapsto yz + x$ does so. Represent it by a determinant-one matrix after multiplying numerator and denominator by a suitable scalar. □

Problem 33.44 (Stabilizer problem). Solve $A(i) = i$ for $A \in SL(2, \mathbb{R})$.

Solution. Equating real and imaginary parts gives $d = a$, $c = -b$, and $a^2 + b^2 = 1$, producing the rotation matrices. $\square$

Problem 33.45 (Geodesic problem). Show real Möbius transformations send half-plane geodesics to geodesics.

Solution. They preserve the real boundary, generalized circles, and angles; hence boundary-orthogonal generalized circles remain boundary-orthogonal. $\square$

Problem 33.46 (Cross-ratio problem). Prove cross-ratio invariance.

Solution. Use $M(z_i) - M(z_j) = (ad - bc)(z_i - z_j)/((cz_i + d)(cz_j + d))$; all matrix and denominator factors cancel. $\square$

Problem 33.47 (Disk-action problem). Derive the disk automorphism form by conjugating a half-plane isometry.

Solution. Conjugate $PSL(2, \mathbb{R})$ with the Cayley map. Normalizing the image of a to 0 yields $(z - a)/(1 - \bar{a}z)$, followed by a unit rotation. $\square$

Problem 33.48 (Boundary problem). Explain why a real Möbius map extends to the ideal boundary.

Solution. The fractional linear map acts continuously on the real projective line, with $cz + d = 0$ mapped to ∞ and ∞ mapped according to a/c . $\square$

Problem 33.49 (Orientation problem). Why are holomorphic Möbius maps orientation preserving?

Solution. A nonzero complex derivative acts locally as a rotation-dilation, which has positive real Jacobian determinant. $\square$

Problem 33.50 (Full-group problem). Why does adjoining one reflection generate all isometries once the orientation-preserving group is known?

Solution. Any orientation-reversing isometry composed with a fixed reflection becomes orientation preserving; therefore every reversing isometry is a product of that reflection with an element of $PSL(2, \mathbb{R})$. $\square$

Problem 33.51 (Homogeneous-space problem). Explain $\mathbb{H} \cong PSL(2, \mathbb{R})/SO(2)$.

Solution. Transitivity identifies points with left cosets of the stabilizer of i , and that stabilizer is $SO(2)$. $\square$

33.16 Challenge problems

Problem 33.52 (Three-point rigidity). Why is a Möbius transformation determined by its values on three distinct points?

Solution. Cross-ratio invariance or direct linear algebra shows that fixing three distinct points forces the identity; composing two transformations with the same three images therefore gives the identity. $\square$

Problem 33.53 (Boundary triple). Show $PSL(2, \mathbb{R})$ acts triply transitively on cyclically ordered boundary triples.

Solution. A unique real Möbius transformation sends any ordered triple of distinct real-projective points to $(0, 1, \infty)$; orientation preservation imposes the cyclic-order condition. $\square$

Problem 33.54 (Distance via cross ratio). How can hyperbolic distance along a geodesic be expressed using its ideal endpoints?

Solution. If a, b are the ideal endpoints and z, w lie on the geodesic, distance is the logarithm of an appropriate positive cross ratio; Möbius invariance reduces the proof to a vertical line. $\square$

Problem 33.55 (Lie-group dimension). Why does $PSL(2, \mathbb{R})$ have real dimension three?

Solution. A real 2×2 matrix has four parameters and the equation $ad - bc = 1$ removes one; quotienting by the discrete subgroup $\{\pm I\}$ does not change dimension. $\square$

Problem 33.56 (Lorentz-group connection). Why should $PSL(2, \mathbb{R})$ be closely related to the identity component of $SO(2, 1)$?

Solution. Both act transitively and isometrically on the hyperbolic plane with the same one-dimensional compact stabilizer; more concretely, the adjoint action of $SL(2, \mathbb{R})$ preserves a quadratic form of signature $(2, 1)$ and descends through $\pm I$. $\square$

Bridge to VII/34

VII/34 studies individual elements of $PSL(2, \mathbb{R})$, classifying them as elliptic, parabolic, or hyperbolic by fixed points, traces, axes, and translation lengths.

Chapter 34

Hyperbolic Isometries

VII/33 identified $PSL(2, \mathbb{R})$ with the orientation-preserving isometry group of the hyperbolic plane. We now classify individual elements by their fixed points and dynamics.

34.1 Fixed-point equation

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{R}).$$

A fixed point satisfies

$$z = \frac{az + b}{cz + d},$$

or

$$cz^2 + (d - a)z - b = 0.$$

Proposition 34.1 (Discriminant). *The discriminant of the fixed-point equation is*

$$(a + d)^2 - 4 = (\operatorname{tr} A)^2 - 4.$$

Thus the absolute trace controls the qualitative dynamics.

34.2 Elliptic, parabolic, hyperbolic

Definition 34.2 (Classification). A nonidentity element represented by $A \in SL(2, \mathbb{R})$ is:

$$\begin{array}{ll} \text{elliptic} & |\operatorname{tr} A| < 2, \\ \text{parabolic} & |\operatorname{tr} A| = 2, \\ \text{hyperbolic} & |\operatorname{tr} A| > 2. \end{array}$$

Because A and $-A$ represent the same element, only the absolute trace is well defined on $PSL(2, \mathbb{R})$.

34.3 Elliptic elements

Theorem 34.3 (Elliptic fixed point). *An elliptic element has one fixed point in $\mathbb{H}$ and no fixed point on $\partial_\infty \mathbb{H}$. It is conjugate to a rotation fixing i .*

Example 34.4 (Rotation about i). Matrices

$$R_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

fix i . Their action is elliptic unless they represent the identity.

34.4 Parabolic elements

Theorem 34.5 (Parabolic normal form). *A parabolic element has exactly one fixed point on the ideal boundary and is conjugate to*

$$z \mapsto z + 1.$$

The unique boundary fixed point is often called a parabolic fixed point.

34.5 Hyperbolic elements

Theorem 34.6 (Hyperbolic normal form). *A hyperbolic element has exactly two fixed points on the ideal boundary and is conjugate to*

$$z \mapsto e^\ell z$$

for a unique number $\ell > 0$.

Definition 34.7 (Axis). The unique geodesic joining the two boundary fixed points of a hyperbolic element is its axis.

Proposition 34.8 (Translation along the axis). *A hyperbolic element preserves its axis and translates every point on it by the same hyperbolic distance ℓ .*

34.6 Translation length

Definition 34.9 (Translation length). For an isometry g , define

$$\tau(g) = \inf_{z \in \mathbb{H}} d(z, gz).$$

Theorem 34.10 (Three cases). *For nonidentity orientation-preserving isometries:*

$$\begin{array}{ll} \text{elliptic:} & \tau(g) = 0, \quad \text{attained in } \mathbb{H}, \\ \text{parabolic:} & \tau(g) = 0, \quad \text{not attained in } \mathbb{H}, \\ \text{hyperbolic:} & \tau(g) > 0, \quad \text{attained precisely on the axis.} \end{array}$$

34.7 Trace and translation length

Theorem 34.11 (Trace-length relation). *If g is hyperbolic and represented by A , then*

$$|\operatorname{tr} A| = 2 \cosh \frac{\tau(g)}{2}.$$

Equivalently,

$$\tau(g) = 2 \operatorname{arcosh} \frac{|\operatorname{tr} A|}{2}.$$

Proof. Conjugate to the normal form

$$z \mapsto e^\ell z.$$

A determinant-one representative has eigenvalues $e^{\ell/2}$ and $e^{-\ell/2}$, so its trace has absolute value

$$e^{\ell/2} + e^{-\ell/2} = 2 \cosh(\ell/2).$$

The axis is the imaginary line and the displacement there is ℓ . □

34.8 Boundary dynamics

Let g be hyperbolic with boundary fixed points

$$g^-, g^+.$$

Proposition 34.12 (North–south dynamics). *For every boundary point $x \neq g^-$,*

$$g^n x \rightarrow g^+ \quad (n \rightarrow +\infty),$$

and for every $x \neq g^+$,

$$g^{-n} x \rightarrow g^-.$$

The point g^+ is attracting and g^- repelling.

34.9 Parabolic dynamics

A parabolic element has one boundary fixed point p . After conjugating p to ∞ , the map is a translation $z \mapsto z + b$. Iterates move toward the same ideal point in both forward and backward directions, though from opposite sides in boundary coordinates.

34.10 Elliptic order

Proposition 34.13 (Finite versus infinite order). *An elliptic element conjugate to a rotation through angle θ has finite order exactly when*

$$\theta/\pi$$

is rational.

34.11 Conjugacy invariants

Proposition 34.14 (Type is conjugacy invariant). *Elliptic, parabolic, and hyperbolic type are invariant under conjugation in $PSL(2, \mathbb{R})$.*

Proof. Conjugation preserves fixed-point structure and preserves the trace of a matrix representative up to the sign ambiguity already absorbed by absolute trace. $\square$

34.12 Orientation-reversing isometries

The full isometry group also contains reflections in hyperbolic geodesics and glide-reflection analogues.

Remark 34.15. The classification above concerns orientation-preserving isometries. The orientation-reversing side can be obtained by composing these elements with a fixed reflection.

34.13 Why this matters for groups

When many hyperbolic isometries are assembled into a discrete subgroup, their fixed points, axes, translation lengths, and interactions control the geometry of the quotient surface. This is the entry point to Fuchsian groups.

34.14 Exercises

Exercise 34.16. Write the fixed-point equation for $A(z) = (az + b)/(cz + d)$.

Hint. Cross multiply. ☐

Solution. $cz^2 + (d - a)z - b = 0$. ☐

Exercise 34.17. What is its discriminant when $\det A = 1$?

Hint. Simplify $(d - a)^2 + 4bc$. ☐

Solution. $(\operatorname{tr} A)^2 - 4$. ☐

Exercise 34.18. Classify an element with $|\operatorname{tr} A| < 2$.

Hint. Use the trace trichotomy. ☐

Solution. It is elliptic. ☐

Exercise 34.19. Classify $|\operatorname{tr} A| = 2$, nonidentity.

Hint. Use the trace trichotomy. ☐

Solution. It is parabolic. ☐

Exercise 34.20. Classify $|\operatorname{tr} A| > 2$.

Hint. Use the trace trichotomy. ☐

Solution. It is hyperbolic. ☐

Exercise 34.21. Why use absolute trace in $PSL(2, \mathbb{R})$?

Hint. A and $-A$ are identical projectively. ☐

Solution. The trace changes sign under $A \mapsto -A$, while its absolute value does not. ☐

Exercise 34.22. Where is the fixed point of an elliptic isometry?

Hint. Look inside the half-plane. ☐

Solution. It has one fixed point in $\mathbb{H}$. ☐

Exercise 34.23. How many ideal fixed points does a parabolic element have?

Hint. Discriminant zero. ☐

Solution. Exactly one. ☐

Exercise 34.24. How many ideal fixed points does a hyperbolic element have?

Hint. Discriminant positive. ☐

Solution. Exactly two. ☐

Exercise 34.25. Give a parabolic normal form.

Hint. Move its fixed point to infinity. ☐

Solution. $z \mapsto z + 1$. ☐

Exercise 34.26. Give a hyperbolic normal form.

Hint. Move its fixed points to $0, \infty$. ☐

Solution. $z \mapsto e^\ell z$, $\ell > 0$. ☐

Exercise 34.27. Define the axis of a hyperbolic isometry.

Hint. Join its two ideal fixed points. ☐

Solution. It is the unique geodesic with those two boundary endpoints. ☐

Exercise 34.28. What does a hyperbolic element do on its axis?

Hint. Use the normal form. ☐

Solution. It translates every point by the same positive distance. ☐

Exercise 34.29. Define translation length.

Hint. Take the infimum displacement. ☐

Solution. $\tau(g) = \inf_z d(z, gz)$. ☐

Exercise 34.30. What is $\tau(g)$ for elliptic g ?

Hint. It has an interior fixed point. ☐

Solution. Zero, attained at the fixed point. ☐

Exercise 34.31. What is $\tau(g)$ for parabolic g ?

Hint. It has no interior fixed point. ☐

Solution. Zero, but the infimum is not attained in $\mathbb{H}$. ☐

Exercise 34.32. What is $\tau(g)$ for hyperbolic g ?

Hint. Use its axis. ☐

Solution. It is positive and attained exactly on the axis. ☐

Exercise 34.33. State the trace–length relation.

Hint. Use $\cosh(\tau/2)$. ☐

Solution. $|\operatorname{tr} A| = 2 \cosh(\tau/2)$. ☐

Exercise 34.34. Which boundary fixed point of a hyperbolic element attracts forward iterates?

Hint. Use north–south dynamics. ☐

Solution. The attracting fixed point g^+ . ☐

Exercise 34.35. What happens under backward iterates?

Hint. Reverse the roles. ☐

Solution. They converge to the repelling fixed point g^- away from g^+ . ☐

Exercise 34.36. When does an elliptic rotation have finite order?

Hint. Require its angle to be rational relative to π . □

Solution. Exactly when $\theta/\pi \in \mathbb{Q}$. □

Exercise 34.37. Is isometry type invariant under conjugation?

Hint. Conjugation preserves fixed points/trace. □

Solution. Yes. □

Exercise 34.38. Give an orientation-reversing hyperbolic isometry type.

Hint. Think reflection. □

Solution. Reflection in a hyperbolic geodesic is the basic example. □

Exercise 34.39. State the bridge to VII/35.

Hint. Pass from single elements to discrete subgroups. □

Solution. VII/35 develops Fuchsian groups, fundamental domains, quotient surfaces, and the geometry encoded by discrete collections of these isometries. □

34.15 Solved problem dossiers

Problem 34.40 (Discriminant problem). Derive the trace discriminant of the fixed-point equation.

Solution. The quadratic discriminant is $(d-a)^2 + 4bc = (a+d)^2 - 4(ad-bc) = (\operatorname{tr} A)^2 - 4$. □

Problem 34.41 (Elliptic problem). Explain why negative discriminant yields one interior fixed point.

Solution. The two algebraic roots are complex conjugates; exactly one lies in the upper half-plane and the other in the lower half-plane. □

Problem 34.42 (Parabolic problem). Show $z \mapsto z + 1$ is parabolic.

Solution. It is represented by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, whose trace is 2, and its unique boundary fixed point is ∞ . □

Problem 34.43 (Hyperbolic problem). Show $z \mapsto e^\ell z$ is hyperbolic.

Solution. Represent it by $\operatorname{diag}(e^{\ell/2}, e^{-\ell/2})$; the trace is $2 \cosh(\ell/2) > 2$, with fixed points $0, \infty$. □

Problem 34.44 (Axis problem). Why is the imaginary axis the axis of $z \mapsto e^\ell z$?

Solution. Its ideal endpoints are 0 and ∞ , and the unique geodesic with those endpoints is the imaginary axis. □

Problem 34.45 (Translation-length problem). Compute the displacement along the axis of $z \mapsto e^\ell z$.

Solution. Using vertical distance, $d(iy, ie^\ell y) = |\log(e^\ell)| = \ell$. □

Problem 34.46 (Trace-length problem). Derive the trace-length formula.

Solution. The determinant-one eigenvalues in normal form are $e^{\ell/2}, e^{-\ell/2}$; add them and identify $\ell = \tau(g)$. $\square$

Problem 34.47 (Parabolic-infimum problem). Why does a parabolic element have zero translation length without a minimizing point?

Solution. For $z \mapsto z + 1$, points high in the half-plane have displacement tending to zero because the metric scale is $1/y$, but no interior point is fixed. $\square$

Problem 34.48 (North-south problem). Prove the boundary dynamics of a hyperbolic normal form.

Solution. For $z \mapsto e^\ell z$, every finite nonzero boundary point tends to ∞ forward and to 0 backward; conjugation transports this behavior to arbitrary fixed points. $\square$

Problem 34.49 (Conjugacy problem). Show each type is preserved by conjugacy.

Solution. Conjugation transports fixed points bijectively and preserves the absolute trace of representatives, so the discriminant sign is unchanged. $\square$

Problem 34.50 (Elliptic-order problem). Determine when a rotation about i has finite order.

Solution. After conjugation to a disk rotation $z \mapsto e^{i\phi} z$, finite order means $e^{im\phi} = 1$ for some integer m , equivalently $\phi/(2\pi) \in \mathbb{Q}$. $\square$

Problem 34.51 (Classification-synthesis problem). Relate trace, fixed points, and translation length in all three cases.

Solution. Elliptic: $|\operatorname{tr}| < 2$, one interior fixed point, zero attained length. Parabolic: $|\operatorname{tr}| = 2$, one ideal fixed point, zero unattained length. Hyperbolic: $|\operatorname{tr}| > 2$, two ideal fixed points, positive attained length. $\square$

34.16 Challenge problems

Problem 34.52 (Eigenvalue classification). Rephrase the three types in terms of eigenvalues of a determinant-one representative.

Solution. Elliptic eigenvalues are complex conjugates on the unit circle; parabolic has repeated eigenvalue ± 1 with nontrivial Jordan form; hyperbolic has distinct real reciprocal eigenvalues λ, λ^{-1} with $|\lambda| > 1$. $\square$

Problem 34.53 (Axis uniqueness). Why is the hyperbolic axis unique?

Solution. The two distinct ideal fixed points determine a unique hyperbolic geodesic. Any invariant geodesic on which the element translates must have those endpoints, hence must be that geodesic. $\square$

Problem 34.54 (Displacement off axis). Why should displacement of a hyperbolic element be strictly larger off its axis?

Solution. Conjugate to a pure dilation. The axis realizes the orthogonal direction of translation; moving away introduces an additional transverse component. The hyperbolic distance formula shows the displacement increases with distance from the axis. $\square$

Problem 34.55 (Parabolic horocycles preview). What Euclidean curves are naturally preserved by $z \mapsto z + 1$?

Solution. Horizontal lines $y = \text{constant}$ are preserved. In hyperbolic geometry these are horocycles centered at the parabolic fixed point ∞ . $\square$

Problem 34.56 (Discrete-group bridge). Why are elliptic finite order, parabolic cusps, and hyperbolic axes the three basic local behaviors in Fuchsian quotients?

Solution. In a discrete subgroup, finite-order elliptics produce orbifold cone behavior, parabolics produce cusp ends, and hyperbolics identify along axes to create closed geodesic behavior. VII/35 organizes these phenomena group-theoretically. $\square$

Bridge to VII/35

VII/35 passes from individual hyperbolic isometries to discrete subgroups: Fuchsian groups, fundamental domains, quotient surfaces, cusps, and closed geodesic structures.

Chapter 35

Fuchsian Groups

35.1 Discrete subgroups of the hyperbolic isometry group

Definition 35.1 (Fuchsian group). A *Fuchsian group* is a discrete subgroup

$$\Gamma < \mathrm{PSL}(2, \mathbb{R}),$$

acting on the upper half-plane $\mathbb{H}$ by orientation-preserving hyperbolic isometries.

Discreteness is a topological condition on the matrix group, not merely an assertion that the abstract group is countable. It forces the identity to be isolated and prevents nontrivial group elements from accumulating at the identity.

Proposition 35.2 (Isolation criterion). *A subgroup $\Gamma < \mathrm{PSL}(2, \mathbb{R})$ is discrete if and only if there is a neighborhood U of the identity such that $U \cap \Gamma = \{1\}$.*

35.2 Proper discontinuity and stabilizers

Theorem 35.3 (Proper discontinuity). *If Γ is Fuchsian, then its action on $\mathbb{H}$ is properly discontinuous: for every compact $K \subset \mathbb{H}$, only finitely many $\gamma \in \Gamma$ satisfy*

$$\gamma K \cap K \neq \emptyset.$$

Corollary 35.4 (Finite stabilizers). *For every $z \in \mathbb{H}$, the stabilizer Γ_z is finite. If Γ is torsion-free, every stabilizer is trivial.*

Remark 35.5. An interior fixed point of a nontrivial orientation-preserving hyperbolic isometry is elliptic. Thus torsion-free Fuchsian groups contain no elliptic elements.

35.3 Quotient surfaces and orbifolds

Proper discontinuity allows local hyperbolic geometry to descend to the quotient.

Theorem 35.6 (Hyperbolic quotient). *If Γ is torsion-free and Fuchsian, then*

$$X = \Gamma \backslash \mathbb{H}$$

is a smooth hyperbolic surface and the projection $\mathbb{H} \rightarrow X$ is a covering map and a local isometry.

If elliptic elements are present, the quotient is naturally a hyperbolic orbifold; elliptic fixed points become cone points.

35.4 Fundamental domains and side pairings

Definition 35.7 (Fundamental domain). A region $F \subset \mathbb{H}$ is a fundamental domain for Γ if its translates cover $\mathbb{H}$, while distinct interiors γF° are disjoint.

Definition 35.8 (Dirichlet domain). For a point $p \in \mathbb{H}$ with trivial stabilizer, the Dirichlet domain centered at p is

$$D_p = \{z \in \mathbb{H} : d(z, p) \leq d(z, \gamma p) \text{ for all } \gamma \in \Gamma\}.$$

Its sides lie on hyperbolic perpendicular bisectors. Side-pairing transformations encode how boundary arcs are identified in the quotient.

35.5 The modular group

The basic arithmetic example is

$$\mathrm{PSL}(2, \mathbb{Z}) < \mathrm{PSL}(2, \mathbb{R}).$$

It is generated by

$$T(z) = z + 1, \quad S(z) = -\frac{1}{z}.$$

A standard fundamental domain is

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\operatorname{Re} z| \leq \frac{1}{2}, |z| \geq 1 \right\}.$$

Proposition 35.9 (Side pairings for the modular domain). *The vertical sides are paired by T , while the two circular boundary arcs on $|z| = 1$ are paired by S .*

Remark 35.10 (Orbifold points). The modular quotient has elliptic orbifold points represented by i and $e^{2\pi i/3}$, together with a cusp represented by ∞ .

35.6 Cusps and parabolic fixed points

Definition 35.11 (Cusp). A cusp of a Fuchsian quotient is an end associated with a Γ -orbit of parabolic fixed points on $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$.

After conjugation, a primitive parabolic stabilizer is generated by a translation $z \mapsto z + \omega$. A sufficiently high horoball

$$\{z : \operatorname{Im} z > Y\}$$

then projects to a standard cusp neighborhood.

35.7 Hyperbolic elements and closed geodesics

Theorem 35.12 (Axis to closed geodesic). *Let $\gamma \in \Gamma$ be hyperbolic. Its invariant axis $A_\gamma \subset \mathbb{H}$ projects to a closed geodesic in $\Gamma \backslash \mathbb{H}$. For a primitive hyperbolic element, the length of that closed geodesic is the translation length $\tau(\gamma)$.*

Conjugate hyperbolic elements determine the same free-homotopy class of closed geodesic. This is the first appearance of the deep dictionary between group conjugacy and quotient geometry.

35.8 Cofinite and cocompact groups

Definition 35.13 (Cofinite and cocompact). A Fuchsian group is *cofinite* if $\Gamma \backslash \mathbb{H}$ has finite hyperbolic area, and *cocompact* if the quotient is compact.

Cocompact groups have no cusps. Cofinite noncocompact groups have finitely many cusp ends.

Example 35.14 (Area of the modular surface). The standard modular fundamental domain has hyperbolic area

$$\text{Area}(\mathcal{F}) = \frac{\pi}{3}.$$

35.9 A first look at limit sets

For $z \in \mathbb{H}$, the orbit Γz has no accumulation point inside $\mathbb{H}$, but it may accumulate on the ideal boundary.

Definition 35.15 (Limit set in dimension two). The limit set $\Lambda(\Gamma) \subset \partial \mathbb{H}$ is the set of accumulation points of one, hence every, orbit Γz in the ideal boundary.

For the modular group the limit set is the entire boundary circle. Boundary dynamics will become central after passing from $\text{PSL}(2, \mathbb{R})$ to $\text{PSL}(2, \mathbb{C})$.

35.10 Bridge to hyperbolic three-space

The group $\text{PSL}(2, \mathbb{R})$ sits naturally inside $\text{PSL}(2, \mathbb{C})$. VII/36 enlarges the upper half-plane to hyperbolic three-space, identifies its ideal boundary with the Riemann sphere, and interprets complex Möbius transformations as all orientation-preserving isometries of $\mathbb{H}^3$.

35.11 Exercises

Exercise 35.16. Define a Fuchsian group.

Hint. State both the ambient group and the discreteness condition. □

Solution. A Fuchsian group is a discrete subgroup $\Gamma < \text{PSL}(2, \mathbb{R})$. □

Exercise 35.17. What does discreteness say near the identity?

Hint. Use the isolation criterion. □

Solution. There is a neighborhood U of the identity with $U \cap \Gamma = \{1\}$. □

Exercise 35.18. State proper discontinuity for the action on $\mathbb{H}$.

Hint. Use compact sets. □

Solution. For compact $K \subset \mathbb{H}$, only finitely many $\gamma \in \Gamma$ satisfy $\gamma K \cap K \neq \emptyset$. □

Exercise 35.19. Why is a point stabilizer finite?

Hint. Combine discreteness with compactness of the rotation stabilizer of a point. □

Solution. The stabilizer lies in a compact conjugate of $\text{SO}(2)$; a discrete subset of a compact group is finite. □

Exercise 35.20. What extra conclusion holds for a torsion-free Fuchsian group?

Hint. A finite subgroup consists of torsion elements. □

Solution. Every interior stabilizer is trivial, so the action on $\mathbb{H}$ is free. □

Exercise 35.21. What is the geometric nature of $\Gamma \backslash \mathbb{H}$ when Γ is torsion-free?

Hint. Use proper discontinuity and freeness. □

Solution. It is a smooth hyperbolic surface, and the quotient map is a local isometry and covering map. □

Exercise 35.22. What happens when elliptic elements are present?

Hint. Look at their fixed points. □

Solution. The quotient is naturally a hyperbolic orbifold with cone points at elliptic fixed-point orbits. □

Exercise 35.23. Define a fundamental domain.

Hint. Require covering and disjoint interiors. □

Solution. Its translates cover $\mathbb{H}$, and distinct translates have disjoint interiors. □

Exercise 35.24. Define the Dirichlet domain centered at p .

Hint. Compare distances to the orbit of p . □

Solution. $D_p = \{z : d(z, p) \leq d(z, \gamma p) \text{ for all } \gamma \in \Gamma\}$. □

Exercise 35.25. Where do sides of a Dirichlet domain lie?

Hint. Set two distances equal. □

Solution. They lie on hyperbolic perpendicular bisectors of segments joining p to γp . □

Exercise 35.26. Give the two standard generators of $\text{PSL}(2, \mathbb{Z})$.

Hint. Translation and inversion. □

Solution. $T(z) = z + 1$ and $S(z) = -1/z$. □

Exercise 35.27. Write a standard modular fundamental domain.

Hint. Use one vertical strip and the exterior of the unit disk. □

Solution. $\mathcal{F} = \{z : |\operatorname{Re} z| \leq 1/2, |z| \geq 1\}$. □

Exercise 35.28. Which transformation pairs the vertical sides of the modular domain?

Hint. Translate by one. □

Solution. The map $T(z) = z + 1$. □

Exercise 35.29. Which transformation pairs the unit-circle arcs?

Hint. Use inversion. □

Solution. The map $S(z) = -1/z$. □

Exercise 35.30. What is a cusp?

Hint. Relate an end to a parabolic ideal fixed point. $\square$

Solution. It is an end of the quotient associated with a Γ -orbit of parabolic fixed points on the ideal boundary. $\square$

Exercise 35.31. What standard map generates a parabolic stabilizer after conjugation?

Hint. Move the fixed point to infinity. $\square$

Solution. A translation $z \mapsto z + \omega$, with $\omega \neq 0$. $\square$

Exercise 35.32. What subset models a cusp neighborhood upstairs?

Hint. Use horoballs based at infinity. $\square$

Solution. A high horoball $\{\operatorname{Im} z > Y\}$. $\square$

Exercise 35.33. How does a hyperbolic element produce a closed geodesic?

Hint. Project its invariant axis. $\square$

Solution. Its axis projects to a closed geodesic in the quotient. $\square$

Exercise 35.34. What is the length of the closed geodesic associated to a primitive hyperbolic element?

Hint. Recall translation length. $\square$

Solution. It equals the translation length $\tau(\gamma)$. $\square$

Exercise 35.35. Define cofinite.

Hint. Use hyperbolic area. $\square$

Solution. Γ is cofinite if $\Gamma \backslash \mathbb{H}$ has finite hyperbolic area. $\square$

Exercise 35.36. Define cocompact.

Hint. Use compactness of the quotient. $\square$

Solution. Γ is cocompact if $\Gamma \backslash \mathbb{H}$ is compact. $\square$

Exercise 35.37. Can a cocompact Fuchsian group have a cusp?

Hint. Cusps give noncompact ends. $\square$

Solution. No. A cusp forces the quotient to be noncompact. $\square$

Exercise 35.38. What is the area of the standard modular fundamental domain?

Hint. Integrate $dx dy/y^2$ over the domain. $\square$

Solution. Its area is $\pi/3$. $\square$

Exercise 35.39. Define the limit set of a Fuchsian group.

Hint. Use orbit accumulation on the ideal boundary. $\square$

Solution. $\Lambda(\Gamma)$ is the set of accumulation points in $\partial\mathbb{H}$ of an orbit Γz . $\square$

35.12 Solved problem dossiers

Problem 35.40 (Discreteness of a cyclic hyperbolic group). Let $g(z) = e^\ell z$ with $\ell > 0$. Show that $\langle g \rangle$ is a discrete subgroup of $\mathrm{PSL}(2, \mathbb{R})$.

Solution. The powers are $g^n(z) = e^{n\ell} z$. A sequence of distinct powers cannot converge to the identity because its translation lengths are $|n|\ell$, bounded away from zero for $n \neq 0$. Hence the identity is isolated and the cyclic subgroup is discrete. $\square$

Problem 35.41 (Torsion and freeness). Suppose Γ is torsion-free and Fuchsian. Prove that no nonidentity element fixes a point of $\mathbb{H}$.

Solution. An orientation-preserving hyperbolic isometry fixing an interior point is elliptic. In a discrete subgroup an elliptic stabilizer is finite, so the element has finite order. Torsion-freeness excludes this unless the element is the identity. $\square$

Problem 35.42 (Modular side pairing). Verify that $T(z) = z + 1$ identifies the two vertical sides $\mathrm{Re} z = \pm 1/2$ of the standard modular domain.

Solution. If $z = -1/2 + iy$, then $Tz = 1/2 + iy$. Thus T maps the left vertical boundary to the right one and preserves the hyperbolic metric. $\square$

Problem 35.43 (The inversion side pairing). Show that $S(z) = -1/z$ preserves the unit semicircle and exchanges its two halves.

Solution. For $|z| = 1$, $|S(z)| = 1/|z| = 1$. Writing $z = e^{i\theta}$ gives $S(z) = -e^{-i\theta} = e^{i(\pi-\theta)}$, which exchanges the two arcs about the imaginary axis. $\square$

Problem 35.44 (Area of the modular domain). Compute the hyperbolic area of $\mathcal{F} = \{|x| \leq 1/2, x^2 + y^2 \geq 1\}$.

Solution. Using $dA = dx dy/y^2$,

$$\mathrm{Area}(\mathcal{F}) = \int_{-1/2}^{1/2} \int_{\sqrt{1-x^2}}^{\infty} \frac{dy dx}{y^2} = \int_{-1/2}^{1/2} \frac{dx}{\sqrt{1-x^2}} = 2 \arcsin(1/2) = \frac{\pi}{3}.$$

$\square$

Problem 35.45 (A standard cusp cylinder). For $T(z) = z + 1$, describe the quotient of $H_Y = \{y > Y\}$ by $\langle T \rangle$.

Solution. A fundamental strip is $0 \leq x \leq 1, y > Y$, with the two vertical sides identified. Topologically the quotient is a cylinder, and metrically it is the standard cusp end whose horizontal circumference is $1/y$, tending to zero as $y \rightarrow \infty$. $\square$

Problem 35.46 (Axis and quotient loop). Let $g(z) = e^\ell z$. Identify its axis and the length of the corresponding quotient geodesic.

Solution. The imaginary axis is invariant and is the unique axis joining the fixed points 0 and ∞ . Since $d(i, e^\ell i) = \ell$, the projected primitive closed geodesic has length ℓ . $\square$

Problem 35.47 (Conjugacy and closed geodesics). Explain why conjugate hyperbolic elements determine the same closed-geodesic geometry up to the quotient action.

Solution. If hgh^{-1} is conjugate to g , then its axis is $h(A_g)$. The two axes project to the same free-homotopy class after changing the lift by the deck transformation h . Translation length is conjugacy invariant. $\square$

Problem 35.48 (Finite stabilizer from discreteness). Give a direct compactness argument that Γ_i is finite.

Solution. The stabilizer of i in $\mathrm{PSL}(2, \mathbb{R})$ is a compact copy of $\mathrm{SO}(2)$. The intersection of a discrete subgroup with a compact set is finite; hence Γ_i is finite. $\square$

Problem 35.49 (Why a cusp prevents compactness). Show geometrically why a quotient containing a standard cusp neighborhood cannot be compact.

Solution. The cusp contains points represented by heights $y \rightarrow \infty$. Their injectivity radii tend to zero and they escape every compact subset of the quotient. Thus the quotient has a noncompact end. $\square$

Problem 35.50 (Orbit accumulation). Why can an orbit of a Fuchsian group have no accumulation point inside $\mathbb{H}$?

Solution. Proper discontinuity implies that every compact neighborhood meets the orbit in only finitely many points. Therefore an infinite orbit cannot accumulate in the interior; any accumulation in the compactification must occur on the ideal boundary. $\square$

Problem 35.51 (Modular limit set). Explain why the rational points together with ∞ already force the modular limit set to be the whole boundary.

Solution. The orbit of ∞ under $\mathrm{PSL}(2, \mathbb{Z})$ is $\mathbb{Q} \cup \{\infty\}$. Since the rationals are dense in the real boundary and the limit set is closed and invariant, its closure is all of $\mathbb{R} \cup \{\infty\}$. $\square$

35.13 Challenge problems

Problem 35.52 (A presentation of the modular group). Using the generators $S(z) = -1/z$ and $T(z) = z + 1$, verify $S^2 = 1$ in $\mathrm{PSL}(2, \mathbb{Z})$ and $(ST)^3 = 1$. Interpret these relations geometrically.

Solution. The matrix of S squares to $-I$, hence to the identity in PSL . A direct multiplication gives $(ST)^3 = -I$, again the identity in PSL . Geometrically these are rotations about the two elliptic orbifold points of orders two and three. They lead to the presentation $\mathrm{PSL}(2, \mathbb{Z}) \cong C_2 * C_3$. $\square$

Problem 35.53 (Dirichlet domains are convex). Show that a Dirichlet domain is an intersection of hyperbolic half-planes and hence geodesically convex.

Solution. Each condition $d(z, p) \leq d(z, \gamma p)$ defines the half-plane bounded by the perpendicular bisector of p and γp . Hyperbolic half-planes are geodesically convex, and arbitrary intersections of convex sets are convex. $\square$

Problem 35.54 (Primitive versus imprimitive geodesics). If $g = h^m$ with $m > 1$ and h primitive hyperbolic, compare the projected geodesics and their lengths.

Solution. The elements have the same axis. The quotient curve associated with g traverses the primitive closed geodesic m times, and $\tau(g) = m\tau(h)$. $\square$

Problem 35.55 (From Fuchsian to Kleinian). Embed a Fuchsian group into $\mathrm{PSL}(2, \mathbb{C})$. Predict what geometric subset of $\mathbb{H}^3$ it should preserve.

Solution. Use the same real matrices, now viewed as complex matrices. Their boundary action preserves the generalized circle $\mathbb{R} \cup \{\infty\}$, so in $\mathbb{H}^3$ they preserve the totally geodesic plane whose ideal boundary is that circle. This plane is naturally a copy of $\mathbb{H}^2$. $\square$

Problem 35.56 (Cusp width and conjugation). Let a parabolic stabilizer be generated by $z \mapsto z + \omega$ with $\omega > 0$. Show that conjugating by a positive dilation can normalize the cusp width to one.

Solution. Conjugate by $h(z) = z/\omega$. Then $h(z + \omega) = z/\omega + 1 = T(h(z))$, so $hgh^{-1}(z) = z + 1$. Thus the numerical cusp width depends on the chosen boundary coordinate, while the parabolic conjugacy type does not. $\square$

Chapter 36

Hyperbolic Three-Space and $\mathrm{PSL}(2, \mathbb{C})$

36.1 The upper half-space model of hyperbolic three-space

Definition 36.1 (Upper half-space). Hyperbolic three-space is

$$\mathbb{H}^3 = \{(z, t) \in \mathbb{C} \times \mathbb{R} : t > 0\}$$

with metric

$$ds^2 = \frac{|dz|^2 + dt^2}{t^2}.$$

The metric is conformal to the Euclidean metric on the upper half-space. Its volume form is

$$dV = \frac{dx dy dt}{t^3}, \quad z = x + iy.$$

36.2 Distance, geodesics, and hyperbolic planes

Theorem 36.2 (Distance formula). For $p = (z, t)$ and $q = (w, s)$,

$$\cosh d(p, q) = 1 + \frac{|z - w|^2 + (t - s)^2}{2ts}.$$

Theorem 36.3 (Geodesics). The geodesics of $\mathbb{H}^3$ are vertical Euclidean lines and Euclidean semicircles orthogonal to the boundary plane $t = 0$.

Proposition 36.4 (Totally geodesic planes). The totally geodesic copies of $\mathbb{H}^2$ are vertical Euclidean half-planes and Euclidean hemispheres orthogonal to $t = 0$.

36.3 Ideal boundary and horospheres

The natural ideal boundary is

$$\partial_\infty \mathbb{H}^3 \cong \widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

Each geodesic has two ideal endpoints. A horosphere based at ∞ is a horizontal Euclidean plane $t = \text{constant}$; horospheres based at finite boundary points are Euclidean spheres tangent to $t = 0$.

Remark 36.5. The induced metric on a horizontal horosphere $t = t_0$ is Euclidean up to the constant scale factor t_0^{-2} .

36.4 The hyperboloid model in signature (3, 1)

Let $\mathbb{R}^{3,1}$ carry the Lorentz form

$$\langle x, y \rangle_{3,1} = x_1y_1 + x_2y_2 + x_3y_3 - x_4y_4.$$

The upper sheet

$$\mathcal{H}^3 = \{x : \langle x, x \rangle_{3,1} = -1, x_4 > 0\}$$

with the induced positive-definite metric is another model of hyperbolic three-space.

Theorem 36.6 (Lorentzian distance). *For $x, y \in \mathcal{H}^3$,*

$$\cosh d(x, y) = -\langle x, y \rangle_{3,1}.$$

36.5 Möbius transformations on the Riemann sphere

Every matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C})$$

acts on $\widehat{\mathbb{C}}$ by

$$z \mapsto \frac{az + b}{cz + d}.$$

The matrices A and $-A$ induce the same map, giving $\mathrm{PSL}(2, \mathbb{C})$.

Theorem 36.7 (Extension theorem). *Every Möbius transformation of $\widehat{\mathbb{C}}$ extends uniquely to an orientation-preserving isometry of $\mathbb{H}^3$, and every orientation-preserving isometry arises this way. Hence*

$$\mathrm{Isom}^+(\mathbb{H}^3) \cong \mathrm{PSL}(2, \mathbb{C}).$$

36.6 Classification of individual isometries

The complex setting adds genuine screw motions.

Definition 36.8 (Loxodromic element). An element of $\mathrm{PSL}(2, \mathbb{C})$ is loxodromic if it has two distinct fixed points on $\widehat{\mathbb{C}}$ and, after conjugation, acts as

$$z \mapsto \lambda z, \quad |\lambda| \neq 1.$$

A loxodromic isometry preserves the geodesic joining its two ideal fixed points. Along that axis it translates and may also rotate.

Definition 36.9 (Complex length). For a loxodromic element, the complex length is

$$L = \ell + i\theta$$

modulo the usual $2\pi i$ ambiguity, where $\ell > 0$ is translation length and θ is rotation angle about the axis.

36.7 Eigenvalues and trace

For a lift $A \in \mathrm{SL}(2, \mathbb{C})$ of a loxodromic element, choose eigenvalues

$$e^{L/2}, \quad e^{-L/2}.$$

Then

$$\mathrm{tr} A = 2 \cosh \frac{L}{2}.$$

This complexifies the trace–translation–length relation from VII/34.

Remark 36.10. In $\mathrm{PSL}(2, \mathbb{C})$, classification by the real number $|\mathrm{tr} A|$ is no longer sufficient. The complex trace, or better $(\mathrm{tr} A)^2$, carries rotational information.

36.8 Real Fuchsian groups inside the complex group

The inclusion

$$\mathrm{PSL}(2, \mathbb{R}) \hookrightarrow \mathrm{PSL}(2, \mathbb{C})$$

preserves the boundary circle $\mathbb{R} \cup \{\infty\}$. Its extension to $\mathbb{H}^3$ preserves the totally geodesic vertical plane with that ideal boundary. Thus every Fuchsian group is a special Kleinian group preserving a hyperbolic plane.

36.9 Bridge to Kleinian groups

VII/37 imposes discreteness on subgroups of $\mathrm{PSL}(2, \mathbb{C})$. The resulting Kleinian groups act simultaneously on $\mathbb{H}^3$ and on its Riemann-sphere boundary. The split between stable boundary action and chaotic accumulation is encoded by the domain of discontinuity and the limit set.

36.10 Exercises

Exercise 36.11. Define the upper half-space model of $\mathbb{H}^3$.

Hint. Give the set and metric. □

Solution. $\mathbb{H}^3 = \{(z, t) \in \mathbb{C} \times \mathbb{R} : t > 0\}$ with $ds^2 = (|dz|^2 + dt^2)/t^2$. □

Exercise 36.12. What is its hyperbolic volume form?

Hint. Take the determinant of the conformal metric. □

Solution. $dV = dx dy dt/t^3$. □

Exercise 36.13. State the distance formula in upper half-space.

Hint. Use the Euclidean separation divided by heights. □

Solution. $\cosh d(p, q) = 1 + (|z - w|^2 + (t - s)^2)/(2ts)$. □

Exercise 36.14. Describe all geodesics.

Hint. Use Euclidean objects orthogonal to the boundary. □

Solution. They are vertical lines and semicircles orthogonal to $t = 0$. □

Exercise 36.15. Describe totally geodesic planes.

Hint. Raise the dimension of the geodesic description by one. ☐

Solution. They are vertical half-planes and hemispheres orthogonal to the boundary plane. ☐

Exercise 36.16. Identify the ideal boundary of $\mathbb{H}^3$.

Hint. Compactify the boundary plane by infinity. ☐

Solution. $\partial_\infty \mathbb{H}^3 \cong \widehat{\mathbb{C}}$. ☐

Exercise 36.17. What is a horosphere based at infinity?

Hint. Use a level set of height. ☐

Solution. A horizontal plane $t = t_0$. ☐

Exercise 36.18. What is a horosphere based at a finite boundary point?

Hint. Think of Euclidean spheres tangent to the boundary. ☐

Solution. A Euclidean sphere tangent to the plane $t = 0$ at that boundary point. ☐

Exercise 36.19. Write the Lorentz form used in the hyperboloid model.

Hint. Use three plus signs and one minus sign. ☐

Solution. $\langle x, y \rangle_{3,1} = x_1y_1 + x_2y_2 + x_3y_3 - x_4y_4$. ☐

Exercise 36.20. Define the hyperboloid model.

Hint. Take the upper unit timelike sheet. ☐

Solution. $\mathcal{H}^3 = \{x : \langle x, x \rangle_{3,1} = -1, x_4 > 0\}$. ☐

Exercise 36.21. State the hyperboloid distance formula.

Hint. Use the Lorentz inner product. ☐

Solution. $\cosh d(x, y) = -\langle x, y \rangle_{3,1}$. ☐

Exercise 36.22. How does $A \in \mathrm{SL}(2, \mathbb{C})$ act on the Riemann sphere?

Hint. Use fractional linear transformation. ☐

Solution. $z \mapsto (az + b)/(cz + d)$. ☐

Exercise 36.23. Why do A and $-A$ define the same Möbius transformation?

Hint. Both numerator and denominator change sign. ☐

Solution. The common sign cancels in the quotient, so the action factors through PSL . ☐

Exercise 36.24. Identify $\mathrm{Isom}^+(\mathbb{H}^3)$.

Hint. Use the extension theorem. ☐

Solution. $\mathrm{Isom}^+(\mathbb{H}^3) \cong \mathrm{PSL}(2, \mathbb{C})$. ☐

Exercise 36.25. How many ideal fixed points does a loxodromic element have?

Hint. Use its normal form. ☐

Solution. Exactly two distinct fixed points. ☐

Exercise 36.26. What geometric object joins those fixed points?

Hint. There is a unique geodesic with given ideal endpoints. ☐

Solution. The invariant geodesic axis. ☐

Exercise 36.27. What two motions can a loxodromic isometry perform along its axis?

Hint. Complex length has real and imaginary parts. ☐

Solution. A translation along the axis and a rotation about it. ☐

Exercise 36.28. Define complex length.

Hint. Separate translation and rotation. ☐

Solution. $L = \ell + i\theta$, with $\ell > 0$ translation length and θ rotation angle modulo 2π . ☐

Exercise 36.29. Give the eigenvalues associated with complex length L .

Hint. Use determinant one. ☐

Solution. They may be chosen as $e^{L/2}$ and $e^{-L/2}$. ☐

Exercise 36.30. Write the complex trace-length relation.

Hint. Add the two eigenvalues. ☐

Solution. $\text{tr } A = 2 \cosh(L/2)$. ☐

Exercise 36.31. Why is $|\text{tr } A|$ alone insufficient in $\text{PSL}(2, \mathbb{C})$?

Hint. Rotational data are complex. ☐

Solution. Different complex traces can have the same modulus while encoding different rotation angles. ☐

Exercise 36.32. What subset of the ideal boundary is preserved by $\text{PSL}(2, \mathbb{R})$?

Hint. Real coefficients preserve the real projective line. ☐

Solution. The circle $\mathbb{R} \cup \{\infty\}$. ☐

Exercise 36.33. What subset of $\mathbb{H}^3$ is then preserved?

Hint. Take the totally geodesic plane with that ideal boundary. ☐

Solution. The vertical totally geodesic copy of $\mathbb{H}^2$. ☐

Exercise 36.34. What is the next step from Fuchsian to Kleinian groups?

Hint. Change the ambient group but keep discreteness. ☐

Solution. Study discrete subgroups of $\text{PSL}(2, \mathbb{C})$ and their action on both $\mathbb{H}^3$ and $\hat{\mathbb{C}}$. ☐

36.11 Solved problem dossiers

Problem 36.35 (Vertical distance). Compute the distance between $(0, t_1)$ and $(0, t_2)$ on a vertical geodesic.

Solution. Restricting the metric gives $ds = dt/t$. Hence

$$d((0, t_1), (0, t_2)) = \left| \int_{t_1}^{t_2} \frac{dt}{t} \right| = \left| \log \frac{t_2}{t_1} \right|.$$

□

Problem 36.36 (Checking the distance formula vertically). Verify the general distance formula when $z = w$.

Solution. The formula gives

$$\cosh d = 1 + \frac{(t-s)^2}{2ts} = \frac{t^2 + s^2}{2ts} = \cosh \left(\log \frac{t}{s} \right),$$

consistent with the vertical-distance computation. □

Problem 36.37 (A totally geodesic plane). Show that the subset $\{(x, t) : x \in \mathbb{R}, t > 0\} \subset \mathbb{H}^3$ is isometric to the usual upper half-plane.

Solution. Set the second horizontal coordinate equal to zero. The induced metric is $(dx^2 + dt^2)/t^2$, exactly the Poincaré upper-half-plane metric. It is therefore a totally geodesic copy of $\mathbb{H}^2$. □

Problem 36.38 (Dilation isometry). Show that $(z, t) \mapsto (az, at)$ for $a > 0$ is an isometry of upper half-space.

Solution. The numerator $|dz|^2 + dt^2$ scales by a^2 , while the denominator t^2 also scales by a^2 . Their ratio is unchanged. □

Problem 36.39 (Boundary action of a dilation). Identify the boundary Möbius map induced by the previous dilation and its fixed points.

Solution. On $\hat{\mathbb{C}}$ it is $z \mapsto az$. Its fixed points are 0 and ∞ , the ideal endpoints of the vertical axis. □

Problem 36.40 (A pure screw motion). Consider $z \mapsto re^{i\theta}z$ with $r > 1$. Describe its extension geometrically.

Solution. It preserves the vertical axis joining 0 to ∞ . It translates by $\log r$ along that axis and rotates through angle θ about it, so its complex length is $\log r + i\theta$ up to convention. □

Problem 36.41 (Trace from complex length). Suppose a lift has eigenvalues $e^{L/2}$ and $e^{-L/2}$. Derive the trace relation.

Solution. Adding the eigenvalues gives

$$\operatorname{tr} A = e^{L/2} + e^{-L/2} = 2 \cosh(L/2).$$

□

Problem 36.42 (Why PSL rather than SL). Explain geometrically why the isometry group is PSL rather than SL.

Solution. The central matrices I and $-I$ induce the same fractional linear transformation on the boundary. By uniqueness of extension, they induce the same interior isometry. Thus the kernel is $\{\pm I\}$. $\square$

Problem 36.43 (Horosphere metric). Compute the induced metric on the horosphere $t = t_0$.

Solution. Setting $dt = 0$ gives $ds^2 = |dz|^2/t_0^2$. This is a Euclidean metric scaled by the constant factor t_0^{-2} . $\square$

Problem 36.44 (Boundary circle and hyperbolic plane). Explain why a generalized circle in $\hat{\mathbb{C}}$ determines a unique totally geodesic plane in $\mathbb{H}^3$.

Solution. If the circle passes through ∞ , it is a Euclidean line and bounds a vertical half-plane. Otherwise it is a Euclidean circle and bounds the hemisphere orthogonal to $t = 0$. These are exactly the totally geodesic planes. $\square$

Problem 36.45 (Fuchsian subgroup in three dimensions). Let $\Gamma < \text{PSL}(2, \mathbb{R})$. Show that its three-dimensional action preserves a copy of $\mathbb{H}^2$.

Solution. Its boundary action preserves $\mathbb{R} \cup \{\infty\}$. The unique totally geodesic plane with this ideal boundary is the vertical plane over the real axis, hence it is Γ -invariant. $\square$

Problem 36.46 (Two endpoints determine an axis). Show that two distinct ideal points determine a unique geodesic in upper half-space.

Solution. If one endpoint is infinity, the geodesic is the vertical line over the finite endpoint. If both are finite, there is a unique Euclidean circle through them with center on the boundary plane and orthogonal to it; its upper semicircle is the required geodesic. $\square$

36.12 Challenge problems

Problem 36.47 (Complex trace ambiguity). Show that $(\text{tr } A)^2$ is well defined for an element of $\text{PSL}(2, \mathbb{C})$, although $\text{tr } A$ itself is only defined up to sign.

Solution. The two lifts are A and $-A$. Their traces differ by a sign, so squaring removes the ambiguity. Thus $(\text{tr } A)^2$ is an invariant of the PSL element. $\square$

Problem 36.48 (Loxodromic dynamics on the boundary). For $g(z) = \lambda z$ with $|\lambda| > 1$, determine the forward and backward limits of every $z \neq 0, \infty$.

Solution. As $n \rightarrow +\infty$, $|\lambda^n z| \rightarrow \infty$, so $g^n z \rightarrow \infty$. As $n \rightarrow -\infty$, $g^{-n} z = \lambda^{-n} z \rightarrow 0$. Thus ∞ is attracting and 0 repelling. $\square$

Problem 36.49 (Recovering the two-dimensional theory). Explain how VII/34 is recovered by restricting to real matrices with real eigenvalues.

Solution. For real hyperbolic matrices the rotation angle is 0 , so complex length reduces to the real translation length $L = \ell$. The trace relation becomes $|\text{tr } A| = 2 \cosh(\ell/2)$, exactly the two-dimensional formula. $\square$

Problem 36.50 (Boundary data determine the interior isometry). Why is uniqueness in the Möbius extension theorem plausible from hyperbolic geometry?

Solution. An orientation-preserving isometry maps geodesics to geodesics, and each geodesic is determined by its two ideal endpoints. If two isometries have the same boundary action, they agree on every geodesic endpoint pair and hence on the geodesic configuration; this forces agreement throughout $\mathbb{H}^3$. $\square$

Problem 36.51 (From one isometry to a discrete group). What new phenomenon must be controlled when replacing a single loxodromic isometry by a subgroup of $\mathrm{PSL}(2, \mathbb{C})$?

Solution. One must control accumulation of infinitely many group elements and orbits. Discreteness gives a good action in the interior, but boundary orbits can accumulate on complicated closed sets. This leads to the Kleinian limit set and domain of discontinuity. $\square$

Chapter 37

Kleinian Groups and Boundary Geometry

37.1 Discrete groups in three-dimensional hyperbolic geometry

Definition 37.1 (Kleinian group). A *Kleinian group* is a discrete subgroup

$$\Gamma < \mathrm{PSL}(2, \mathbb{C}).$$

It acts by orientation-preserving isometries on $\mathbb{H}^3$ and by Möbius transformations on the ideal boundary $\widehat{\mathbb{C}}$.

Interior discreteness implies proper discontinuity on $\mathbb{H}^3$, but the boundary action is typically not properly discontinuous everywhere. The boundary therefore splits into two dynamically different pieces.

37.2 Limit set and domain of discontinuity

Definition 37.2 (Limit set). For an infinite Kleinian group Γ , the limit set $\Lambda(\Gamma) \subset \widehat{\mathbb{C}}$ is the set of accumulation points of an orbit Γp , where $p \in \mathbb{H}^3$.

Definition 37.3 (Domain of discontinuity). The domain of discontinuity is

$$\Omega(\Gamma) = \widehat{\mathbb{C}} \setminus \Lambda(\Gamma).$$

Theorem 37.4 (Boundary proper discontinuity). *The action of Γ on $\Omega(\Gamma)$ is properly discontinuous.*

The limit set is closed, Γ -invariant, and independent of the chosen interior base point.

37.3 Elementary and non-elementary groups

Definition 37.5 (Elementary Kleinian group). A Kleinian group is elementary if its limit set has at most two points. Otherwise it is non-elementary.

A cyclic parabolic group has one-point limit set. A cyclic loxodromic group has a two-point limit set consisting of the fixed points of its generator. Non-elementary groups have much richer dynamics; their limit sets are infinite and perfect.

37.4 Loxodromic fixed points and density

Theorem 37.6 (Density of loxodromic fixed points). *For a non-elementary Kleinian group, the fixed points of loxodromic elements are dense in $\Lambda(\Gamma)$.*

This theorem makes the one-element north–south dynamics from VII/36 into a global tool for understanding the group action on the limit set.

37.5 Fuchsian groups as special Kleinian groups

A Fuchsian group, viewed inside $\mathrm{PSL}(2, \mathbb{C})$, preserves a generalized circle $C \subset \widehat{\mathbb{C}}$ and the corresponding totally geodesic plane in $\mathbb{H}^3$. Its limit set lies in C .

Example 37.7 (First kind). For a cofinite Fuchsian group such as $\mathrm{PSL}(2, \mathbb{Z})$, the limit set is the entire invariant circle.

37.6 Convex hulls and convex cores

Let $\mathrm{Hull}(\Lambda) \subset \mathbb{H}^3$ denote the smallest closed hyperbolically convex set whose ideal boundary contains $\Lambda(\Gamma)$.

Definition 37.8 (Convex core). The convex core of the quotient $M = \Gamma \backslash \mathbb{H}^3$ is

$$\mathrm{Core}(M) = \Gamma \backslash \mathrm{Hull}(\Lambda(\Gamma)).$$

The convex core packages the part of the quotient directly controlled by recurrent group dynamics. Outside it, geometry heads toward conformal boundary components when $\Omega(\Gamma) \neq \emptyset$.

37.7 Quotient three-manifolds and orbifolds

Theorem 37.9 (Hyperbolic quotient). *If Γ is torsion-free and Kleinian, then*

$$M = \Gamma \backslash \mathbb{H}^3$$

is an orientable hyperbolic three-manifold. With torsion, the natural quotient is a hyperbolic orbifold.

Parabolic subgroups generate cusp ends, while loxodromic elements generate closed geodesics, now possibly accompanied by rotational holonomy around their axes.

37.8 Conformal boundary

Because the action on $\Omega(\Gamma)$ is properly discontinuous, one may form the quotient

$$\partial_c M = \Gamma \backslash \Omega(\Gamma).$$

When nonempty, this is the *conformal boundary* of the hyperbolic three-manifold or orbifold. It is naturally a Riemann surface or Riemann orbifold because Möbius transformations are conformal.

Warning 37.10. The conformal boundary is not the same object as the topological ideal boundary $\widehat{\mathbb{C}}$, and it may be empty when $\Lambda(\Gamma) = \widehat{\mathbb{C}}$.

37.9 Schottky groups as a geometric model

A classical Schottky group is generated by loxodromic transformations pairing disjoint Jordan curves (classically, circles) on the Riemann sphere. Its domain of discontinuity is nonempty and its limit set is a Cantor-type fractal.

Remark 37.11. Schottky groups provide a clean model in which algebraic freeness, hyperbolic quotient geometry, conformal boundary, and fractal limit sets can all be seen at once.

37.10 Why the boundary matters

The limit set records asymptotic orbit accumulation, while the domain of discontinuity records where the boundary action remains geometrically tame. Together they convert a discrete subgroup of matrices into a global dynamical object.

For non-elementary groups, $\Lambda(\Gamma)$ is the smallest nonempty closed Γ -invariant subset of $\widehat{\mathbb{C}}$. This minimality is one of the basic organizing principles of Kleinian dynamics.

37.11 Bridge to discrete geodesic problems

Chapters VII/31–VII/37 built a continuous geometric arc from hyperbolic models through discrete isometry groups and quotient geometry. VII/38 changes viewpoint: geodesics are no longer sought on smooth model spaces alone, but on triangulated and computational representations. The next part studies discrete geodesic problems and the algorithms used to approximate or compute them.

37.12 Exercises

Exercise 37.12. Define a Kleinian group.

Hint. Change the ambient group from $\mathrm{PSL}(2, \mathbb{R})$ to $\mathrm{PSL}(2, \mathbb{C})$. □

Solution. A Kleinian group is a discrete subgroup $\Gamma < \mathrm{PSL}(2, \mathbb{C})$. □

Exercise 37.13. On which two spaces does a Kleinian group naturally act?

Hint. Use the interior and its ideal boundary. □

Solution. It acts on $\mathbb{H}^3$ by isometries and on $\widehat{\mathbb{C}}$ by Möbius transformations. □

Exercise 37.14. Define the limit set.

Hint. Use accumulation points of an interior orbit. □

Solution. $\Lambda(\Gamma)$ is the set of accumulation points on $\widehat{\mathbb{C}}$ of an orbit Γp , $p \in \mathbb{H}^3$. □

Exercise 37.15. Define the domain of discontinuity.

Hint. Take the boundary complement of the limit set. □

Solution. $\Omega(\Gamma) = \widehat{\mathbb{C}} \setminus \Lambda(\Gamma)$. □

Exercise 37.16. How does Γ act on $\Omega(\Gamma)$?

Hint. This is the tame part of the boundary. □

Solution. Properly discontinuously. ☐

Exercise 37.17. Is the limit set dependent on the chosen interior base point?

Hint. Use bounded distance between two base-point orbits. ☐

Solution. No. For an infinite Kleinian group it is independent of the chosen point in $\mathbb{H}^3$. ☐

Exercise 37.18. State two basic properties of $\Lambda(\Gamma)$.

Hint. Think topology and invariance. ☐

Solution. It is closed and Γ -invariant. ☐

Exercise 37.19. Define an elementary Kleinian group.

Hint. Count limit-set points. ☐

Solution. It is elementary when $|\Lambda(\Gamma)| \leq 2$. ☐

Exercise 37.20. What is the limit set of an infinite cyclic parabolic group?

Hint. All iterates approach the unique parabolic fixed point in one time direction. ☐

Solution. The single parabolic fixed point. ☐

Exercise 37.21. What is the limit set of an infinite cyclic loxodromic group?

Hint. Use attracting and repelling fixed points. ☐

Solution. The two fixed points of the loxodromic generator. ☐

Exercise 37.22. What can be said about the limit set of a non-elementary Kleinian group?

Hint. It is much larger than two points. ☐

Solution. It is infinite; in fact it is perfect and contains a dense set of loxodromic fixed points. ☐

Exercise 37.23. Where are loxodromic fixed points dense?

Hint. Use the density theorem. ☐

Solution. In $\Lambda(\Gamma)$ for non-elementary Γ . ☐

Exercise 37.24. How is a Fuchsian group viewed as a Kleinian group?

Hint. Include real matrices among complex matrices. ☐

Solution. Via $\mathrm{PSL}(2, \mathbb{R}) \subset \mathrm{PSL}(2, \mathbb{C})$. ☐

Exercise 37.25. What boundary set does such a Fuchsian group preserve?

Hint. Use the real projective line. ☐

Solution. A generalized circle, in the standard model $\mathbb{R} \cup \{\infty\}$. ☐

Exercise 37.26. Define the convex hull of the limit set informally.

Hint. Use hyperbolic convexity. ☐

Solution. It is the smallest closed hyperbolically convex subset of $\mathbb{H}^3$ whose ideal boundary contains $\Lambda(\Gamma)$. $\square$

Exercise 37.27. Define the convex core.

Hint. Take the quotient of the hull. $\square$

Solution. $\text{Core}(M) = \Gamma \backslash \text{Hull}(\Lambda(\Gamma))$. $\square$

Exercise 37.28. What is $\Gamma \backslash \mathbb{H}^3$ when Γ is torsion-free?

Hint. Use free proper action. $\square$

Solution. An orientable hyperbolic three-manifold. $\square$

Exercise 37.29. What changes when torsion is present?

Hint. Fixed points create singular quotient structure. $\square$

Solution. The natural quotient is a hyperbolic orbifold. $\square$

Exercise 37.30. What group elements produce cusp ends?

Hint. Use ideal fixed points with zero translation length. $\square$

Solution. Parabolic elements and their cusp subgroups. $\square$

Exercise 37.31. What group elements produce closed geodesics?

Hint. Project invariant axes. $\square$

Solution. Loxodromic elements. $\square$

Exercise 37.32. Define the conformal boundary.

Hint. Quotient the domain of discontinuity. $\square$

Solution. $\partial_c M = \Gamma \backslash \Omega(\Gamma)$, when this quotient is nonempty. $\square$

Exercise 37.33. Why is the conformal boundary a Riemann surface or orbifold?

Hint. Use the nature of boundary maps. $\square$

Solution. Because Möbius transformations are conformal, so the complex structure descends. $\square$

Exercise 37.34. What kind of limit set occurs for a classical Schottky group?

Hint. Think free loxodromic dynamics. $\square$

Solution. Typically a Cantor-type fractal limit set. $\square$

Exercise 37.35. What topic begins in VII/38?

Hint. The smooth hyperbolic arc ends here. $\square$

Solution. Discrete geodesic problems on computational and triangulated geometries. $\square$

37.13 Solved problem dossiers

Problem 37.36 (Cyclic loxodromic limit set). Let $g(z) = 2z$. Determine $\Lambda(\langle g \rangle)$ and $\Omega(\langle g \rangle)$.

Solution. For every $z \neq 0, \infty$, forward iterates tend to ∞ and backward iterates tend to 0. Thus $\Lambda = \{0, \infty\}$ and $\Omega = \widehat{\mathbb{C}} \setminus \{0, \infty\} = \mathbb{C}^*$. $\square$

Problem 37.37 (Cyclic parabolic limit set). Let $g(z) = z + 1$. Determine its limit set.

Solution. The unique fixed point on the sphere is ∞ . For every finite z , $g^n(z) = z + n \rightarrow \infty$ as $|n| \rightarrow \infty$ in the spherical topology. Hence $\Lambda = \{\infty\}$. $\square$

Problem 37.38 (Why the limit set is invariant). Prove that $\Lambda(\Gamma)$ is Γ -invariant.

Solution. If $x = \lim \gamma_n p$ and $h \in \Gamma$, then by continuity on the compactification, $hx = \lim h\gamma_n p$. Since $h\gamma_n \in \Gamma$, the point hx is again a limit point. Applying the same argument to h^{-1} gives equality. $\square$

Problem 37.39 (Base-point independence). Explain why the limit set does not depend on the chosen interior point p .

Solution. For $p, q \in \mathbb{H}^3$, isometry invariance gives $d(\gamma p, \gamma q) = d(p, q)$ for all γ . Two sequences staying at bounded hyperbolic distance converge to the same ideal point, so the orbit accumulation sets coincide. $\square$

Problem 37.40 (Fuchsian limit set inside a circle). Let $\Gamma < \mathrm{PSL}(2, \mathbb{R})$ be viewed as Kleinian. Show $\Lambda(\Gamma) \subset \mathbb{R} \cup \{\infty\}$.

Solution. The group preserves the totally geodesic plane $P \cong \mathbb{H}^2$ whose ideal boundary is $\mathbb{R} \cup \{\infty\}$. Choosing the orbit base point inside P , every orbit point lies in P , so all ideal accumulation points lie in $\partial_\infty P$. $\square$

Problem 37.41 (Closed geodesic from a loxodromic). Let Γ be torsion-free and contain a primitive loxodromic g . Explain the resulting closed geodesic in $M = \Gamma \backslash \mathbb{H}^3$.

Solution. The loxodromic preserves its unique axis and translates along it by $\ell > 0$. Quotienting the axis by $\langle g \rangle$ gives a circle of length ℓ , which maps locally isometrically to a closed geodesic in M . $\square$

Problem 37.42 (Conformal boundary of a cyclic loxodromic group). For $g(z) = qz$, $|q| > 1$, describe $\Omega/\langle g \rangle$.

Solution. Here $\Omega = \mathbb{C}^*$. Passing to logarithmic coordinate $u = \log z$, multiplication by q becomes translation by $\log q$, while $u \sim u + 2\pi i$. The quotient is therefore a complex torus when the two periods are independent over $\mathbb{R}$. $\square$

Problem 37.43 (When conformal boundary is empty). What happens if $\Lambda(\Gamma) = \widehat{\mathbb{C}}$?

Solution. Then $\Omega(\Gamma) = \emptyset$, so there is no conformal-boundary quotient. The boundary action is dynamically recurrent everywhere. $\square$

Problem 37.44 (Elementary versus non-elementary dynamics). Contrast a cyclic loxodromic group with a non-elementary Kleinian group.

Solution. The cyclic loxodromic group has exactly two limit points and completely organized north-south dynamics. A non-elementary group has an infinite perfect limit set, with loxodromic fixed points dense and many competing attracting-repelling pairs. $\square$

Problem 37.45 (Convex hull for two points). What is the convex hull in $\mathbb{H}^3$ of a two-point limit set?

Solution. The smallest closed convex set with those two ideal endpoints is the unique geodesic joining them. Thus the convex core of a cyclic loxodromic quotient is the closed geodesic obtained from its axis. $\square$

Problem 37.46 (Why Schottky groups are non-elementary). A rank-two classical Schottky group has a Cantor limit set. Explain why it is non-elementary.

Solution. A Cantor set is infinite, so the limit set has more than two points. By definition the group is therefore non-elementary. $\square$

Problem 37.47 (Minimality of the limit set). Suppose Γ is non-elementary and $C \subset \hat{\mathbb{C}}$ is nonempty, closed, and Γ -invariant. Explain why one expects $\Lambda(\Gamma) \subset C$.

Solution. The attracting and repelling fixed points of loxodromic elements lie in every nonempty closed invariant set once one iterates suitable points. Since those fixed points are dense in $\Lambda(\Gamma)$, closedness forces $\Lambda(\Gamma) \subset C$. $\square$

37.14 Challenge problems

Problem 37.48 (A fractal versus a circle). Compare the qualitative boundary geometry of a cofinite Fuchsian group and a classical Schottky group.

Solution. A cofinite Fuchsian group has limit set equal to its invariant circle, leaving two disk components in the sphere when regarded in three dimensions. A classical Schottky group has a Cantor-type limit set, so its domain of discontinuity is connected in the standard classical situation and has a highly nontrivial quotient. The contrast shows how discreteness alone permits radically different boundary geometry. $\square$

Problem 37.49 (Convex-core viewpoint). Why is the convex core often a better finite geometric object than the whole quotient $\Gamma \backslash \mathbb{H}^3$?

Solution. The whole quotient may contain infinite ends extending toward components of the conformal boundary. The convex core removes those geometrically redundant outer regions and retains the quotient of the convex hull of the recurrent limit-set dynamics. For geometrically finite groups it is controlled by finite-type data up to cusp structure. $\square$

Problem 37.50 (Parabolic degeneration intuition). Explain qualitatively how a loxodromic element can degenerate toward a parabolic one in a family of Kleinian groups.

Solution. As translation length tends to zero, the two loxodromic fixed points may coalesce into one ideal fixed point. In the limit the invariant axis disappears as a two-ended object and the isometry becomes parabolic. This mechanism underlies cusp formation in deformation theory. $\square$

Problem 37.51 (Boundary quotient as complex geometry). Why does Kleinian theory naturally connect hyperbolic three-geometry to Riemann surfaces?

Solution. The interior quotient is hyperbolic three-dimensional, while the tame boundary region Ω carries the standard complex structure of the Riemann sphere. Since the group acts there by Möbius transformations, the quotient Ω/Γ inherits a Riemann-surface or orbifold structure. Thus one group controls both three-dimensional hyperbolic and two-dimensional conformal geometry. $\square$

Problem 37.52 (Transition to computation). Give a precise conceptual reason that discrete geodesic algorithms form a natural next part after Kleinian groups.

Solution. The preceding chapters show how global geometry can be encoded by discrete algebraic data acting on smooth spaces. Computational geometry makes a complementary discretization: the space itself is represented by a mesh or graph, and geodesic questions become finite optimization or PDE problems. The transition preserves the central theme of recovering continuous geometric structure from discrete information. $\square$

Part VII

Computational Geometry

Chapter 38

Discrete Geodesic Problems

A smooth geodesic is defined by a differential equation, but a triangular mesh is only piecewise smooth. The computational problem is therefore not to discretize the geodesic equation blindly. One first decides what metric the mesh carries, what class of paths is admissible, and what notion of accuracy is required.

38.1 Triangulated piecewise-Euclidean surfaces

Definition 38.1 (Triangulated surface). A triangulated surface is a two-dimensional simplicial complex $M = (V, E, F)$ whose triangles are assigned Euclidean edge lengths compatible on shared edges. Each face is therefore a Euclidean triangle, while curvature is concentrated at vertices.

The metric is *intrinsic*: once all edge lengths are fixed, distances can be measured without referring to an embedding in $\mathbb{R}^3$. Two bent realizations of the same intrinsically isometric mesh have the same intrinsic distance function.

Definition 38.2 (Piecewise-Euclidean path length). For a rectifiable path γ crossing finitely many faces, its length is the sum of the Euclidean lengths of its facewise pieces. The intrinsic distance is

$$d_M(p, q) = \inf_{\gamma} L(\gamma),$$

where the infimum runs over all admissible surface paths from p to q .

Proposition 38.3 (Metric property). *On a connected piecewise-Euclidean surface, d_M is a metric. If boundary is present, paths are constrained to remain on the surface, including its boundary.*

38.2 Continuous geodesics versus edge-graph paths

A crucial distinction is between the continuous surface distance and the shortest path in the mesh edge graph.

Definition 38.4 (Edge-graph distance). For vertices $u, v \in V$, the edge-graph distance $d_E(u, v)$ is the minimum sum of edge lengths over edge paths from u to v .

Every edge path is a valid surface path, hence

$$d_M(u, v) \leq d_E(u, v).$$

The inequality can be strict because a shortest surface path may cut through triangle interiors. Thus graph shortest paths provide an upper bound, not in general the exact polyhedral distance.

Example 38.5 (A diagonal shortcut). If two vertices lie opposite one another across a nearly flat strip of triangles, a surface geodesic can cross face interiors almost as a straight Euclidean segment, while an edge-only path zigzags along the triangulation.

38.3 Local structure away from vertices

Inside one face, a locally shortest path is a straight segment. Across an interior edge, the corresponding statement is obtained by unfolding.

Theorem 38.6 (Unfolding condition). *Suppose a locally shortest path crosses an interior edge at a point that is not a vertex. Rotate one incident triangle into the plane of the other around the shared edge. The two facewise path segments become one straight segment in the unfolded plane.*

Proof. If the unfolded segments met with a nonzero turning angle, replacing them by the straight chord would shorten the path while remaining in the same two faces. $\square$

This gives the polyhedral analogue of zero geodesic curvature away from mesh vertices.

38.4 Vertices, cone angles, and discrete curvature

For an interior vertex v , let

$$\Theta_v = \sum_{f \ni v} \theta_f(v)$$

be the total incident angle.

Definition 38.7 (Angle defect). The discrete Gaussian curvature at an interior vertex is the angle defect

$$K_v = 2\pi - \Theta_v.$$

A vertex is called positive, flat, or saddle according as $\Theta_v < 2\pi$, $\Theta_v = 2\pi$, or $\Theta_v > 2\pi$.

The local geometry is that of a Euclidean cone. Shortest paths generally avoid positive-curvature cone tips. Saddle vertices can play a different role: multiple locally shortest continuations may emanate from them after sufficient angle is available on both sides of the path.

Warning 38.8 (Do not impose a smooth geodesic equation at vertices). Christoffel symbols are not defined at a cone singularity. Vertex behavior must be analyzed by cone geometry and unfolding, not by differentiating a smooth metric that the mesh does not possess.

38.5 Source and query models

Algorithms differ substantially depending on the query.

Definition 38.9 (Geodesic query types). Common tasks include:

- (i) point-to-point distance and path recovery;
- (ii) single-source distances to all vertices or all surface points;

- (iii) multi-source distance $d_S(x) = \min_{s \in S} d_M(s, x)$;
- (iv) repeated queries after preprocessing;
- (v) a full distance field for downstream optimization or geometry processing.

Sources and targets need not be vertices. A point in a triangle may be stored by barycentric coordinates, and an edge point by one scalar parameter. Forcing all queries to the nearest vertex changes the problem before the solver even starts.

38.6 Wavefronts and the cut locus

For a fixed source s , level sets

$$W_r = \{x \in M : d_M(s, x) = r\}$$

behave like expanding wavefronts. On a flat region they are locally circular. They bend when they encounter cone curvature and can split or collide because of topology.

Definition 38.10 (Cut locus, computational viewpoint). The cut locus of s is the set where the minimizing geodesic from s ceases to be locally unique or ceases to remain minimizing beyond that point.

Distance is continuous everywhere but is generally not differentiable on the cut locus. This matters numerically: a distance algorithm may have small scalar error while the recovered path direction changes abruptly near a cut-locus branch.

38.7 Exact and approximate problem statements

Definition 38.11 (Additive and relative error). For an estimate $\tilde{d}$ of a true distance $d > 0$, common scalar errors are

$$|\tilde{d} - d|, \quad \frac{|\tilde{d} - d|}{d}.$$

A multiplicative $(1 + \varepsilon)$ -upper approximation satisfies

$$d \leq \tilde{d} \leq (1 + \varepsilon)d.$$

Path error is subtler than distance error. Two geometrically different paths can have almost the same length, especially near the cut locus. A useful benchmark therefore reports both length error and a geometric discrepancy such as Hausdorff distance after suitable sampling.

38.8 Triangulation dependence and convergence

The piecewise-Euclidean metric is itself a geometric model. Refining a mesh does not automatically guarantee convergence to a smooth surface metric; the vertices, edge lengths, and shape quality must approximate the intended smooth geometry.

Proposition 38.12 (Intrinsic invariance). *If two triangulated surfaces are related by an intrinsic simplicial isometry that preserves every face edge length, then all piecewise-Euclidean geodesic distances correspond exactly, even if the two surfaces have different embeddings.*

Remark 38.13. Skinny triangles can amplify numerical error in gradients, unfoldings, and discrete differential operators. Mesh quality is therefore both a geometric and an algorithmic issue.

38.9 Algorithmic design axes

A geodesic solver should be described along several independent axes:

- (1) exact versus approximate;
- (2) edge-restricted versus continuous-on-faces;
- (3) point-to-point versus single-source versus multi-source;
- (4) preprocessing cost versus per-query cost;
- (5) scalar distances only versus explicit path recovery;
- (6) worst-case guarantees versus practical performance.

No single method dominates every axis. Edge-graph algorithms are simple and robust, exact polyhedral algorithms preserve the continuous mesh metric, and PDE methods trade exactness for smooth global computations and sparse linear algebra.

38.10 Exercises

Exercise 38.14. Why is the metric of a triangle mesh called intrinsic?

Hint. Ask which data determine lengths after the mesh is bent in space. □

Solution. Because the piecewise-Euclidean metric is determined by the face edge lengths and gluings, not by the particular embedding in ambient space. □

Exercise 38.15. Define the intrinsic distance $d_M(p, q)$.

Hint. Use an infimum over surface paths. □

Solution. $d_M(p, q) = \inf_{\gamma} L(\gamma)$, where γ ranges over admissible surface paths joining p to q . □

Exercise 38.16. Why is a shortest path inside one face a straight segment?

Hint. Compare any bent planar curve with its chord. □

Solution. Each face is Euclidean, and the straight chord is no longer than any other curve with the same endpoints. □

Exercise 38.17. State the unfolding condition at an interior edge.

Hint. Flatten the two incident triangles into one plane. □

Solution. After one incident face is rotated into the plane of the other, the two path segments of a locally shortest crossing form one straight segment. □

Exercise 38.18. For vertices u, v , compare $d_M(u, v)$ and $d_E(u, v)$.

Hint. Every graph path is also a surface path. □

Solution. $d_M(u, v) \leq d_E(u, v)$. □

Exercise 38.19. Why can the graph inequality be strict?

Hint. An intrinsic path is not required to follow edges. □

Solution. A continuous shortest path may cross triangle interiors and cut across a zigzagging edge graph. □

Exercise 38.20. Define the total angle Θ_v at an interior vertex.

Hint. Sum all incident face angles. □

Solution. $\Theta_v = \sum_{f \ni v} \theta_f(v)$. □

Exercise 38.21. Define the angle defect at an interior vertex.

Hint. Compare the total angle with 2π . □

Solution. $K_v = 2\pi - \Theta_v$. □

Exercise 38.22. What is a saddle vertex in the piecewise-Euclidean sense?

Hint. Look for more than one full turn of incident angle. □

Solution. It is an interior vertex with $\Theta_v > 2\pi$, equivalently negative angle defect. □

Exercise 38.23. Why should one not write the smooth geodesic equation at a cone vertex?

Hint. The metric is not smooth there. □

Solution. The cone tip is a metric singularity, so smooth Christoffel-symbol calculus does not apply at the vertex. □

Exercise 38.24. What is a point-to-point geodesic query?

Hint. Specify both output types that may be requested. □

Solution. It asks for the intrinsic distance between two specified surface points and often also an explicit minimizing path. □

Exercise 38.25. What is a single-source query?

Hint. Fix one source and vary the target. □

Solution. It computes distances from one fixed source to many or all target locations, often producing a distance field. □

Exercise 38.26. Define multi-source distance.

Hint. Take the nearest source. □

Solution. For a source set S , $d_S(x) = \min_{s \in S} d_M(s, x)$. □

Exercise 38.27. Why are barycentric coordinates useful for geodesic queries?

Hint. A source may lie in a face interior. □

Solution. They represent arbitrary points inside a triangle without snapping them to mesh vertices. □

Exercise 38.28. What is a distance wavefront?

Hint. Use a level set of the distance function. □

Solution. It is $W_r = \{x : d_M(s, x) = r\}$ for a fixed source s . □

Exercise 38.29. What computational phenomenon occurs at the cut locus?

Hint. Think about uniqueness of minimizing paths. □

Solution. The minimizing path can cease to be unique, and the distance field can lose differentiability there. □

Exercise 38.30. Give an additive distance error.

Hint. Subtract estimate and truth. □

Solution. $|\tilde{d} - d|$. □

Exercise 38.31. Give a relative distance error.

Hint. Normalize the additive error. □

Solution. $|\tilde{d} - d|/d$ for $d > 0$. □

Exercise 38.32. What does a $(1 + \varepsilon)$ -upper approximation guarantee?

Hint. It should never underestimate. □

Solution. $d \leq \tilde{d} \leq (1 + \varepsilon)d$. □

Exercise 38.33. Why can path geometry be unstable even when distance error is tiny?

Hint. Consider nearly equal competing geodesics. □

Solution. Near the cut locus, distinct paths can have almost equal lengths, so a small scalar perturbation can switch which path is selected. □

Exercise 38.34. What does intrinsic isometry preserve for a triangulated surface?

Hint. Preserve the piecewise-Euclidean metric. □

Solution. It preserves every intrinsic path length and hence every intrinsic geodesic distance. □

Exercise 38.35. Why can poor triangle quality hurt a numerical geodesic method?

Hint. Think of conditioning and geometric reconstruction. □

Solution. Very skinny triangles can make gradients, unfoldings, and discrete operators ill-conditioned or sensitive to roundoff. □

Exercise 38.36. Name two distinct algorithmic design axes for a geodesic solver.

Hint. Examples include exactness and query model. □

Solution. For example: exact versus approximate, and point-to-point versus single-source; preprocessing and path recovery are additional independent axes. □

Exercise 38.37. What is the main conceptual distinction that VII/39 will exploit?

Hint. Separate graph restriction from continuous face geometry. □

Solution. VII/39 compares edge-graph shortest paths with algorithms that compute exact continuous geodesics on a polyhedral mesh. □

38.11 Solved problem dossiers

Problem 38.38 (A shortcut across two triangles). Two coplanar triangles form a square of side 1 split by a diagonal. Compute the intrinsic distance between opposite square corners and compare it with a boundary-edge path.

Solution. The intrinsic surface is the Euclidean square, so the distance is the diagonal length $\sqrt{2}$. A boundary-edge path has length 2. This simple example shows why restricting paths to selected mesh edges can overestimate the continuous distance. $\square$

Problem 38.39 (Unfolding an edge crossing). A path crosses the common edge of two triangles. Explain how to test whether the crossing can be locally shortest.

Solution. Rigidly rotate one triangle around the shared edge until both triangles lie in one plane. If the two path pieces do not form one straight segment after unfolding, replacing them by the straight segment shortens the path. Straightness in the unfolded plane is therefore the local optimality test. $\square$

Problem 38.40 (Positive cone curvature). An interior vertex has total angle $3\pi/2$. Compute its angle defect and classify the vertex.

Solution. The defect is $2\pi - 3\pi/2 = \pi/2 > 0$. It is a positive-curvature cone vertex. $\square$

Problem 38.41 (A saddle cone). An interior vertex has total angle $5\pi/2$. Compute the angle defect and explain why the local geometry differs from a convex cone tip.

Solution. The defect is $2\pi - 5\pi/2 = -\pi/2$. There is more than a full 2π of angle around the vertex, so the cone is saddle-like; shortest-path continuations can have angular room unavailable at a positive-curvature tip. $\square$

Problem 38.42 (A face-interior source). A solver accepts only vertex sources, but the desired source lies at barycentric coordinates $(1/4, 1/4, 1/2)$ in a triangle. Why is snapping to the nearest vertex a modeling error rather than merely an implementation detail?

Solution. Snapping changes the source point and therefore changes the distance function being computed. The resulting error is not the numerical error of the geodesic algorithm; it is an error introduced before the intended problem is solved. $\square$

Problem 38.43 (Disconnected input). A mesh has two connected components and the source and target lie on different components. What should a geodesic-distance routine report?

Solution. No admissible surface path joins the points, so the intrinsic distance is $+\infty$ or an explicit unreachable status. Returning a large finite number would confuse topology with distance. $\square$

Problem 38.44 (Graph upper bound). A graph solver returns length 8.4 between two vertices. An exact continuous solver returns 7.9. Is this ordering theoretically consistent?

Solution. Yes. Edge paths form a subset of all surface paths, so the graph optimum cannot be smaller than the continuous optimum. Here $7.9 \leq 8.4$, exactly as expected. $\square$

Problem 38.45 (Multi-source Voronoi distance). For sources s_1, s_2, s_3 , define the scalar field used to assign each point to its nearest source and describe where assignment ambiguity occurs.

Solution. Use $d_S(x) = \min_i d_M(s_i, x)$. Ambiguity occurs on loci where two or more source distances tie; these loci form intrinsic Voronoi boundaries. $\square$

Problem 38.46 (Cut-locus sensitivity). Two minimizing geodesics from s to x have equal length. What should one expect from a small perturbation of x ?

Solution. The tie can break immediately, selecting one branch or the other. The distance remains continuous, but its gradient need not be continuous or even uniquely defined at the original point. $\square$

Problem 38.47 (Embedding versus metric). A triangulated sheet is bent isometrically in $\mathbb{R}^3$ without stretching any triangle. What happens to its intrinsic geodesic distances?

Solution. They remain unchanged. The bending modifies extrinsic geometry but preserves all face edge lengths and gluings, hence the intrinsic piecewise-Euclidean metric. $\square$

Problem 38.48 (Relative-error interpretation). The true distance is 20 and an approximation returns 20.3. Compute the additive and relative errors.

Solution. The additive error is 0.3. The relative error is $0.3/20 = 0.015$, or 1.5%. $\square$

Problem 38.49 (Choosing a solver). A pipeline needs thousands of approximate single-source distance fields on one fixed mesh and can tolerate small error. Why might an exact point-to-point method be a poor default?

Solution. The required output is a global field and the query is repeated. Methods based on sparse factorization or other reusable preprocessing can amortize setup cost, whereas an exact path algorithm optimized for isolated queries may spend unnecessary work recovering combinatorial path structure. $\square$

38.12 Challenge problems

Problem 38.50 (Local vertex criterion). Unfold a neighborhood of a cone vertex to a planar sector of angle Θ . Explain why a path through the tip can be locally minimizing only if each side of the path has unfolded angle at least π .

Solution. Cut the cone along the candidate path and develop the two sides into planar sectors. If one side has angle $< \pi$, then in that sector a small shortcut joining points on the incoming and outgoing rays avoids the tip and is shorter. Thus each side must contribute at least π . In particular a positive-curvature vertex with total angle $< 2\pi$ cannot satisfy the condition. $\square$

Problem 38.51 (Distance versus path convergence). Construct the qualitative argument for why convergence of scalar distances under mesh refinement does not by itself imply convergence of a chosen minimizing path near a cut locus.

Solution. Near a cut locus there are at least two competing geodesic branches with equal or nearly equal length. A sequence of refined meshes can approximate the common distance increasingly well while alternating which branch is microscopically shorter. Scalar values converge, yet the selected paths can jump between macroscopically different branches. $\square$

Problem 38.52 (A benchmark protocol). Design a minimal benchmark for a mesh geodesic solver that separates modeling error, numerical distance error, and path-recovery error.

Solution. Use a mesh with a known piecewise-Euclidean reference solution and keep the query points fixed without snapping. Report the reference metric and mesh resolution first; then compare scalar distance error against an exact or certified solution; finally compare recovered paths by length and a geometric discrepancy such as sampled Hausdorff distance. Repeating under controlled refinement separates discretization/modeling effects from solver error. $\square$

Problem 38.53 (Refinement does not automatically help). Explain why subdividing every triangle by inserting poorly placed vertices can fail to improve a geodesic computation in practice even though the mesh has more triangles.

Solution. Triangle count alone is not a convergence hypothesis. Poorly placed vertices can create extreme aspect ratios and ill-conditioned local geometry; if the intrinsic metric is also perturbed, the modeled surface may not approach the intended target. Useful refinement controls both geometric approximation and element quality. $\square$

Problem 38.54 (Solver decision tree). Give a principled decision tree for choosing among edge-graph, exact polyhedral, and PDE-style geodesic methods.

Solution. First decide whether the continuous piecewise-Euclidean metric or an edge-restricted approximation is required. If exact continuous distances or certified paths are essential, choose an exact polyhedral method. If robustness and simplicity dominate and graph error is acceptable, use graph search. If many smooth global distance fields are needed and approximation is acceptable, a PDE method is attractive, especially when sparse factorizations can be reused. Then refine the choice by source type, preprocessing budget, path-recovery need, and error tolerance. $\square$

38.13 Bridge to the next chapter

VII/39 turns the problem taxonomy into algorithms. It begins with weighted edge graphs, Dijkstra's algorithm, and admissible A^* heuristics, then returns to the continuous polyhedral metric through unfolding, face sequences, and exact wavefront propagation.

Chapter 39

Graph and Exact Mesh Geodesics

The first computational reduction of a mesh geodesic problem is to a weighted graph. The reduction is powerful because shortest paths on nonnegative weighted graphs are algorithmically mature. It is also a geometric approximation because the true piecewise-Euclidean shortest path is not generally constrained to mesh edges.

39.1 Weighted edge graphs

Let $G = (V, E, w)$ be the mesh one-skeleton with

$$w_{ij} = \|v_i - v_j\|$$

equal to the intrinsic length of the edge (i, j) .

Definition 39.1 (Graph geodesic). A graph geodesic from s to t is a minimum-weight edge path

$$s = v_{i_0}, v_{i_1}, \dots, v_{i_k} = t, \quad L = \sum_{r=0}^{k-1} w_{i_r i_{r+1}}.$$

For vertex endpoints,

$$d_M(s, t) \leq d_G(s, t),$$

because every graph path is a valid surface path.

39.2 Dijkstra's algorithm

Theorem 39.2 (Dijkstra correctness). *For a graph with nonnegative edge weights, Dijkstra's algorithm permanently settles vertices in nondecreasing shortest-path distance from the source. When a vertex u is settled, its tentative label equals the true graph distance $d_G(s, u)$.*

The proof is a greedy cut argument: if a shorter unsettled route existed, the first edge crossing from settled to unsettled vertices would have produced a smaller tentative label.

With a binary heap, a standard implementation runs in

$$O((|V| + |E|) \log |V|),$$

which is $O(|V| \log |V|)$ for a triangular manifold mesh because $|E| = O(|V|)$.

39.3 Multi-source propagation and Voronoi labels

Multiple sources can be handled by initializing every source with distance zero.

Proposition 39.3 (Multi-source Dijkstra). *If all sources $S \subset V$ are inserted into the priority queue with label zero, Dijkstra computes*

$$d_G(S, v) = \min_{s \in S} d_G(s, v)$$

for all vertices v .

Recording which source first attains each optimum yields a graph Voronoi partition.

39.4 A^* search

For a fixed target t , A^* orders states by

$$f(v) = g(v) + h(v),$$

where $g(v)$ is the best known source-to- v cost and $h(v)$ estimates the remaining cost to t .

Definition 39.4 (Admissible and consistent heuristics). A heuristic is admissible if

$$0 \leq h(v) \leq d_G(v, t).$$

It is consistent if for every edge (u, v) ,

$$h(u) \leq w_{uv} + h(v).$$

A consistent heuristic is admissible when $h(t) = 0$, and it makes the f -value nondecreasing along an optimal search frontier.

Example 39.5 (Euclidean chord heuristic). For a mesh embedded in Euclidean space,

$$h(v) = \|v - t\|_{\mathbb{R}^3}$$

is a lower bound on any surface or edge path length, hence is admissible for the edge graph.

39.5 Limits of edge restriction

Increasing graph-search sophistication does not remove geometric bias introduced by the edge graph. Dijkstra and A^* solve the same discrete graph problem exactly; they differ only in how much of the graph they explore.

Warning 39.6 (Exact graph search is not exact surface geodesy). An exact answer for the edge graph can still be a poor approximation to the continuous polyhedral distance if the triangulation directions force a zigzagging path.

Adding Steiner points or denser directional connections can reduce this bias, but also enlarges the graph.

39.6 Face sequences and unfolding

To compute continuous polyhedral geodesics, one must allow a path to cross face interiors. Suppose a candidate path crosses a sequence of faces

$$f_1, f_2, \dots, f_k$$

without passing through a mesh vertex. Successively unfold these faces into the plane.

Theorem 39.7 (Straight line in an unfolded strip). *A locally shortest path whose combinatorial face sequence is fixed unfolds to a straight line segment across the planar strip of unfolded triangles.*

Thus, if the face sequence were known, the geometric path could be recovered by a planar straight-line construction and folded back to the surface. The hard part is discovering which face sequences can contain a globally shortest path.

39.7 Vertex events and pseudo-sources

At a saddle vertex, shortest-path wavefronts can create new branches. Exact algorithms account for such events rather than treating every path as a simple sequence of generic edge crossings.

Definition 39.8 (Pseudo-source, conceptual). A pseudo-source is a mesh vertex at which an exact wavefront representation creates new locally valid propagation branches, typically because the total incident angle permits shortest paths to pass through the vertex.

Positive-curvature vertices do not admit a locally minimizing path through the tip under the two-sided angle criterion from VII/38.

39.8 Windows on edges

Exact polyhedral methods propagate not merely one scalar label per vertex but continuous distance information along edge intervals.

Definition 39.9 (Window). A window is an interval on a mesh edge together with sufficient data to express the distance to points of that interval through one unfolded source image or pseudo-source branch.

Inside one unfolded configuration, the distance typically has the Euclidean form

$$d(x) = d_0 + \|x - \tilde{s}\|,$$

where $\tilde{s}$ is an unfolded source image and d_0 is the already accumulated distance to a pseudo-source when present.

Competing windows can overlap. Dominated portions are trimmed because only the lower envelope can contribute to the exact distance field.

39.9 Exact wavefront algorithms

Classical exact algorithms such as the Mitchell–Mount–Papadimitriou framework and the Chen–Han framework differ in data structures and propagation details, but share core geometry:

- (i) propagate continuous intervals across edges;
- (ii) unfold local face configurations;
- (iii) keep distance formulas attached to source images or pseudo-sources;
- (iv) prune dominated information;
- (v) backtrack through the winning propagation chain to recover a path.

The important conceptual contrast with Dijkstra is that an exact continuous solver stores enough information to represent where the wavefront crosses an edge, not only which mesh vertex has been reached.

39.10 Boundary, degeneracy, and robustness

A shortest path may run along a boundary edge, hit an edge endpoint, or encounter nearly simultaneous window events. Exact combinatorial decisions can therefore be sensitive to floating-point error.

Warning 39.10 (Robust predicates). A mathematically exact algorithm is not automatically numerically robust. Implementations need consistent orientation tests, tolerances or exact predicates, and deterministic tie handling for degenerate configurations.

39.11 Graph versus exact polyhedral methods

Feature	Edge graph	Exact polyhedral
Path locations	mesh edges	face interiors allowed
Core state	vertex labels	edge intervals/windows
Typical implementation	simple	substantially more complex
Distance on mesh metric	upper approximation	exact in principle
Path recovery	predecessor vertices	unfolded propagation chain

For many interactive applications the graph method is an excellent baseline. When exact intrinsic distances matter, however, continuous propagation is the correct reference.

39.12 Exercises

Exercise 39.11. What weights should the mesh edge graph use for intrinsic shortest paths?

Hint. Use the intrinsic length of each edge. ☐

Solution. Assign each edge (i, j) the piecewise-Euclidean edge length w_{ij} . ☐

Exercise 39.12. What assumption on edge weights is needed by Dijkstra's algorithm?

Hint. Negative edges break the greedy settling argument. ☐

Solution. All edge weights must be nonnegative. ☐

Exercise 39.13. What does it mean when Dijkstra settles a vertex?

Hint. Its tentative label becomes final. ☐

Solution. The settled label equals the true shortest graph distance from the source. ☐

Exercise 39.14. Give the standard binary-heap complexity of Dijkstra.

Hint. Include both vertices and edges. □

Solution. $O((|V| + |E|) \log |V|)$. □

Exercise 39.15. Why is this $O(|V| \log |V|)$ on a triangular manifold mesh?

Hint. Use sparsity of the one-skeleton. □

Solution. Such meshes have $|E| = O(|V|)$, so the edge term is linear in the number of vertices. □

Exercise 39.16. How do you initialize multi-source Dijkstra?

Hint. All sources start equally close. □

Solution. Set every source label to zero and insert all sources into the priority queue initially. □

Exercise 39.17. What additional data gives a graph Voronoi partition?

Hint. Track provenance. □

Solution. Store the source label or source identifier from which each winning shortest path originates. □

Exercise 39.18. Define the A^* priority key.

Hint. Add known and estimated remaining cost. □

Solution. $f(v) = g(v) + h(v)$. □

Exercise 39.19. What is an admissible heuristic?

Hint. It must not overestimate. □

Solution. A heuristic satisfying $0 \leq h(v) \leq d_G(v, t)$ for every v . □

Exercise 39.20. What is a consistent heuristic?

Hint. Use the edge triangle inequality. □

Solution. It satisfies $h(u) \leq w_{uv} + h(v)$ on every edge. □

Exercise 39.21. Why is ambient Euclidean distance to the target admissible?

Hint. A surface path cannot be shorter than its straight spatial chord. □

Solution. Every edge path has length at least the Euclidean chord distance between its endpoints. □

Exercise 39.22. Do Dijkstra and A^* compute different graph distances?

Hint. Assume an admissible heuristic and correct implementations. □

Solution. No. They compute the same optimal graph distance; A^* can explore fewer states for a fixed target. □

Exercise 39.23. Why can exact graph search overestimate the surface distance?

Hint. The admissible path class is smaller. □

Solution. The graph solver only uses edge paths, whereas the continuous surface solver may cross face interiors. □

Exercise 39.24. What is a face sequence of a surface path?

Hint. List the triangles traversed in order. □

Solution. It is the ordered sequence of mesh faces crossed by the path. □

Exercise 39.25. How does a locally shortest path look after unfolding a fixed face sequence?

Hint. Use VII/38's edge-crossing condition repeatedly. □

Solution. It becomes one straight line segment through the unfolded planar strip. □

Exercise 39.26. Why is knowing the face sequence not enough for a complete algorithm?

Hint. There are many possible sequences. □

Solution. The global problem must discover which combinatorial sequence or propagation branch actually realizes the shortest path. □

Exercise 39.27. What geometric role can a saddle vertex play?

Hint. Recall the two-sided angle criterion. □

Solution. It can act as a branching event or pseudo-source for valid shortest-path propagation. □

Exercise 39.28. Define an edge window conceptually.

Hint. It represents a continuous interval, not one vertex. □

Solution. A window is an edge interval carrying a distance formula and propagation data for one unfolded source or pseudo-source branch. □

Exercise 39.29. Give the typical distance formula inside one unfolded window.

Hint. Use an unfolded source image. □

Solution. $d(x) = d_0 + \|x - \tilde{s}\|$. □

Exercise 39.30. Why can overlapping windows be trimmed?

Hint. Only the lower envelope can be shortest. □

Solution. Where one window gives strictly larger distance than another, that dominated portion cannot contribute to the exact distance field. □

Exercise 39.31. What extra information does an exact continuous solver store compared with Dijkstra?

Hint. Think edge intervals. □

Solution. It stores continuous wavefront information along edges, including source-image or pseudo-source geometry, rather than only scalar vertex labels. □

Exercise 39.32. Why are degeneracies difficult in exact mesh geodesic codes?

Hint. Events can coincide or nearly coincide. □

Solution. Floating-point error can change ordering, intersection, and dominance decisions for nearly simultaneous or boundary events. □

Exercise 39.33. What is path recovery in a graph solver?

Hint. Store one predecessor. □

Solution. Backtrack predecessor vertices from the target to the source. □

Exercise 39.34. What does VII/40 trade away relative to exact polyhedral propagation?

Hint. It uses PDE solves. □

Solution. The heat method gives a fast smooth approximation rather than exact continuous polyhedral distances. □

39.13 Solved problem dossiers

Problem 39.35 (Dijkstra on a four-vertex graph). Edges have weights $s-a=1$, $s-b=4$, $a-b=1$, and $b-t=2$. Compute $d_G(s, t)$.

Solution. Dijkstra settles a at distance 1, improves b to 2, then reaches t with distance 4. The shortest path is $s \rightarrow a \rightarrow b \rightarrow t$. □

Problem 39.36 (Multi-source initialization). Sources s_1, s_2 both receive label zero. A vertex v is reached from s_1 with cost 5 and from s_2 with cost 3. What distance and source label should remain?

Solution. The distance is 3, and the provenance label is s_2 . Multi-source Dijkstra computes the lower envelope of all source distance functions. □

Problem 39.37 (An admissible A^* heuristic). A target is embedded in $\mathbb{R}^3$. Explain why $h(v) = \|v - t\|$ cannot overestimate the remaining graph distance.

Solution. Each graph path from v to t is a polygonal spatial curve. Its length is at least the straight Euclidean chord between its endpoints. Therefore $\|v - t\| \leq d_G(v, t)$. □

Problem 39.38 (Graph versus continuous distance). A graph path has length 12, while unfolding reveals a valid continuous face-interior path of length 10.7. What can be concluded about the exact intrinsic distance?

Solution. It is at most 10.7. The graph value 12 remains an upper bound. Additional continuous candidates could reduce the exact value below 10.7. □

Problem 39.39 (Two-face exact crossing). Two adjacent triangles are unfolded and the source and target images are joined by a straight segment crossing the common edge in its interior. Why is this the shortest path constrained to those two faces?

Solution. After unfolding, all paths through those faces have ordinary Euclidean length. The straight segment minimizes Euclidean length, and folding preserves lengths. □

Problem 39.40 (Invalid unfolded candidate). An unfolded source-target segment crosses the extension of the shared edge outside the actual edge interval. Can it represent a path with the proposed two-face sequence?

Solution. No. Folding the segment would leave the specified triangles. A valid candidate must cross every shared edge inside the corresponding edge segment. $\square$

Problem 39.41 (A convex vertex). At a vertex the total incident angle is 1.8π . Can a locally minimizing path pass through the tip with at least π of angle on both sides?

Solution. No. The two sides would require total available angle at least 2π , but only 1.8π is available. A path through this positive-curvature tip can therefore be shortened locally. $\square$

Problem 39.42 (A saddle pseudo-source). A vertex has total angle 2.6π . Explain why an exact propagation method may need to create new wavefront branches there.

Solution. There is more than 2π of angular room, so some through-vertex paths can satisfy the local two-sided π condition. The vertex can therefore transmit shortest-path information in multiple directions, motivating pseudo-source branches. $\square$

Problem 39.43 (Window dominance). Two windows overlap on an edge interval. Window A gives distance $d_A(x)$ and B gives $d_B(x)$. On a subinterval $d_A(x) < d_B(x)$ everywhere. What should an exact lower-envelope implementation do?

Solution. Discard or trim the dominated part of B on that subinterval, because it can never be the shortest arrival there. $\square$

Problem 39.44 (Boundary path). A target lies on a boundary edge and the shortest route approaches along the boundary. Why must a robust solver treat boundary edges differently from ordinary interior edges?

Solution. An interior edge has two incident faces and supports propagation across both sides. A boundary edge has only one surface side, so propagation cannot continue through it as though a second triangle existed. $\square$

Problem 39.45 (Graph refinement). Steiner points are inserted on every edge and connected within faces. What is the intended effect on graph geodesics?

Solution. The enriched graph offers more directions and more opportunities to cut across the original coarse edge pattern, so its shortest paths can better approximate continuous surface geodesics, at the cost of a larger graph. $\square$

Problem 39.46 (Choosing A^* or Dijkstra). A single target is known and a good admissible spatial heuristic is available. Which search is likely preferable, and when would Dijkstra still be attractive?

Solution. A^* is likely preferable for one target because the heuristic focuses exploration. Dijkstra is attractive for complete single-source fields, multi-source propagation, or when no useful heuristic is available. $\square$

39.14 Challenge problems

Problem 39.47 (Consistency implies monotone f). Show that if h is consistent, then along an edge relaxation from u to v on an optimal-prefix path, the A^* key cannot decrease.

Solution. Consistency gives $h(u) \leq w_{uv} + h(v)$. If $g(v) = g(u) + w_{uv}$ on the path, then $f(u) = g(u) + h(u) \leq g(u) + w_{uv} + h(v) = g(v) + h(v) = f(v)$. Thus f is nondecreasing along that path. $\square$

Problem 39.48 (Unfolding a triangle strip). Give an algorithm for testing a proposed face sequence $f_1, \dots, f_k$ for a source-target geodesic that avoids vertices.

Solution. Successively rigidly unfold the faces into one plane, preserving each shared edge. Join the unfolded source to the unfolded target by a straight segment. Check that the segment crosses the shared edges in the prescribed order and inside their edge intervals. If it does, folding back yields the unique shortest path within that fixed sequence; otherwise the sequence is invalid. $\square$

Problem 39.49 (Why one label per vertex is insufficient). Explain geometrically why an exact polyhedral single-source method cannot in general summarize its whole state by one scalar distance per mesh vertex.

Solution. A shortest wavefront crosses edges at continuously varying points, and different source images can be optimal on different subintervals of the same edge. Vertex distances do not encode these crossing locations or the lower envelope between vertices. Exact propagation therefore needs interval-valued state such as windows. $\square$

Problem 39.50 (Numerical exactness). An algorithm has a proof of exactness over real arithmetic. Explain why a floating-point implementation can nevertheless return the wrong combinatorial path.

Solution. The proof assumes exact comparisons of event times, intersections, and window dominance. Near degeneracies, floating-point rounding can reverse such comparisons, clip a valid interval, or choose the wrong branch. Robust predicates, controlled tolerances, and deterministic tie rules are needed to preserve the intended combinatorics. $\square$

Problem 39.51 (Hybrid validation strategy). Design a practical workflow that uses graph, exact, and approximate methods together rather than treating them as competitors.

Solution. Use Dijkstra or A^* as a cheap robust baseline and upper bound. Use an exact polyhedral solver on a smaller benchmark set to establish reference distances and difficult path cases. Then tune or validate an approximate global method such as the heat method against those references. This separates implementation bugs from approximation error and gives useful speed-accuracy tradeoffs. $\square$

39.15 Bridge to the next chapter

Exact wavefront propagation follows the metric faithfully but carries substantial combinatorial state. VII/40 takes a different route: diffuse heat for a short time, normalize the resulting gradient to estimate distance directions, and recover a scalar distance by one Poisson solve.

Chapter 40

The Heat Method

The heat method replaces explicit shortest-path combinatorics by two sparse elliptic solves and one local vector-field normalization. Its central insight is that very short-time heat flow contains the direction of geodesic distance even though the heat values themselves are not distances.

40.1 Short-time heat and distance

On a Riemannian surface, let $k_t(x, s)$ be the heat kernel. Short-time asymptotics have the schematic form

$$k_t(x, s) \approx C(t, x, s) \exp\left(-\frac{d(x, s)^2}{4t}\right).$$

Consequently,

$$-4t \log k_t(x, s) \longrightarrow d(x, s)^2 \quad (t \downarrow 0)$$

away from technical singularities. The heat method does not directly use this logarithmic formula. Instead it extracts the direction of increasing distance.

40.2 Step 1: diffuse heat

Fix the sign convention that the smooth Laplace–Beltrami operator $\Delta = \operatorname{div} \nabla$ is nonpositive on closed manifolds. A backward-Euler heat step solves

$$(I - t\Delta)u = \delta_s.$$

For a finite-element mesh, let M be a mass matrix and let L be the positive-semidefinite stiffness matrix representing $-\Delta$ weakly. Then a standard discrete system is

$$(M + tL)u = M\delta_s.$$

Remark 40.1 (Sign conventions). Some libraries define their discrete Laplacian with the opposite sign. The implementation equations must be changed consistently; copying formulas without checking the sign convention is a common source of errors.

40.3 Step 2: normalize the negative heat gradient

Immediately after diffusion, heat decreases away from the source. Therefore $-\nabla u$ points approximately in the direction of increasing distance.

Definition 40.2 (Normalized direction field). On each face where $\nabla u \neq 0$, define

$$X = -\frac{\nabla u}{\|\nabla u\|}.$$

For piecewise-linear finite elements, u is affine on each triangle, so ∇u is constant on each face. In practice one regularizes the normalization when $\|\nabla u\|$ is extremely small.

Warning 40.3 (Do not use heat magnitude as distance). The scalar u decays roughly exponentially with squared distance and time. It is the normalized gradient direction, not the raw heat value, that the standard heat method feeds into the reconstruction step.

40.4 Step 3: integrate the direction field

The normalized field is generally not exactly a gradient because of discretization and finite time. The method therefore finds the scalar function whose gradient best matches X in a least-squares sense.

Theorem 40.4 (Poisson reconstruction). *A minimizer of*

$$E(\phi) = \frac{1}{2} \int_M \|\nabla \phi - X\|^2 dA$$

satisfies, in the weak interior sense,

$$\Delta \phi = \operatorname{div} X.$$

Proof. Vary $\phi \mapsto \phi + \varepsilon \eta$, differentiate at zero, and integrate the term $\langle X, \nabla \eta \rangle$ by parts. The resulting Euler–Lagrange equation is the Poisson equation. $\square$

With the positive stiffness matrix L , define

$$b_i = \int_M \langle \nabla \varphi_i, X \rangle dA.$$

Then the discrete Poisson equation is

$$L\phi = b,$$

up to the usual additive constant.

40.5 Gauge fixing

On a connected closed mesh,

$$L\mathbf{1} = 0.$$

Therefore ϕ is determined only up to an additive constant.

Definition 40.5 (Distance normalization). After solving the Poisson equation, fix the constant by imposing

$$\phi(s) = 0$$

for a vertex source, or by subtracting the minimum value when a source set is used.

Pinning one degree of freedom or solving in the mean-zero subspace are standard linear-algebra choices.

40.6 Choosing the heat time

A common mesh-scale choice is

$$t = mh^2,$$

where h is a representative edge length and m is a dimensionless constant of order one.

Too small a time makes the discrete heat field highly localized and sensitive to mesh resolution and roundoff. Too large a time oversmooths directional information and introduces global bias.

Remark 40.6 (Time-scale invariance). The square scaling is dimensionally natural: diffusion distance grows like $\sqrt{t}$, so choosing $t \propto h^2$ diffuses over a length comparable with the local mesh scale.

40.7 Multiple sources

For a source set S , initialize heat on all sources and perform the same three steps. The recovered scalar approximates distance to the nearest source.

The method naturally computes a global field rather than an isolated path. A path can be recovered afterward by descending the distance field from target to source, with suitable handling near vertices and nondifferentiable regions.

40.8 Boundary conditions

On a surface with boundary, the heat and Poisson solves require boundary conditions. Homogeneous Neumann conditions keep flux normal to the boundary zero and are common when geodesics are meant to remain on the surface without prescribed boundary values. Dirichlet conditions solve a different problem and should not be introduced accidentally.

40.9 Sparse factorization reuse

For fixed mesh geometry and fixed t , the matrices

$$M + tL \quad \text{and} \quad L$$

do not depend on the source. Their sparse factorizations can therefore be reused for many source queries.

Proposition 40.7 (Amortized query structure). *After matrix assembly and factorization, changing the source mainly changes the right-hand sides, the facewise gradient normalization, and the back substitutions.*

This is one reason the heat method is attractive for interactive geometry processing and repeated distance-field computation.

40.10 Accuracy and failure modes

The heat method is approximate. Error depends on mesh resolution, element quality, Laplacian discretization, time step, boundary treatment, and the normalization of small gradients.

Distance values can be accurate even when the field is less reliable very close to the source or near the cut locus. As in VII/38, path identity is a more sensitive diagnostic than scalar distance.

Warning 40.8 (Disconnected meshes). A source only heats its connected component. Solvers should process components explicitly; otherwise null spaces and unreachable vertices can be mistaken for numerical failures.

40.11 Spectral viewpoint

Let generalized eigenpairs satisfy

$$L\psi_k = \lambda_k M\psi_k, \quad 0 = \lambda_0 \leq \lambda_1 \leq \cdots.$$

The backward-Euler heat solve damps mode k by approximately

$$\frac{1}{1 + t\lambda_k}.$$

High-frequency modes are damped more strongly. This explains both the numerical stability of diffusion and the oversmoothing that appears when t is chosen too large.

40.12 Relation to exact geodesics

The method never enumerates face sequences or edge windows. It replaces combinatorial wavefront tracking by globally coupled linear algebra:

$$\text{heat solve} \longrightarrow \text{normalized directions} \longrightarrow \text{Poisson solve}.$$

The price is approximation; the benefit is simplicity, parallel local operations, and reusable sparse factorizations.

40.13 Exercises

Exercise 40.9. What short-time quantity of the heat kernel encodes squared distance?

Hint. Take a logarithm. □

Solution. $-4t \log k_t(x, s)$ tends to $d(x, s)^2$ as $t \downarrow 0$, under the usual short-time assumptions. □

Exercise 40.10. With $\Delta = \text{div } \text{grad}$ nonpositive, write one backward-Euler heat step.

Hint. Move the Laplacian to the left. □

Solution. $(I - t\Delta)u = \delta_s$. □

Exercise 40.11. Write the standard finite-element heat system with positive stiffness matrix L .

Hint. Include the mass matrix. □

Solution. $(M + tL)u = M\delta_s$. □

Exercise 40.12. Why must an implementation check its Laplacian sign convention?

Hint. Libraries differ. □

Solution. Using a formula derived for the opposite sign can turn a positive heat solve into the wrong operator or reverse the Poisson reconstruction. □

Exercise 40.13. Define the heat-method direction field X .

Hint. Normalize the negative gradient. □

Solution. $X = -\nabla u / \|\nabla u\|$ wherever the gradient is nonzero. □

Exercise 40.14. Why is $ablau$ constant on each triangle for piecewise-linear finite elements?

Hint. An affine function has constant gradient. □

Solution. The nodal interpolant is affine on each face, hence its gradient is a constant face vector. □

Exercise 40.15. Why should very small $\|ablau\|$ be regularized?

Hint. Division amplifies noise. □

Solution. Normalization can become unstable or undefined when the gradient magnitude is near zero. □

Exercise 40.16. Why is raw heat u not the distance?

Hint. Recall the heat-kernel asymptotic. □

Solution. Heat decays roughly exponentially with squared distance divided by time, so its magnitude is not linearly proportional to distance. □

Exercise 40.17. What energy is minimized in the integration step?

Hint. Match a scalar gradient to X . □

Solution. $E(\phi) = \frac{1}{2} \int_M \|\nabla \phi - X\|^2 dA$. □

Exercise 40.18. What Poisson equation follows from that energy?

Hint. Use the Euler–Lagrange equation. □

Solution. $\Delta \phi = \operatorname{div} X$. □

Exercise 40.19. With positive stiffness L , define the standard load entry b_i .

Hint. Test against a basis gradient. □

Solution. $b_i = \int_M \langle \nabla \varphi_i, X \rangle dA$. □

Exercise 40.20. What linear system reconstructs ϕ ?

Hint. Use the stiffness matrix. □

Solution. $L\phi = b$, together with a gauge condition. □

Exercise 40.21. Why is the Poisson solution not unique on a connected closed mesh?

Hint. Constants have zero gradient. □

Solution. Because $L\mathbf{1} = 0$; adding a constant to ϕ leaves its gradient unchanged. □

Exercise 40.22. Give one way to fix the additive constant.

Hint. Pin the source. □

Solution. Impose $\phi(s) = 0$, or equivalently pin one degree of freedom and then normalize. $\square$

Exercise 40.23. What common time-step scaling is used on a mesh?

Hint. Diffusion length scales like square root of time. $\square$

Solution. $t = mh^2$, where h is a representative edge length. $\square$

Exercise 40.24. What happens if t is too small?

Hint. The heat remains extremely localized. $\square$

Solution. The direction field becomes sensitive to discretization, resolution, and roundoff near the source. $\square$

Exercise 40.25. What happens if t is too large?

Hint. Diffusion smooths too broadly. $\square$

Solution. Directional information is oversmoothed and the recovered distance can acquire global bias. $\square$

Exercise 40.26. How is a multi-source heat query initialized?

Hint. Heat every source simultaneously. $\square$

Solution. Place the initial impulse or source weight on all sources and perform the same heat, normalization, and Poisson steps. $\square$

Exercise 40.27. How can a path be recovered after computing a distance field?

Hint. Move downhill. $\square$

Solution. Trace a curve from target toward decreasing ϕ , with mesh-aware handling of face crossings and singular regions. $\square$

Exercise 40.28. Why are homogeneous Neumann boundary conditions natural in many surface-distance applications?

Hint. They impose no normal flux. $\square$

Solution. They keep the PDE on the surface without prescribing artificial distance values along the boundary. $\square$

Exercise 40.29. Which matrices can be factorized once and reused for many sources?

Hint. Geometry and t stay fixed. $\square$

Solution. $M + tL$ and the gauge-fixed Poisson matrix derived from L . $\square$

Exercise 40.30. What changes when the source changes after factorization?

Hint. Mostly right-hand sides and local field operations. $\square$

Solution. The source vector changes, then heat back substitution, face gradients, normalization, divergence assembly, and Poisson back substitution are recomputed. $\square$

Exercise 40.31. What does the spectral factor $1/(1 + t\lambda_k)$ do to large λ_k ?

Hint. Large eigenvalues are high frequency. $\square$

Solution. It strongly damps high-frequency modes. □

Exercise 40.32. What topic does VII/41 supply in detail?

Hint. The heat method relies on one central discrete operator. □

Solution. VII/41 develops discrete Laplacians, especially the cotangent stiffness operator and mass matrices used in these solves. □

40.14 Solved problem dossiers

Problem 40.33 (One-dimensional sign check). Suppose Δ is nonpositive. Explain why $I - t\Delta$ is positive for $t > 0$ on every Laplacian eigenmode.

Solution. If $\Delta\psi = -\lambda\psi$ with $\lambda \geq 0$, then $(I - t\Delta)\psi = (1 + t\lambda)\psi$. Every eigenvalue is at least 1, so the backward-Euler operator is positive and invertible. □

Problem 40.34 (Mesh matrix sign check). The stiffness matrix satisfies $v^T Lv \geq 0$. Which smooth operator does L represent under the convention of this chapter?

Solution. It represents $-\Delta$ weakly. This is why the heat matrix is $M + tL$, matching $I - t\Delta$. □

Problem 40.35 (Facewise gradient). On one triangle, $u = \sum_{i=1}^3 u_i \varphi_i$. Express its gradient.

Solution. Because the basis functions are affine on the face, $\nabla u = \sum_i u_i \nabla \varphi_i$, and each $\nabla \varphi_i$ is constant on the triangle. Hence ∇u is constant there. □

Problem 40.36 (Normalization safeguard). A face has $\|\nabla u\| = 10^{-15}$. Why is blindly dividing by this value dangerous, and what is a standard policy?

Solution. Roundoff in the gradient direction would be magnified enormously. One can skip or regularize the face, for example normalize by $\max(\|\nabla u\|, \varepsilon)$ with a scale-aware ε , while documenting the policy. □

Problem 40.37 (Poisson gauge). A numerical solver reports that $L\phi = b$ is singular on a closed connected mesh. Is this necessarily a bug?

Solution. No. Constants lie in the null space of the Laplacian. Pinning one vertex, imposing a mean-zero constraint, or using a compatible null-space solver removes the expected gauge freedom. □

Problem 40.38 (Time-step dimensional analysis). If all mesh lengths are scaled by a factor c , how should a mesh-scale heat time $t = mh^2$ change?

Solution. The representative length becomes ch , so the time becomes $m(ch)^2 = c^2t$. This matches diffusion's length-squared scaling. □

Problem 40.39 (Too-large heat time). Why can an extremely large t produce a smooth but inaccurate distance field?

Solution. Large t suppresses many spatial modes before the direction field is extracted. The resulting gradient reflects broad global diffusion rather than the short-time local geometry tied closely to geodesic distance. □

Problem 40.40 (Multiple sources). Two source sets are activated. What geometric scalar should the final field approximate after normalization?

Solution. It should approximate distance to the nearest activated source, i.e. the lower envelope of their individual geodesic distance functions. $\square$

Problem 40.41 (Factorization reuse). A mesh is fixed and 10,000 source queries are required. Explain where the main amortization comes from.

Solution. The expensive sparse factorizations of $M + tL$ and the gauge-fixed Poisson matrix are performed once. Each new query then uses new right-hand sides plus triangular solves and local face operations. $\square$

Problem 40.42 (Disconnected components). A source lies on component A and component B has no source. What should the final distance on B mean?

Solution. Geodesic distance to the source is infinite on B . The algorithm should mark the component unreachable rather than interpret an arbitrary Poisson null-space value as a finite distance. $\square$

Problem 40.43 (Spectral damping). For $t = 0.01$, compare the backward-Euler factors of modes with $\lambda = 1$ and $\lambda = 1000$.

Solution. The factors are $1/(1.01) \approx 0.9901$ and $1/11 \approx 0.0909$. The high-frequency mode is damped far more strongly. $\square$

Problem 40.44 (Exact versus heat method). An application needs a visually smooth distance field every frame but only occasionally needs a certified shortest path. Propose a division of labor.

Solution. Use the heat method for frequent interactive global fields and an exact polyhedral solver for occasional validation or certified path queries. This uses each method where its computational strengths matter most. $\square$

40.15 Challenge problems

Problem 40.45 (Derive the Poisson equation). Starting from $E(\phi) = \frac{1}{2} \int \|\nabla\phi - X\|^2 dA$, derive the Euler-Lagrange equation including the sign carefully.

Solution. For a variation $\phi + \varepsilon\eta$, $\frac{dE}{d\varepsilon}|_0 = \int \langle \nabla\phi - X, \nabla\eta \rangle dA$. Integration by parts gives $-\int (\Delta\phi - \operatorname{div} X) \eta dA$ plus the boundary term. Under compatible natural boundary conditions, stationarity for all η yields $\Delta\phi = \operatorname{div} X$. $\square$

Problem 40.46 (Discrete weak right-hand side). Under the positive stiffness convention $L_{ij} = \int \langle \nabla\varphi_i, \nabla\varphi_j \rangle dA$, show why the Poisson load can be assembled as $b_i = \int \langle \nabla\varphi_i, X \rangle dA$.

Solution. Testing $\Delta\phi = \operatorname{div} X$ by φ_i and integrating both sides by parts gives $-\int \langle \nabla\varphi_i, \nabla\phi \rangle = -\int \langle \nabla\varphi_i, X \rangle$. Cancelling the minus signs yields $\sum_j L_{ij}\phi_j = b_i$ with the stated b_i . $\square$

Problem 40.47 (Scaling experiment). Predict what should happen if a mesh is uniformly scaled by c , the heat time is scaled by c^2 , and the same dimensionless algorithm is rerun.

Solution. The recovered distances should scale approximately by c , while normalized direction fields should be unchanged after identifying corresponding points. The heat-time scaling preserves the diffusion length relative to the mesh. $\square$

Problem 40.48 (Failure near a cut locus). Explain why the heat method can return an accurate scalar field but a gradient-descent path can still be unstable near a cut-locus branch.

Solution. Distance is continuous across the cut locus but its minimizing direction is multi-valued or discontinuous there. A small approximation error can tilt the numerical gradient toward a different valid branch, changing the recovered path substantially while barely changing scalar distance. $\square$

Problem 40.49 (Production implementation checklist). Give a compact checklist for a robust repeated-query heat-method implementation on triangle meshes.

Solution. Validate connectivity and source locations; assemble a consistent mass matrix and positive stiffness matrix; choose and document $t = mh^2$; factor $M + tL$ and a gauge-fixed Poisson system; regularize tiny heat gradients; assemble divergence with the matching sign convention; normalize the final field at the source set; handle boundaries and unreachable components explicitly; and compare against graph and exact references on regression meshes. $\square$

40.16 Bridge to the next chapter

The heat method has deliberately treated L and M as trusted geometric operators. VII/41 opens that black box: it derives the cotangent stiffness matrix, mass matrices, boundary behavior, null spaces, energy identities, and the numerical conditions under which a discrete Laplacian is a faithful geometric operator.

Chapter 41

Discrete Laplacians

The Laplace–Beltrami operator is the central second-order operator of Riemannian geometry. On a triangle mesh it becomes a sparse matrix assembled from local geometry. This chapter derives the cotangent finite-element operator, explains the role of mass matrices, and separates three notions that are often conflated in software: a stiffness matrix, a metric-dependent Laplacian, and a graph Laplacian.

41.1 From the smooth operator to a weak form

We keep the convention from VII/40,

$$\Delta f = \operatorname{div}(\nabla f),$$

so Δ is nonpositive on a closed Riemannian manifold. For smooth functions f, g on a closed surface,

$$-\int_M g \Delta f \, dA = \int_M \langle \nabla g, \nabla f \rangle \, dA.$$

The right-hand side is the Dirichlet bilinear form. It is symmetric and nonnegative when $f = g$, which is why finite elements naturally produce a positive-semidefinite matrix for $-\Delta$.

Definition 41.1 (Dirichlet energy). For a scalar function f ,

$$E(f) = \frac{1}{2} \int_M \|\nabla f\|^2 \, dA.$$

41.2 Piecewise-linear functions on a triangle mesh

Let $\mathcal{T} = (V, E, F)$ be a nondegenerate triangle mesh. For each vertex i , let φ_i be the standard hat function: it equals one at vertex i , zero at all other vertices, and is affine on every incident triangle. A piecewise-linear function is

$$f_h = \sum_i f_i \varphi_i.$$

On each face, $\nabla \varphi_i$ and hence ∇f_h are constant.

Remark 41.2 (Intrinsic data suffice). The stiffness entries depend only on triangle edge lengths and angles. The mesh may be embedded in $\mathbb{R}^3$, but the scalar cotangent Laplacian is an intrinsic operator once the triangulated metric is fixed.

41.3 The stiffness matrix

Definition 41.3 (Finite-element stiffness matrix). Define

$$K_{ij} = \int_M \langle \nabla \varphi_i, \nabla \varphi_j \rangle dA.$$

Then

$$f^T K f = \int_M \|\nabla f_h\|^2 dA \geq 0.$$

Thus K is symmetric positive semidefinite. Because $\sum_i \varphi_i = 1$, one has

$$K \mathbf{1} = 0.$$

On a connected mesh this is the expected constant null mode, subject to the usual boundary-condition qualifications.

41.4 Cotangent weights

Consider an interior edge ij shared by two triangles. Let α_{ij} and β_{ij} be the angles opposite that edge. Define

$$w_{ij} = \frac{1}{2}(\cot \alpha_{ij} + \cot \beta_{ij}).$$

For a boundary edge there is only one opposite angle and the corresponding weight is $\frac{1}{2} \cot \alpha_{ij}$.

Theorem 41.4 (Cotangent stiffness formula). *For $i \neq j$,*

$$K_{ij} = -w_{ij}$$

when ij is an edge and $K_{ij} = 0$ otherwise. The diagonal is

$$K_{ii} = \sum_{j \sim i} w_{ij}.$$

Consequently every row sums to zero.

Proof. On one triangle, the gradients of barycentric basis functions are constant. Their inner products can be written in terms of the angle opposite the corresponding edge, giving a local off-diagonal contribution $-\frac{1}{2} \cot \theta$. Summing the one or two incident face contributions produces $-w_{ij}$; the identity $\sum_j K_{ij} = 0$ then gives the diagonal. $\square$

Proposition 41.5 (Edge-energy form). *The Dirichlet energy can be written*

$$\frac{1}{2} f^T K f = \frac{1}{2} \sum_{ij \in E} w_{ij} (f_i - f_j)^2,$$

with each unoriented edge counted once.

41.5 Why a mass matrix is needed

The stiffness matrix represents the bilinear form for $-\Delta$, not yet a pointwise operator. The L^2 inner product of finite-element functions is represented by the mass matrix

$$M_{ij} = \int_M \varphi_i \varphi_j dA.$$

For a triangle of area A , its consistent local mass matrix is

$$M_T = \frac{A}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Definition 41.6 (Lumped barycentric mass). A common diagonal approximation assigns

$$M_{ii}^{\text{lump}} = \frac{1}{3} \sum_{T \ni i} \text{Area}(T).$$

Diagonal masses make application of M^{-1} cheap. Consistent masses more faithfully represent the finite-element L^2 product. Mixed or Voronoi areas are also common in geometry processing; the choice must be stated because it changes the discrete pointwise operator and the generalized spectrum.

41.6 Three related matrices and a sign convention

Under the convention of this volume, K discretizes $-\Delta$ weakly. A pointwise-like positive Laplacian is

$$L_+ = M^{-1}K,$$

whereas the approximation to the smooth nonpositive operator is

$$\Delta_h = -M^{-1}K.$$

The generalized formulation

$$Kf = Mg$$

is often preferable to explicitly forming $M^{-1}K$.

Warning 41.7 (Library sign conventions). Many graphics libraries call either K , $M^{-1}K$, or its negative simply “the Laplacian.” Before combining formulas, determine whether eigenvalues are expected to be nonnegative or nonpositive and whether the mass matrix has already been absorbed.

41.7 Harmonic functions and Poisson equations

A discrete harmonic function satisfies the interior equation

$$(Kf)_i = 0.$$

Equivalently, when the relevant weights are positive,

$$f_i = \frac{\sum_{j \sim i} w_{ij} f_j}{\sum_{j \sim i} w_{ij}}.$$

A finite-element Poisson problem takes the weak form

$$Kf = b,$$

with boundary conditions or a gauge condition supplying uniqueness.

Proposition 41.8 (Variational characterization). *With Dirichlet values fixed on the boundary, the discrete harmonic extension is the unique minimizer of the discrete Dirichlet energy among all functions with those boundary values, provided the free system has no remaining null mode.*

41.8 Boundary conditions

Dirichlet conditions are imposed by fixing boundary degrees of freedom and solving for the interior unknowns. Natural finite-element boundary conditions are Neumann-type: boundary flux enters through the weak-form boundary term. On a closed connected mesh, the pure Poisson equation requires compatibility with the constant null space; for example, a load must have zero total integral in the consistent weak sense.

41.9 Delaunay geometry and weight signs

For an interior edge,

$$\cot \alpha + \cot \beta = \frac{\sin(\alpha + \beta)}{\sin \alpha \sin \beta}.$$

Hence an intrinsic Delaunay edge, characterized by $\alpha + \beta \leq \pi$, has $w_{ij} \geq 0$.

Theorem 41.9 (Delaunay positivity of cotangent edge weights). *If every interior edge is intrinsically Delaunay, then all interior cotangent weights are nonnegative.*

Positive off-diagonal weights support a graph-like averaging interpretation and are important for discrete maximum-principle behavior. On a non-Delaunay mesh, individual cotangent weights may be negative even though the assembled finite-element stiffness matrix still represents a nonnegative Dirichlet energy.

Warning 41.10 (Maximum principle is not automatic). Positive semidefiniteness of K does not by itself imply that every interior harmonic value is a convex combination of its neighbors. That stronger local property depends on the weight signs and boundary configuration.

41.10 Intrinsic Delaunay triangulations

Because the cotangent operator depends on intrinsic angles, one can flip edges intrinsically until the Delaunay condition is satisfied, without changing the underlying piecewise-flat metric. This can improve weight signs and numerical behavior while preserving intrinsic distances. Degenerate cocircular cases may admit more than one Delaunay connectivity but produce compatible intrinsic geometry.

41.11 The generalized spectrum

For a closed mesh, solve

$$K\phi_k = \lambda_k M\phi_k.$$

With positive-definite M ,

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots.$$

The constant function spans the first eigenspace on a connected mesh. The eigenvectors can be chosen M -orthonormal:

$$\phi_i^T M \phi_j = \delta_{ij}.$$

These modes drive spectral filtering, diffusion, shape signatures, and many model-reduction techniques.

41.12 Connection with the heat method

VII/40 solved

$$(M + tK)u = M\delta_s$$

under the present positive-stiffness convention, followed by a Poisson solve with K . The generalized eigenmode ϕ_k is damped by

$$\frac{1}{1 + t\lambda_k}.$$

Thus the mass and stiffness choices made here directly control heat-method accuracy.

41.13 The Laplacian of an embedding

Let $X = (x, y, z)$ collect the vertex coordinates of a triangulated surface. Applying the operator coordinatewise gives a discrete mean-curvature-normal quantity. With the smooth convention

$$\Delta X = -2H\mathbf{n},$$

one expects

$$M^{-1}KX \approx 2H\mathbf{n},$$

up to discretization, mass choice, orientation, and the convention for scalar mean curvature.

Remark 41.11 (Curvature is more delicate than scalar diffusion). The cotangent Laplacian is robust for many scalar PDEs, but pointwise curvature estimates amplify mesh noise and irregularity. VII/42 therefore combines principal-direction geometry with explicit estimation and scale controls rather than treating a single raw Laplacian value as a complete feature detector.

41.14 Assembly and numerical diagnostics

A production implementation should assemble face contributions locally, sum into sparse global matrices, verify symmetry, check row sums of K , check positive vertex masses, and test the constant null mode. Meshes with zero-area faces or nearly degenerate angles require preprocessing because cotangents can become extremely large.

Under a uniform geometric scaling by c , triangle areas and M scale by c^2 , while cotangent weights and K are dimensionless. Therefore $M^{-1}K$ scales by c^{-2} , exactly like a second-order differential operator.

41.15 Exercises

Exercise 41.12. State the sign convention for the smooth Laplace–Beltrami operator used here.

Hint. Recall VII/40. □

Solution. $\Delta = \operatorname{div} \nabla$ is nonpositive on a closed manifold; the positive weak operator is $-\Delta$. □

Exercise 41.13. Write the weak identity relating $-\Delta$ to the Dirichlet bilinear form on a closed surface.

Hint. Integrate by parts. □

Solution. $-\int g \Delta f \, dA = \int \langle \nabla g, \nabla f \rangle \, dA.$ □

Exercise 41.14. What is a piecewise-linear hat function φ_i ?

Hint. Specify values at vertices and behavior on faces. □

Solution. It is one at vertex i , zero at every other vertex, and affine on each triangle. □

Exercise 41.15. Define the finite-element stiffness matrix.

Hint. Use gradients of basis functions. □

Solution. $K_{ij} = \int \langle \nabla \varphi_i, \nabla \varphi_j \rangle \, dA.$ □

Exercise 41.16. Why is K positive semidefinite?

Hint. Test it with a coefficient vector. □

Solution. For any f , $f^T K f = \int \|\nabla f_h\|^2 \, dA \geq 0.$ □

Exercise 41.17. Why does $K \mathbf{1} = 0$?

Hint. Use the partition of unity. □

Solution. Because $\sum_i \varphi_i = 1$, its gradient vanishes, so the constant finite-element function has zero Dirichlet energy and zero stiffness image. □

Exercise 41.18. Give the cotangent weight for an interior edge ij .

Hint. Use the two opposite angles. □

Solution. $w_{ij} = \frac{1}{2}(\cot \alpha_{ij} + \cot \beta_{ij}).$ □

Exercise 41.19. What is the off-diagonal stiffness entry on an edge?

Hint. Watch the sign. □

Solution. For $i \neq j$ joined by an edge, $K_{ij} = -w_{ij}.$ □

Exercise 41.20. How is the diagonal K_{ii} obtained?

Hint. Use zero row sum. □

Solution. $K_{ii} = \sum_{j \sim i} w_{ij}.$ □

Exercise 41.21. Write the edge-based Dirichlet-energy identity.

Hint. Use squared differences. □

Solution. $\frac{1}{2} f^T K f = \frac{1}{2} \sum_{ij \in E} w_{ij} (f_i - f_j)^2$, counting each unoriented edge once. □

Exercise 41.22. Define the consistent mass matrix.

Hint. Use the L^2 inner product. □

Solution. $M_{ij} = \int \varphi_i \varphi_j \, dA.$ □

Exercise 41.23. What is the local consistent mass matrix of a triangle of area A ?

Hint. The diagonal entries are twice the off-diagonal entries. □

Solution. $M_T = \frac{A}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$ □

Exercise 41.24. Give the lumped barycentric mass at vertex i .

Hint. Give one third of each incident face. □

Solution. $M_{ii}^{\text{lump}} = \frac{1}{3} \sum_{T \ni i} \text{Area}(T).$ □

Exercise 41.25. Under this chapter's convention, which matrix approximates the smooth nonpositive Δ ?

Hint. Distinguish the positive stiffness from the signed operator. □

Solution. $\Delta_h = -M^{-1}K.$ □

Exercise 41.26. Write the generalized positive eigenproblem.

Hint. Avoid explicitly inverting the mass matrix. □

Solution. $K\phi = \lambda M\phi$ with $\lambda \geq 0.$ □

Exercise 41.27. What is the discrete harmonic equation at a free interior vertex?

Hint. Use the stiffness row. □

Solution. $(Kf)_i = 0.$ □

Exercise 41.28. When positive weights are available, express a harmonic vertex value as a weighted average.

Hint. Solve the stiffness row for $f_i.$ □

Solution. $f_i = \frac{\sum_j w_{ij} f_j}{\sum_j w_{ij}}.$ □

Exercise 41.29. What geometric condition guarantees a nonnegative cotangent weight on an interior edge?

Hint. Use the sum of opposite angles. □

Solution. The intrinsic Delaunay condition $\alpha_{ij} + \beta_{ij} \leq \pi.$ □

Exercise 41.30. Can a non-Delaunay cotangent weight be negative?

Hint. The finite-element energy and local weight signs are different issues. □

Solution. Yes. An individual weight may be negative even though the assembled stiffness matrix still represents a nonnegative Dirichlet energy. □

Exercise 41.31. How are Dirichlet boundary conditions imposed algebraically?

Hint. Separate fixed and free degrees of freedom. □

Solution. Fix the prescribed boundary values, move their contributions to the right-hand side, and solve the reduced system for the free vertices. □

Exercise 41.32. What is the constant-mode compatibility issue for a closed-mesh Poisson problem?

Hint. The stiffness matrix is singular. □

Solution. The load must be orthogonal to the constant null mode in the appropriate weak sense, and a gauge such as zero mean or one pinned value is needed for uniqueness. □

Exercise 41.33. How does $M^{-1}K$ scale if all lengths are multiplied by c ?

Hint. Cotan weights are dimensionless but areas are not. □

Solution. It scales by c^{-2} , since K is unchanged and M scales by c^2 . □

Exercise 41.34. What two matrices from this chapter enter the heat method directly?

Hint. Recall $(M + tK)u$. □

Solution. The mass matrix M and the positive stiffness matrix K . □

Exercise 41.35. What topic does VII/42 add to this operator-level picture?

Hint. Move from scalar PDEs to directional surface features. □

Solution. VII/42 develops principal-direction line fields, curvature lines, and ridge/valley feature extraction, including mesh estimation and scale issues. □

41.16 Solved problem dossiers

Problem 41.36 (Single triangle cotangent sign). A triangle has an angle 120° opposite edge ij . What sign does that face's contribution $\frac{1}{2} \cot 120^\circ$ have, and what does this illustrate?

Solution. Since $\cot 120^\circ = -1/\sqrt{3}$, the contribution is negative. Obtuse geometry can therefore create negative cotangent contributions; weight positivity is not automatic on an arbitrary triangulation. □

Problem 41.37 (Row-sum diagnostic). A code assembles a closed-mesh stiffness matrix but obtains $\|K\mathbf{1}\|$ comparable to ordinary matrix entries. What does this indicate?

Solution. It indicates an assembly or orientation/indexing error, or a degenerate/preprocessing issue. The finite-element partition of unity requires exact zero row sums in exact arithmetic. □

Problem 41.38 (Two-triangle Delaunay test). An interior edge has opposite angles 70° and 80° . Determine the sign of its cotangent weight without evaluating either cotangent numerically.

Solution. Their sum is $150^\circ < 180^\circ$. Since $\cot \alpha + \cot \beta = \sin(\alpha + \beta)/(\sin \alpha \sin \beta)$ and the sines in the denominator are positive, the weight is positive. □

Problem 41.39 (Non-Delaunay test). Opposite angles are 100° and 100° . What does the intrinsic Delaunay criterion say about the edge weight?

Solution. The angle sum is $200^\circ > 180^\circ$, so the edge is non-Delaunay and $\sin(200^\circ) < 0$; hence the cotangent sum and the edge weight are negative. □

Problem 41.40 (Mass of a single triangle). For a single triangle of area A , what total mass is obtained by summing all entries of the consistent local mass matrix?

Solution. The three diagonal entries contribute 6 and the six off-diagonal entries contribute 6, so the coefficient matrix has total sum 12. Multiplying by $A/12$ gives total mass A , matching $\int_T 1 dA$. $\square$

Problem 41.41 (Lumped versus consistent). Why can replacing a consistent mass matrix by a lumped diagonal mass change eigenvalues even though both represent area?

Solution. They define different discrete L^2 inner products. The generalized Rayleigh quotient $f^T K f / (f^T M f)$ therefore changes, so the discrete spectrum changes even if both masses integrate constants correctly. $\square$

Problem 41.42 (Dirichlet minimizer). A mesh has fixed boundary values. Explain why solving the interior stiffness equations finds the minimum-energy extension.

Solution. The discrete Dirichlet energy is a convex quadratic $\frac{1}{2} f^T K f$. Differentiating with respect to each free interior coefficient yields the corresponding equation $(K f)_i = 0$; with the boundary fixed and no free null mode, this stationary point is the unique minimizer. $\square$

Problem 41.43 (Poisson gauge). On a closed connected mesh, $K f = b$ has a solution and f is one solution. Describe all solutions.

Solution. Because the null space is the constants, every $f + c\mathbf{1}$ is also a solution. Under the standard one-dimensional null-space assumption these are all solutions. $\square$

Problem 41.44 (Heat-system positivity). Why is $M + tK$ nonsingular for $t > 0$ when M is positive definite and K positive semidefinite?

Solution. For nonzero v , $v^T (M + tK) v = v^T M v + t v^T K v > 0$. Hence the matrix is symmetric positive definite and invertible. $\square$

Problem 41.45 (Scaling a mesh). A mesh is uniformly enlarged by factor 5. How should nonzero generalized eigenvalues λ of $K\phi = \lambda M\phi$ scale?

Solution. The stiffness matrix is unchanged while the mass matrix scales by 25, so eigenvalues scale by $1/25$, consistent with the inverse-length-squared dimension of the Laplacian. $\square$

Problem 41.46 (Embedding Laplacian on a sphere). For a well-resolved unit sphere with outward normal and $\Delta X = -2H\mathbf{n}$, what vector should $M^{-1} K X$ approximate?

Solution. For the unit sphere $H = 1$, so $M^{-1} K X \approx 2\mathbf{n}$. The sign follows because K approximates $-\Delta$. $\square$

Problem 41.47 (Production regression). Give four cheap algebraic checks that should accompany cotangent-matrix assembly before it is trusted by a heat-method solver.

Solution. Check symmetry of K , near-zero row sums, positive diagonal masses, and nonnegative quadratic energies $v^T K v$ on test vectors. Also check the expected number of constant null modes against mesh connectivity. $\square$

41.17 Challenge problems

Problem 41.48 (Derive the Delaunay sign identity). Prove $\cot \alpha + \cot \beta = \sin(\alpha + \beta) / (\sin \alpha \sin \beta)$, and use it to characterize the sign of an interior cotangent weight for $0 < \alpha, \beta < \pi$.

Solution. Using $\cot \theta = \cos \theta / \sin \theta$, the numerator becomes $\cos \alpha \sin \beta + \sin \alpha \cos \beta = \sin(\alpha + \beta)$. The denominator is positive. Thus the weight is positive, zero, or negative according as $\alpha + \beta$ is less than, equal to, or greater than π . $\square$

Problem 41.49 (Rayleigh quotient). Show that generalized eigenvalues of $K\phi = \lambda M\phi$ are stationary values of $R(f) = f^T K f / (f^T M f)$, and explain why they are nonnegative.

Solution. Differentiate the quotient subject to the normalization $f^T M f = 1$. A Lagrange multiplier gives $Kf = \lambda Mf$, with $\lambda = f^T K f$. Since K is positive semidefinite and M positive definite, $R(f) \geq 0$ for every nonzero f . $\square$

Problem 41.50 (Discrete maximum principle). Assume every free interior vertex has strictly positive incident weights. Prove that a nonconstant discrete harmonic function cannot attain a strict maximum at a free interior vertex.

Solution. At a harmonic vertex, f_i is a positive weighted average of neighboring values. If it were a strict maximum, every neighbor would be smaller and the weighted average would be strictly smaller than f_i , a contradiction. Equality forces all positively weighted neighbors to share the same value and propagation handles connected regions. $\square$

Problem 41.51 (Intrinsic flip thought experiment). Two triangles form a convex intrinsic quadrilateral whose shared diagonal is non-Delaunay. Explain why flipping to the other diagonal can improve local weight signs without changing the intrinsic metric of the quadrilateral.

Solution. The quadrilateral is built from the same four boundary edge lengths and the same flat intrinsic region; the diagonal is only a triangulation choice. The alternate diagonal satisfies the local Delaunay criterion, making its cotangent weight nonnegative, while geodesic distances intrinsic to the unchanged piecewise-flat quadrilateral are preserved. $\square$

Problem 41.52 (Operator validation suite). Design a compact validation suite for M and K that tests geometry, scaling, PDE behavior, and spectrum rather than only matrix dimensions.

Solution. Use a planar patch to test linear finite-element consistency; a closed connected mesh to test one constant null mode; uniform rescaling to verify K invariance, $M \sim c^2$, and eigenvalues $\sim c^{-2}$; a Dirichlet harmonic problem to test averaging/maximum behavior on Delaunay meshes; and a heat solve to verify symmetric positive definiteness of $M + tK$ and convergence under refinement. $\square$

41.18 Bridge to the next chapter

Discrete Laplacians turn scalar fields into a computational language for surface geometry. The final chapter changes viewpoint from scalar diffusion to directional shape structure. VII/42 returns to the shape operator and its principal directions, then develops curvature lines, ridges, valleys, and their stable extraction on noisy triangle meshes.

Chapter 42

Curvature Lines, Ridges and Valleys

The smooth theory of VII/16–VII/17 associates to every nonumbilic point two principal directions and two principal curvatures. Those directions form line fields whose integral curves are curvature lines. In applications, derivatives of the principal curvatures along these lines identify ridge and valley features. This final chapter joins the smooth definitions to the estimation, scale selection, and tracing problems that arise on triangle meshes.

42.1 Principal directions revisited

Let $S : T_p M \rightarrow T_p M$ be the shape operator of an oriented smooth surface. At a nonumbilic point it has distinct eigenvalues k_1, k_2 and orthogonal unit eigenvectors e_1, e_2 :

$$S e_i = k_i e_i, \quad \langle e_1, e_2 \rangle = 0.$$

We locally order the eigenvalues as $k_1 > k_2$ when convenient.

Warning 42.1 (Directions are lines, not arrows). The eigenvectors e_i and $-e_i$ represent the same principal direction. A principal-direction field is therefore naturally a line field. Numerical algorithms that treat it as an ordinary oriented vector field can introduce spurious sign flips.

42.2 Euler's normal-curvature formula

If a unit tangent vector makes angle θ with e_1 , Euler's formula is

$$k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta.$$

Thus the principal directions are exactly the directions in which normal curvature is extremal. This gives a geometric interpretation of the eigenvectors that does not depend on a particular parameterization.

42.3 Curvature lines

Definition 42.2 (Line of curvature). A regular surface curve γ is a line of curvature if its tangent is a principal direction at every nonumbilic point of the curve:

$$\gamma'(s) \parallel e_i(\gamma(s))$$

for one principal line field $i \in \{1, 2\}$.

Away from umbilics and points where the chosen field loses regularity, ordinary differential-equation theory gives local integral curves. Since $e_1 \perp e_2$, the two curvature-line families cross orthogonally wherever the principal curvatures are distinct.

Proposition 42.3 (Normal derivative along a curvature line). *If γ is a unit-speed curvature line with tangent e_i , then, under the convention $S = -dN$,*

$$\frac{d}{ds}N(\gamma(s)) = -k_i e_i.$$

42.4 Umbilics and singular line fields

At an umbilic,

$$k_1 = k_2, \quad S = kI,$$

so every tangent direction is principal and no distinguished principal line field exists. Umbilics are therefore genuine singularities for curvature-line tracing, not merely numerical nuisances.

On a round sphere every point is umbilic. On a circular cylinder there are no umbilics: one principal family follows the generators and the other follows the circles.

Remark 42.4 (Topology enters through singularities). On a closed surface, principal line fields cannot generally be made globally nonsingular. The arrangement and indices of umbilic singularities are tied to global topology, foreshadowing the homotopy and homology viewpoint of Volume VIII.

42.5 Directional derivatives of principal curvature

For a principal curvature k_i , define

$$D_{e_i}k_i = \langle \nabla k_i, e_i \rangle.$$

A zero of this derivative means that k_i is stationary to first order when moving in its own principal direction. This is the basic differential signal used by many ridge and valley definitions.

42.6 A declared ridge–valley convention

Terminology varies across geometry processing and computer vision, so we make a specific local convention.

Definition 42.5 (Principal ridge candidate). On an oriented patch with $k_1 > k_2$, a k_1 -ridge candidate is a nonumbilic point satisfying

$$D_{e_1}k_1 = 0.$$

It is a strict ridge point for this convention when additionally

$$D_{e_1}^2k_1 < 0.$$

Definition 42.6 (Principal valley candidate). A k_2 -valley candidate satisfies

$$D_{e_2}k_2 = 0,$$

and is strict when

$$D_{e_2}^2k_2 > 0.$$

These definitions encode maxima of the larger principal curvature along its principal direction and minima of the smaller principal curvature along its principal direction.

Warning 42.7 (Orientation and convention dependence). Reversing the surface normal changes S to $-S$ and flips principal curvature signs. It can also change how an ordering such as $k_1 > k_2$ is named. A ridge/valley implementation must therefore document its normal, ordering, and sign conventions. Some applications instead use absolute curvature or other orientation-insensitive feature measures.

42.7 Ridges are not curvature lines

A curvature line is an integral curve of a principal direction field. A ridge or valley is a locus where a directional derivative of principal curvature vanishes and a second-derivative test selects an extremum. A ridge curve may be traced across the surface, but its tangent need not itself equal the principal direction used in the defining derivative.

Warning 42.8 (Common conceptual error). Do not identify “ridge curve” with “line of curvature.” The first is an implicit feature locus; the second is an integral curve of an eigenline field. They coincide only under additional geometry.

42.8 Estimating curvature on triangle meshes

A triangle mesh has no classical pointwise second fundamental form at a vertex. One therefore estimates smooth local geometry. A common pipeline is:

neighborhood $\rightarrow$ normal/tangent frame $\rightarrow$ local surface fit $\rightarrow$ shape operator $\rightarrow (k_i, e_i)$.

For example, in a local tangent frame (x, y, z) , fit a Monge patch

$$z(x, y) \approx \frac{1}{2}(ax^2 + 2bxy + cy^2) + \text{lower/higher-order terms.}$$

At small slope, the Hessian

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

approximates the shape operator in that frame. More complete fits account for first derivatives, metric effects, and weighting.

42.9 Neighborhood and scale selection

Curvature is a second-derivative quantity and is therefore sensitive to noise. A one-ring fit may be too noisy, while a very large neighborhood can erase narrow features. Robust implementations expose a physical or intrinsic scale, use weighted least squares, reject degenerate neighborhoods, and compare features across multiple scales.

Definition 42.9 (Umbilic tolerance). A practical tracer treats a vertex as numerically near-umbilic when

$$|k_1 - k_2| < \varepsilon_k$$

for a scale-aware tolerance ε_k .

Near this set, principal directions are ill-conditioned: a small perturbation of the estimated shape operator can rotate the eigenvectors substantially.

42.10 Cohering a principal line field

Since e and $-e$ are equivalent, neighboring estimates should be compared by their unoriented angle. When a temporary orientation is required for interpolation, choose the sign that maximizes the dot product with a previously oriented neighbor. This creates local coherence but must not be mistaken for a global orientation across line-field singularities.

42.11 Extracting ridge and valley loci

Suppose a scalar estimate

$$r_1 = D_{e_1} k_1$$

is available at mesh vertices. A standard first step is to detect sign changes of r_1 along edges and linearly interpolate candidate zero crossings within triangles. Candidates are then filtered by the second-directional-derivative criterion, curvature strength, confidence, and proximity to umbilics.

Remark 42.10 (Zero crossing is only a candidate test). A sign change in a noisy first derivative can create a false feature. Stable feature extraction generally combines derivative tests with strength and persistence thresholds and with geometric continuity across neighboring faces.

42.12 Tracing curvature lines on a mesh

To trace a curvature line, start from a seed and integrate an estimated principal line field in the tangent plane. Within a triangle, interpolate a representation that respects the sign ambiguity, advance to the next edge, transport the local frame as needed, and stop at boundaries, singular regions, low-confidence zones, or after closing a loop.

Naively interpolating signed eigenvectors can cause cancellation when two vertices store opposite representatives of the same line. Line-field-aware interpolation or local sign alignment is essential.

42.13 Canonical examples

Example 42.11 (Round sphere). For a round sphere, $k_1 = k_2$ everywhere. Every direction is principal, so a principal-direction tracer has no unique field to follow. The sphere is a useful regression test for correct umbilic detection.

Example 42.12 (Circular cylinder). For a cylinder of radius R , the principal curvatures are $1/R$ and 0 up to orientation. Their principal directions are circumferential and axial. Both curvatures are constant, so the directional derivative ridge tests vanish everywhere; this illustrates why derivative zero alone is insufficient for selecting salient ridges.

Example 42.13 (Noisy scan). On a noisy triangulated scan, raw one-ring curvature derivatives can oscillate in sign at nearly every edge. Increasing the fit scale or smoothing the estimated shape operator can reveal persistent feature curves that survive across a range of scales.

42.14 Validation and regression geometry

Feature-line code should be tested on analytic surfaces with known principal geometry before it is trusted on arbitrary meshes. Useful tests include a plane, sphere, cylinder, torus

patches, and synthetic Monge surfaces. Numerical checks should distinguish errors in normals, curvature magnitudes, principal directions, directional derivatives, and final line extraction.

For a direction estimate, compare lines rather than oriented vectors. If e is estimated and e_* exact, a suitable error is

$$\theta_{\text{line}} = \arccos(|\langle e, e_* \rangle|), \quad 0 \leq \theta_{\text{line}} \leq \frac{\pi}{2}.$$

42.15 Relationship to the discrete Laplacian

VII/41 showed that $M^{-1}KX$ supplies a mean-curvature-normal estimate. Principal curvatures require more information: mean curvature alone gives only $k_1 + k_2$. Recovering directions and separating k_1 from k_2 needs a shape-operator or equivalent second-order estimate. The Laplacian is therefore one useful ingredient, not a complete ridge detector.

42.16 A production feature-line pipeline

A robust pipeline typically performs the following conceptual steps:

- mesh validation and scale choice
- normal and tangent-frame estimation
- local shape-operator fitting
- principal curvature/direction estimation
- line-field coherence and umbilic masking
- directional curvature derivatives
- zero-crossing candidates and extremum tests
- connectivity, pruning, and multiscale validation.

Each stage should expose confidence diagnostics rather than silently passing ill-conditioned data downstream.

42.17 Exercises

Exercise 42.14. What are principal directions at a nonumbilic point?

Hint. Use the shape operator. □

Solution. They are the two orthogonal eigenlines of the self-adjoint shape operator S , with eigenvalues equal to the principal curvatures. □

Exercise 42.15. Why is a principal direction naturally a line rather than an oriented vector?

Hint. Compare e and $-e$. □

Solution. If $Se = ke$, then $S(-e) = k(-e)$; the two vectors represent the same one-dimensional eigenspace. □

Exercise 42.16. State Euler's formula for normal curvature.

Hint. Measure angle from e_1 . □

Solution. $k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta$. □

Exercise 42.17. Define a line of curvature.

Hint. Its tangent follows an eigendirection. □

Solution. It is a regular surface curve whose tangent is parallel to one principal direction at each nonumbilic point. □

Exercise 42.18. How do the two curvature-line families meet at a nonumbilic point?

Hint. The shape operator is self-adjoint. □

Solution. They meet orthogonally because eigenvectors for distinct eigenvalues of a self-adjoint operator are orthogonal. □

Exercise 42.19. What happens to principal directions at an umbilic?

Hint. The eigenvalues coincide. □

Solution. The shape operator is a scalar multiple of the identity, so every tangent direction is principal and no distinguished principal line field exists. □

Exercise 42.20. Give an analytic surface that is umbilic everywhere.

Hint. Think constant positive curvature. □

Solution. A round sphere. □

Exercise 42.21. Give the two principal-direction families on a circular cylinder.

Hint. One bends; one does not. □

Solution. The circumferential circles and the axial generators. □

Exercise 42.22. Define $D_{e_i}k_i$.

Hint. It is a directional derivative. □

Solution. $D_{e_i}k_i = \langle \nabla k_i, e_i \rangle$. □

Exercise 42.23. Under the convention of this chapter, what first-order equation defines a k_1 -ridge candidate?

Hint. Stationarity along the first principal direction. □

Solution. $D_{e_1}k_1 = 0$. □

Exercise 42.24. What second-order sign selects a strict k_1 -ridge in this convention?

Hint. A ridge is a local maximum along e_1 . □

Solution. $D_{e_1}^2 k_1 < 0$. □

Exercise 42.25. What equations select a strict k_2 -valley in this convention?

Hint. Use a local minimum along e_2 . □

Solution. $D_{e_2}k_2 = 0$ and $D_{e_2}^2 k_2 > 0$. □

Exercise 42.26. Why must ridge/valley code document its normal convention?

Hint. Reverse the normal. □

Solution. Reversing the normal flips the shape operator and principal-curvature signs, which can change signed ridge/valley naming and curvature ordering. □

Exercise 42.27. Are ridge curves necessarily lines of curvature?

Hint. Compare an implicit locus with an integral curve. □

Solution. No. A ridge is a zero/extremum locus of a directional curvature derivative; its tangent need not equal the principal direction used in that derivative. □

Exercise 42.28. What local polynomial model is commonly used to estimate curvature on a mesh?

Hint. Use a tangent-frame height graph. □

Solution. A local Monge patch, for example $z \approx \frac{1}{2}(ax^2 + 2bxy + cy^2) + \dots$. □

Exercise 42.29. Why is one-ring curvature estimation often noisy?

Hint. Curvature uses second derivatives. □

Solution. Second derivatives amplify positional and normal noise, and a small irregular neighborhood provides little averaging or conditioning. □

Exercise 42.30. Give a practical numerical test for a near-umbilic vertex.

Hint. Compare the two estimated principal curvatures. □

Solution. Mark it near-umbilic when $|k_1 - k_2| < \varepsilon_k$ for a scale-aware tolerance. □

Exercise 42.31. Why are principal directions ill-conditioned near an umbilic?

Hint. Eigenvalues nearly coincide. □

Solution. When the eigenvalue gap is small, small perturbations of the estimated shape operator can cause large rotations of its eigenvectors. □

Exercise 42.32. How can neighboring representatives of the same principal line be locally sign-aligned?

Hint. Use a dot product. □

Solution. Replace one vector by its negative when necessary so its dot product with the chosen neighboring representative is nonnegative or maximal. □

Exercise 42.33. How can a mesh locate first-order ridge candidates from a sampled derivative scalar r_1 ?

Hint. Look along edges. □

Solution. Detect sign changes of r_1 across edges and interpolate zero crossings within incident triangles. □

Exercise 42.34. Why is a derivative zero crossing alone insufficient for a stable ridge?

Hint. Noise can create many sign changes. □

Solution. It must be filtered by the appropriate second-derivative sign, feature strength, confidence, scale persistence, and singularity handling. $\square$

Exercise 42.35. Give a line-field-aware angular error between estimated and exact unit directions.

Hint. Remove the sign ambiguity. $\square$

Solution. $\theta = \arccos(|\langle e, e_* \rangle|)$. $\square$

Exercise 42.36. Why does mean curvature alone not determine principal directions?

Hint. Mean curvature is only one scalar combination. $\square$

Solution. It supplies essentially $(k_1 + k_2)/2$ but not the eigenvectors or the separation of the two principal curvatures; a fuller shape-operator estimate is required. $\square$

Exercise 42.37. What global subject naturally follows the singular line-field phenomena seen here?

Hint. The next volume studies global deformation and invariants. $\square$

Solution. Algebraic topology: homotopy and homology provide tools for understanding global topological constraints beyond local differential geometry. $\square$

42.18 Solved problem dossiers

Problem 42.38 (Euler extrema). Using Euler's formula with $k_1 > k_2$, show that the maximum and minimum normal curvatures occur in the principal directions.

Solution. Write $k_n(\theta) = k_2 + (k_1 - k_2) \cos^2 \theta$. Since $0 \leq \cos^2 \theta \leq 1$, the maximum k_1 occurs for $\theta = 0$ modulo π , and the minimum k_2 for $\theta = \pi/2$ modulo π . $\square$

Problem 42.39 (Sphere regression). A curvature estimator on a round sphere reports stable, highly anisotropic principal directions. Why is this suspicious?

Solution. A sphere is umbilic everywhere, so the principal eigenvalue gap should vanish. The directions are mathematically undetermined; stable anisotropy is likely coming from triangulation bias, noise, or the estimator rather than the smooth surface. $\square$

Problem 42.40 (Cylinder ridge test). On an ideal cylinder, $k_1 = 1/R$ and $k_2 = 0$ are constant. Why does the first-order ridge equation hold everywhere, and why should a feature extractor not label the entire cylinder a sharp ridge?

Solution. All directional derivatives of the constant principal curvatures vanish, so the stationarity equations hold identically. But the second derivative also vanishes and there is no strict local extremum; strength and strictness criteria prevent the whole cylinder from becoming a ridge. $\square$

Problem 42.41 (Normal reversal). Suppose an algorithm uses outward normals and finds principal curvatures k_1, k_2 . What happens after all normals are reversed?

Solution. With $S = -dN$, reversing N changes S to $-S$, so the same principal lines remain but both eigenvalues change sign. If the code reorders eigenvalues afterward, their labels may also swap. $\square$

Problem 42.42 (Line-angle metric). An exact principal direction is $e_* = (1, 0)$, while an estimator returns $e = (-\cos 5^\circ, -\sin 5^\circ)$. What is the line-field angular error?

Solution. The sign is irrelevant. The absolute dot product is $\cos 5^\circ$, so $\theta_{\text{line}} = 5^\circ$, not 175° . $\square$

Problem 42.43 (Near-umbilic conditioning). Two estimated principal curvatures are 1.0000 and 0.9999. Explain why a principal-direction feature may be unreliable even if the curvature values themselves look accurate.

Solution. The eigenvalue gap is only 10^{-4} . Eigenvectors of a symmetric matrix become highly sensitive to perturbations when eigenvalues nearly coincide, so the direction can rotate substantially under tiny fitting errors. $\square$

Problem 42.44 (Monge patch eigenvectors). A small-slope quadratic fit has Hessian $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$. What are the approximate principal curvatures and directions in the local frame?

Solution. The eigenvalues are approximately 3 and 1, with eigenvectors along the local x - and y -axes, respectively. $\square$

Problem 42.45 (Sign-coherent interpolation). Two adjacent vertices store eigenvectors e and approximately $-e$ for the same smooth principal line. What goes wrong if they are linearly interpolated without sign alignment?

Solution. They can cancel near the midpoint, creating an artificial near-zero vector and an unstable normalized direction. Flipping one representative first avoids the spurious cancellation locally. $\square$

Problem 42.46 (Zero-crossing interpolation). Along an edge, a ridge derivative estimate has values $r(a) = -2$ and $r(b) = 6$. Under linear interpolation, at what fraction from a does the zero occur?

Solution. Solve $-2 + t(6 - (-2)) = 0$, giving $t = 2/8 = 1/4$. The candidate lies one quarter of the way from a to b . $\square$

Problem 42.47 (Multiscale persistence). A feature appears at one-ring scale but disappears for every moderately larger fitting radius. How should this affect confidence?

Solution. Confidence should decrease. A feature that exists only at the smallest, noise-sensitive scale is less likely to represent persistent surface geometry than one that survives across a reasonable scale interval. $\square$

Problem 42.48 (Laplacian versus principal data). A program knows only a reliable mean-curvature normal vector at each vertex. Can it reconstruct both principal directions uniquely?

Solution. Not in general. Mean curvature gives only the sum of principal curvatures (with normal direction) and lacks the anisotropic second-fundamental-form information needed to recover the two eigenlines. $\square$

Problem 42.49 (Tracing stop condition). A curvature-line tracer approaches a region where $|k_1 - k_2|$ falls below tolerance. Why is stopping or reducing confidence preferable to forcing a direction through it?

Solution. The principal line becomes ill-conditioned near an umbilic. Forcing a direction lets numerical noise choose an arbitrary continuation and can create visually plausible but geometrically unsupported curves. $\square$

42.19 Challenge problems

Problem 42.50 (Orthogonality of curvature lines). Prove that at a nonumbilic point the two curvature-line families meet orthogonally, starting only from self-adjointness of the shape operator.

Solution. Let $Se_1 = k_1e_1$ and $Se_2 = k_2e_2$ with $k_1 \neq k_2$. Self-adjointness gives $k_1\langle e_1, e_2 \rangle = \langle Se_1, e_2 \rangle = \langle e_1, Se_2 \rangle = k_2\langle e_1, e_2 \rangle$. Hence $(k_1 - k_2)\langle e_1, e_2 \rangle = 0$, so the inner product vanishes. $\square$

Problem 42.51 (Why ridge tangents differ). Let $r = D_{e_1}k_1$. At a regular ridge point where $r = 0$ and $\nabla r \neq 0$, what determines the tangent to the ridge locus, and why need it not be e_1 ?

Solution. The ridge is locally the level set $r = 0$, so its tangent vectors are orthogonal to ∇r . The defining principal direction e_1 appears inside the scalar r , but there is no general reason for e_1 to be orthogonal to ∇r ; hence the ridge tangent need not equal e_1 . $\square$

Problem 42.52 (Perturbation near an umbilic). Consider symmetric matrices $S_\varepsilon = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & 1 \end{pmatrix}$. Compare their eigenvectors with those of $S_0 = I$ and explain the numerical lesson for principal-direction tracing.

Solution. For every nonzero ε , the eigenvectors are the diagonal directions $(1, 1)$ and $(1, -1)$, while for $\varepsilon = 0$ every direction is an eigenvector. An arbitrarily small perturbation selects specific directions out of a completely degenerate eigenspace, illustrating why directions near umbilics are intrinsically unstable. $\square$

Problem 42.53 (Orientation-invariant feature proposal). Design an orientation-insensitive scalar candidate for strong anisotropic bending using the principal curvatures, and explain one advantage and one limitation compared with signed ridge/valley labels.

Solution. One example is $|k_1 - k_2|$, or a normalized variant such as $|k_1 - k_2|/(|k_1| + |k_2| + \varepsilon)$. It is unchanged when the normal is reversed and highlights anisotropy, but it does not distinguish maxima from minima along principal directions and therefore is not itself a signed ridge/valley definition. $\square$

Problem 42.54 (End-to-end validation design). Design an end-to-end regression suite for curvature-line and ridge extraction that separates estimator error from tracer error and from feature-selection error.

Solution. First use analytic surfaces to compare estimated k_i and principal lines against ground truth using sign-invariant angular error. Next feed exact or synthetic principal line fields to the tracer to isolate integration and face-crossing behavior. Then feed analytic curvature and derivative samples to zero-crossing/selection code to test ridge logic. Finally run the complete pipeline over meshes at multiple resolutions, noise levels, and scales, checking sphere umbilics, cylinder line families, known torus/Monge features, continuity, and persistence. $\square$

42.20 Volume VII synthesis

Volume VII began with topological and smooth manifolds, built tangent and cotangent geometry, developed bundles and differential forms, passed through the classical geometry of curves

and surfaces, and then constructed Riemannian, Lorentzian, and hyperbolic geometry. The final computational arc translated geodesic distance, heat flow, Laplace–Beltrami operators, and curvature features into algorithms on meshes.

The recurring theme is that local differential data and global structure must be kept distinct but compatible. Metrics produce lengths and geodesics; connections produce parallel transport and curvature; shape operators produce principal geometry; discrete operators approximate these objects only after choices of mesh, mass, scale, and numerical convention are made explicit.

42.21 Bridge to Volume VIII

Principal line fields become singular at umbilics, geodesics interact with nontrivial topology, and global geometry cannot be understood from local coordinates alone. Volume VIII begins with homotopies of maps and develops the algebraic-topological invariants that measure global deformation, loops, cells, and homology. The passage is therefore natural: differential geometry describes local geometric structure, while algebraic topology organizes the global constraints that local geometry must obey.